Advanced Communication Networks

Course Syllabus

شبکه‌های ارتباطی پیشرفته -- طرح درس

1 Course description

This course follows mobile communication networks from the second to the sixth generation. It explains the radio access, core network, signalling, mobility, security and quality-of-service mechanisms of GSM/GPRS, UMTS, LTE/EPC and 5G NR/5GC, and the software-driven technologies (SDN, NFV, service-based architecture) that transformed them. Each lecture combines architecture, procedures and quantitative models with worked examples and graded exercises. All material is available in English and Persian with identical structure, notation, figures and numbering.

2 Learning outcomes

On completing the course students will be able to:

- **LO1** **Describe** the architecture and functions of 2G, 3G, 4G and 5G networks and map functions across generations.
- **LO2** **Trace** the main procedures (attach/registration, authentication, session setup, handover, paging) in sequence diagrams.
- **LO3** **Apply** quantitative models – traffic and queueing, link budgets, scheduling, capacity, latency and signalling load – to dimension networks.
- **LO4** **Explain** SDN, NFV and the service-based architecture and their role in 5G slicing and QoS.
- **LO5** **Read** 3GPP and ITU specifications and **assess** emerging 6G technologies critically.

3 Prerequisites

Probability and basic queueing, signals and systems, digital communications (modulation, Shannon capacity) and computer networking (TCP/IP).

4 Course structure

The 26 lectures are grouped into five parts. Each lecture is a standalone document with fourteen sections: objectives, prerequisites, introduction, fundamentals, technical detail, mathematical models, architecture, procedures, worked examples, technology comparison, key takeaways, exercises at three levels, solutions and further reading.

Table 1. Course modules.

#	Lecture	Generation
1	GSM, GPRS and EDGE	2G
2	2G Circuit Switching	2G
3	2G/3G Core Networks	3G
4	Mobility and Handover	4G
5	4G LTE Architecture	4G
6	E-UTRAN: the LTE Radio Access Network	4G
7	LTE Scheduling	4G
8	Random Access Channel (RACH)	4G
9	Evolved Packet Core (EPC)	4G
10	Diameter Protocol	4G
11	Home Subscriber Server (HSS)	4G
12	Mobility Management Entity (MME)	4G
13	Serving and PDN Gateways (S-GW and P-GW)	4G
14	5G Networks: Introduction	5G
15	Software-Defined Networking (SDN)	5G
16	Network Functions Virtualisation (NFV)	5G
17	3GPP Standardisation	5G
18	5G Service-Based Architecture (SBA)	5G
19	5G Core Network Architecture	5G
20	5G System Procedures	5G
21	Network Slicing	5G
22	Quality of Service (QoS) in 5G	5G
23	The 5G User Plane	5G
24	The 5G Control Plane	5G
25	Network Evolution: a Grand Synthesis	5G
26	Future 6G Technologies	6G

- **Part I – 2G/3G foundations** (Lectures 1–4).
- **Part II – 4G access and core** (Lectures 5–13).
- **Part III – Software-driven architectures** (Lectures 15–18).
- **Part IV – 5G system and procedures** (Lectures 14, 19–22).
- **Part V – Planes and outlook** (Lectures 23–26).

5 Suggested schedule

For a 13-week semester, two lectures per week in numerical order; each lecture needs about 6–8 hours of study (3 hours of reading and 3–5 hours of exercises). Weeks 1–2 cover Part I, weeks 3–7 Part II and Lecture 14, weeks 8–9 Part III, weeks 10–11 the rest of Part IV, and weeks 12–13 Part V.

6 Suggested assessment

The following weighting is a suggestion that instructors may adapt: weekly exercise sets 30 %, a mid-term examination on Parts I–II 25 %, a project (for example, dimensioning or simulating a part of a 5G network) 15 %, and a final examination on the whole course 30 %.

7 Materials

The lecture notes, figures and bibliography of this repository; the 3GPP and ITU documents cited in each lecture; and the textbooks listed under further reading, in particular Dahlman et al. (NR), Sesia et al. (LTE) and Rommer et al. (5G core).

8 Language

Both editions are equivalent; students may study in either language. Technical abbreviations are kept in their standard Latin form in both.

01

Lecture 01

GSM, GPRS and EDGE

جی‌اس‌ام، جی‌پی‌آر‌اس و اج

2G — 3G — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Introductory to intermediate

PREREQUISITES
Decibels, basic digital modulation, TDMA/FDMA, the OSI model

In this lecture

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-01-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the GSM reference architecture, name every network element (MS, BTS, BSC, MSC, VLR, GMSC, HLR, AuC, EIR) and state the interface between each pair (Um, Abis, A, C, D, E, F).
- **LO2** **Derive** the GSM radio timing (bit period, burst, TDMA frame and 26-multiframe) from the 270.833 kbit/s gross rate and **compute** the 22.8 kbit/s gross rate of a full-rate traffic channel.
- **LO3** **Explain** the GSM challenge–response authentication with the triplet (RAND, SRES, K_c) and identify its security weaknesses.
- **LO4** **Describe** the GPRS overlay (PCU, SGSN, GGSN, Gb, Gn, Gi, Gr) and the PDP context, and **trace** a PDP context activation step by step.
- **LO5** **Calculate** per-user GPRS/EDGE throughput from the coding scheme and the number of timeslots, and **evaluate** the trade-off between robustness and rate.
- **LO6** **Relate** cluster size, path-loss exponent and co-channel interference through the first-tier C/I model.

2 Prerequisite knowledge

The lecture assumes the decibel arithmetic of the prerequisites review ($x_{\mathrm{dB}} = 10\log_{10} x$), the notion of a modulation symbol and of bandwidth, and the difference between frequency-division (FDMA) and time-division (TDMA) multiple access. If you are unsure about any of these, revise them before continuing; every quantitative result below is built from them.

Key terms. The table lists the vocabulary used throughout the lecture with its Persian equivalent, so that both editions can be read side by side.

Term	Persian	Meaning in this lecture
Mobile Station (MS)	ایستگاه موبایل	Handset = mobile equipment (ME) + SIM card.
Base Transceiver Station (BTS)	ایستگاه فرستنده‌گیرندهٔ پایه	Radio site: transmitters, receivers and antennas of one or more cells.
Base Station Controller (BSC)	کنترل‌گر ایستگاه پایه	Manages radio resources and handovers of many BTSs.
Mobile Switching Centre (MSC)	مرکز سوئیچ موبایل	Circuit switch for calls; also runs mobility signalling.
Home Location Register (HLR)	ثبات مکان خانگی	Permanent subscriber database of the home operator.
Visitor Location Register (VLR)	ثبات مکان مهمان	Temporary copy of data for subscribers in an MSC area.
Authentication Centre (AuC)	مرکز احراز هویت	Stores the secret key K_i and generates triplets.
Timeslot (TS)	شیار زمانی	One of the eight TDMA positions on a carrier.
Burst	بُرست	The radio waveform transmitted in one timeslot.
PDP context	زمینهٔ PDP	Packet session state: address, APN, QoS, tunnel IDs.
SGSN / GGSN	گره‌های پشتیبان GPRS	Serving and gateway packet nodes of the GPRS core.
Coding scheme (CS/MCS)	طرح کدگذاری	Combination of modulation and channel-code rate.

3 Introduction and motivation

In the 1980s Western Europe operated several incompatible analogue cellular systems. A handset bought in one country was useless across the border, every system needed its own equipment industry, and analogue FM voice offered no protection against eavesdropping. The engineering problem that GSM[1] set out to solve was therefore not only a radio problem; it was a *system* problem:

- **Interoperability.** Standardised, open interfaces would let an operator buy base stations from one vendor and switches from another.
- **Roaming.** A subscriber identity independent of the handset (the SIM) and a home database reachable from any visited network.
- **Spectral efficiency and capacity.** Digital speech coding, channel coding and TDMA let several users share one carrier and tolerate more interference, so frequencies can be reused more

[1] Originally *Groupe Spécial Mobile*, later *Global System for Mobile communications*.

tightly.

- **Security.** Authentication of the subscriber and encryption of the radio link.

GSM was designed for circuit-switched voice. When data demand grew in the 1990s, the same radio network was extended with a packet-switched overlay: *GPRS* (General Packet Radio Service), and later *EDGE* (Enhanced Data rates for GSM Evolution), which raised the bit rate per timeslot with a higher-order modulation. Studying these three steps in one lecture shows a pattern that recurs throughout the course: **each generation reuses what it can and adds the minimum new machinery needed for the new problem.**

4 Fundamental concepts

4.1 Cells, frequency reuse and multiple access

A mobile network divides its service area into *cells*, each served by one set of antennas. Because spectrum is scarce, the same carrier frequencies are *reused* in cells that are far enough apart. GSM combines two multiple-access methods:

- **FDMA:** the band is divided into carriers spaced 200 kHz apart. In the primary GSM-900 band, the uplink occupies 890 MHz–915 MHz and the downlink 935 MHz–960 MHz, giving 124 usable carrier pairs with a duplex spacing of 45 MHz [2].
- **TDMA:** each carrier is divided in time into frames of eight timeslots; a full-rate voice user occupies one timeslot in every frame.

Definition 1 • Physical and logical channel

A *physical channel* is a timeslot number on a carrier (or a hopping sequence of carriers). A *logical channel* is a type of information — traffic (TCH) or control (BCCH, CCCH, SDCCH, SACCH, FACCH) — that is mapped onto a physical channel according to a fixed multiframe structure.

4.2 Separating identity from equipment

GSM separates the *subscriber* from the *device*. The subscriber is identified by the IMSI (International Mobile Subscriber Identity: mobile country code, mobile network code and a subscriber number, at most 15 digits) stored on the SIM together with a 128-bit secret key K_i. The device is identified by its IMEI. To avoid sending the IMSI over the air, the VLR assigns a temporary identity, the TMSI, which is used instead whenever possible [6].

4.3 Circuit switching versus packet switching

Voice in GSM is *circuit switched*: when a call is set up, a timeslot on the radio, a channel on the Abis and A links and a path through the MSC are reserved for the whole call, whether or not anybody is speaking. Data in GPRS is *packet switched*: radio blocks are assigned only when the user has data to send, and many users share the same timeslots. This difference is the reason GPRS needs new nodes rather than just new software in the MSC, and it is analysed in depth in Lecture 2.

5 Detailed technical explanation

5.1 The GSM network elements

Mobile Station (MS). The MS is the mobile equipment plus the SIM. The SIM runs the authentication algorithm A3 and the key-generation algorithm A8 so that K_i never leaves the card.

Base Transceiver Station (BTS). The BTS contains the radio transceivers (TRX) of one or more cells. It performs modulation (GMSK), channel coding, interleaving, encryption with A5 and the radio measurements that feed handover decisions. It is intentionally simple: decisions are taken by the BSC.

Base Station Controller (BSC). One BSC controls tens to hundreds of BTSs. It allocates radio channels, manages power control and decides most handovers between its own cells. Together the BTSs and their BSC form the *Base Station Subsystem* (BSS). Speech transcoding between the 13 kbit/s radio codec and the 64 kbit/s PCM format of the core is performed in a transcoder unit (TRAU), usually located at the MSC site to save transmission capacity.

Mobile Switching Centre (MSC) and VLR. The MSC is a telephone exchange extended with mobility functions: it sets up and releases calls, controls inter-BSC handovers, and runs location updating and authentication. The VLR (in practice integrated in the MSC) caches the subscription data and security triplets of every subscriber currently in its area.

Gateway MSC (GMSC). Calls arriving from the fixed network (PSTN/ISDN) are routed to a GMSC, which interrogates the HLR to find where the called subscriber is.

HLR, AuC and EIR. The HLR stores the permanent subscription profile and the address of the VLR currently serving the subscriber. The AuC holds K_i and computes authentication triplets. The EIR holds lists of white, grey and black-listed IMEIs.

5.2 Interfaces

Table 1. Main GSM interfaces and their protocols [5].

Interface	Between	Main protocol	Role
Um	MS – BTS	LAPDm, RR, MM, CC	Radio interface, TDMA/FDMA, GMSK
Abis	BTS – BSC	LAPD over E1/T1	Transport of traffic and BTS control
A	BSC – MSC	BSSAP over SS7	Call and mobility signalling, 64 kbit/s circuits
B	MSC – VLR	MAP	Internal in practice (MSC/VLR combined)
C	GMSC – HLR	MAP	Routing interrogation for incoming calls
D	VLR – HLR	MAP	Location updating, subscriber data
E	MSC – MSC	MAP + ISUP	Inter-MSC handover, call routing
F	MSC – EIR	MAP	IMEI check

The value of an open interface is economic as well as technical: because the A interface is fully standardised, an operator can connect any vendor's BSS to any vendor's MSC. The Abis interface was standardised too, but in practice it often contains vendor-specific operation and maintenance

extensions.

5.3 The radio interface Um

GSM uses Gaussian Minimum Shift Keying (GMSK) with a bandwidth–time product of $BT = 0.3$. GMSK is a constant-envelope modulation, so the handset power amplifier can run near saturation, which saves battery. One bit is carried per symbol, at a gross rate of $13/48\,\text{MHz} \approx 270.833\,\text{kbit/s}$. The time structure derived from this rate is drawn in Figure 4 and computed in Section 6.

A normal burst carries two blocks of 57 data bits, a 26-bit training sequence in the middle (the receiver uses it to estimate the multipath channel for its equaliser), two stealing flags, tail bits and a guard period of 8.25 bit durations that absorbs timing uncertainty and power ramping. Frequency hopping (up to 217 hops per second, one hop per TDMA frame) averages fading and interference across carriers.

5.4 Mobility: location areas and the TMSI

Cells are grouped into *location areas* (LAs). An idle MS listens to the broadcast channel and performs a *location update* only when it enters a new LA, or periodically. When a call arrives, the network pages the MS in all cells of its current LA. The size of an LA is a trade-off between update signalling (small LAs) and paging load (large LAs); this trade-off is quantified in Lectures 4 and 12.

5.5 Security: authentication and ciphering

GSM authentication is a *challenge–response* protocol [7]. The AuC draws a 128-bit random number RAND and computes

$$\text{SRES} = \text{A3}(K_i, \text{RAND}), \qquad (32\ \text{bits}) \tag{1a}$$

$$K_c = \text{A8}(K_i, \text{RAND}), \qquad (64\ \text{bits}) \tag{1b}$$

and sends the triplet $(\text{RAND}, \text{SRES}, K_c)$ to the VLR. The VLR sends only RAND to the MS; the SIM repeats the computation of Eq. (1a) and returns its result. If it matches SRES, the subscriber is authentic and both sides now share K_c, which keys the A5 stream cipher on the radio link. Note what this design achieves and what it does not:

- K_i never travels over any interface, and RAND is useless without K_i.
- Authentication is *one-way*: the network is not authenticated to the MS, so a false base station can impersonate the network.
- Encryption ends at the BTS; Abis, A and the core links are not protected by A5.
- K_c has only 64 bits, and the A5/1 and A5/2 ciphers have published cryptanalytic attacks; A5/2 has been withdrawn and A5/3 (based on KASUMI) was added later.

UMTS (Lecture 3) and LTE (Lectures 9 and 11) fix these weaknesses with mutual authentication, integrity protection and longer keys.

5.6 GPRS: a packet overlay

GPRS adds packet switching while reusing the GSM radio carriers, sites and subscriber databases [1]. Three new functions are needed:

- **Packet Control Unit (PCU)** in the BSS, which implements the packet-oriented RLC/MAC layer and schedules radio blocks among users.
- **Serving GPRS Support Node (SGSN)**, the packet counterpart of the MSC: mobility management (routing areas), authentication, ciphering (in GPRS, ciphering runs between MS and SGSN at the LLC layer) and charging.
- **Gateway GPRS Support Node (GGSN)**, the border router to external packet data networks: it allocates the user IP address and anchors the session.

The SGSN connects to the BSS over *Gb* (originally Frame Relay), to the GGSN over *Gn* using the GPRS Tunnelling Protocol (GTP), and to the HLR over *Gr* using MAP. The GGSN connects to the Internet or a corporate network over *Gi*. The idea of a *tunnel* between a serving node and a gateway node is the single most durable concept introduced by GPRS: it survives as the S5/S8 interface of LTE and the N9 interface of 5G.

On the radio side, GPRS defines packet data channels (PDCH). A PDCH is an ordinary timeslot dedicated to packet traffic; data is sent in *radio blocks* of four consecutive bursts (one block every 20 ms on average). Four coding schemes trade protection for rate: CS-1 (9.05 kbit/s), CS-2 (13.4), CS-3 (15.6) and CS-4 (21.4 kbit/s) per timeslot. A multislot-capable MS can use several timeslots at once.

5.7 GPRS mobility management states

A GPRS MS is in one of three mobility-management states: **IDLE** (not attached; the network does not know the MS), **STANDBY** (attached; location known per routing area; must be paged before data can be delivered) and **READY** (location known per cell; data can be sent immediately). A timer moves the MS from READY back to STANDBY after a period of inactivity. The same idea — keep the device reachable cheaply while idle, and spend signalling only when traffic flows — reappears as ECM-IDLE/CONNECTED in LTE and CM-IDLE/CONNECTED in 5G.

5.8 EDGE: higher-order modulation on the same carrier

EDGE keeps the GSM symbol rate (270.833 ksymbol/s), the 200 kHz carrier and the TDMA structure, but introduces 8-PSK, which carries three bits per symbol instead of one. Enhanced GPRS (EGPRS) defines nine modulation and coding schemes: MCS-1 to MCS-4 use GMSK and MCS-5 to MCS-9 use 8-PSK, from 8.8 kbit/s up to 59.2 kbit/s per timeslot. Two mechanisms make the higher rates usable:

- **Link adaptation:** the network selects the MCS from channel-quality measurements, moving to a robust scheme when the user is at the cell edge.
- **Incremental redundancy:** if a block is received in error, the retransmission carries different puncturing of the same coded bits, and the receiver combines both copies. This is an early form of the hybrid ARQ used in HSPA, LTE and NR.

6 Mathematical models

6.1 GSM radio timing

Engineering problem. Every timing parameter of GSM — the burst length, the frame duration, the maximum cell radius — follows from one number, the gross bit rate. We derive them to see where the familiar values come from.

System model and assumptions. One carrier, GMSK with one bit per symbol, gross rate $R = f_{\text{ref}}/48$ with reference clock $f_{\text{ref}} = 13\,\text{MHz}$. A timeslot lasts 156.25 bit periods and a TDMA frame contains eight timeslots.

Derivation. The bit period and the derived durations are

$$T_{\text{b}} = \frac{1}{R} = \frac{48}{13 \times 10^6}\ \text{s} \approx 3.692\,\mu\text{s}, \tag{2}$$

$$T_{\text{slot}} = 156.25\,T_{\text{b}} = \frac{15}{26}\ \text{ms} \approx 576.9\,\mu\text{s}, \tag{3}$$

$$T_{\text{frame}} = 8\,T_{\text{slot}} = \frac{60}{13}\ \text{ms} \approx 4.615\,\text{ms}, \tag{4}$$

$$T_{26} = 26\,T_{\text{frame}} = 120\,\text{ms}. \tag{5}$$

Gross rate of a full-rate traffic channel. In each 26-multiframe a full-rate TCH uses 24 frames for traffic, one for the SACCH and leaves one idle. Each traffic burst carries $2 \times 57 = 114$ coded bits, so

$$R_{\text{TCH/F}} = \frac{114\ \text{bit} \times 24}{T_{26}} = \frac{2736\ \text{bit}}{0.120\ \text{s}} = 22.8\,\text{kbit/s}. \tag{6}$$

The 13 kbit/s speech codec output is protected by channel coding that expands it to exactly this 22.8 kbit/s.

Physical interpretation. The ratio $R_{\text{TCH/F}}/R = 22.8/270.833 \approx 8.4\%$ is close to $1/8$ minus the share of the frame spent on training sequence, tail bits, guard time and signalling. GSM therefore spends roughly a quarter of its gross capacity on overhead — a figure to remember when comparing it with OFDM systems in Lectures 6 and 14.

Spectral efficiency. With $B = 200\,\text{kHz}$ the gross spectral efficiency is $\eta = R/B \approx 1.35\,\text{bit/s/Hz}$. The Shannon bound [10]

$$C = B \log_2(1 + \text{SNR}) \tag{7}$$

with C the capacity in bit/s, B the bandwidth in Hz and SNR the linear signal-to-noise ratio, shows that 1.35 bit/s/Hz could in principle be carried at $\text{SNR} = 2^{1.354} - 1 \approx 1.56$ ($\approx 1.9\,\text{dB}$). GSM operates at a far higher SNR (around 9 dB carrier-to-interference) because it uses simple, robust modulation and coding rather than capacity-approaching codes. Equation (7) assumes an ideal band-limited additive white Gaussian noise (AWGN) channel and unlimited coding complexity; it is an upper bound, not a prediction of throughput.

6.2 Timing advance and the maximum cell radius

Problem. The BTS expects each burst to arrive inside its timeslot. A distant MS sees the downlink late and its uplink arrives late again, so it must transmit early by the round-trip delay.

Model. Line-of-sight propagation at speed c; the timing advance (TA) is signalled in steps of one bit period, $\text{TA} \in \{0, 1, \ldots, 63\}$.

Derivation. A round-trip delay of $2d/c$ must be compensated by $\mathrm{TA} \cdot T_\mathrm{b}$, so one TA step corresponds to a distance

$$\Delta d = \frac{c\,T_\mathrm{b}}{2} = \frac{2.998 \times 10^8\,\mathrm{m/s} \times 3.692\,\mu\mathrm{s}}{2} \approx 553\,\mathrm{m}, \qquad d_\mathrm{max} = 63\,\Delta d \approx 34.9\,\mathrm{km}. \tag{8}$$

Interpretation and limits. The familiar "35 km" limit of a GSM cell is a consequence of the 6-bit TA field, not of the link budget. Extended-range cells overcome it by using two adjacent timeslots per user, at the cost of capacity. The same reasoning, with a finer step, gives the LTE timing advance in Lecture 8.

6.3 Co-channel interference and the cluster size

Problem. How tightly can carriers be reused? Smaller reuse distance means more carriers per cell but stronger interference.

Model and assumptions. Hexagonal cells of radius R_c, clusters of $N = i^2 + ij + j^2$ cells (i, j non-negative integers), power-law path loss $P_\mathrm{rx} \propto d^{-n}$ with exponent n, a user at the cell edge, and only the six first-tier co-channel cells at reuse distance $D = R_\mathrm{c}\sqrt{3N}$ [9].

Derivation. The wanted signal is $C \propto R_\mathrm{c}^{-n}$ and each of the six interferers contributes approximately D^{-n}, hence

$$\frac{C}{I} \approx \frac{R_\mathrm{c}^{-n}}{6\,D^{-n}} = \frac{1}{6}\left(\frac{D}{R_\mathrm{c}}\right)^{n} = \frac{\left(\sqrt{3N}\right)^{n}}{6}. \tag{9}$$

Interpretation. Equation (9) grows with both N and n: a large cluster puts interferers further away, and a steep path loss attenuates them faster. Since GSM tolerates a C/I of roughly 9 dB, Figure 1 suggests that $N = 4$ (13.8 dB for $n = 4$) is comfortable, and $N = 3$ is feasible for $n = 4$ but marginal for $n = 3.5$. In practice operators combined tight reuse with frequency hopping and power control, which average and reduce interference.

Limitations. The model ignores shadow fading (which varies C/I by several dB), sectorisation (which removes some interferers), and the fact that not all interferers are active at once. It is a planning first estimate, not a guarantee.

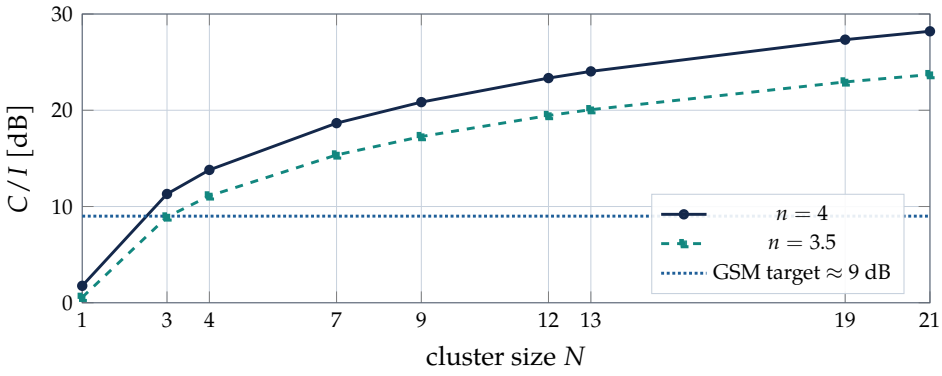


Figure 1. First-tier C/I from Eq. (9) versus cluster size for two path-loss exponents. The horizontal line marks the approximate GSM protection ratio.

6.4 Packet throughput with multiple timeslots

For a user allocated n_{TS} timeslots with per-timeslot rate r_{CS}, the peak radio-link-control (RLC) data rate is

$$R_{\mathrm{user}} = n_{\mathrm{TS}}\, r_{\mathrm{CS}}\, (1 - p_{\mathrm{BLER}}), \tag{10}$$

where p_{BLER} is the block error ratio and the factor $(1 - p_{\mathrm{BLER}})$ assumes every erroneous block is retransmitted once successfully (a simple ARQ model). Higher coding schemes increase r_{CS} but, at a given C/I, also increase p_{BLER}; the best scheme maximises the product. Figure 2 shows the peak values.

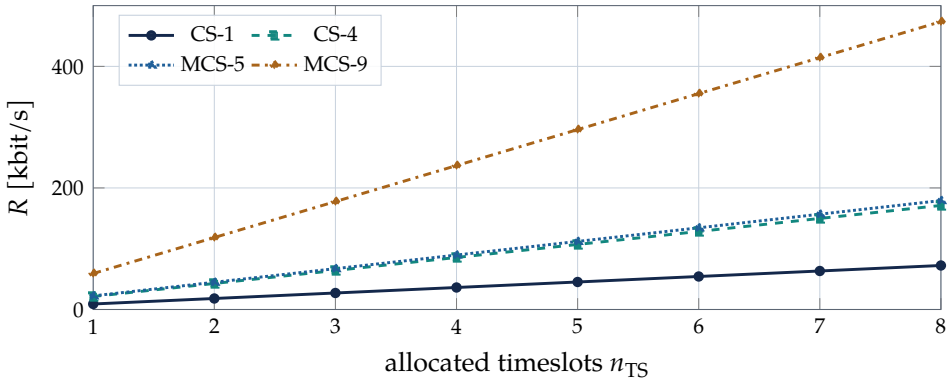


Figure 2. Peak per-user RLC data rate versus number of timeslots ($p_{\mathrm{BLER}} = 0$) for two GPRS and two EGPRS schemes. Real throughput is lower because of retransmissions, sharing among users and higher-layer overhead.

7 Network architecture and diagrams

Figure 3 combines the circuit-switched GSM network and the GPRS packet overlay. Two observations are worth making. First, the BSS is shared: the same BTS carries voice timeslots towards the MSC and packet timeslots towards the SGSN, and the BSC splits them. Second, the HLR is shared: a single subscription record serves both domains, which is why GPRS could be launched without re-provisioning subscribers.

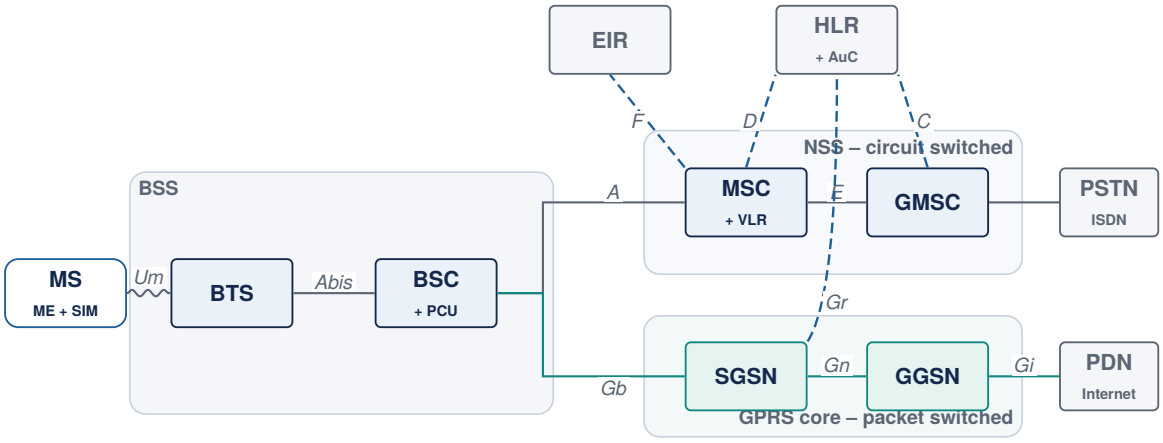


Figure 3. GSM network with the GPRS overlay. Solid grey: transport links; teal: packet-switched links; dashed blue: MAP signalling towards the HLR and EIR.

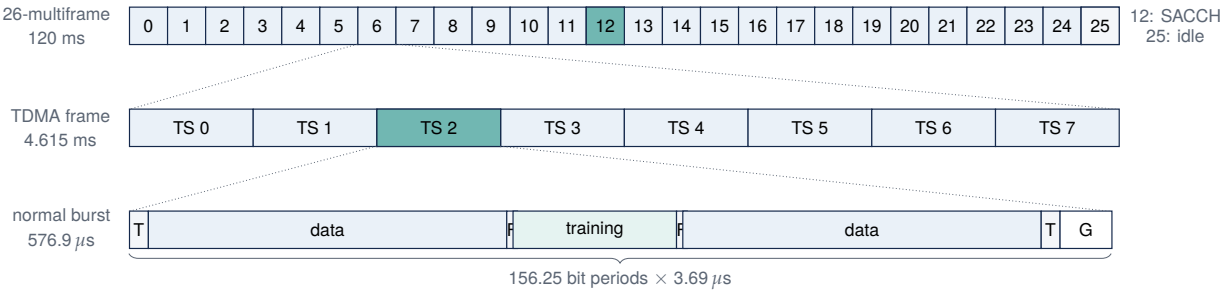


Figure 4. GSM time structure. A 26-multiframe (120 ms) contains 24 traffic frames, one SACCH frame and one idle frame; each TDMA frame contains eight timeslots; a normal burst carries 114 data bits.

8 Step-by-step procedures

Procedure 1 • Authentication and start of ciphering

1. The MS sends a Location Updating Request (or call set-up request) identifying itself with its TMSI, or with its IMSI if no TMSI is valid.
2. If the VLR has no unused triplet, it requests a batch from the HLR/AuC with MAP Send Authentication Info; the AuC returns several triplets.
3. The MSC/VLR sends Authentication Request containing only RAND.
4. The SIM computes SRES′ and K_c from Eq. (1) and the MS returns SRES′.
5. The VLR compares SRES′ with SRES. On a match it orders ciphering by sending K_c to the BSS in a BSSMAP Cipher Mode Command.
6. The BTS sends Ciphering Mode Command; the MS starts A5 with K_c and answers Ciphering Mode Complete. From now on Um is encrypted.
7. The VLR may allocate a new TMSI, sent encrypted, so the IMSI is protected next time.

Procedure 2 • GPRS attach and PDP context activation

1. **GPRS attach.** The MS sends Attach Request to the SGSN. The SGSN authenticates the MS (triplets from the HLR over Gr), registers itself in the HLR with Update GPRS Location, receives the subscription with Insert Subscriber Data and returns Attach Accept with a P-TMSI. The MS is now in READY state but has no IP address.
2. **Session request.** The MS sends Activate PDP Context Request with an NSAPI (identifier of the context), the requested access point name (APN) and QoS.
3. **Gateway selection.** The SGSN checks the subscription and resolves the APN with DNS to the address of a GGSN.
4. **Tunnel set-up.** The SGSN sends GTP-C Create PDP Context Request; the GGSN allocates an IP address (or obtains one from a RADIUS/DHCP server) and returns its tunnel end-point identifier (TEID).
5. **Completion.** The SGSN sends Activate PDP Context Accept with the IP address. User packets now flow: MS–SGSN over LLC/SNDCP, SGSN–GGSN inside a GTP-U tunnel

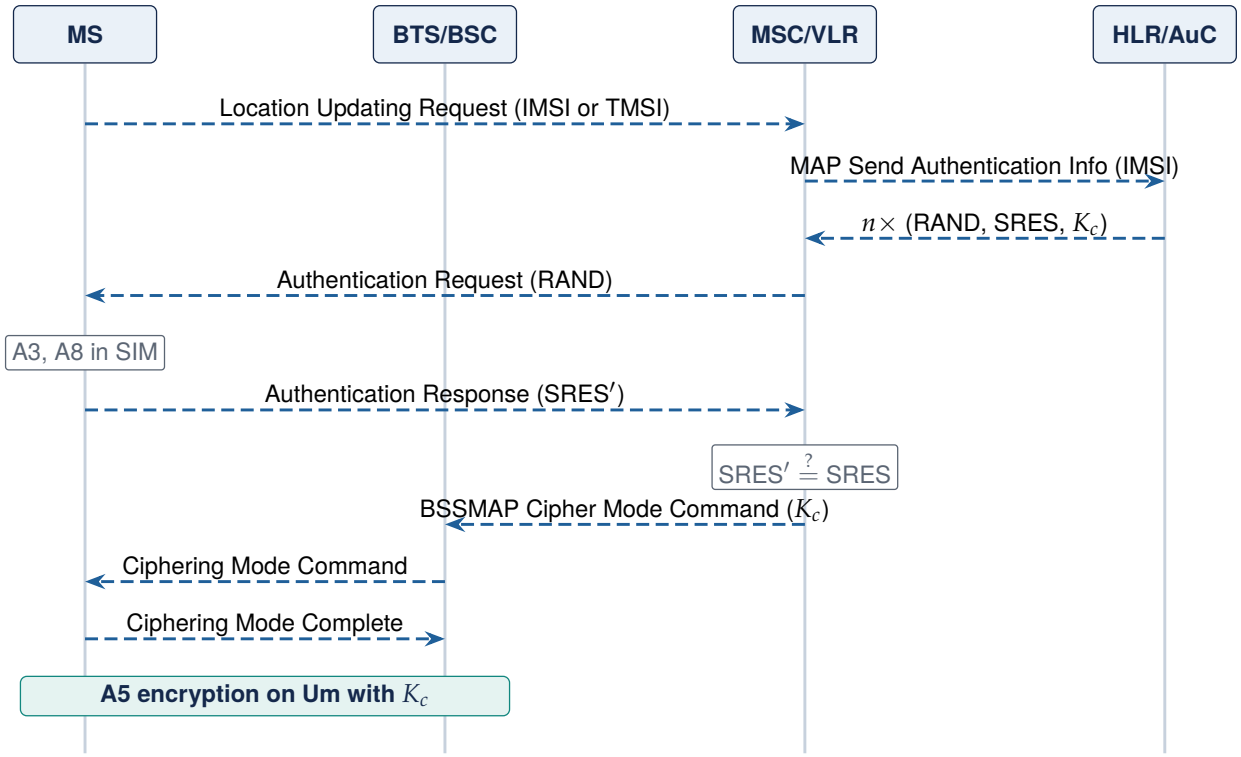


Figure 5. Message sequence of GSM authentication and ciphering (Procedure 1).

> identified by TEIDs.

The procedure illustrates a principle that the course will meet again in LTE (the default bearer, Lecture 9) and 5G (the PDU session, Lecture 20): *being attached* (known to the network) and *having connectivity* (an IP address and a tunnel) are separate states, established by separate procedures. LTE later merged them for the first PDN connection to reduce delay.

9 Worked examples

Worked example 1 • Capacity of a GSM-900 operator

Problem. An operator owns 10 MHz of paired GSM-900 spectrum and plans with cluster size $N = 4$. How many full-rate voice channels does a cell offer?

Solution. Number of carriers: $\lfloor 10\,\text{MHz}/200\,\text{kHz} \rfloor = 50$; one is kept as a guard, so 49 are usable. Per cell: $\lfloor 49/4 \rfloor = 12$ carriers, i.e. $12 \times 8 = 96$ timeslots. Reserving one timeslot for the broadcast/common control channel and one for SDCCH signalling leaves about 94 traffic channels.

Interpretation. 94 simultaneous calls per cell is the hard limit of circuit switching. How much *traffic* (in erlangs) these channels carry at a given blocking probability is computed with the Erlang-B formula in Lecture 2.

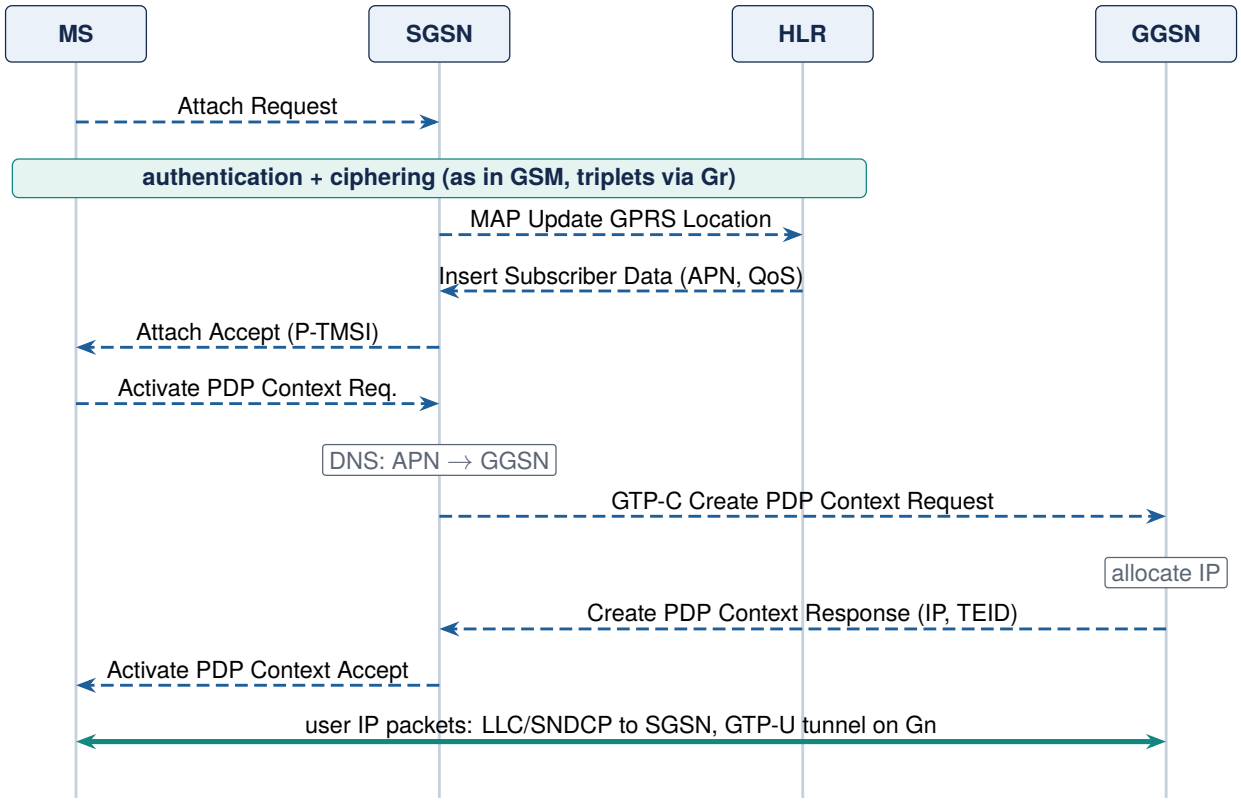


Figure 6. GPRS attach followed by PDP context activation (Procedure 2). Dashed blue: signalling; solid teal: user plane.

Worked example 2 • Timing advance

Problem. The BTS measures that an MS's bursts arrive 55 µs late. Which TA value is signalled and how far is the MS?

Solution. The round trip lasts 55 µs, so TA $= \text{round}(55/3.692) = 15$ and $d \approx 15 \times 553\,\text{m} \approx 8.3\,\text{km}$. (Directly: $d = c\,\tau/2 = 3 \times 10^8 \times 55 \times 10^{-6}/2 = 8.25\,\text{km}$.)

Interpretation. The TA quantisation error is up to half a step, ≈ 277 m; TA therefore also gives the network a coarse estimate of the user's distance, the basis of early cell-ID+TA positioning.

Worked example 3 • Choosing a GPRS coding scheme

Problem. At a given C/I a user observes $p_{\text{BLER}} = 0.05$ with CS-2 and $p_{\text{BLER}} = 0.45$ with CS-4. With four timeslots, which scheme gives the higher throughput according to Eq. (10)?

Solution.

$$R_{\text{CS-2}} = 4 \times 13.4 \times (1 - 0.05) = 50.9\,\text{kbit/s},$$
$$R_{\text{CS-4}} = 4 \times 21.4 \times (1 - 0.45) = 47.1\,\text{kbit/s}.$$

Interpretation. CS-2 wins despite its lower nominal rate. Link adaptation is precisely the selection of the scheme that maximises this product; the same logic reappears with CQI and MCS tables in LTE (Lecture 7).

10 Technology comparison

Table 2. GSM, GPRS and EDGE compared. Rates are peak RLC rates per timeslot or per user with eight timeslots; achievable user rates are lower.

	GSM (circuit data)	GPRS	EDGE (EGPRS)
Switching	Circuit	Packet	Packet
Modulation	GMSK	GMSK	GMSK and 8-PSK
Bits per symbol	1	1	1 or 3
Rate per timeslot	9.6 or 14.4 kbit/s	9.05–21.4 kbit/s	8.8–59.2 kbit/s
Peak (8 TS)	–	171.2 kbit/s	473.6 kbit/s
Resource use	Reserved for the call	Shared, on demand	Shared, on demand
Retransmission	None (FEC only)	ARQ in RLC	ARQ + incremental redundancy
Core nodes	MSC, GMSC	SGSN, GGSN	SGSN, GGSN (unchanged)
Charging basis	Time	Volume	Volume

Trade-offs. GPRS made data cheap (volume-based charging, no idle circuit) but introduced queuing delay and variable throughput; round-trip times of several hundred milliseconds were typical. EDGE tripled the peak rate with a software and transceiver upgrade, but 8-PSK is not constant-envelope, so it needs more linear power amplifiers, and its high schemes are only usable close to the BTS. Neither changed the fundamental limit: data rides on 200 kHz carriers designed for voice. Breaking that limit required a new radio interface, WCDMA in 3G (Lecture 3).

11 Key takeaways

Key takeaways

- GSM is a *system* standard: open interfaces (A, Um) and a separable identity (SIM/IMSI) enabled multi-vendor networks and roaming.
- All GSM timing follows from $R = 270.833\,\text{kbit/s}$: bit 3.69 µs, slot 577 µs, frame 4.615 ms, 26-multiframe 120 ms; a full-rate TCH carries 22.8 kbit/s gross.
- The 6-bit timing advance limits a normal cell to about 35 km.
- Authentication uses triplets (RAND, SRES, K_c); it is one-way, and encryption stops at the BTS.
- GPRS adds PCU, SGSN and GGSN; the PDP context and the GTP tunnel are the ancestors of the LTE bearer and the 5G PDU session.
- EDGE raises the bit rate per symbol (8-PSK) and adds incremental redundancy, an early hybrid ARQ.
- The first-tier model $C/I \approx (\sqrt{3N})^n/6$ links reuse, path loss and interference; it is a first planning estimate.

12 Exercises

Exercise 1 • Level 1 – conceptual • Architecture

Explain why the HLR stores the address of the VLR but not the cell in which the subscriber is located. What would be the signalling consequence of storing the cell?

Exercise 2 • Level 1 – conceptual • Security

List two attacks made possible by the one-way authentication of GSM and say which later mechanism (Lecture 11) prevents each.

Exercise 3 • Level 2 – numerical • Half-rate channel

A half-rate traffic channel uses a timeslot in every second traffic frame. Using Eq. (6), compute its gross bit rate, and explain how it doubles the voice capacity of the cell in Example 1.

Exercise 4 • Level 2 – numerical • Reuse planning

Using Eq. (9), find the smallest valid cluster size N that guarantees $C/I \geq 12\,\text{dB}$ for (a) $n = 4$ and (b) $n = 3$.

Exercise 5 • Level 2 – numerical • EDGE throughput

A user has five timeslots. With MCS-9 the block error ratio is 0.30; with MCS-7 (44.8 kbit/s per timeslot) it is 0.08. Which scheme should link adaptation choose, and what is the expected throughput?

Exercise 6 • Level 3 – analytical • Overhead budget

Starting from Eqs. (2) and (4), compute what fraction of the gross bit rate of one timeslot is used by (i) guard period, (ii) tail bits and stealing flags, (iii) training sequence, (iv) SACCH and idle frames. Compare the total with the cyclic-prefix overhead of LTE ($\approx 7\%$) and discuss why the comparison is not entirely fair.

Exercise 7 • Level 3 – analytical • Generalising the C/I model

Extend Eq. (9) to a cell divided into three 120° sectors, where only two first-tier interferers fall into the main lobe of the sector antenna. Derive the new expression and the gain in dB, and discuss which assumptions of the original model the result inherits.

Solutions

Solution to Exercise 1

The cell changes every few minutes for a moving user; if the HLR tracked it, every cell change of every subscriber in the world would generate signalling to the home network, often internationally. Storing only the VLR address limits HLR updates to changes of MSC area, and the location area kept in the VLR limits radio paging to a group of cells. This two-level hierarchy

(HLR → VLR → LA) is the core of GSM mobility management.

Solution to Exercise 2

(1) A false base station can request the IMSI in clear and track the user (IMSI catching); 5G conceals the permanent identity with SUCI, and LTE/UMTS reduce the exposure. (2) A false base station can accept the MS without authentication and order no ciphering or a weak cipher, then relay traffic (man-in-the-middle). UMTS/LTE mutual authentication with the AUTN token (Lecture 11) lets the USIM verify that the challenge was produced by its home network, and integrity protection of signalling prevents tampering with the security-mode command.

Solution to Exercise 3

A half-rate channel gets 12 of the 24 traffic frames per 26-multiframe, so $R_{\mathrm{TCH/H}} = 114 \times 12/0.120 = 11.4\,\mathrm{kbit/s}$. Two half-rate users share one timeslot, so the 94 timeslots of Example 1 can carry up to 188 calls, at the price of a lower-rate (lower-quality) speech codec.

Solution to Exercise 4

We need $(\sqrt{3N})^n/6 \geq 10^{1.2} = 15.85$, i.e. $(3N)^{n/2} \geq 95.1$. (a) $n = 4$: $(3N)^2 \geq 95.1 \Rightarrow N \geq 3.25$; the smallest valid N is 4 ($C/I = 13.8\,\mathrm{dB}$). (b) $n = 3$: $(3N)^{1.5} \geq 95.1 \Rightarrow 3N \geq 20.8 \Rightarrow N \geq 6.94$; the smallest valid N is 7 ($C/I = 10\log_{10}(21^{1.5}/6) = 12.05\,\mathrm{dB}$). Valid sizes are $N = i^2 + ij + j^2 \in \{1, 3, 4, 7, 9, 12, \dots\}$.

Solution to Exercise 5

$R_{\text{MCS-9}} = 5 \times 59.2 \times 0.70 = 207.2\,\mathrm{kbit/s}$ and $R_{\text{MCS-7}} = 5 \times 44.8 \times 0.92 = 206.1\,\mathrm{kbit/s}$. The two are practically equal; MCS-9 is marginally better on throughput, but MCS-7 gives fewer retransmissions and therefore lower delay and jitter. With incremental redundancy the effective MCS-9 throughput would be higher than the simple model predicts, because failed blocks still contribute soft information.

Solution to Exercise 6

Per 156.25-bit slot: guard $8.25/156.25 = 5.3\%$; tail and flags $(6+2)/156.25 = 5.1\%$; training $26/156.25 = 16.6\%$; data $114/156.25 = 73.0\%$. Of the 26 frames, 2 do not carry traffic, so the useful fraction is $0.730 \times 24/26 = 67.4\%$: $270.833/8 \times 0.674 = 22.8\,\mathrm{kbit/s}$, consistent with Eq. (6). Total overhead $\approx 33\%$. LTE's cyclic prefix costs about 7%, but LTE also spends resources on reference signals, control region and guard bands; a fair comparison must count all of them (Lecture 7).

Solution to Exercise 7

With two interferers, $C/I \approx (\sqrt{3N})^n/2$, which is three times (4.8 dB) better than Eq. (9). For $N = 4$, $n = 4$: $10\log_{10}(144/2) = 18.6\,\mathrm{dB}$. The result still assumes ideal hexagons, equal powers, no shadowing and a worst-case user at the cell edge; in addition it assumes ideal antenna patterns with no back-lobe leakage. Sectorisation trades trunking efficiency (each sector has fewer channels) for interference reduction.

13 Further reading

The architecture of all 3GPP systems is summarised in [5]; the GSM radio structure is specified in [2] and security functions in [7]. GPRS stage-2 procedures are in [1] and layer-3 messages in [4]. The textbook by Eberspächer et al. [8] remains the most complete single description of GSM and GPRS, and Rappaport [9] derives the cellular reuse model used in Section 6.3. MAP is specified in [3]; identities in [6].

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02

Lecture 02

2G Circuit Switching

سوئیچ مداری در نسل دوم

2G — 3G — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lecture 1 (GSM architecture); basic probability (Poisson process, exponential distribution)

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
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10 Technology comparison . . . 11
11 Key takeaways . . . 12
12 Exercises . . . 12
13 Further reading . . . 14

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-02-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** what a circuit is, which resources a GSM call reserves and why reservation guarantees quality but wastes capacity.
- **LO2** **Describe** the SS7 protocol stack (MTP, SCCP, TCAP, MAP, ISUP, BSSAP) and the role of signalling points and signal transfer points.
- **LO3** **Trace** a mobile-originated and a mobile-terminated call, including HLR interrogation and the use of the MSRN.
- **LO4** **Derive** the Erlang-B formula from the $M/M/m/m$ model and **dimension** a cell or trunk group for a target blocking probability.
- **LO5** **Quantify** trunking efficiency and the cost of splitting a channel pool.
- **LO6** **Analyse** roaming signalling and call routing, and **evaluate** why circuit switching became a limitation for data.

2 Prerequisite knowledge

You need the GSM architecture and identities of Lecture 1 (MSC, VLR, HLR, GMSC, IMSI, TMSI) and elementary probability: a Poisson arrival process with rate λ, and an exponentially distributed holding time with mean h. We use the word *traffic* in the telephony sense: the average number of simultaneously busy circuits.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Circuit switching	سوئیچ مداری	A dedicated path is reserved end to end for the whole call.
Signalling System No. 7 (SS7)	سامانهٔ سیگنالینگ شمارهٔ ۷	Out-of-band signalling network between exchanges.
ISUP	بخش کاربر ISDN	SS7 protocol for setting up and releasing trunk circuits.
MAP	بخش کاربرد موبایل	SS7 application protocol for mobility (HLR, VLR, MSC).
MSISDN	شمارهٔ تلفن مشترک	The dialled telephone number of a subscriber.
MSRN	شمارهٔ رومینگ ایستگاه موبایل	Temporary number used to route a call to the serving MSC.
Erlang (E)	ارلانگ	Unit of traffic: one circuit busy continuously.
Blocking probability	احتمال انسداد	Probability that a call attempt finds all circuits busy.
Grade of service	درجهٔ خدمت	Target blocking probability used for dimensioning.
Trunking efficiency	بازده ترانکینگ	Carried traffic per circuit; grows with pool size.
Roaming	رومینگ	Use of a visited network with a home subscription.

3 Introduction and motivation

Telephony was built on one simple promise: once the called party answers, the voice path behaves like a pair of wires, with constant delay, no loss and no competition from other users. The engineering way to keep that promise is to *reserve* resources for the whole call. GSM inherited this model from ISDN, because its first job was to be a mobile extension of the fixed telephone network and because 1990s exchanges, trunks and billing systems were all built around 64 kbit/s circuits.

Two problems had to be solved on top of the classic telephone network. First, *signalling* had to carry much more than digits: authentication, location updates, handover, roaming between countries. Second, the network had to *find* a subscriber whose location changes, before any circuit can be set up. GSM solved both with the Signalling System No. 7 (SS7) and its Mobile Application Part (MAP). This lecture follows a call from dial to hang-up, then asks the quantitative question every operator faces: *how many circuits are enough?*

4 Fundamental concepts

4.1 What a circuit reserves

For a GSM voice call the network reserves, for its entire duration:

- one traffic channel (TCH) on the radio interface (a timeslot on a carrier);
- one 16 kbit/s sub-channel on the Abis link and one 64 kbit/s timeslot on the A interface after transcoding;
- a connection through the switching matrix of the MSC;
- a 64 kbit/s trunk on each inter-exchange link (one ISUP circuit per link).

The reservation is held even during silence. In a typical conversation each direction is active only about 40% of the time, so most of the reserved capacity carries nothing.

Definition 1 • Traffic intensity

If calls arrive at rate λ (calls per second) and each lasts on average h seconds, the offered traffic is $A = \lambda h$, a dimensionless number measured in erlangs. One erlang is the traffic that keeps one circuit busy all the time.

4.2 In-band versus common-channel signalling

Early telephone networks signalled with tones on the voice circuit itself. This was slow, limited in vocabulary and insecure (tones could be imitated by users). Common-channel signalling separates the two: signalling messages travel over a dedicated packet network, while voice circuits carry only voice. SS7 is the ITU-T standard for this signalling network [6]. It is the first example in this course of the *separation of control plane and user plane*, an idea that will return in every generation.

4.3 Signalling points

Nodes of an SS7 network are *signalling points* (SPs), each identified by a point code. A *service switching point* (SSP) is an exchange such as an MSC. A *signal transfer point* (STP) is a packet router of the SS7 network: it forwards messages between SPs without terminating them, and STPs are deployed in mated pairs for reliability. A *service control point* (SCP) hosts databases and service logic, for example the intelligent-network logic used for prepaid charging.

5 Detailed technical explanation

5.1 The SS7 protocol stack

Figure 1 shows the stack as used in GSM.

- **MTP 1–3** (Message Transfer Part) carries messages reliably between adjacent nodes (MTP 2: framing, error detection and retransmission) and routes them across the signalling network using point codes (MTP 3).
- **SCCP** (Signalling Connection Control Part) adds connection-less and connection-oriented services and, crucially for mobility, *global title translation*: a message can be addressed with

a telephone-number-like global title (for example derived from an IMSI) and the network finds the point code of the HLR.

- **TCAP** (Transaction Capabilities Application Part) structures request–response dialogues between applications.
- **MAP** [3] runs over TCAP and defines mobility operations: Update Location, Insert Subscriber Data, Send Authentication Info, Send Routing Information (SRI), Provide Roaming Number (PRN), Cancel Location.
- **ISUP** (ISDN User Part) runs directly over MTP 3 and sets up and releases trunk circuits between exchanges with messages such as IAM, ACM, ANM, REL and RLC.
- **BSSAP** runs over SCCP on the A interface. Its BSSMAP part carries messages between MSC and BSC (assignment, handover, ciphering); its DTAP part transparently carries messages between MSC and MS (call control and mobility management).

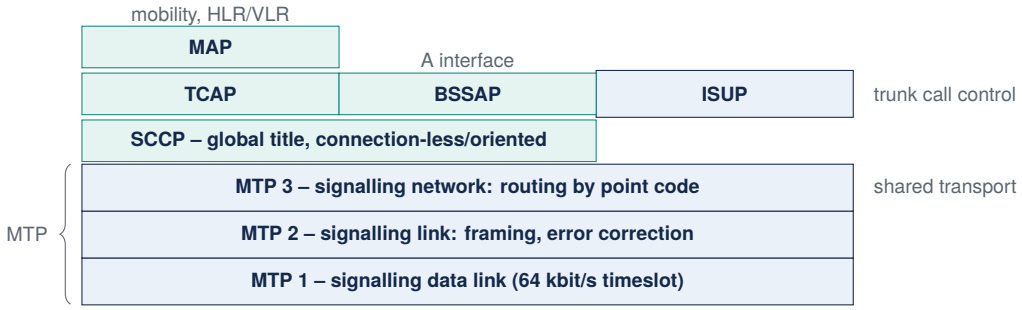


Figure 1. SS7 stack in a GSM core network. MTP provides transport; ISUP controls trunk circuits; SCCP and TCAP support the MAP mobility dialogues and BSSAP on the A interface.

5.2 Layer-3 protocols between MS and MSC

On the radio side, layer 3 is divided into three sublayers [4]: *radio resource management* (RR, terminated in the BSS: channel assignment, handover), *mobility management* (MM: location updating, authentication, TMSI reallocation) and *connection management* (CM, with call control CC: Setup, Alerting, Connect, Disconnect). MM and CC messages are carried transparently through the BSS in DTAP.

5.3 Numbers and identities used for routing

Three numbers must not be confused:

- **MSISDN** – the number people dial; it points to the subscriber's home network and is used by the GMSC to find the HLR.
- **IMSI** – the subscription identity used inside the mobile network (MAP addressing, authentication); never dialled.
- **MSRN** – a temporary E.164 number allocated by the visited VLR for one incoming call; it routes the call through the fixed network to the serving MSC.

The HLR maps MSISDN to IMSI and stores which VLR serves the subscriber; the VLR maps the MSRN back to the IMSI when the call arrives.

5.4 Location updating

Three cases trigger a location update [2]:

- **Normal** – the MS reads a new location area identity (LAI) on the broadcast channel. If the new LA belongs to another VLR, the new VLR fetches the subscriber data from the HLR with Update Location; the HLR sends Cancel Location to the old VLR.
- **Periodic** – the MS re-registers when timer T3212 expires, so the network can purge subscribers that disappeared without detaching (battery removed, coverage lost).
- **IMSI attach** – at power-on in the same LA. The mirror procedure, **IMSI detach**, tells the VLR that the MS is off, so incoming calls are not paged in vain.

5.5 Roaming

A roaming subscriber registers in a visited network (VPLMN). The visited VLR addresses the home HLR (HPLMN) over the international SS7 network, using global titles derived from the IMSI, and receives the subscription data and authentication triplets. Roaming agreements define charging and which services are allowed. Two consequences are worth noting:

- **Authentication stays with the home network**: only the home AuC knows K_i.
- **Tromboning**: a call to a roamer is always routed first to the home GMSC (the MSISDN belongs to the home network) and then internationally to the visited MSC. A local caller in the visited country therefore pays for two international legs, unless optimal routing is implemented.

5.6 Call release

Either party can release. The mobile sends Disconnect; the MSC replies Release and the MS confirms Release Complete. Towards the fixed network the MSC sends ISUP REL and receives RLC, which frees the trunk. Finally the MSC orders the BSS to release the radio channel (BSSMAP Clear Command). Billing records are closed when the circuit is released, which is why circuit-switched charging is naturally time-based.

6 Mathematical models

6.1 The loss system and the Erlang-B formula

Engineering problem. A cell has m traffic channels. Calls that arrive when all are busy are *blocked* (lost); they do not wait. How large must m be so that no more than, say, 2% of attempts are blocked?

System model and assumptions. (i) Call arrivals form a Poisson process of rate λ from a large population; (ii) holding times are independent and exponentially distributed with mean $h = 1/\mu$; (iii) blocked calls are cleared (no retry); (iv) all m channels are fully accessible. This is the $M/M/m/m$ queue.

Derivation. Let p_k be the steady-state probability that k channels are busy. In state k the system moves up at rate λ and down at rate $k\mu$. Balancing the flow between neighbouring states gives

$$\lambda\, p_{k-1} = k\mu\, p_k \quad\Longrightarrow\quad p_k = \frac{A}{k}\, p_{k-1} = \frac{A^k}{k!}\, p_0, \qquad A = \frac{\lambda}{\mu} = \lambda h. \tag{1}$$

Normalising $\sum_{k=0}^{m} p_k = 1$ yields $p_0 = \left(\sum_{i=0}^{m} A^i/i!\right)^{-1}$. Because Poisson arrivals see time averages (the PASTA property), the probability that an arriving call is blocked equals the probability that all m channels are busy:

$$P_{\mathrm{B}} = B(A,m) = \frac{A^m/m!}{\displaystyle\sum_{i=0}^{m} \frac{A^i}{i!}}. \tag{2}$$

Symbols and units. A: offered traffic [E]; m: number of channels; λ: arrival rate [calls/s]; h: mean holding time [s]; P_{B}: blocking probability (dimensionless). The carried traffic is $A_{\mathrm{c}} = A(1-P_{\mathrm{B}})$ and the utilisation of each channel is $\rho = A_{\mathrm{c}}/m$.

Numerically stable form. Direct evaluation of Eq. (2) overflows for large m. Dividing numerator and denominator by $A^m/m!$ gives the recursion

$$B(A,0) = 1, \qquad B(A,k) = \frac{A\,B(A,k-1)}{k + A\,B(A,k-1)}, \quad k = 1,\dots,m, \tag{3}$$

used in `scripts/acnmath.py` to produce Figure 2 and every number below.

Interpretation. Equation (2) depends only on the *mean* holding time (it is insensitive to the holding-time distribution), which is why it worked well for telephony for a century. It assumes, however, that blocked users go away; in reality many retry, which increases the offered traffic and the true blocking (Exercise 5).

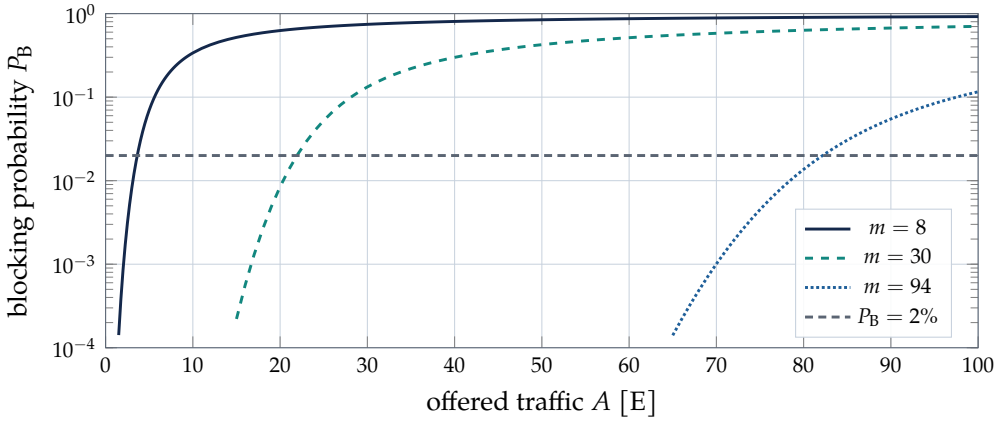


Figure 2. Erlang-B blocking probability versus offered traffic for $m = 8$, 30 and 94 channels. For the same 2% grade of service, 8 channels carry 3.6 E (0.45 E per channel) whereas 94 channels carry 82.2 E (0.87 E per channel).

6.2 Trunking efficiency

Problem. Is it better to have one pool of $2m$ channels or two pools of m?

Result. For a fixed grade of service, large pools carry more traffic per channel. This is the *trunking gain* (or pooling gain) and follows from Eq. (2): the relative fluctuation of the number of active calls, $\sqrt{A}/A$, shrinks as A grows, so less spare capacity is needed. Table 1 quantifies it. It explains why GSM operators avoid splitting channels between sectors unless interference requires it, and why statistical multiplexing in packet networks is so attractive.

Table 1. Traffic carried at 2% blocking (computed with Eq. (3)).

Channels m	Offered traffic A [E]	Carried traffic per channel [E]	Utilisation
8	3.63	0.44	44%
15	9.01	0.59	59%
30	21.93	0.72	72%
47	37.46	0.78	78%
94	82.17	0.86	86%

6.3 Waste of a reserved circuit

If the voice activity factor is α (fraction of time a speaker is active), the useful load of a reserved circuit is α times its capacity. With $\alpha \approx 0.4$ in each direction, roughly 60% of every reserved circuit-second carries silence. Packet switching recovers this capacity statistically, at the price of variable delay – the fundamental trade-off between the two switching paradigms, revisited in Lectures 3 and 5.

7 Network architecture and diagrams

The SS7 network is a separate packet network overlaid on the circuit network. Every MSC, GMSC and HLR is a signalling point; STPs route between them. Trunks between MSCs and the PSTN carry voice, while MAP and ISUP messages travel on signalling links that may follow completely different physical routes. Figure 4 shows the interplay for the hardest case, an incoming call, and Figure 3 the outgoing case.

8 Step-by-step procedures

Procedure 1 • Mobile-originated call (MOC)

1. The MS sends a channel request on the random access channel (RACH) and receives a signalling channel (SDCCH) on the access grant channel (AGCH).
2. It sends CM Service Request; the MSC/VLR authenticates the MS and starts ciphering (Lecture 1).
3. The MS sends Setup with the called number. The MSC checks the subscription (barring, prepaid credit) and answers Call Proceeding.
4. The MSC orders the BSS to assign a traffic channel (BSSMAP Assignment Request); the

MS moves from SDCCH to a TCH.

5. The MSC analyses the number and sends ISUP IAM towards the destination; the remote exchange answers ACM when the called phone rings, and the MSC sends Alerting to the MS (ring-back tone).
6. When the called party answers, ISUP ANM arrives; the MSC sends Connect, the MS acknowledges, and charging starts.

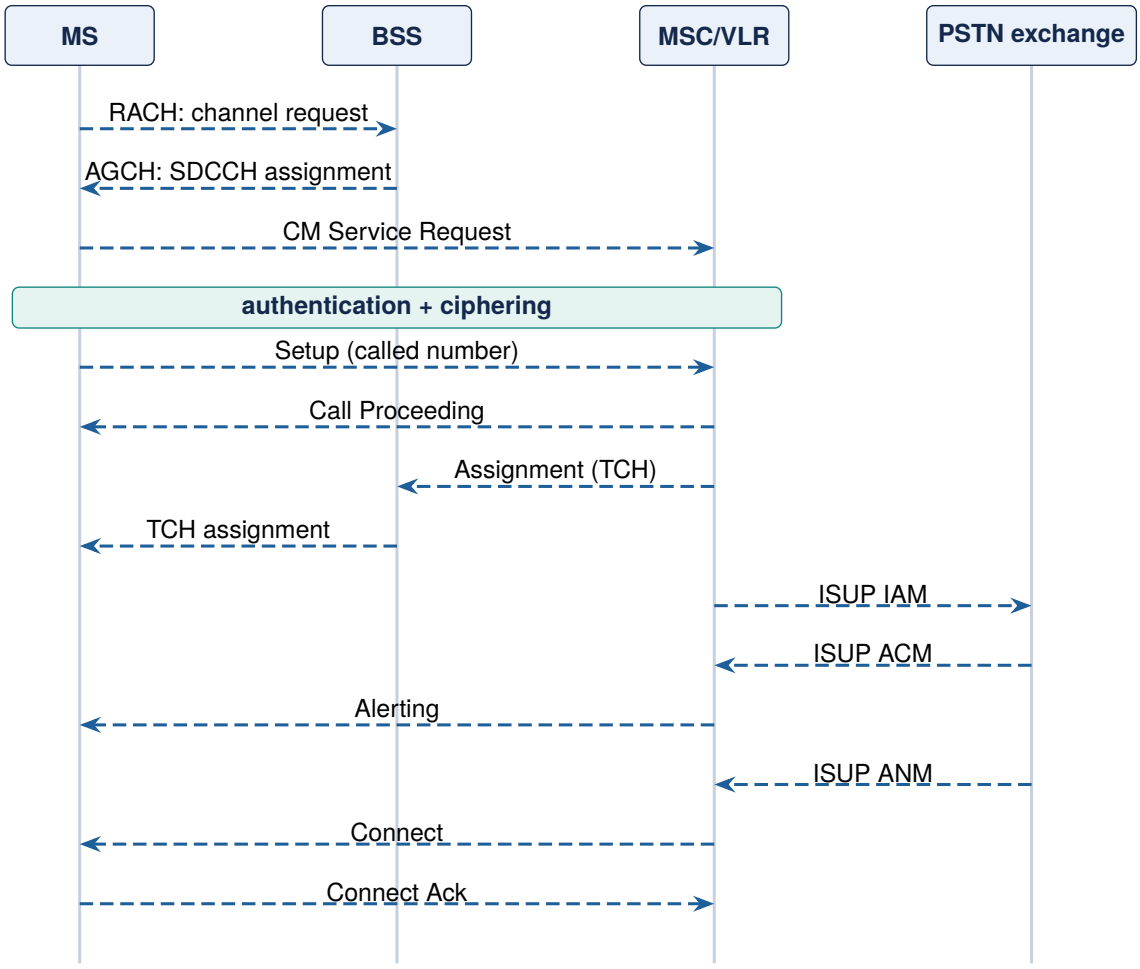


Figure 3. Mobile-originated call (Procedure 1).

Procedure 2 • Mobile-terminated call (MTC)

1. A caller in the PSTN dials the MSISDN. The PSTN routes the call (ISUP IAM) to a GMSC of the home network.
2. The GMSC sends MAP Send Routing Information to the HLR.
3. The HLR finds the serving VLR and requests a roaming number with MAP Provide Roaming Number; the VLR allocates an MSRN and returns it.
4. The HLR returns the MSRN to the GMSC, which routes the call with a new IAM addressed to the MSRN – the fixed network delivers it to the serving MSC.
5. The MSC asks its VLR for the subscriber context; the VLR releases the MSRN and returns the location area; the MSC pages the MS in all cells of that LA.

6 The MS answers the page, is authenticated, receives Setup and a TCH; ringing and answer are signalled back with ACM and ANM.

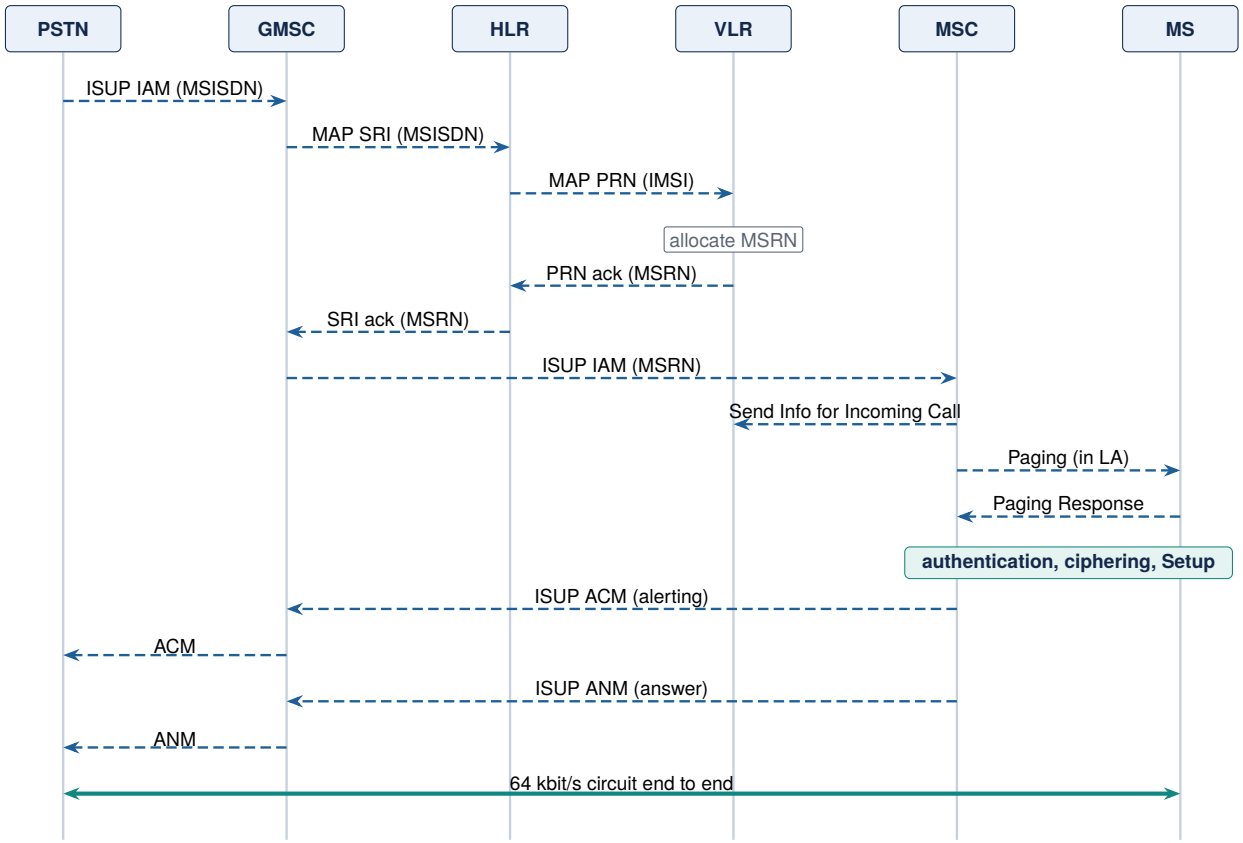


Figure 4. Mobile-terminated call: HLR interrogation, MSRN allocation, routing and paging (Procedure 2). Dashed: signalling; solid: the reserved circuit.

9 Worked examples

Worked example 1 • Dimensioning a cell

Problem. The cell of Lecture 1, Example 1, has 94 traffic channels. Subscribers generate 25 mE each during the busy hour. How many subscribers can the cell serve at a 2% grade of service?
Solution. Solving $B(A, 94) = 0.02$ with Eq. (3) gives $A = 82.2$ E. The number of subscribers is $N = A/a = 82.2/0.025 \approx 3290$. Carried traffic $A_c = 82.2 \times 0.98 = 80.5$ E, i.e. 86% channel utilisation.
Interpretation. If the same 94 channels were split into two sectors of 47, each sector would carry 37.5 E, 74.9 E in total – about 9% fewer subscribers for the same hardware (Table 1).

Worked example 2 • Erlang-B by hand

Problem. A small site has $m = 3$ channels and $A = 2$ E. Compute P_B.

Solution.

$$P_B = \frac{2^3/3!}{1 + 2 + 2^2/2! + 2^3/3!} = \frac{1.333}{6.333} = 0.211.$$

The recursion gives the same: $B_1 = 2/3$, $B_2 = (2 \cdot \frac{2}{3})/(2 + \frac{4}{3}) = 0.4$, $B_3 = 0.8/(3 + 0.8) = 0.211$.

Interpretation. One call in five is lost: a 3-channel site is useless at 2 E. Small pools need very low load for an acceptable grade of service.

Worked example 3 • Busy-hour arrival rate

Problem. Mean call duration is 90 s. What arrival rate produces the 82.2 E of Example 1, and how many call set-ups per second must the MSC handle in that cell?

Solution. $\lambda = A/h = 82.2/90 = 0.913$ calls/s, i.e. about 3290 call attempts per hour in one cell. Each attempt generates tens of signalling messages (RR, MM, CC, BSSMAP and ISUP), so signalling load scales with λ, not with A.

Interpretation. Short calls (or SMS and location updates, which use SDCCH) stress signalling capacity, which must be dimensioned separately (Exercise 4).

10 Technology comparison

Table 2. Circuit switching versus packet switching.

	Circuit switching (GSM voice)	Packet switching (GPRS and later)
Resource allocation	Reserved at set-up for the whole call	Per packet or per scheduling interval
Delay	Constant, low	Variable (queuing)
Loss mechanism	Blocking at set-up	Packet loss or delay under load
Efficiency	Low for bursty sources	High (statistical multiplexing)
Dimensioning model	Erlang-B ($M/M/m/m$)	Queueing models ($M/M/1$, etc.)
Charging	Time	Volume or flat rate
Signalling	SS7: ISUP + MAP	GTP, later Diameter and HTTP/2

Why circuit switching was right for voice. Voice is a constant-rate, delay-sensitive stream; reserving 13 kbit/s guarantees quality without complex scheduling, and the Erlang model allows precise dimensioning.

Why it became a limitation. Data is bursty: a web page needs a high rate for a second and nothing for a minute. Reserving a circuit for that is wasteful, and a single 9.6 kbit/s circuit is far too slow. Running two core networks (circuit and packet) also doubled operational cost – the motivation for the all-IP core of LTE (Lecture 5), where voice itself becomes packets (VoLTE).

11 Key takeaways

Key takeaways

- A GSM call reserves a radio timeslot, Abis/A channels, a switch path and trunks for its whole duration – even during silence.
- SS7 separates signalling from voice: MTP transports, SCCP addresses (global titles), TCAP structures dialogues, MAP handles mobility, ISUP controls trunks.
- An incoming call needs the HLR (SRI) and the visited VLR (PRN) to obtain an MSRN before it can be routed and paged.
- Erlang-B, $B(A, m)$, dimensions loss systems; compute it with the recursion Eq. (3).
- Larger pools are more efficient (trunking gain); splitting pools costs capacity.
- Circuit switching wastes the silent part of every call and cannot serve bursty data efficiently; this drove the move to packet switching.

12 Exercises

Exercise 1 • Level 1 – conceptual • Signalling

Why does ISUP run directly over MTP 3, while MAP needs SCCP? Relate your answer to how the destination of each message is known.

Exercise 2 • Level 1 – conceptual • Numbers

Why can the GMSC not route an incoming call using the IMSI? Why is the MSRN released as soon as the call reaches the serving MSC?

Exercise 3 • Level 2 – numerical • Dimensioning

A cell has 30 traffic channels. (a) What traffic can it offer at 1% and at 5% blocking? (b) With 20 mE per subscriber, how many more subscribers are served at 5% than at 1%?

Exercise 4 • Level 2 – numerical • SDCCH capacity

Location updates, SMS and call set-ups each hold an SDCCH for about 3 s. A cell sees 1.5 such transactions per second in the busy hour. How many SDCCH sub-channels are needed for a blocking of at most 0.5%?

Exercise 5 • Level 3 – analytical • Retries

Assume every blocked caller retries once, immediately and independently. Show that the offered traffic becomes approximately $A' = A(1 + P'_{\mathrm{B}})$ where $P'_{\mathrm{B}} = B(A', m)$, solve this fixed-point equation numerically for $m = 30$, $A = 21.93$ E, and comment on the result.

Exercise 6 • Level 3 – analytical • Tromboning

A subscriber of an Iranian operator roams in Turkey and is called by a Turkish fixed-line user. Draw the circuit path and list which networks carry each leg. Propose a signalling change that would avoid the international legs, and identify what information the Turkish network would need.

Solutions

Solution to Exercise 1

ISUP messages travel between adjacent exchanges along the circuit being set up; the next exchange and its point code are known from the routing tables, so MTP 3 routing is enough. MAP messages are addressed to databases identified by subscriber identities (e.g. "the HLR of IMSI 432..."), possibly in another country; SCCP global title translation converts such an identity into a point code, hop by hop.

Solution to Exercise 2

The PSTN routes by E.164 numbers; an IMSI is not a dialable number and is deliberately not exposed outside the mobile network. The MSRN is a scarce temporary number taken from the visited MSC's pool; once the call reaches that MSC the IMSI is retrieved from the VLR and the MSRN can be reused for another call.

Solution to Exercise 3

(a) $B(A,30) = 0.01 \Rightarrow A = 20.3\,\mathrm{E}$; $B(A,30) = 0.05 \Rightarrow A = 24.8\,\mathrm{E}$. (b) Subscribers: $20.3/0.02 \approx 1017$ versus $24.8/0.02 \approx 1240$, i.e. about 220 more (22%) at the price of five times more blocking.

Solution to Exercise 4

$A = \lambda h = 1.5 \times 3 = 4.5\,\mathrm{E}$. Iterating Eq. (3): $B(4.5,10) = 0.0105$, $B(4.5,11) = 0.0043$. Eleven sub-channels are needed; in practice operators configure two SDCCH/8 channels (16 sub-channels), which also leaves room for peaks.

Solution to Exercise 5

A fraction P'_{B} of the λ first attempts generate a second attempt, so the total attempt rate is $\lambda(1 + P'_{\mathrm{B}})$ and $A' = A(1 + P'_{\mathrm{B}})$ (treating retries as Poisson, which is an approximation). Iterating from $P'_{\mathrm{B}} = 0.02$: $A' = 22.37$, $B = 0.0236$; $A' = 22.45$, $B = 0.0243$; converging to $A' \approx 22.47\,\mathrm{E}$, $P'_{\mathrm{B}} \approx 0.0244$. The blocking seen by an attempt grows by about 22%; the fraction of *users* who finally fail is $P'^{2}_{\mathrm{B}} \approx 0.06\%$, but the network carries more signalling. Retries are a positive feedback that can destabilise an overloaded network.

Solution to Exercise 6

Path: Turkish caller → Turkish PSTN → international leg to the home GMSC in Iran (the MSISDN is Iranian) → HLR interrogation returns a Turkish MSRN → second international leg back to the visited MSC in Turkey → the roamer. Optimal routing lets the Turkish gateway query the home HLR directly (MAP SRI towards the HPLMN) and route to the MSRN inside Turkey; it requires the visited network to recognise the number as belonging to a roaming

subscriber and an agreement allowing the HLR query, and it changes charging, which is why it was rarely deployed.

13 Further reading

SS7 is introduced in [6]; MAP is specified in [3], location management in [2] and call handling in [1]. Layer-3 call control and mobility messages are in [4]. Kleinrock [7] derives the loss formula rigorously, and Rappaport [8] applies it to cellular dimensioning. Eberspächer et al. [5] describe GSM call flows in detail.

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[1] 3GPP, *Basic call handling; technical realization*, TS 23.018, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23018.htm.

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[8] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002.

03

Lecture 03

2G/3G Core Networks

شبکه‌های هستهٔ نسل دوم و سوم

2G — **3G** — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 1–2; spread spectrum basics; decibels

In this lecture

1 Learning objectives . . . 2
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-03-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the requirements that drove 3G (IMT-2000) and why a new radio interface, WCDMA, was needed.
- **LO2** **Draw** the UMTS architecture – UE, NodeB, RNC, CS and PS core domains – and name the interfaces Uu, Iub, Iur, Iu-CS and Iu-PS.
- **LO3** **Compute** processing gain, the uplink load factor and the noise rise of a WCDMA cell, and **estimate** its voice pole capacity.
- **LO4** **Describe** the HSPA evolution (Releases 5–8) and **derive** the HSDPA peak rate from codes, spreading factor and modulation.
- **LO5** **Trace** an RRC connection and radio access bearer (RAB) set-up with RANAP.
- **LO6** **Explain** the role of IMS and **evaluate** why LTE removed the CS/PS split.

2 Prerequisite knowledge

This lecture builds directly on the GSM/GPRS architecture of Lecture 1 and the circuit-switched core of Lecture 2. You should also recall that a spread-spectrum signal occupies a bandwidth much larger than its data rate, and that a receiver correlating with the right code recovers the signal while other codes look like noise.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
UMTS	سامانهٔ جهانی مخابرات موبایل	The 3GPP 3G system: WCDMA radio + evolved GSM core.
WCDMA	دسترسی چندگانهٔ تقسیم کد پهن‌باند	3G radio interface: 5 MHz carrier, 3.84 Mchip/s.
NodeB	نود بی	3G base station: radio transmission, spreading, inner-loop power control.
RNC	کنترل‌گر شبکهٔ رادیویی	Controls NodeBs: radio resources, handover, ciphering.
Spreading factor (SF)	ضریب گسترش	Chips per symbol; trades rate against protection.
Processing gain	بهرهٔ پردازش	W/R: how much despreading lifts the wanted signal.
Soft handover	تحویل نرم	UE connected to several cells at once (make-before-break).
Noise rise	افزایش نویز	Increase of uplink interference over thermal noise.
RAB	حامل دسترسی رادیویی	Bearer between UE and core network, set up by RANAP.
HSPA	دسترسی بستهٔ پرسرعت	Shared-channel evolution of WCDMA (HSDPA + HSUPA).
IMS	زیرسامانهٔ چندرسانه‌ای IP	SIP-based control of IP multimedia services.

3 Introduction and motivation

By the late 1990s GSM had proved that mobile telephony could be global, but it could not become a data network. EDGE squeezed about 59 kbit/s out of a timeslot, and one 200 kHz carrier can never give a single user megabits per second. The ITU's IMT-2000 programme asked for up to 2 Mbit/s indoors, 384 kbit/s for pedestrians and 144 kbit/s for vehicles, flexible multiplexing of services with different rates on one connection, and global roaming.

3GPP's answer, UMTS, combined two decisions. On the radio side it adopted a completely new interface, Wideband CDMA, with a 5 MHz carrier. On the network side it did the opposite: it *reused* the GSM/GPRS core – MSC, HLR, SGSN, GGSN – and connected it to the new radio network through a new interface, Iu. This lecture studies that hybrid: a new radio access network attached to an evolved 2G core with two parallel domains, and the strains that eventually led LTE to abandon the circuit-switched domain altogether.

4 Fundamental concepts

4.1 Code division multiple access

In WCDMA all users of a cell transmit at the same time on the same 5 MHz carrier. Each user's symbols are multiplied by a code running at the chip rate $W = 3.84\,\text{Mchip/s}$. The receiver multiplies again by the same code (despreading): the wanted signal is concentrated back into its narrow data bandwidth, while signals with other codes remain spread and act like noise.

Definition 1 • Spreading factor and processing gain

The spreading factor is the number of chips per symbol, $\text{SF} = W/R_\text{s}$. For a service with information rate R, the processing gain is $G_\text{p} = W/R$. It is the factor by which despreading increases the signal-to-interference ratio.

Two families of codes are used. *Channelisation codes* (orthogonal variable spreading factor, OVSF) separate the channels sent by one transmitter; *scrambling codes* separate transmitters (cells in the downlink, UEs in the uplink). OVSF codes form a tree (Figure 4); a code can be used only if no ancestor or descendant is in use, so assigning a short (low-SF, high-rate) code blocks a whole branch.

4.2 Interference-limited capacity

Because everyone shares one carrier, WCDMA capacity is limited by interference, not by a fixed number of channels: every additional user raises the interference seen by all the others. Two consequences follow. First, *power control* is essential: an uplink user close to the NodeB must not drown distant users (the near–far problem), so the NodeB adjusts each UE's power 1500 times per second. Second, capacity is *soft*: the operator can trade a little quality for more users, and the cell coverage shrinks as the load grows (cell breathing).

4.3 Soft handover

Since neighbouring cells use the same frequency, a UE near a cell border can be connected to two or three cells simultaneously. In the uplink the RNC selects the best frame; in the downlink the UE combines the signals of all cells in its active set. Soft handover avoids the gap of a hard handover and provides macro-diversity, at the cost of extra NodeB and Iub resources. It is analysed in Lecture 4.

5 Detailed technical explanation

5.1 UTRAN: NodeB and RNC

The UMTS Terrestrial Radio Access Network (UTRAN) [6] has two node types. The **NodeB** performs the physical layer: channel coding, spreading, modulation, RF and the fast inner-loop power control. The **RNC** (Radio Network Controller) is much more powerful than a GSM BSC. It terminates the radio protocols RLC, MAC (for dedicated channels) and RRC, performs admission control, handover decisions, outer-loop power control, macro-diversity combining, and – unlike GSM, where ciphering ended at the BTS – ciphering and integrity protection. An RNC plays different roles for a given UE:

- **SRNC** (serving RNC): terminates the Iu connection and the RRC signalling of that UE;
- **DRNC** (drift RNC): lends radio resources of its cells when the UE is in soft handover with cells of another RNC; data passes over *Iur* to the SRNC;
- **CRNC** (controlling RNC): the RNC that controls a given NodeB's resources.

5.2 Interfaces

Table 1. UMTS interfaces.

Interface	Between	Control protocol	Notes
Uu	UE – NodeB/RNC	RRC	WCDMA radio interface
Iub	NodeB – RNC	NBAP	Transport originally ATM, later IP
Iur	RNC – RNC	RNSAP	Soft handover across RNCs; no GSM equivalent
Iu-CS	RNC – MSC	RANAP	Replaces the A interface; AMR speech
Iu-PS	RNC – SGSN	RANAP	User plane in GTP-U; replaces Gb

A key design change is that the Iu-PS user plane uses GTP-U all the way from the RNC, so the SGSN no longer handles LLC/SNDCP towards the terminal; the radio-specific protocols end in the RNC. RANAP [5] carries the signalling for RAB management, relocation of the SRNC, paging and security mode control.

5.3 The core network: CS and PS domains

The UMTS core keeps the two domains of GSM/GPRS:

- The **circuit-switched (CS) domain** (MSC/VLR, GMSC) serves voice and video calls with Erlang-dimensioned circuits. From Release 4 the MSC is split into an *MSC Server* (call control and mobility, the control plane) and a *Media Gateway* (MGW, the user plane that switches and transcodes the speech). This allows the CS core transport to run over IP or ATM – an early example of control/user plane separation.
- The **packet-switched (PS) domain** (SGSN, GGSN) serves IP traffic with PDP contexts, now with QoS classes (conversational, streaming, interactive, background).

The HLR/AuC is shared; the AuC now produces *quintets* (RAND, XRES, CK, IK, AUTN) for mutual authentication (*UMTS AKA*, [1]). AUTN lets the USIM verify that the challenge came from its home network and is fresh; IK protects signalling integrity. Lecture 11 derives these values.

5.4 From Release 99 to HSPA

Release 99 carried packet data on dedicated channels with fixed spreading factors and 10–80 ms transmission time intervals (TTIs). Allocating a dedicated code to a bursty user wasted code space, and scheduling in the RNC was slow. HSPA moved intelligence to the NodeB:

- **Release 5 – HSDPA** [2]: a shared downlink channel (HS-DSCH) using up to 15 codes of SF 16, a 2 ms TTI, adaptive modulation (QPSK/16QAM), fast hybrid ARQ and channel-dependent scheduling in the NodeB, driven by CQI reports from the UE every 2 ms.
- **Release 6 – HSUPA** (enhanced uplink): an enhanced dedicated channel (E-DCH) with NodeB scheduling and HARQ, up to 5.76 Mbit/s.

- **Release 7 – HSPA+**: 64QAM in the downlink, 2×2 MIMO, 16QAM uplink, and a flat-architecture option.
- **Release 8**: dual-carrier HSDPA (two adjacent 5 MHz carriers).

The philosophical shift matters more than the numbers: the network stopped allocating fixed resources to users and started scheduling a shared pipe every 2 ms according to channel quality – exactly the paradigm LTE adopts from the start (Lecture 7).

5.5 IMS: IP multimedia over the PS domain

The IP Multimedia Subsystem (IMS), introduced in Release 5 [3], defines how to offer voice, video and messaging as IP services controlled by SIP [8]. Its core elements are the call session control functions: the **P-CSCF** (proxy, the UE's first contact point), the **I-CSCF** (interrogating, entry point of the home network) and the **S-CSCF** (serving, which registers the user and executes service logic), plus the HSS, the evolution of the HLR. In 3G, IMS was rarely used for mass-market voice; its real role came with LTE, where it is the only way to make calls (VoLTE, Lecture 5).

6 Mathematical models

6.1 Processing gain and the required SINR

Problem. What signal-to-interference ratio must a WCDMA receiver see before despreading for a given service quality?

Model. Required energy per bit to noise-plus-interference density $(E_\mathrm{b}/N_0)_\mathrm{req}$, chip rate W, service bit rate R.

Derivation. Despreading accumulates W/R chips per bit, so

$$\frac{E_\mathrm{b}}{N_0} = \frac{W}{R} \cdot \mathrm{SINR}_\mathrm{wideband} \quad \Longrightarrow \quad \mathrm{SINR}_\mathrm{req} = \frac{(E_\mathrm{b}/N_0)_\mathrm{req}}{G_\mathrm{p}}, \qquad G_\mathrm{p} = \frac{W}{R}. \tag{1}$$

Example and interpretation. AMR voice at $R = 12.2\,\mathrm{kbit/s}$: $G_\mathrm{p} = 3.84 \times 10^6 / 12.2 \times 10^3 = 315$ (25.0 dB). With $(E_\mathrm{b}/N_0)_\mathrm{req} = 5\,\mathrm{dB}$, the wideband SINR may be as low as $5 - 25 = -20\,\mathrm{dB}$: the signal is buried 20 dB below the interference and still decoded. A 384 kbit/s data user has only $G_\mathrm{p} = 10\,\mathrm{dB}$ and needs roughly 15 dB more received power – which is why high-rate users consume a large share of the cell's capacity.

6.2 Uplink load factor, noise rise and pole capacity

Problem. How many simultaneous users can a WCDMA cell carry in the uplink?

Assumptions. N users of the same service, perfect power control, received powers adjusted so that each user just meets its E_b/N_0; voice activity factor ν; other-cell interference equal to a fraction i of own-cell interference; thermal noise power P_N [7].

Derivation. Let I_{tot} be the total received wideband power and P_j the received power of user j. Its requirement is

$$\left(\frac{E_\text{b}}{N_0}\right)_j = \frac{W}{\nu_j R_j}\,\frac{P_j}{I_{\text{tot}} - P_j} \quad\Longrightarrow\quad P_j = L_j\, I_{\text{tot}}, \qquad L_j = \frac{1}{1 + \dfrac{W}{(E_\text{b}/N_0)_j\, R_j\, \nu_j}}. \tag{2}$$

Summing own-cell users and adding other-cell interference, $I_{\text{tot}} = (1+i)\sum_j P_j + P_\text{N}$, which defines the uplink load factor

$$\eta_\text{UL} = (1+i)\sum_{j=1}^{N} L_j \qquad \text{and} \qquad \frac{I_{\text{tot}}}{P_\text{N}} = \frac{1}{1-\eta_\text{UL}}. \tag{3}$$

The ratio $I_{\text{tot}}/P_\text{N}$ is the *noise rise*. It diverges as $\eta_\text{UL} \to 1$; the number of users at which this happens is the *pole capacity*

$$N_\text{pole} = \frac{1}{(1+i)L} = \frac{1 + \dfrac{W}{(E_\text{b}/N_0)R\nu}}{1+i}. \tag{4}$$

Interpretation. Figure 1 shows the noise rise: 3 dB at 50% load, 6 dB at 75%, 10 dB at 90%. Each decibel of noise rise is a decibel of lost link budget – coverage shrinks as the cell fills up. Operators therefore plan for 50–75% load, not for the pole.

Limitations. The model assumes perfect power control, a single service and a fixed i; real systems need margins for power-control errors and fast fading.

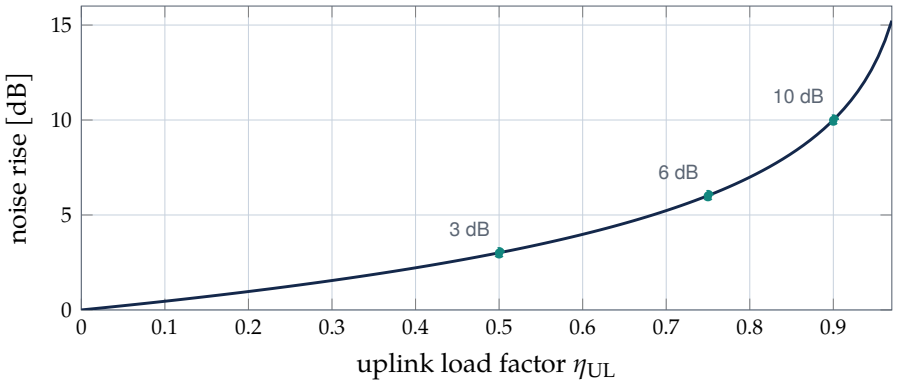


Figure 1. Uplink noise rise $-10\log_{10}(1-\eta_\text{UL})$ from Eq. (3).

6.3 HSDPA peak rate

With n_c codes of spreading factor SF and b bits per symbol, the symbol-level rate of the HS-DSCH is

$$R_\text{peak} = n_\text{c}\,\frac{W}{\text{SF}}\,b\,r_\text{c}, \tag{5}$$

where r_c is the effective code rate. For $n_\text{c} = 15$, SF $= 16$, 16QAM ($b = 4$) and $r_\text{c} \to 1$: $15 \times 240\,\text{ksymbol/s} \times 4 = 14.4\,\text{Mbit/s}$; the highest standardised UE category reaches 14.0 Mbit/s. 64QAM gives 21.6 Mbit/s (21.1 in practice), and two MIMO layers or two carriers double it (Figure 2). These are peak values for one user alone in a cell with excellent SINR; average cell throughput is several times lower.

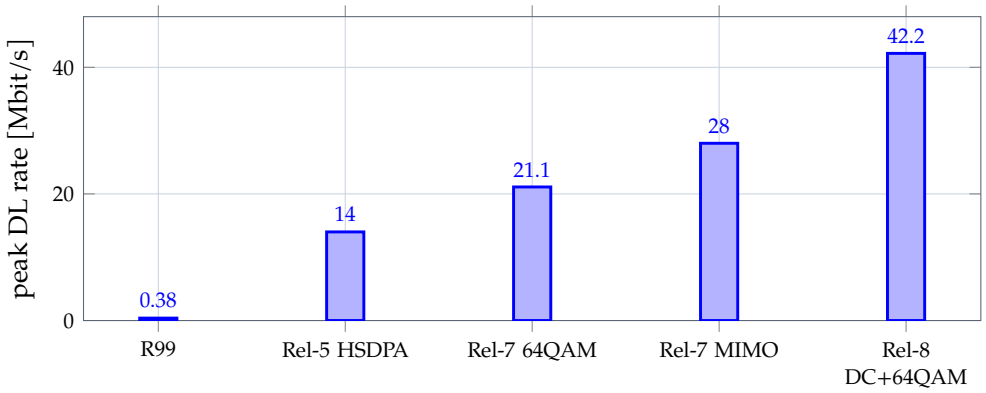


Figure 2. Peak downlink rates along the WCDMA/HSPA evolution (R99 shown with its practical 384 kbit/s; categories with the highest standardised rate).

7 Network architecture and diagrams

Figure 3 shows the UMTS network. Compared with GSM (Lecture 1, Figure 3) three things changed: the RAN uses new nodes and interfaces (NodeB, RNC, Iub, Iur); the core attaches to the RAN only through Iu, which has a CS and a PS flavour; and the PS user plane is a GTP-U tunnel that starts at the RNC. The core network itself is recognisably the GSM/GPRS core.

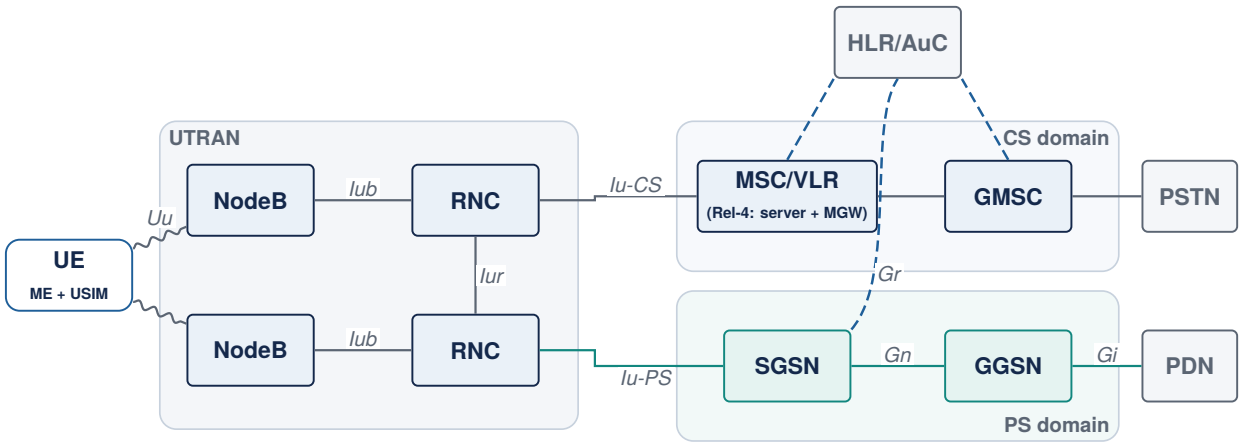


Figure 3. UMTS architecture with the UTRAN, the circuit-switched and the packet-switched core domains. Dashed: MAP signalling to the HLR.

8 Step-by-step procedures

Procedure 1 • PS service request and RAB establishment

1. The idle UE sends RRC Connection Request on the random access channel.
2. The RNC performs admission control, sets up a radio link in the NodeB (NBAP) and answers RRC Connection Setup; the UE confirms with RRC Connection Setup Complete.
3. The UE sends a NAS Service Request to the SGSN, carried in an RRC Initial Direct Transfer

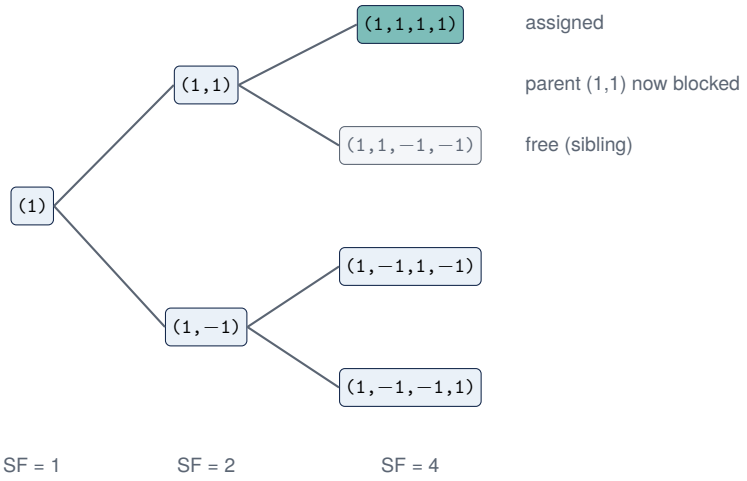


Figure 4. OVSF code tree. Assigning $(1,1,1,1)$ blocks its parent $(1,1)$ and the root, but its sibling $(1,1,-1,-1)$ remains usable; all codes of one SF are mutually orthogonal.

and the RANAP Initial UE Message.

4. The SGSN may run UMTS AKA; it then sends RANAP Security Mode Command with CK and IK to the RNC, which starts ciphering and integrity protection towards the UE.
5. The SGSN requests the bearer with RANAP RAB Assignment Request, including QoS and its GTP-U tunnel endpoint.
6. The RNC configures the radio bearer with RRC Radio Bearer Setup; the UE answers Radio Bearer Setup Complete; the RNC returns RAB Assignment Response with its own tunnel endpoint. User data now flows over the radio bearer and the Iu-PS tunnel.

Note the layering: NAS messages (Service Request) travel transparently from UE to SGSN; access-stratum signalling (RRC) ends in the RNC; RANAP connects the two worlds. This split between *non-access stratum* and *access stratum* remains in LTE and 5G.

9 Worked examples

Worked example 1 • Voice pole capacity

Problem. AMR voice, $R = 12.2\,\text{kbit/s}$, $(E_b/N_0)_{\text{req}} = 5\,\text{dB}$, activity $\nu = 0.67$, other-cell interference ratio $i = 0.65$. Find the pole capacity and the number of users at 50% load.

Solution. $W/((E_b/N_0)R\nu) = 3.84\times10^6/(3.162\times12\,200\times0.67) = 148.6$. Equation (4): $N_{\text{pole}} = (1+148.6)/1.65 = 90.6$. Each user loads the cell by $(1+i)L = 1.65/149.6 = 1.10\%$, so 50% load corresponds to about 45 users, with a noise rise of 3 dB.

Interpretation. 45 voice users per 5 MHz carrier and sector, i.e. 9 per MHz – a substantial gain over GSM's roughly 94 channels in 10 MHz with reuse (Lecture 1).

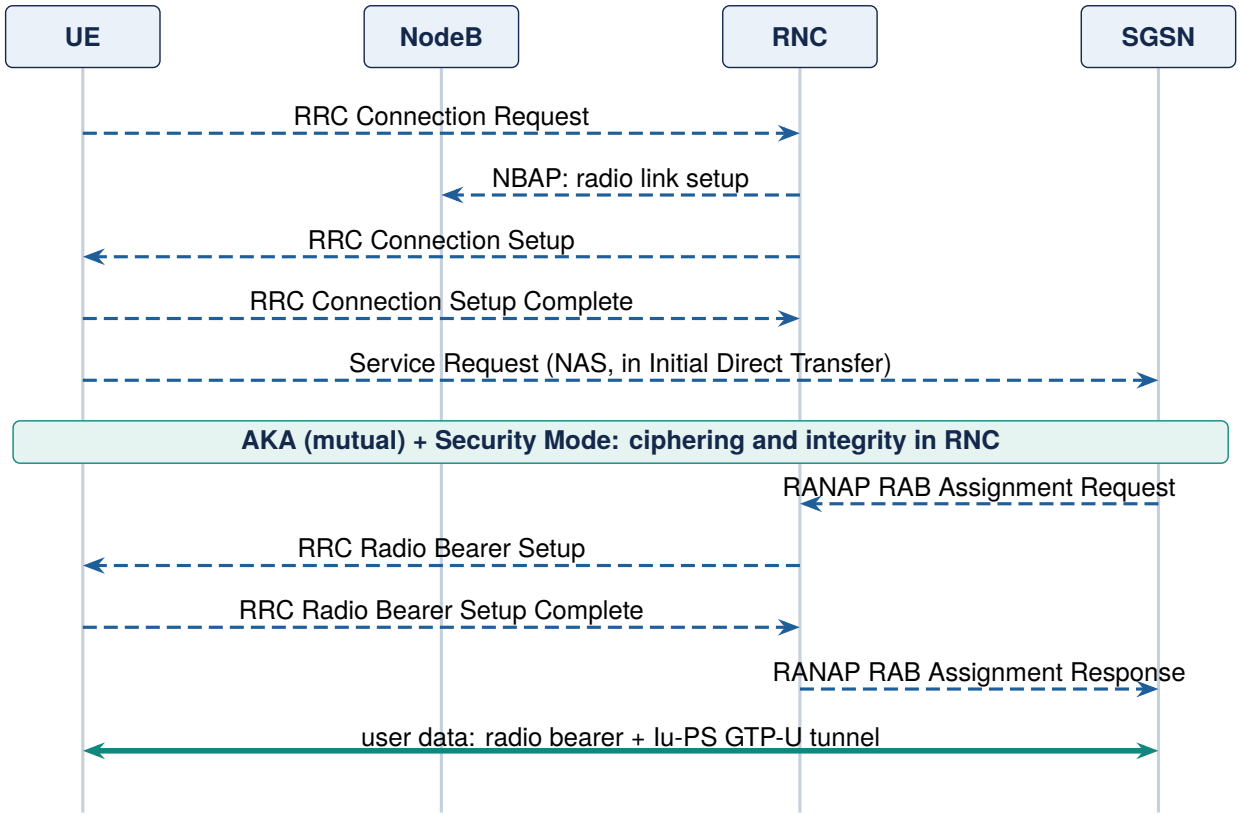


Figure 5. RRC connection and RAB set-up for a packet service (Procedure 1).

Worked example 2 • Mixing services

Problem. In the cell of Example 1, one 384 kbit/s data user needs $(E_b/N_0) = 1.5\,\text{dB}$ with $\nu = 1$. How many voice users does it replace?

Solution. $W/((E_b/N_0)R\nu) = 3.84 \times 10^6/(1.413 \times 384\,000) = 7.08$, so $L = 1/8.08 = 0.124$ and its load contribution is $1.65 \times 0.124 = 20.4\%$. One voice user contributes 1.10%: the data user replaces about 18 voice users.

Interpretation. Resource consumption in CDMA scales with $R \cdot E_b/N_0$, which is why high-rate packet data on dedicated channels was so costly in Release 99.

Worked example 3 • OVSF code budget

Problem. HSDPA reserves 15 codes of SF 16. What fraction of the downlink code tree is left for everything else?

Solution. SF 16 has 16 codes; 15 are used, leaving one SF 16 branch, i.e. $1/16 = 6.25\%$ of the tree, which must hold common channels (pilot, broadcast, paging), the HS-SCCH control channels and any Release 99 dedicated channels.

Interpretation. In practice operators configure 5–10 HS-PDSCH codes and keep the rest for voice – code space, not power, often limits the downlink.

10 Technology comparison

Table 2. Mobile generations up to 3.5G (indicative peak values).

	GSM/GPRS	EDGE	UMTS R99	HSPA (Rel-5/6)
Multiple access	TDMA/FDMA	TDMA/FDMA	WCDMA	WCDMA + shared channel
Carrier	200 kHz	200 kHz	5 MHz	5 MHz
Peak DL	171 kbit/s	474 kbit/s	384 kbit/s (2 Mbit/s theor.)	14 Mbit/s
Handover	Hard	Hard	Soft + hard	Soft (DCH); serving-cell change (HS-DSCH)
Scheduling	BSC/PCU	BSC/PCU	RNC, 10–80 ms	NodeB, 2 ms
Core	CS + PS	CS + PS	CS + PS (Iu)	CS + PS (Iu)
Authentication	One-way (triplet)	One-way	Mutual AKA (quintet)	Mutual AKA

The CS/PS split and why LTE removed it. Running two core domains means two sets of nodes, two mobility management procedures (location area and routing area updates), two charging systems and two sets of interconnection agreements. Every innovation (for example a new QoS scheme) had to be done twice. Once HSPA proved that a shared packet channel can carry voice-quality traffic, the economic case for a single, packet-only core became overwhelming: LTE's EPC (Lecture 9) has no CS domain, and voice is carried by IMS over IP. The price is complexity elsewhere – QoS bearers must now guarantee what a circuit guaranteed for free, and legacy voice requires fallback procedures (CS fallback, SRVCC) during the transition.

11 Key takeaways

Key takeaways

- UMTS = new radio (WCDMA, 5 MHz, 3.84 Mchip/s) + evolved GSM/GPRS core, joined by Iu.
- The RNC terminates RLC/MAC/RRC, ciphering and integrity; SRNC/DRNC roles enable soft handover across RNCs over Iur.
- CDMA capacity is interference-limited: $\eta_{\mathrm{UL}} = (1+i)\sum L_j$, noise rise $1/(1-\eta_{\mathrm{UL}})$, plan for 50–75% load.
- Processing gain W/R lets low-rate users operate at negative SINR; high-rate users cost proportionally more.
- HSPA moved scheduling and HARQ to the NodeB with a 2 ms TTI – the template for LTE.
- Mutual authentication (AKA) and integrity protection fixed the main GSM security gaps.

- Two core domains doubled cost and complexity; LTE kept only the packet domain and moved voice to IMS.

12 Exercises

Exercise 1 • Level 1 – conceptual • RNC versus BSC

List three functions performed by the UMTS RNC that the GSM BSC did not perform, and explain which design choice of WCDMA made each necessary.

Exercise 2 • Level 1 – conceptual • Why Iur?

GSM has no interface between BSCs. Explain why UMTS needs Iur, and what would happen to soft handover at an RNC border without it.

Exercise 3 • Level 2 – numerical • Load budget

Using the parameters of Example 1, how many voice users can be admitted if the noise rise must not exceed 4 dB? How does the answer change if i drops to 0.4 (better isolated cells)?

Exercise 4 • Level 2 – numerical • HSDPA rate

A cell configures 10 HS-PDSCH codes. Compute the symbol-level peak rate for QPSK, 16QAM and 64QAM, and the rate with code rate $r_c = 0.75$ for 16QAM.

Exercise 5 • Level 3 – analytical • Cell breathing

Assume a path-loss exponent of 3.5. By what factor does the cell radius shrink when the load grows from 50% to 75%, if the UE transmit power is the limiting factor? Derive the general expression for the radius ratio as a function of the two load factors.

Exercise 6 • Level 3 – analytical • Mixed traffic

A cell must carry N_v voice users (parameters of Example 1) and N_d users at 64 kbit/s with $E_b/N_0 = 2$ dB and $\nu = 1$, at a total load of 60%. Derive the linear constraint between N_v and N_d and plot (or tabulate) the admissible region.

Solutions

Solution to Exercise 1

(1) Macro-diversity combining and soft-handover control – needed because all cells use the same frequency. (2) Outer-loop power control (setting the SIR target from frame error rates) – needed because CDMA is interference-limited and power control is the main resource control. (3) Ciphering and integrity protection, terminating RLC/MAC/RRC – moved from BTS/SGSN to keep security and retransmission in one place despite soft handover across many NodeBs. Admission control based on load (Eq. (3)) is a fourth, because capacity is soft.

Solution to Exercise 2

In soft handover a UE may be connected to cells of two RNCs. Without Iur the SRNC could not receive the uplink frames decoded by the other RNC's NodeB, nor send downlink data through it; the UE would have to perform a hard handover at every RNC border, losing macro-diversity exactly where links are weakest. Iur lets the DRNC forward frames to the SRNC, which keeps the Iu connection unchanged until a relocation is worthwhile.

Solution to Exercise 3

Noise rise 4 dB: $1-\eta = 10^{-0.4} = 0.398$, so $\eta_{\max} = 0.602$. With $(1+i)L = 1.10\%$, $N = 0.602/0.0110 = 54$ users. With $i = 0.4$: $(1+i)L = 1.4/149.6 = 0.936\%$, $N = 64$ users – better cell isolation (sectorisation, antenna tilt) translates directly into capacity.

Solution to Exercise 4

Per code: $W/16 = 240$ ksymbol/s; 10 codes: 2.4 Msymbol/s. QPSK: 4.8 Mbit/s; 16QAM: 9.6 Mbit/s; 64QAM: 14.4 Mbit/s. 16QAM with $r_c = 0.75$: $9.6 \times 0.75 = 7.2$ Mbit/s.

Solution to Exercise 5

The extra noise rise between loads η_1 and η_2 is $\Delta = 10\log_{10}\frac{1-\eta_1}{1-\eta_2}$ dB. With a fixed UE power the tolerable path loss drops by Δ; for $L_p \propto d^n$, $d_2/d_1 = 10^{-\Delta/(10n)} = \left(\frac{1-\eta_2}{1-\eta_1}\right)^{1/n}$. For $\eta_1 = 0.5$, $\eta_2 = 0.75$: $\Delta = 3.01$ dB and $d_2/d_1 = 0.5^{1/3.5} = 0.820$: the radius shrinks by 18% and the area by 33%.

Solution to Exercise 6

Data user: $W/((E_b/N_0)R) = 3.84\times10^6/(1.585\times64\,000) = 37.9$, $L_d = 1/38.9 = 0.0257$. Constraint: $1.65\,(N_v \times 0.00668 + N_d \times 0.0257) \le 0.60$, i.e. $0.0110\,N_v + 0.0424\,N_d \le 0.60$. Corner points: $(N_v, N_d) = (54, 0)$, $(0, 14)$; e.g. 30 voice users leave room for 6 data users. One 64 kbit/s user costs about 3.9 voice users.

13 Further reading

Holma and Toskala [7] is the standard reference for WCDMA and HSPA radio planning, including the load equations of Section 6.2. The UTRAN architecture is in [6], RANAP in [5], HSDPA in [2] and 3G security in [1]. IMS is specified in [3] on top of SIP [8]; the general 3GPP architecture is in [4].

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[7] H. Holma and A. Toskala, *WCDMA for UMTS: HSPA Evolution and LTE*, 5th ed. Wiley, 2010.

[8] J. Rosenberg, H. Schulzrinne, G. Camarillo, *et al.*, "SIP: Session initiation protocol", IETF, RFC 3261, 2002. [Online]. Available: https://www.rfc-editor.org/rfc/rfc3261.

04

Lecture 04

Mobility and Handover

تحرک و تحویل

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 1–3; log-distance path loss; Gaussian distribution and the Q-function

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
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5 Detailed technical explanation . . . 5
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-04-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Distinguish** idle-mode mobility (cell reselection, location registration) from connected-mode mobility (handover) and explain who decides in each case.
- **LO2** **Compare** location areas, routing areas, tracking-area lists and RAN notification areas, and **optimise** the size of a registration area with a cost model.
- **LO3** **Classify** handovers (intra-/inter-frequency, inter-RAT, hard/soft) and state the measurement quantities RSRP and RSRQ.
- **LO4** **Apply** the LTE A3 entering condition with hysteresis, offsets and time-to-trigger, and **compute** where a handover is triggered.
- **LO5** **Trace** an X2-based LTE handover message by message, including data forwarding and path switch.
- **LO6** **Diagnose** too-late, too-early, wrong-cell and ping-pong handovers and relate them to parameter settings.

2 Prerequisite knowledge

You need the location-area concept of Lectures 1–2, the soft handover of WCDMA (Lecture 3), and the log-distance path-loss model with log-normal shadowing: $L_{\mathrm{p}}(d) = L_{\mathrm{p}}(d_0) + 10n\log_{10}(d/d_0) + X_\sigma$, where X_σ is a zero-mean Gaussian variable (in dB) with standard deviation σ. The tail function is $Q(x) = \Pr\{Z > x\}$ for a standard Gaussian Z.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Mobility management	مدیریت تحرک	Keeping a moving user reachable and connected.
Cell reselection	انتخاب مجدد سلول	UE-controlled change of camping cell in idle mode.
Handover (HO)	تحویل	Network-controlled transfer of an active connection to another cell.
Tracking area (TA)	ناحیهٔ ردیابی	LTE/5G registration area; the paging scope.
RSRP	توان دریافتی سیگنال مرجع	Average power of one cell-specific reference-signal resource element.
RSRQ	کیفیت دریافتی سیگنال مرجع	$N \cdot \mathrm{RSRP}/\mathrm{RSSI}$: a load- and interference-aware quality.
Hysteresis	پسماند	Margin that prevents repeated triggering around a threshold.
Time-to-trigger (TTT)	زمان تا راه‌اندازی	Duration a condition must hold before a report is sent.
Ping-pong	پینگ‌پنگ	Unnecessary handover followed quickly by a return.
Radio link failure (RLF)	شکست پیوند رادیویی	Loss of the connection detected by timers (e.g. T310).

3 Introduction and motivation

A fixed telephone is always at the same socket; a mobile user is not. Mobility management answers two different questions:

(a) **Reachability** – when the user is idle, where should the network look for it when a call or a packet arrives?

(b) **Session continuity** – when the user is active, how can the connection follow the user from cell to cell without dropping or interrupting it?

Both questions involve a trade-off between *signalling cost* and *performance*. If the network knew the exact cell of every idle user, every cell change would cost an update; if it knew nothing, every incoming call would have to be paged everywhere. If handovers are triggered eagerly, users bounce between cells (ping-pong); if they are triggered late, links fail. This lecture develops both sides across 2G to 5G and gives each trade-off a quantitative form.

4 Fundamental concepts

4.1 Idle mode versus connected mode

In **idle mode** the UE has no radio connection. It *camps* on a cell, listens to its broadcast and paging channels, and chooses by itself when to move to a better cell (cell reselection). The network only knows the UE's registration area. In **connected mode** the UE has a dedicated radio connection; the network decides the serving cell, based on measurement reports from the UE (network-controlled, mobile-assisted handover).

4.2 Registration areas across generations

- **Location area (LA)** – GSM/UMTS CS domain; updated towards the VLR.
- **Routing area (RA)** – GPRS/UMTS PS domain; a subset of an LA, updated towards the SGSN. Two parallel procedures were needed because the two domains were separate.
- **Tracking area (TA)** – LTE and 5G. The MME (AMF in 5G) gives the UE a *list* of TAs; the UE updates only when it enters a TA outside the list. Lists can be tailored per UE, which removes the ping-pong of updates at a fixed border.
- **RAN notification area (RNA)** – 5G RRC_INACTIVE state: the core considers the UE connected, the RAN keeps its context and pages it within a RAN-configured area.

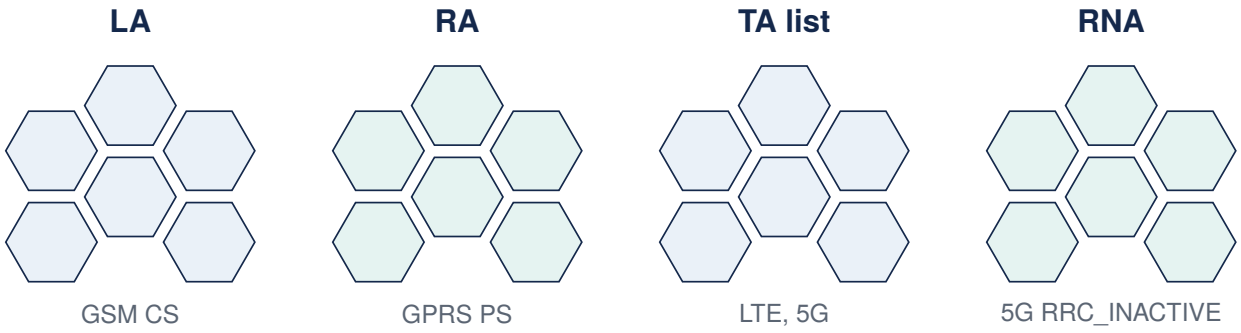


Figure 1. Registration areas of successive generations.

4.3 Measurement quantities

LTE defines two basic measurements on the cell-specific reference signals [1]:

Definition 1 • RSRP and RSRQ

RSRP is the linear average of the power of the resource elements that carry cell-specific reference signals within the measured bandwidth (range about -140 to -44 dBm). *RSRQ* is

$$\mathrm{RSRQ} = \frac{N \cdot \mathrm{RSRP}}{\mathrm{RSSI}}, \tag{1}$$

where N is the number of resource blocks in the RSSI measurement bandwidth and RSSI is the total received power (wanted signal, interference and noise) over that bandwidth.

RSRP measures coverage; RSRQ also reflects load and interference, because RSSI grows when neighbours are busy. NR uses the analogous SS-RSRP and SS-RSRQ on synchronisation signal

blocks.

5 Detailed technical explanation

5.1 Cell reselection in idle mode

In LTE a cell is *suitable* if it satisfies the S-criterion [3]:

$$S_{\text{rxlev}} = Q_{\text{rxlevmeas}} - (Q_{\text{rxlevmin}} + Q_{\text{rxlevminoffset}}) - P_{\text{compensation}} > 0, \tag{2}$$

where $Q_{\text{rxlevmeas}}$ is the measured RSRP, Q_{rxlevmin} the broadcast minimum level and $P_{\text{compensation}}$ a penalty for UEs whose maximum power is below the level assumed by the operator. Among suitable intra-frequency cells the UE ranks the serving and neighbour cells with

$$R_{\text{s}} = Q_{\text{meas,s}} + Q_{\text{hyst}}, \qquad R_{\text{n}} = Q_{\text{meas,n}} - Q_{\text{offset}}, \tag{3}$$

and reselects a neighbour only if it is ranked best for a time $T_{\text{reselection}}$. Inter-frequency and inter-RAT reselection use absolute *priorities*: a UE always moves to a higher-priority layer that is good enough, and leaves to a lower-priority layer only when its current layer becomes poor. Priorities let operators steer idle users towards, for example, a capacity layer or a 5G carrier.

5.2 Handover classification

- By **frequency**: *intra-frequency* (same carrier; measurable without gaps), *inter-frequency* (another carrier; the UE needs measurement gaps unless it has a second receiver) and *inter-RAT* (another technology, e.g. LTE to UMTS).
- By **connection continuity**: *hard* handover (break-before-make; GSM, LTE, NR) and *soft* handover (make-before-break; WCDMA, where several cells transmit the same data simultaneously).
- By **network scope**: within one controller (intra-BSC, intra-eNB), between base stations (X2/Xn handover) or involving the core (inter-MSC in GSM, S1-based in LTE, N2-based in 5G).

Handover in GSM. The BTS measures the uplink and the MS reports the six strongest neighbours every 480 ms on the SACCH. The BSC decides; inter-BSC handovers involve the MSC, and inter-MSC handovers keep the first MSC as an *anchor* for the call [4].

Soft handover in WCDMA. The UE maintains an *active set* of cells; events add a cell that comes within a reporting range of the best cell, remove a cell that falls out of it, or replace the weakest member. The cost is extra downlink power and Iub capacity, typically 30–40% of users being in soft handover.

Handover in LTE. LTE removed the RNC and soft handover. The eNB decides alone, based on measurement *events* configured through RRC [2]:

5.3 Handover in 5G NR

NR keeps the LTE principles (eNB→gNB, X2→Xn, S1→N2) and adds features aimed at the interruption and the failures of hard handover [5]:

Table 1. LTE intra-RAT measurement events (thresholds and offsets are configured by RRC).

Event	Entering condition (simplified)	Typical use
A1	Serving becomes better than a threshold	Stop inter-frequency measurements (remove gaps)
A2	Serving becomes worse than a threshold	Start inter-frequency/inter-RAT measurements
A3	Neighbour becomes offset better than serving (PCell)	Intra-frequency handover
A4	Neighbour becomes better than a threshold	Load-based or coverage-layer handover
A5	Serving worse than threshold 1 and neighbour better than threshold 2	Inter-frequency handover
A6	Neighbour becomes offset better than a secondary cell	Carrier-aggregation management
B1/B2	Inter-RAT neighbour better than a threshold (and serving worse, B2)	LTE to UMTS/GSM, LTE to NR

- **Conditional handover (CHO)**: the network prepares one or more target cells in advance and sends the UE a handover command with an execution condition; the UE executes it when the condition is met, avoiding the race in which the command arrives over a failing link.
- **Dual active protocol stack (DAPS)**: the UE keeps receiving from the source while connecting to the target, approaching zero interruption.
- **Beam management** inside a cell: many beam changes are handled at layers 1/2 without RRC handover.

5.4 Handover failures

A UE declares *radio link failure* (RLF) when, after $N310$ consecutive out-of-sync indications, timer $T310$ expires without recovery; it then attempts RRC re-establishment. Self-organising networks classify failures to tune parameters (mobility robustness optimisation, MRO):

- **Too late** – RLF in the source cell before (or during) the handover; the UE re-establishes in the target. Remedy: earlier triggering (lower offset/TTT).
- **Too early** – RLF shortly after a successful handover; the UE re-establishes in the *source*. Remedy: later triggering.
- **Wrong cell** – RLF shortly after handover; re-establishment in a third cell. Remedy: adjust the cell-individual offsets of the two neighbours.
- **Ping-pong** – a successful handover followed by a return to the source within a short time; wastes signalling and causes interruptions. Remedy: more hysteresis or TTT.

Too-late and ping-pong pull the parameters in opposite directions: this is the central tuning dilemma of handover.

6 Mathematical models

6.1 The A3 condition and the handover point

Engineering problem. Where along a road between two cells does a handover actually happen, and how do the parameters move that point?

The A3 entering condition. In LTE [2] the condition, with all quantities in dB or dBm, is

$$M_{\mathrm{n}} + O_{\mathrm{fn}} + O_{\mathrm{cn}} - \mathrm{Hys} \;>\; M_{\mathrm{p}} + O_{\mathrm{fp}} + O_{\mathrm{cp}} + \mathrm{Off}, \tag{4}$$

where M_{n}, M_{p} are the (layer-3 filtered) measurements of the neighbour and the primary serving cell, $O_{\mathrm{f\cdot}}$ frequency-specific offsets, $O_{\mathrm{c\cdot}}$ cell-individual offsets, Hys the hysteresis and Off the event offset. The condition must hold continuously for T_{TTT} before the UE reports. With zero frequency and cell offsets it reduces to $M_{\mathrm{n}} - M_{\mathrm{p}} > H$ with $H = \mathrm{Off} + \mathrm{Hys}$.

System model. Two eNBs at distance D with equal transmit power; the UE moves along the line joining them; deterministic path loss with exponent n (no shadowing). At distance x from the serving eNB,

$$M_{\mathrm{n}} - M_{\mathrm{p}} = 10n \log_{10} \frac{x}{D - x}. \tag{5}$$

Derivation. Setting $10n \log_{10}\big(x/(D-x)\big) = H$ and solving,

$$x_{\mathrm{HO}} = D\,\frac{r}{1+r}, \qquad r = 10^{H/(10n)}. \tag{6}$$

The UE then travels a further $v\,T_{\mathrm{TTT}}$ before reporting.

Interpretation. For $D = 1000\,\mathrm{m}$, $n = 3.5$ and $H = 3\,\mathrm{dB}$: $r = 1.218$ and $x_{\mathrm{HO}} = 549\,\mathrm{m}$, i.e. 49 m past the cell border; $H = 6\,\mathrm{dB}$ moves it to 597 m. At 30 m/s a TTT of 320 ms adds another 9.6 m (Figure 2). Every decibel of margin is paid for with signal quality on the old cell, where the link is already the weakest – the origin of too-late failures at high speed.

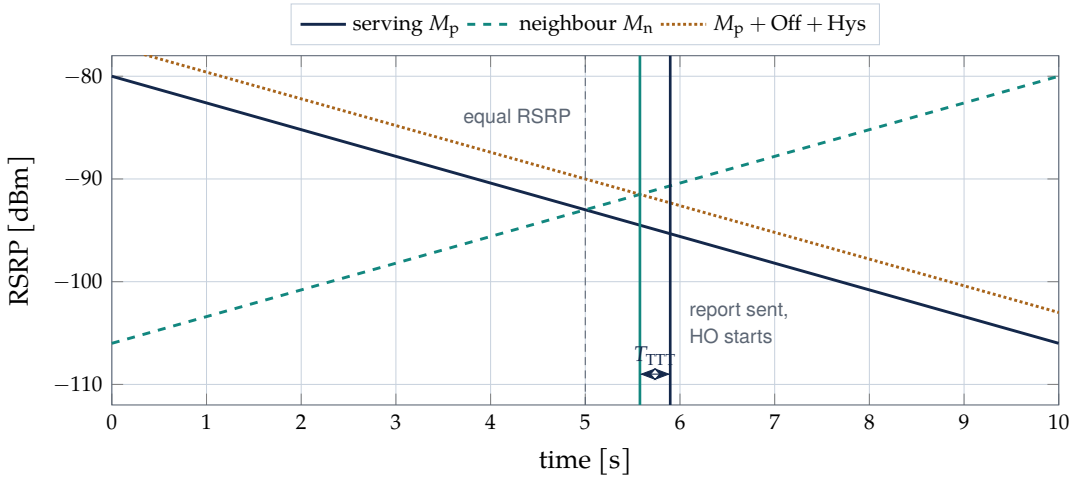


Figure 2. A3 triggering: the neighbour must exceed the serving cell by $\mathrm{Off} + \mathrm{Hys} = 3\,\mathrm{dB}$ (dotted) for $T_{\mathrm{TTT}} = 320\,\mathrm{ms}$ before the measurement report is sent.

6.2 Shadowing and ping-pong

Model. Add independent log-normal shadowing $X_{\mathrm{s}}, X_{\mathrm{n}} \sim \mathcal{N}(0, \sigma^2)$ (dB) to both cells. At the cell border the mean difference is zero, and $M_{\mathrm{n}} - M_{\mathrm{p}} \sim \mathcal{N}(0, 2\sigma^2)$.

Result. The probability that a single sample at the border exceeds the margin H is

$$P_{\text{trig}} = Q\left(\frac{H}{\sqrt{2}\,\sigma}\right). \tag{7}$$

With $\sigma = 8$ dB and $H = 3$ dB, $P_{\text{trig}} = Q(0.265) = 0.40$; even with $H = 6$ dB it is still 0.30. Hysteresis alone cannot suppress spurious triggers caused by large-scale fading; TTT and layer-3 filtering are needed. The layer-3 filter

$$F_k = (1-a)\,F_{k-1} + a\,M_k, \qquad a = 2^{-K/4}, \tag{8}$$

with filter coefficient K, reduces the variance of *independent* fluctuations (fast fading, measurement error) by the factor $a/(2-a)$ – one third for $K = 4$ – but it cannot remove shadowing, which is correlated over tens of metres. The model is pessimistic in one respect (it ignores the correlation of shadowing between the two cells) and optimistic in another (it looks at one point only); simulation along trajectories is the usual next step.

6.3 Sizing a registration area

Problem. How many cells should a location or tracking area contain?

Model and assumptions. Square cells of side a; a registration area of N cells forms a square of side $a\sqrt{N}$ and perimeter $L = 4a\sqrt{N}$. Users are uniformly distributed and move in uniformly random directions at mean speed v (the *fluid-flow* model [6]): the rate at which users of an area S cross its perimeter L is $\rho v L/\pi$, where ρ is the user density. Each user receives λ_c calls per unit time, each paged in all N cells. One update costs c_u and paging one cell costs c_p.

Derivation. Per user, the update rate is $\rho v L/(\pi\rho S) = vL/(\pi S)$ with $S = Na^2$, so

$$C(N) = c_u\,\frac{4v}{\pi a\sqrt{N}} + c_p\,\lambda_c\,N. \tag{9}$$

The first term decreases and the second increases with N. Setting $\mathrm{d}C/\mathrm{d}N = 0$,

$$-\frac{2c_u v}{\pi a}N^{-3/2} + c_p\lambda_c = 0 \quad\Longrightarrow\quad N^{*} = \left(\frac{2\,c_u\,v}{\pi\,a\,c_p\,\lambda_c}\right)^{2/3}. \tag{10}$$

Interpretation. Fast users (v large) favour large areas; users who receive many calls (λ_c large) favour small ones. Because the optimum depends on the user, LTE's per-UE tracking-area lists can approach it user by user, which a fixed LA plan cannot.

7 Network architecture and diagrams

The LTE handover architecture is flat: the source and target eNBs negotiate directly over X2, and the MME and S-GW are involved only at the end to move the downlink path (*path switch*). If no X2 exists, or if the MME or S-GW must change, the same steps run through the MME over S1 (*S1-based handover*). In 5G the roles map to gNB, Xn, AMF/SMF and UPF (Lecture 20). GSM and UMTS involve their controllers (BSC/RNC) and, across controller borders, the MSC/SGSN.

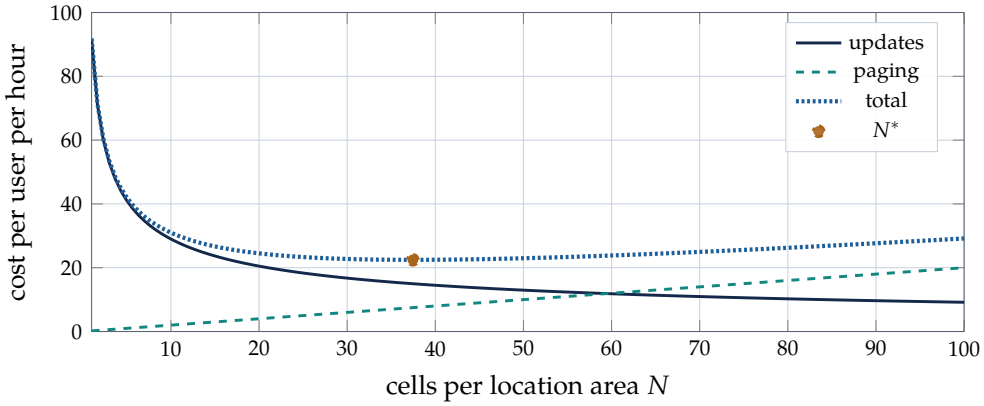


Figure 3. Update, paging and total signalling cost of Eq. (9) for $c_u = 1$, $c_p = 0.1$, $v = 10\,\mathrm{m/s}$, $a = 500\,\mathrm{m}$, $\lambda_c = 2$ calls per hour (Example 3).

8 Step-by-step procedures

Procedure 1 • LTE X2-based handover

1. **Preparation trigger.** The UE sends a Measurement Report after the A3 condition held for TTT. The source eNB decides to hand over.
2. **Preparation.** The source sends X2AP Handover Request with the UE context (bearers, security, RRC configuration). The target performs admission control, reserves resources (including a dedicated RACH preamble) and answers Handover Request Acknowledge containing the RRC message for the UE.
3. **Execution.** The source sends RRC Connection Reconfiguration with mobility control information, stops scheduling the UE, sends SN Status Transfer and starts forwarding buffered and in-transit downlink packets to the target over X2.
4. **Access.** The UE synchronises to the target, performs contention-free random access (Lecture 8), receives a timing advance and an uplink grant, and sends RRC Connection Reconfiguration Complete. The interruption ends.
5. **Completion.** The target asks the MME to move the path with S1AP Path Switch Request; the MME sends Modify Bearer Request to the S-GW, which switches the downlink tunnel and sends *end marker* packets on the old path so that the target knows when forwarded data is finished.
6. **Release.** After Path Switch Request Acknowledge, the target sends UE Context Release; the source frees its resources.

The user-plane interruption lasts from the moment the UE leaves the source until it can be scheduled in the target; it is of the order of a few tens of milliseconds in LTE and is the quantity that RACH-less handover, make-before-break and DAPS reduce.

9 Worked examples

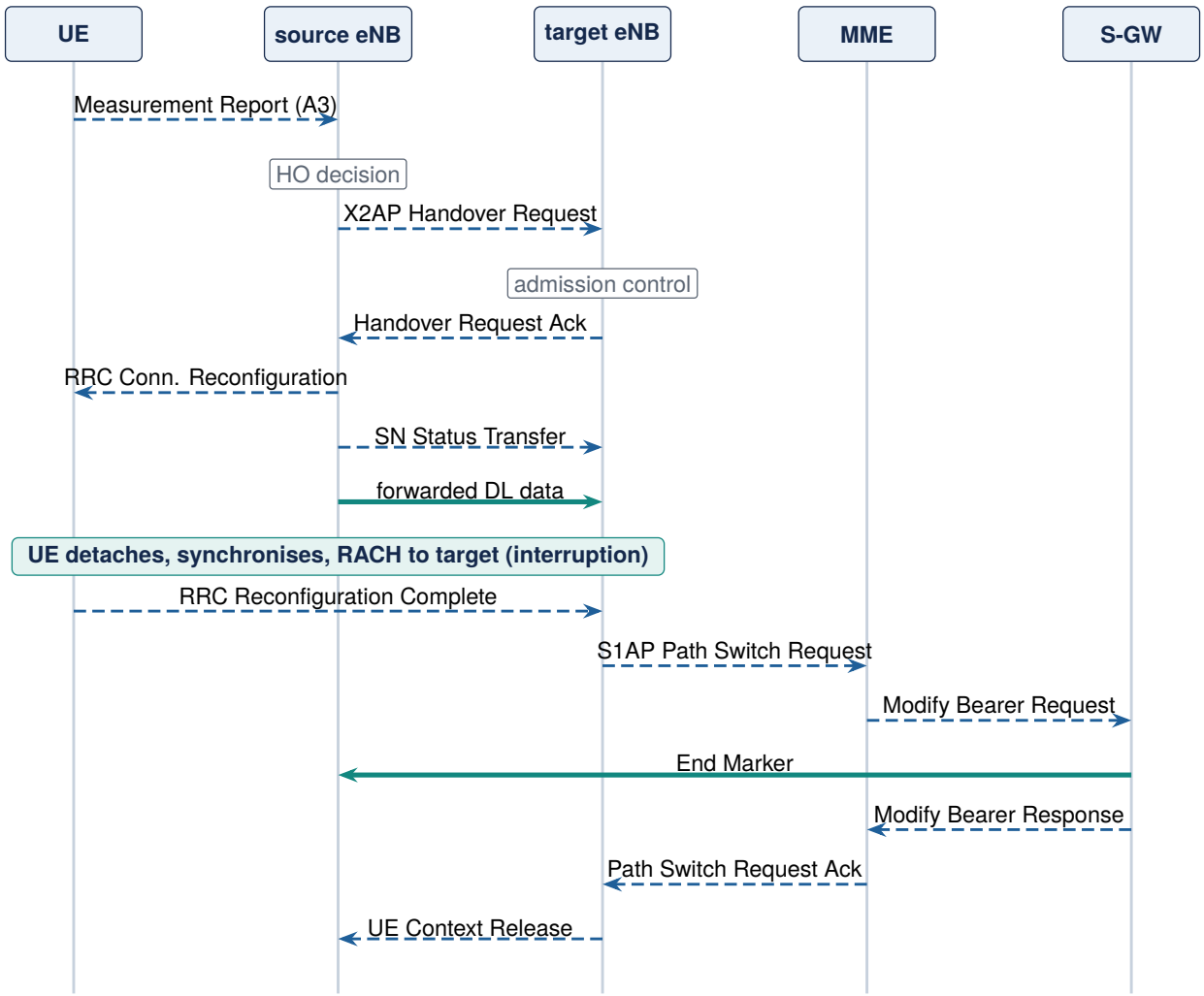


Figure 4. X2-based handover in LTE (Procedure 1). Teal arrows: user-plane data forwarding and end marker.

Worked example 1 • Handover point and speed

Problem. Inter-site distance 1000 m, $n = 3.5$, Off + Hys = 3 dB, $T_{\mathrm{TTT}} = 320$ ms. Where is the measurement report sent for a pedestrian (1.5 m/s) and for a car on a motorway (120 km/h)?
Solution. From Eq. (6), $x_{\mathrm{HO}} = 549$ m. The pedestrian adds $1.5 \times 0.32 = 0.5$ m; the car adds $33.3 \times 0.32 = 10.7$ m, reporting at ≈ 560 m.
Interpretation. Then preparation and execution take further tens of milliseconds, during which the car moves another 1–3 m per 30 ms. Motorway cells are therefore often given a lower offset or TTT, and speed-dependent scaling of TTT is part of the standard.

Worked example 2 • RSRQ under load

Problem. A UE measures RSRP $= -90$ dBm over $N = 50$ resource blocks. Compute RSRQ when the RSSI is -65 dBm (lightly loaded neighbours) and -60 dBm (heavily loaded).
Solution. RSRQ $= 10 \log_{10} 50 +$ RSRP $-$ RSSI $= 17.0 - 90 + 65 = -8.0$ dB, and $17.0 - 90 + 60 = -13.0$ dB.
Interpretation. Same coverage, 5 dB worse quality: RSRQ sees the load of the neighbours, RSRP does not. Using RSRQ for A3 spreads load, at the cost of more volatile reports.

Worked example 3 • Tracking-area size

Problem. With the parameters of Figure 3, find the optimal number of cells per area and the cost per user and hour.
Solution. $\lambda_c = 2/3600\,\mathrm{s}^{-1}$. Equation (10): $N^* = \left(2 \times 1 \times 10/(\pi \times 500 \times 0.1 \times 2/3600)\right)^{2/3} = (229.2)^{2/3} = 37.5$. A 6×6 area ($N = 36$) gives $C = 3600\,[4 \times 10/(\pi \times 500 \times 6) + 0.1 \times (2/3600) \times 36] = 15.28 + 7.20 = 22.5$ cost units per user-hour, against 91.9 for $N = 1$ and 29.2 for $N = 100$.
Interpretation. The optimum is flat: anything between about 20 and 60 cells costs within 10% of the minimum. Operators can therefore choose area borders for practical reasons (along low-traffic lines, away from busy roads) without losing much.

10 Technology comparison

Hard versus soft. Soft handover gives diversity gain and no interruption, but needs same-frequency neighbours, a central combining point (the RNC) and extra resources. LTE chose a flat architecture without a central RAN node, so soft handover was impossible; it compensated with fast hard handover, data forwarding and, later, DAPS. The trade-off – *centralised combining versus flat, fast decisions* – reappears in 5G with the CU/DU split (Lecture 14).

11 Key takeaways

Table 2. Connected-mode mobility from 2G to 5G.

	GSM	UMTS (WCDMA)	LTE	NR (5G)
Type	Hard	Soft (same freq.) + hard	Hard	Hard; CHO and DAPS options
Decision	BSC	RNC	Source eNB	Source gNB
Measurements	RXLEV, RXQUAL	CPICH RSCP, E_c/N_0	RSRP, RSRQ, SINR	SS-RSRP, SS-RSRQ, SS-SINR, CSI-RS
Inter-node link	A (via MSC)	Iur	X2 (or S1)	Xn (or N2)
Core involvement	MSC anchor	SRNC relocation	Path switch (MME, S-GW)	Path switch (AMF, SMF, UPF)
Idle area	LA (RA for PS)	LA + RA	TA list	TA list + RNA

Key takeaways

- Idle mobility is UE-controlled (reselection, area updates); connected mobility is network-controlled and mobile-assisted (measurement reports, handover).
- Area size trades update cost against paging cost: $N^* = (2c_u v/(\pi a c_p \lambda_c))^{2/3}$; TA lists let the network tailor it per UE.
- RSRP measures coverage, RSRQ $= N$ RSRP/RSSI adds load and interference.
- A3: $M_n - M_p > \text{Off} + \text{Hys}$ (plus offsets) for T_{TTT}; the handover point moves to $x = Dr/(1+r)$, $r = 10^{H/(10n)}$.
- With 8 dB shadowing a 3 dB margin is crossed by chance 40% of the time at the border: TTT and filtering are essential.
- X2 handover: report → request/ack → command → forwarding → RACH → complete → path switch → release.
- Too-late and ping-pong pull parameters in opposite directions; MRO, CHO and DAPS address the dilemma.

12 Exercises

Exercise 1 • Level 1 – conceptual • Who decides?

For each of the following, state whether the UE or the network takes the decision and on which information: (a) LTE cell reselection; (b) LTE handover; (c) WCDMA active-set update; (d) 5G conditional handover execution.

Exercise 2 • Level 1 – conceptual • Two update procedures

Explain why a GPRS-capable GSM phone could perform both a location area update and a routing area update at the same border, and how LTE avoided this duplication.

Exercise 3 • Level 2 – numerical • Offsets

In Example 1 the operator sets a cell-individual offset $O_{cn} = +2\,\text{dB}$ for the neighbour (to off-load the serving cell). Where is the A3 condition now met? Which failure type becomes more likely?

Exercise 4 • Level 2 – numerical • Filtering

Measurements are taken every 40 ms with independent errors of standard deviation 4 dB. Using Eq. (8), compute the standard deviation after filtering for $K = 4$ and $K = 8$, and the filter's time constant (number of samples for the impulse response to fall to $1/\text{e}$).

Exercise 5 • Level 3 – analytical • Optimal area with two classes

A tracking area serves 70% pedestrians ($v = 1.5\,\text{m/s}$, 1 call/hour) and 30% vehicles ($v = 15\,\text{m/s}$, 3 calls/hour). Using Eq. (9) with the other parameters of Example 3, find the single N that minimises the average cost, and the optimal N if each class had its own TA list. How much does per-class tailoring save?

Exercise 6 • Level 3 – analytical • Too late

Assume RLF occurs if the serving RSRP falls 8 dB below the neighbour before the handover command is received, and that preparation plus command delivery take 80 ms after the report. For the geometry of Example 1 without shadowing, find the maximum speed for which no too-late failure occurs with $H = 3\,\text{dB}$ and $T_{\text{TTT}} = 320\,\text{ms}$. Discuss which parameter you would change for a high-speed train line.

Solutions

Solution to Exercise 1

(a) UE, using broadcast parameters (Q_{rxlevmin}, Q_{hyst}, priorities) and its own measurements. (b) Source eNB, using UE measurement reports and its own load/admission knowledge. (c) RNC, using UE event reports (1A/1B/1C). (d) The UE executes, but only towards targets prepared by the network and when the network-configured condition holds – a shared decision.

Solution to Exercise 2

LA and RA belonged to two separate core domains (MSC/VLR and SGSN) with separate procedures. Unless the optional Gs interface allowed the SGSN to update the MSC on the phone's behalf (combined updates), the phone signalled twice. LTE has a single packet core with one mobility entity (MME) and one area concept (TA); CS-related registration, where needed for CS fallback, is performed by the MME over SGs.

Solution to Exercise 3

The condition becomes $M_n + 2 - M_p > 3$, i.e. $H_{\text{eff}} = 1\,\text{dB}$: $r = 10^{1/35} = 1.068$, $x_{\text{HO}} = 1000 \times 1.068/2.068 = 516\,\text{m}$. Handovers happen earlier; ping-pong and too-early failures become more likely, too-late less likely.

Solution to Exercise 4

Variance factor $a/(2-a)$. $K=4$: $a=0.5$, factor $1/3$, $\sigma_F = 4/\sqrt{3} = 2.31$ dB. $K=8$: $a=0.25$, factor $1/7$, $\sigma_F = 1.51$ dB. Impulse response $(1-a)^k$ falls to $1/\mathrm{e}$ after $k = -1/\ln(1-a)$: 1.44 samples (58 ms) for $K=4$, 3.48 samples (139 ms) for $K=8$. Better smoothing costs reaction time.

Solution to Exercise 5

Per-user cost $C_i(N) = 4c_\mathrm{u}v_i/(\pi a\sqrt{N}) + c_\mathrm{p}\lambda_i N$. The average has the same form with $\bar{v} = 0.7\times 1.5 + 0.3\times 15 = 5.55$ m/s and $\bar{\lambda} = 0.7\times 1 + 0.3\times 3 = 1.6$ calls/h, so $N^* = (2\times 5.55/(\pi\times 500\times 0.1\times 1.6/3600))^{2/3} = 29.3$, cost ≈ 14.1 units per user-hour. Separately: pedestrians $N^* = 16.8$ (cost 5.0), vehicles $N^* = 37.5$ (cost 33.7), average $0.7\times 5.0 + 0.3\times 33.7 = 13.6$. Tailoring saves only about 3% here, because the cost curve is flat near its optimum; it pays off more when the classes differ much more (e.g. stationary IoT devices versus trains). (The script `scripts/validate_examples.py` reproduces these values.)

Solution to Exercise 6

The RLF point is where $M_\mathrm{n} - M_\mathrm{p} = 8$ dB: $r = 10^{8/35} = 1.693$, $x_\mathrm{RLF} = 1000\times 1.693/2.693 = 629$ m. The report is triggered at 549 m and the command arrives after $T_\mathrm{TTT} + 80$ ms $= 0.40$ s. The UE must not travel more than $629 - 549 = 80$ m in that time: $v_\mathrm{max} = 80/0.40 = 200$ m/s. Without shadowing there is ample margin; with shadowing, fading of the serving cell can cut this distance drastically. For a train line, one would reduce TTT (speed-dependent scaling), lower the offset, deploy cells along the track with larger overlap, or use conditional handover.

13 Further reading

LTE mobility is described in [1], idle-mode procedures in [3] and the measurement events in [2]; GSM/UMTS handover in [4]; NR mobility, CHO and DAPS in [5]. Akyildiz et al. [6] survey location management and the fluid-flow model. Sesia et al. [8] and Holma and Toskala [7] discuss handover parameter tuning in LTE.

References

[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA) and evolved universal terrestrial radio access network (E-UTRAN); overall description; stage 2*, TS 36.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36300.htm.

[2] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); radio resource control (RRC); protocol specification*, TS 36.331, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36331.htm.

[3] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); user equipment (UE) procedures in idle mode*, TS 36.304, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36304.htm.

[4] 3GPP, *Handover procedures*, TS 23.009, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23009.htm.

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[6] I. F. Akyildiz, J. McNair, J. S. M. Ho, H. Uzunalioğlu, and W. Wang, "Mobility management in next-generation wireless systems", *Proceedings of the IEEE*, vol. 87, no. 8, pp. 1347–1384, 1999.

[7] H. Holma and A. Toskala, *LTE for UMTS: Evolution to LTE-Advanced*, 2nd ed. Wiley, 2011.

[8] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. Wiley, 2011.

05

Lecture 05

4G LTE Architecture

معماری LTE نسل چهارم

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 1–4; OFDM basics; IP networking

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-05-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the limitations of 3G that motivated LTE and list the main LTE requirements.
- **LO2** **State** the LTE design principles – flat, all-IP, shared channels, no soft handover – and the engineering reason for each.
- **LO3** **Draw** the Evolved Packet System (E-UTRAN + EPC) and **name** every reference point (S1-MME, S1-U, X2, S5/S8, S6a, S11, SGi, Gx, Rx).
- **LO4** **Derive** the LTE downlink peak rate from the resource grid and **build** a user-plane latency budget.
- **LO5** **Describe** how voice is carried in an all-IP system (IMS, VoLTE, CS fallback, SRVCC).
- **LO6** **Explain** control/user plane separation (CUPS) and its role as a bridge to 5G.

2 Prerequisite knowledge

You should know the UMTS architecture and its CS/PS split (Lecture 3), the GTP tunnel of GPRS (Lecture 1) and the basic idea of OFDM: a wideband channel divided into many narrow, orthogonal subcarriers, each protected from inter-symbol interference by a cyclic prefix.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
LTE	تکامل بلندمدت	3GPP radio access of 4G (E-UTRA), from Release 8.
EPS	سامانهٔ بستهٔ تکامل‌یافته	Complete 4G system: E-UTRAN + EPC.
eNB (eNodeB)	ایستگاه پایهٔ LTE	Base station incl. radio resource management; no RNC.
EPC	هستهٔ بستهٔ تکامل‌یافته	All-IP core: MME, S-GW, P-GW, HSS, PCRF.
MME	موجودیت مدیریت تحرک	Control-plane node: NAS, mobility, security, bearer control.
S-GW / P-GW	دروازهٔ خدمت‌دهنده / دروازهٔ PDN	User-plane anchors: mobility anchor / IP anchor.
EPS bearer	حامل EPS	End-to-end QoS pipe between UE and P-GW.
OFDMA / SC-FDMA	دسترسی چندگانهٔ متعامد / تک‌حامل	LTE downlink / uplink multiple access.
VoLTE	صدا روی LTE	IMS-based voice over the EPS bearer of QCI 1.
CUPS	جدایی صفحهٔ کنترل و کاربر	Rel-14 split of S/P-GW into C and U parts (PFCP).

3 Introduction and motivation

By 2004 HSPA was delivering mobile broadband, but its foundations were showing their age. WCDMA's 5 MHz carrier could not grow to the 10–20 MHz needed for higher rates without complex equalisers; the hierarchy UE–NodeB–RNC–SGSN–GGSN added latency and cost; two parallel cores (CS and PS) doubled operations. Meanwhile competing technologies (WiMAX) were promising OFDM-based, all-IP broadband.

3GPP's Long Term Evolution study set ambitious targets [8]: peak rates of 100 Mbit/s downlink and 50 Mbit/s uplink in 20 MHz; user-plane latency below 5 ms; idle-to-active transition below 100 ms; 2–4 times the spectral efficiency of Release-6 HSPA; scalable bandwidth from 1.4 to 20 MHz; and support for speeds up to 350 km/h. The *System Architecture Evolution* (SAE) defined a matching core, the EPC. Together they form the Evolved Packet System (EPS), first specified in Release 8 (2008) [3], [4].

4 Fundamental concepts

4.1 Design principles and their reasons

- **Flat RAN.** The RNC is removed; its functions move to the eNB (radio resource control, scheduling, ciphering, handover decision) or to the core. Fewer nodes means lower latency and cost. The price is that no node can combine signals from several cells – hence no soft handover (Lecture 4).
- **All-packet core.** There is no circuit-switched domain. Voice must be carried as IP packets (VoLTE over IMS), or temporarily handed to 2G/3G (CS fallback).
- **Shared channels only.** All user data uses shared channels scheduled every 1 ms by the eNB; there are no dedicated traffic channels.
- **OFDMA downlink, SC-FDMA uplink.** OFDM scales gracefully with bandwidth and makes frequency-selective scheduling and MIMO simple. In the uplink, single-carrier FDMA keeps a low peak-to-average power ratio for the handset's amplifier (Lecture 7).
- **Separation of control and user planes** in the core: the MME never touches user packets; the gateways never terminate NAS signalling.
- **Self-organising networks (SON).** Automatic neighbour relations, mobility robustness and load balancing reduce manual planning in a network of many small, flat nodes.

4.2 The EPS bearer

Definition 1 • EPS bearer

An EPS bearer is a logical pipe between the UE and the P-GW with a single QoS treatment. It is the concatenation of a *radio bearer* (UE–eNB), an *S1 bearer* (eNB–S-GW, a GTP-U tunnel) and an *S5/S8 bearer* (S-GW–P-GW, another GTP-U tunnel). Every UE gets a *default bearer* at attach; *dedicated bearers* are added for traffic needing different QoS, such as VoLTE (QCI 1).

Unlike GPRS, where connectivity required a separate PDP context activation after attach, LTE establishes the default bearer *during* attach: an LTE device is always on with an IP address. Bearers and QoS classes are developed in Lectures 9 and 22.

5 Detailed technical explanation

5.1 Network elements

UE. Mobile equipment plus UICC (with the USIM application). LTE UEs are classified by *categories* specifying peak rates, MIMO layers and buffer sizes.

eNB. The only node of the E-UTRAN [3]. It runs the full radio stack (PHY, MAC, RLC, PDCP, RRC), schedules uplink and downlink every millisecond, performs admission control, handover decisions, header compression and ciphering of user data, routes user data to the S-GW and selects the MME for a UE at attach. eNBs connect to each other over X2 and to the core over S1.

MME. The control-plane brain of the EPC: terminates NAS signalling and its security, authenticates the UE with vectors from the HSS, manages tracking areas and paging, selects S-GW and P-GW, and sets up, modifies and releases bearers through the S-GW. It is the subject of Lecture 12.

S-GW. Terminates S1-U and is the *local mobility anchor*: during inter-eNB handover only the S1-U tunnel changes. It buffers downlink data for idle UEs and triggers paging through the MME, and it anchors mobility to 2G/3G.

P-GW. The *IP anchor*: allocates the UE's IP address, is the point of attachment to a packet data network (identified by an APN), enforces policy and charging (PCEF), performs deep packet inspection and lawful interception. A UE may have connections to several P-GWs for different APNs (e.g. `internet` and `ims`). Lecture 13 studies both gateways.

HSS. The subscriber database, evolved from the HLR: subscription profile, AuC functions, current MME. Lecture 11.

PCRF. The policy and charging rules function [7]: decides QoS and charging rules per service data flow and pushes them to the P-GW over Gx; application functions such as the IMS P-CSCF request resources over Rx.

5.2 Reference points

Table 1. Main EPS reference points [4].

Reference point	Between	Protocol	Purpose
LTE-Uu	UE – eNB	RRC, PDCP/RLC/MAC/PHY	Radio interface
S1-MME	eNB – MME	S1AP over SCTP	UE context, bearers, paging, handover
S1-U	eNB – S-GW	GTP-U over UDP/IP	User data per bearer
X2	eNB – eNB	X2AP / GTP-U	Handover, data forwarding, interference coordination
S11	MME – S-GW	GTPv2-C	Session and bearer management
S5/S8	S-GW – P-GW	GTPv2-C + GTP-U	S5 within a network, S8 between home and visited
S6a	MME – HSS	Diameter	Authentication vectors, location update, subscription
SGi	P-GW – PDN	IP	To the Internet, IMS or corporate networks
Gx	PCRF – P-GW	Diameter	Policy and charging rules
Rx	AF – PCRF	Diameter	Application requests for QoS (e.g. VoLTE)
Gy/Gz	P-GW – OCS/OFCS	Diameter / file	Online / offline charging

Two patterns stand out. First, every control interface of the core is either GTPv2-C (between nodes that manage tunnels) or Diameter (towards databases and policy) – SS7 and MAP have disappeared from the EPC. Second, S8 is the *roaming* variant of S5: with *home routing* the visited S-GW tunnels traffic back to the home P-GW, so the home operator keeps control of IP addressing and charging.

5.3 Voice in an all-IP system

Without a CS domain, LTE offers three voice solutions:

- **VoLTE** – voice as an IMS service [6]. The UE registers with IMS over a default bearer to the `ims` APN (QCI 5 for SIP signalling). At call set-up the P-CSCF asks the PCRF over Rx for resources; the PCRF instructs the P-GW over Gx to create a dedicated GBR bearer with QCI 1 for the speech packets (Lecture 10).
- **CS fallback (CSFB)** – the UE is registered in the 2G/3G CS domain via the MME (SGs interface); for a call it is redirected to 2G/3G [2]. Simple, but call set-up is slower and data speed drops during the call.
- **SRVCC** – single radio voice call continuity: an ongoing VoLTE call is handed over to the CS domain when the UE leaves LTE coverage [9].

5.4 Control and user plane separation (CUPS)

In Release 8 the S-GW and P-GW each contained both control logic (GTP-C, policy, IP allocation) and packet forwarding. Release 14 split them [1]: SGW-C and PGW-C keep the control logic and program SGW-U and PGW-U through the *Sx* interface with the Packet Forwarding Control Protocol (PFCP) [5]. The user planes can then be scaled independently and placed close to the radio for low latency (edge computing), while control remains centralised. CUPS is the direct ancestor of the 5G SMF/UPF split (Lectures 13 and 19).

6 Mathematical models

6.1 Downlink peak rate from the resource grid

Problem. Where does the "300 Mbit/s" of a Category-5 LTE device come from?

System model. Bandwidth 20 MHz, normal cyclic prefix, FDD. The resource grid (Lecture 7) has $N_{\mathrm{RB}} = 100$ resource blocks of 12 subcarriers at $\Delta f = 15\,\mathrm{kHz}$, and 14 OFDM symbols per 1 ms subframe, i.e. $12 \times 14 = 168$ resource elements (REs) per RB per subframe. Assumptions: one OFDM symbol for the control region, four cell-specific reference-signal ports, 64QAM ($Q_m = 6$), $\nu = 4$ MIMO layers, and we neglect the synchronisation and broadcast channels, which occupy a few subframes only.

Derivation. Four CRS ports use 24 REs per RB pair, four of which lie in the control symbol; the data REs per RB per subframe are therefore

$$N_{\mathrm{RE}}^{\mathrm{data}} = 168 - \underbrace{12}_{\text{control}} - \underbrace{20}_{\text{CRS}} = 136. \tag{1}$$

The peak rate is

$$R_{\mathrm{peak}} = \nu\, Q_m\, r_{\mathrm{c}}\, \frac{N_{\mathrm{RB}}\, N_{\mathrm{RE}}^{\mathrm{data}}}{T_{\mathrm{sf}}} = 4 \times 6 \times r_{\mathrm{c}} \times \frac{100 \times 136}{1\,\mathrm{ms}} = 326.4\, r_{\mathrm{c}}\ \mathrm{Mbit/s}, \tag{2}$$

where r_{c} is the channel-code rate and T_{sf} the subframe duration. With the highest standardised transport block ($r_{\mathrm{c}} \approx 0.92$) this gives ≈ 300 Mbit/s, consistent with the 299.6 Mbit/s of UE Category 5.

Interpretation. The reference signals and control region cost about 19% of the REs; the cyclic prefix (6.7% of time, Exercise 3) is already built into the 14-symbol subframe. Peak rates need four receive

antennas, excellent SINR and an empty cell – they are an upper bound of the air interface, not a user experience.

6.2 User-plane latency budget

Problem. Can LTE meet the 5 ms one-way user-plane latency target?

Model. One-way latency from the arrival of a packet at the eNB's PDCP to its delivery at the UE's IP layer, unloaded cell, decomposed as

$$D_{\mathrm{UP}} = \underbrace{\tfrac{1}{2}T_{\mathrm{TTI}}}_{\text{alignment}} + T_{\mathrm{TTI}} + D_{\mathrm{eNB}} + D_{\mathrm{UE}} + p_{\mathrm{re}}\, T_{\mathrm{HARQ}} + D_{\mathrm{tr}}, \tag{3}$$

where the alignment term is the mean wait for the next TTI boundary, D_{eNB} and D_{UE} are processing delays, p_{re} is the probability of one HARQ retransmission, $T_{\mathrm{HARQ}} = 8$ ms the FDD HARQ round trip and D_{tr} the residual transport delay.

Numerical example. With $T_{\mathrm{TTI}} = 1$ ms, $D_{\mathrm{eNB}} = D_{\mathrm{UE}} = 1$ ms, $p_{\mathrm{re}} = 0.1$ and $D_{\mathrm{tr}} = 0.5$ ms (illustrative values, Figure 1): $D_{\mathrm{UP}} = 0.5 + 1 + 1 + 1 + 0.8 + 0.5 = 4.8$ ms.

Interpretation and limits. The 1 ms TTI and the 8 ms HARQ loop dominate: these are the levers that 5G pulls (shorter slots, mini-slots, faster HARQ, Lecture 14). The model ignores queueing (zero load), scheduling-request delay in the uplink (a UE without a grant must first ask for one, adding several milliseconds) and core and Internet delay. Measured end-to-end ping times in LTE networks are therefore typically tens of milliseconds even though the radio target is 5 ms.

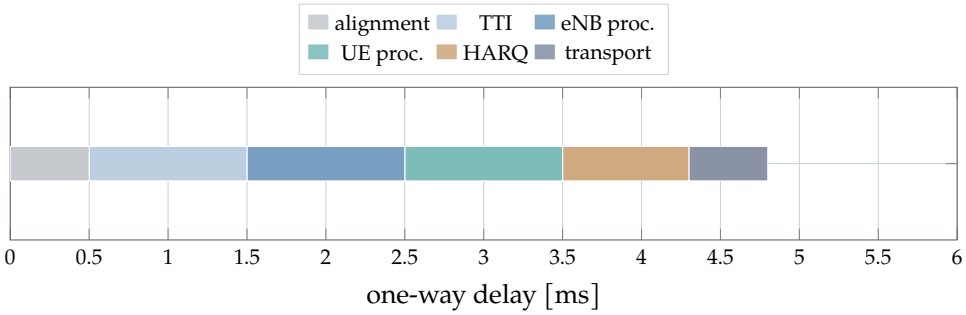


Figure 1. Illustrative one-way downlink user-plane latency budget of Eq. (3) (4.8 ms).

6.3 Encapsulation overhead

Every user packet crossing S1-U or S5 is carried in GTP-U/UDP/IPv4, adding $h = 8 + 8 + 20 = 36$ bytes. For a payload of L bytes the transport efficiency is

$$\eta_{\mathrm{tr}} = \frac{L}{L + h}. \tag{4}$$

A 1400-byte web packet has $\eta_{\mathrm{tr}} = 97.5\%$; a 60-byte VoLTE packet (speech frame plus RTP/UDP/IP headers; header compression acts only on the radio link) only 62.5%. Backhaul dimensioning for voice must account for this (Exercise 4); IPv6 transport adds a further 20 bytes.

7 Network architecture and diagrams

Figure 2 shows the EPS. Compared with UMTS (Lecture 3, Figure 3) the RAN has one node type, the core has no CS domain, and the user plane crosses only two core nodes (S-GW and P-GW), both of which can be co-located. Control (dashed) and user (solid teal) planes use different interfaces everywhere except the radio. Figure 3 shows the user-plane stack: radio protocols end in the eNB, and from there GTP-U tunnels carry the user's IP packets.

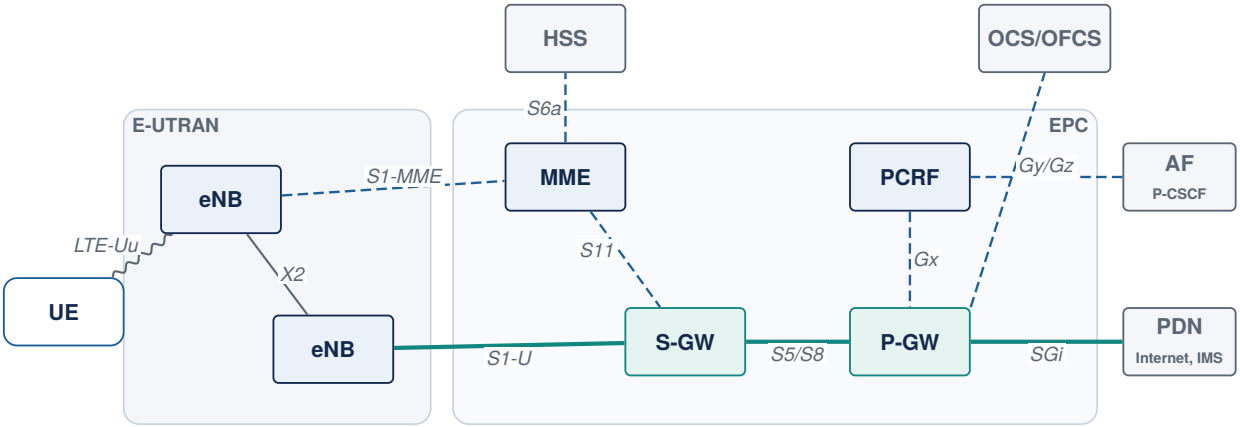


Figure 2. The Evolved Packet System. Dashed blue: control plane; solid teal: user plane.

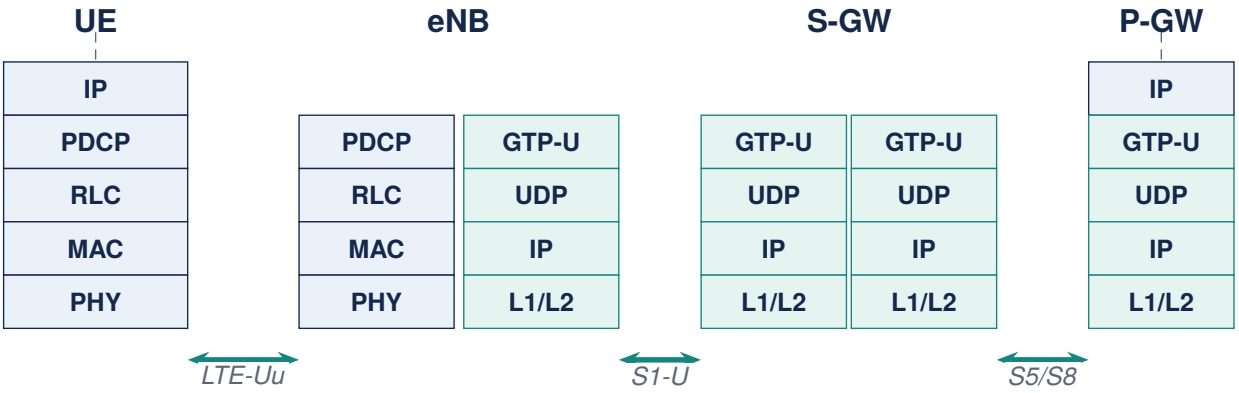


Figure 3. LTE user-plane protocol stack. The UE's IP packet is carried by PDCP/RLC/MAC over the radio and by GTP-U tunnels on S1-U and S5/S8; the P-GW is the IP anchor.

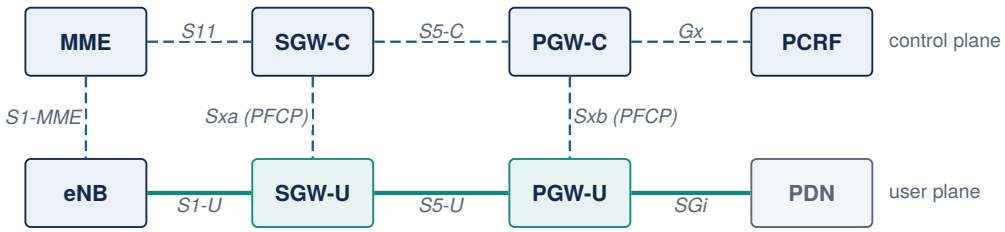


Figure 4. Control and user plane separation of the EPC gateways (Release 14).

8 Step-by-step procedures

Procedure 1 • LTE initial attach (overview)

1. The UE performs random access and sets up an RRC connection with the eNB (Lectures 6 and 8).
2. It sends a NAS Attach Request with a PDN Connectivity Request; the eNB selects an MME and forwards it in S1AP Initial UE Message.
3. The MME authenticates the UE with EPS-AKA (vectors from the HSS over S6a), activates NAS security, and registers itself in the HSS (Update Location), receiving the subscription.
4. The MME selects an S-GW and a P-GW and sends GTPv2-C Create Session Request; the P-GW allocates an IP address, obtains policy from the PCRF and creates the default bearer.
5. The MME sends S1AP Initial Context Setup Request with Attach Accept; the eNB activates AS security and configures the radio bearer with RRC Connection Reconfiguration.
6. The UE answers Attach Complete; the MME sends Modify Bearer Request so that the S-GW learns the eNB's tunnel endpoint. Uplink and downlink data can flow.

The detailed message flow, with every information element that matters, is built up in Lectures 9, 11 and 12. Note already that one procedure delivers registration, security *and* an IP address – three separate GPRS procedures merged into one.

9 Worked examples

Worked example 1 • Peak rate of a 10 MHz cell

Problem. A 10 MHz LTE carrier (50 RBs) uses two CRS ports, two control symbols and 2×2 MIMO with 64QAM. Estimate the peak rate for $r_c = 0.9$.
Solution. Two CRS ports use 16 REs per RB pair, of which 4 lie in the first (control) symbol; with two control symbols, $N_{\mathrm{RE}}^{\mathrm{data}} = 168 - 24 - 12 = 132$. $R = 2 \times 6 \times 0.9 \times 50 \times 132/10^{-3} = 71.3\,\mathrm{Mbit/s}$.
Interpretation. Half the bandwidth and half the layers give roughly a quarter of the Category-5 figure; the control region and RS overhead are similar ($\approx 21\%$).

Worked example 2 • Hops in the user plane

Problem. Count the network nodes that process a user packet between the radio and the Internet in UMTS R99 and in LTE, and estimate the saving if each node adds 0.5 ms of processing and queuing.
Solution. UMTS: NodeB, RNC, SGSN, GGSN (4 nodes). LTE: eNB, S-GW, P-GW (3 nodes; 2 if S-GW and P-GW are combined). Saving: 0.5–1 ms one way, 1–2 ms round trip.
Interpretation. The node count matters less than the RNC's long TTIs and slow scheduling, which the eNB replaces by 1 ms scheduling – the main reason LTE's latency is a fraction of R99's.

Worked example 3 • VoLTE capacity of a cell

Problem. A VoLTE call needs on average 2 RBs every 20 ms in each direction (one speech frame per 20 ms, robust MCS). How many calls can a 10 MHz cell carry if 70% of RBs may be used for voice?

Solution. Per call: 2 RBs per 20 subframes = 0.1 RB per subframe. Available: $0.7 \times 50 = 35$ RBs per subframe. Calls $= 35/0.1 = 350$ per direction (in practice the PDCCH control channel limits this earlier, which is why *semi-persistent scheduling* is used for VoLTE).

Interpretation. Compare with about 45 WCDMA voice users per 5 MHz (Lecture 3): packet voice with header compression and scheduling is far more spectrum-efficient than circuit voice.

10 Technology comparison

Table 2. 3G (UMTS/HSPA) versus 4G (LTE/EPC).

	UMTS / HSPA	LTE / EPC
Radio access	WCDMA, 5 MHz	OFDMA DL / SC-FDMA UL, 1.4–20 MHz
RAN nodes	NodeB + RNC	eNB only (flat)
Handover	Soft + hard	Hard (with X2 forwarding)
Core domains	CS + PS	PS only
Voice	CS	VoLTE (IMS); CSFB/SRVCC for interworking
Core signalling	MAP (SS7), GTPv1-C, RANAP	Diameter, GTPv2-C, S1AP
TTI	2 ms (HSPA), 10–80 ms (R99)	1 ms
Always-on IP	No (PDP context on demand)	Yes (default bearer at attach)
Peak DL (typical device)	14–42 Mbit/s	150–300 Mbit/s (Rel-8)

Trade-offs. The flat, all-IP design simplified the network but moved complexity elsewhere: QoS for voice must now be engineered with bearers and policy (PCRF), the eNB carries more functions and state, mobility without soft handover relies on fast hard handovers, and the transition period needed CSFB and SRVCC. The control/user plane separation already present in the EPC (MME versus gateways) and completed by CUPS foreshadows the service-based 5G core.

11 Key takeaways

Key takeaways

- LTE requirements: 100/50 Mbit/s peak in 20 MHz, <5 ms user-plane latency, scalable bandwidth, packet-only.
- EPS = E-UTRAN (eNB only) + EPC (MME, S-GW, P-GW, HSS, PCRF).

- Control interfaces use S1AP, GTPv2-C and Diameter; user data uses GTP-U on S1-U and S5/S8.
- S-GW = mobility anchor; P-GW = IP anchor and policy enforcement; MME = signalling only.
- A default EPS bearer is created at attach: always-on IP.
- Peak rate $\approx \nu Q_m r_c N_{\mathrm{RB}} N_{\mathrm{RE}}^{\mathrm{data}} / T_{\mathrm{sf}}$; the 1 ms TTI and 8 ms HARQ loop dominate the radio latency.
- Voice: VoLTE over IMS with a QCI 1 bearer; CSFB and SRVCC for interworking with 2G/3G.
- CUPS splits the gateways into C and U parts controlled with PFCP – the bridge to 5G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Why two gateways?

Explain why the EPC has both an S-GW and a P-GW. Give one deployment where they are combined and one where they must be separate.

Exercise 2 • Level 1 – conceptual • Where did the RNC go?

For each of the following RNC functions state where it lives in LTE: RRC termination, ciphering of user data, handover decision, outer-loop power control, mobility anchoring towards the core.

Exercise 3 • Level 2 – numerical • Cyclic-prefix overhead

In LTE with normal CP, a 0.5 ms slot contains 7 OFDM symbols of $2048\,T_{\mathrm{s}}$ with $T_{\mathrm{s}} = 1/(15000 \times 2048)$ s; the first CP lasts $160\,T_{\mathrm{s}}$ and the others $144\,T_{\mathrm{s}}$. Compute the CP durations in microseconds and the fraction of time spent on CP. What is the maximum delay spread tolerated without inter-symbol interference?

Exercise 4 • Level 2 – numerical • VoLTE backhaul

A VoLTE packet carries a 33-byte speech frame every 20 ms with 12 bytes RTP, 8 bytes UDP and 20 bytes IPv4. On S1-U it is encapsulated in GTP-U/UDP/IPv4. Compute the bit rate per call on the backhaul and the transport efficiency Eq. (4). How many calls fit in a 100 Mbit/s link loaded at 70%?

Exercise 5 • Level 3 – analytical • Uplink latency

Extend Eq. (3) to the uplink of a UE that has no grant: it must wait for its next scheduling-request (SR) opportunity (period T_{SR}), send the SR, receive a grant after k_1 ms and transmit k_2 ms after the grant. Derive the mean latency and evaluate it for $T_{\mathrm{SR}} = 10$ ms, $k_1 = 3$ ms, $k_2 = 4$ ms, other values as in Section 6.2.

Exercise 6 • Level 3 – analytical • Home routing versus local breakout

A subscriber from Tehran roams in Istanbul and browses a server in Istanbul. Compare the user-plane path and the round-trip propagation delay for home-routed roaming (S8 to the home P-GW) and local breakout (visited P-GW). Assume 2000 km between the cities, fibre at 2×10^8 m/s and a route factor of 1.5. What does each option mean for charging and lawful interception?

Solutions

Solution to Exercise 1

They anchor different things. The S-GW anchors mobility between eNBs (and to 2G/3G) and buffers for idle UEs; there is one per UE at a time and it may change with location. The P-GW anchors the IP address and the connection to a PDN and enforces policy; it must not change during a session. Combined: a small national operator, one site, no roaming. Separate: roaming with home routing (visited S-GW, home P-GW over S8), or a UE with an Internet PDN on a local P-GW and an IMS PDN on a central P-GW.

Solution to Exercise 2

RRC termination: eNB. User-data ciphering: eNB (PDCP). Handover decision: source eNB. Outer-loop power control: not needed in the same form – LTE uses scheduled, orthogonal resources; uplink power control (open loop plus closed-loop corrections) is in the eNB. Mobility anchoring towards the core: S-GW (the RNC's Iu anchor role).

Solution to Exercise 3

$T_s = 32.55$ ns. First CP: $160\,T_s = 5.21\,\mu$s; others: $144\,T_s = 4.69\,\mu$s; useful symbol $2048\,T_s = 66.67\,\mu$s. CP fraction: $(160 + 6 \times 144)/15360 = 6.67\%$. Delay spreads up to about $4.7\,\mu$s (about 1.4 km of path difference) are absorbed; the extended CP ($512\,T_s = 16.7\,\mu$s) serves large cells and broadcast.

Solution to Exercise 4

IP packet: $33 + 12 + 8 + 20 = 73$ bytes; on S1-U: $73 + 36 = 109$ bytes every 20 ms $= 43.6$ kbit/s (plus Ethernet framing in practice). Efficiency relative to the speech payload: $33/109 = 30\%$; relative to the user IP packet Eq. (4): $73/109 = 67\%$. Calls: $0.7 \times 100 \times 10^6/43\,600 \approx 1600$ per direction.

Solution to Exercise 5

Mean wait for an SR opportunity $T_{SR}/2$, then SR transmission (1 ms), grant after k_1, data k_2 after the grant, then the downlink-like terms: $D_{UL} = T_{SR}/2 + T_{TTI} + k_1 + k_2 + T_{TTI} + D_{eNB} + p_{re}T_{HARQ} + D_{tr} = 5 + 1 + 3 + 4 + 1 + 1 + 0.8 + 0.5 = 16.3$ ms. The scheduling-request cycle dominates; pre-scheduling or short SR periods reduce it, and 5G adds configured grants that remove it.

Solution to Exercise 6

Home routing: Istanbul UE $\rightarrow$ visited S-GW $\rightarrow$ S8 to Tehran P-GW $\rightarrow$ back to the Istanbul server: two traversals of $1.5 \times 2000 = 3000$ km each way, i.e. 12 000 km per round trip, or 60 ms

of propagation. Local breakout: essentially local, $\ll 1$ ms. Home routing keeps charging, policy and lawful interception in the home network (simple for the home operator); local breakout requires the visited network to apply the home operator's policy (via S9 / roaming agreements) and to perform interception locally.

13 Further reading

The E-UTRAN overall description is [3] and the EPC procedures [4]; the LTE requirements are in [8]. Policy and charging control is specified in [7], CUPS in [1] with PFCP in [5], CSFB in [2] and SRVCC in [9]. Sesia et al. [11] and Olsson et al. [10] are the recommended textbooks for the radio and core sides respectively.

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06

Lecture 06

E-UTRAN: the LTE Radio Access Network

E-UTRAN: شبکهٔ دسترسی رادیویی LTE

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

PREREQUISITES
Lecture 5; the OSI layering; basic ARQ

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 7
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 9
9 Worked examples . . . 9
10 Technology comparison . . . 10
11 Key takeaways . . . 11
12 Exercises . . . 11
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-06-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the functions of the eNB and explain why the flat E-UTRAN could absorb the RNC.
- **LO2** **Draw** the LTE control-plane and user-plane protocol stacks and state the role of PHY, MAC, RLC, PDCP and RRC.
- **LO3** **Map** logical channels to transport and physical channels in both directions.
- **LO4** **Explain** the S1 and X2 interfaces, their protocol stacks and main procedures.
- **LO5** **Describe** the RRC state machine, DRX and the NR RRC_INACTIVE extension, and **quantify** the power/latency trade-off of DRX.
- **LO6** **Size** PDCP sequence numbers and RLC windows for a target data rate.

2 Prerequisite knowledge

This lecture zooms into the left half of the EPS architecture of Lecture 5. You need the idea of protocol layering (each layer offers a service to the one above and adds a header), the notion of automatic repeat request (ARQ) with sequence numbers and windows, and the EPS bearer concept.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
E-UTRAN	شبکهٔ دسترسی رادیویی LTE	The LTE RAN: a mesh of eNBs, no controller.
RRC	کنترل منابع رادیویی	Layer-3 control between UE and eNB: connection, measurements, security.
PDCP	پروتکل همگرایی دادهٔ بسته‌ای	Header compression, ciphering, integrity, in-order delivery.
RLC	کنترل پیوند رادیویی	Segmentation and ARQ in TM, UM or AM mode.
MAC	کنترل دسترسی به رسانه	Multiplexing, scheduling, HARQ, random access.
Logical channel	کانال منطقی	*What* is carried (control or traffic).
Transport channel	کانال انتقال	*How* it is carried (format, coding) between MAC and PHY.
Physical channel	کانال فیزیکی	The actual time–frequency resources.
S1AP / X2AP	پروتکل‌های کاربردی S1 و X2	Signalling eNB–MME and eNB–eNB over SCTP.
DRX	دریافت ناپیوسته	Sleep cycles that save battery at the cost of latency.

3 Introduction and motivation

In UMTS the radio protocols were split across two nodes: the NodeB handled the physical layer, the RNC everything above it. Every scheduling decision, every retransmission of RLC and every handover therefore crossed the Iub transport network, adding delay and making the RNC a bottleneck. HSPA already moved fast scheduling and HARQ to the NodeB. LTE completed the move: the eNB terminates *all* radio protocols, so the round trip between a radio event and a decision stays inside one box.

This lecture opens that box. We follow user data and signalling down the protocol stack, see how the different kinds of information are organised into channels, and study the two network interfaces of the eNB: S1 towards the core and X2 towards its neighbours. We end with the RRC state machine, which decides how much the UE and the network must keep in memory and how fast an idle phone can start sending data.

4 Fundamental concepts

4.1 Why no RNC?

The RNC existed for three reasons: soft-handover combining, central radio resource management of many NodeBs, and termination of protocols that were too complex for 1990s base stations. LTE removed each reason: OFDMA uses hard handover, so no combining is needed; scheduling every 1 ms must be done next to the radio; and processing power had become cheap. Inter-cell coordination that the RNC used to provide is replaced by direct eNB-to-eNB signalling over X2 and by self-organising network functions.

4.2 Layering and the bearer service

Each layer of the radio stack offers a well-defined service upwards:

- **PHY** offers *transport channels* to MAC.
- **MAC** offers *logical channels* to RLC.
- **RLC** offers *RLC channels* to PDCP.
- **PDCP** offers *radio bearers* to IP (data radio bearers, DRBs) and to RRC (signalling radio bearers, SRBs).

A data radio bearer is the radio segment of an EPS bearer (Lecture 5); SRB0 carries RRC messages before security is active, SRB1 and SRB2 carry RRC and NAS messages afterwards.

5 Detailed technical explanation

5.1 eNB functions

The eNB performs [1]:

- **Radio resource management**: radio bearer control, radio admission control, connection mobility control (handover), dynamic allocation of resources to UEs in uplink and downlink (scheduling), inter-cell interference coordination (ICIC), load balancing.
- **Header compression and encryption** of user data (PDCP).
- **MME selection** at attach (NAS node selection function) and routing of user data towards the S-GW.
- **Paging** and **system information broadcast**, originating from the MME and O&M respectively.
- **Measurement configuration** for mobility and scheduling.

5.2 The protocol layers

Physical layer (PHY). OFDMA in the downlink and SC-FDMA in the uplink with 15 kHz subcarriers; turbo coding for data; QPSK, 16QAM and 64QAM (256QAM in later releases); up to 4×4 (later 8×8) MIMO; HARQ soft combining; cell search with primary and secondary synchronisation signals (PSS/SSS). The resource grid and its numerology are the subject of Lecture 7.

MAC. Maps logical channels to transport channels, multiplexes several logical channels into one transport block, performs the *scheduling* decision every TTI, runs HARQ (8 parallel stop-and-wait processes in FDD), handles logical-channel priorities, random access (Lecture 8), timing advance and buffer status and power headroom reports [2].

RLC. Has three modes [4]:

- **Transparent mode (TM)**: no header, no segmentation – used for broadcast, paging and SRB0.
- **Unacknowledged mode (UM)**: segmentation/concatenation and reordering, no retransmission – used for delay-sensitive traffic such as VoLTE.
- **Acknowledged mode (AM)**: adds ARQ with status reports – used for TCP traffic and signalling, where reliability matters more than delay.

RLC adapts IP packets to the transport-block size chosen by the scheduler: it may cut a packet into segments or pack several into one PDU.

PDCP. [3] performs robust header compression (ROHC) of IP/UDP/RTP headers, ciphering of user and control data, integrity protection of control data (SRBs), and in-order, duplicate-free delivery across handover (using the SN status transfer of Lecture 4). Each PDCP PDU carries a sequence number (SN); together with a hyper-frame number (HFN) kept at both ends it forms the 32-bit *COUNT*, an input to the ciphering algorithm that must never repeat under the same key.

RRC. [5] broadcasts system information (MIB on PBCH, SIBs on PDSCH), manages the RRC connection (setup, reconfiguration, re-establishment, release), configures measurements and receives reports, activates access-stratum security, and transfers NAS messages.

5.3 Channels

Figure 1 shows the main mapping. The logical channels are the broadcast (BCCH), paging (PCCH), common control (CCCH, before an RRC connection exists), dedicated control (DCCH) and dedicated traffic (DTCH) channels. Transport channels group them by transmission format: the downlink and uplink shared channels (DL-SCH, UL-SCH) carry almost everything. Physical channels include the shared data channels PDSCH and PUSCH, control channels without transport channels (PDCCH for scheduling assignments, PCFICH for the size of the control region, PHICH for uplink HARQ acknowledgements, PUCCH for uplink control) and reference signals (CRS, DMRS, SRS) used for measurements and demodulation.

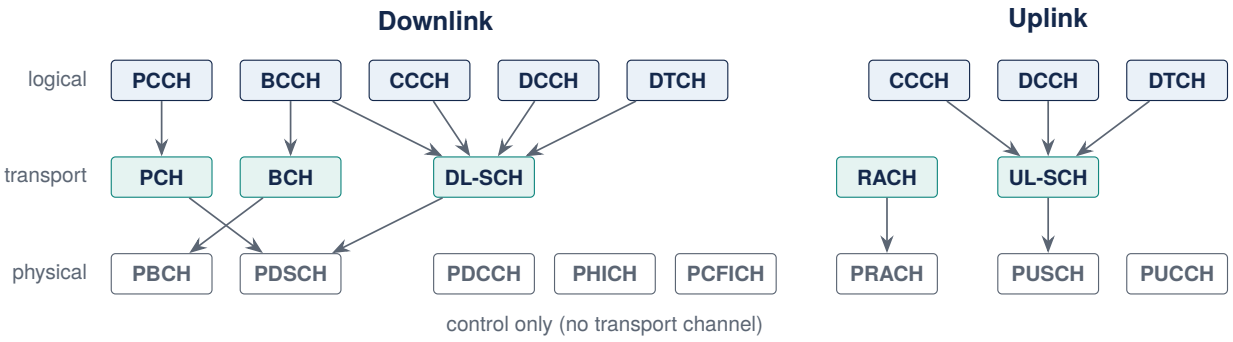


Figure 1. Main LTE channel mapping. Almost all logical channels converge on the shared channels; PDCCH, PHICH, PCFICH and PUCCH carry layer-1/2 control without a transport channel.

5.4 The S1 interface

S1 connects each eNB to one or more MMEs (S1-MME) and S-GWs (S1-U) – a many-to-many relation called *S1-flex* that allows MME pooling (Lecture 12).

- **S1-MME**: S1AP over SCTP over IP [6]. SCTP provides reliable, message-oriented transport with multi-homing and multiple streams, avoiding head-of-line blocking between UEs.
- **S1-U**: GTP-U over UDP over IP; one tunnel per E-RAB (the S1 part of an EPS bearer).

Main S1AP procedures: *S1 Setup* (eNB and MME exchange identities and served tracking areas), *Initial UE Message* (first NAS message of a UE), *Initial Context Setup* (MME installs the UE context: security key, bearers, capabilities), *E-RAB Setup/Modify/Release*, *Paging*, *Handover Preparation/Resource Allocation* (S1 handover), *Path Switch*, *UE Context Release* and *Downlink/Uplink NAS Transport*.

5.5 The X2 interface

X2 connects neighbouring eNBs [7]: X2AP over SCTP for control, GTP-U for forwarding user data during handover. Main procedures: *X2 Setup* (exchange of served cells, enabling automatic neighbour relations), *Handover Preparation*, *SN Status Transfer*, *UE Context Release*, *Load Indication* (interference information for ICIC) and *Resource Status Reporting* (load for mobility load balancing). X2 is optional: without it, handovers go through S1 and the MME.

5.6 RRC states and DRX

LTE has two RRC states (Figure 2):

- **RRC_IDLE**: no RRC connection, no context in the eNB; the UE performs cell reselection, reads system information and monitors paging once per paging cycle (idle DRX); the MME knows it at tracking-area level (ECM-IDLE).
- **RRC_CONNECTED**: the eNB holds the UE context; the UE can be scheduled, reports measurements and is handed over by the network; to save power it may use connected-mode DRX, waking for an *on-duration* once per DRX cycle.

Moving from IDLE to CONNECTED requires random access, RRC connection setup, a NAS service request, security activation and bearer set-up – tens of milliseconds and a dozen messages. NR therefore adds **RRC_INACTIVE** [8]: the UE and the gNB keep the context (security, bearers), the core still sees the UE as connected, and resumption needs only a short exchange.

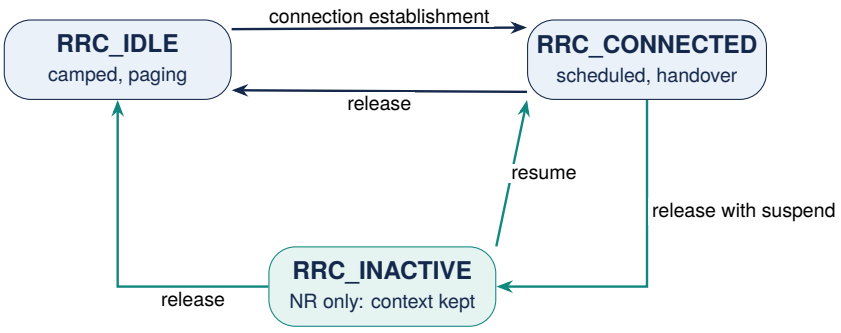


Figure 2. RRC state machine. LTE has IDLE and CONNECTED; NR adds INACTIVE, in which the context is suspended rather than released.

6 Mathematical models

6.1 Header compression gain

Problem. How much radio capacity does ROHC save for VoLTE?

Model. A speech frame of L_p bytes is carried with H bytes of RTP/UDP/IP headers; ROHC reduces H to a compressed header of H_c bytes in steady state. The radio efficiency of the packet is

$$\eta = \frac{L_p}{L_p + H}, \qquad G = \frac{L_p + H}{L_p + H_c} \text{ (capacity gain)}. \tag{1}$$

Example. $L_p = 32$ bytes, $H = 40$ bytes (IPv4 20 + UDP 8 + RTP 12), $H_c = 3$ bytes: η rises from 44% to 91% and $G = 72/35 = 2.06$. With IPv6 ($H = 60$) the gain is $92/35 = 2.6$. ROHC roughly doubles the number of VoLTE users a cell can carry; it is not applied on S1-U, where the headers are carried in full (Lecture 5).

Limits. The steady-state compressed size is reached only after a context is established; packet loss can desynchronise the decompressor, so ROHC profiles trade compression against robustness.

6.2 Sequence-number space and wrap-around

Problem. How many PDCP SN bits are needed at a given rate?

Model. A window-based protocol with an n-bit SN can have at most 2^{n-1} PDUs outstanding (half the space), otherwise old and new PDUs become ambiguous. With a PDU rate λ_p (PDUs per second), the window covers the time

$$T_{\text{win}} = \frac{2^{n-1}}{\lambda_p}, \qquad \lambda_p = \frac{R}{8L}, \tag{2}$$

for a bit rate R and PDUs of L bytes. T_{win} must exceed the longest reordering or retransmission delay, typically including HARQ, RLC ARQ and handover data forwarding.

Example. $R = 300$ Mbit/s, $L = 1500$ bytes: $\lambda_p = 25\,000$ PDU/s. A 12-bit SN gives $T_{\text{win}} = 2048/25\,000 = 82$ ms – uncomfortably short during handover. This is why LTE added a 15-bit (655 ms) and NR an 18-bit PDCP SN.

6.3 RLC acknowledged-mode throughput

For a window of W PDUs of L bytes and a status round-trip time T_{RTT}, a window-limited sender cannot exceed

$$R_{\max} = \frac{8\,W\,L}{T_{\text{RTT}}}. \tag{3}$$

With the LTE RLC AM 10-bit SN, $W = 512$; for $L = 1500$ bytes and $T_{\text{RTT}} = 20$ ms, $R_{\max} = 512 \times 12\,000/0.02 = 307$ Mbit/s: enough for Release 8, the reason for longer SNs in later releases with higher rates (Exercise 3).

6.4 DRX: power versus latency

Problem. How long should a connected UE sleep?

Model. DRX cycle T_{DRX} with on-duration T_{on}; power P_{on} when awake and P_{sleep} when asleep; downlink packets arrive at random instants (no traffic during the sleep period otherwise).

Derivation. With duty cycle $d = T_{\mathrm{on}}/T_{\mathrm{DRX}}$,

$$\bar{P} = d\,P_{\mathrm{on}} + (1-d)\,P_{\mathrm{sleep}}, \qquad \bar{D}_{\mathrm{DRX}} \approx \frac{T_{\mathrm{DRX}}}{2}, \tag{4}$$

because a packet arriving at a uniformly random time waits on average half a cycle for the next on-duration.

Example and interpretation. $T_{\mathrm{on}} = 10\,\mathrm{ms}$, $P_{\mathrm{on}} = 500\,\mathrm{mW}$, $P_{\mathrm{sleep}} = 10\,\mathrm{mW}$ (illustrative): a 40 ms cycle gives 132.5 mW and 20 ms extra delay; a 320 ms cycle 25.3 mW and 160 ms; 2560 ms gives 11.9 mW but 1.28 s (Figure 3). Power falls like $1/T_{\mathrm{DRX}}$ until P_{sleep} dominates, while latency grows linearly: beyond a few hundred milliseconds longer cycles buy little energy at a large latency cost. Real devices also stay awake for an inactivity timer after each packet, which the model ignores.

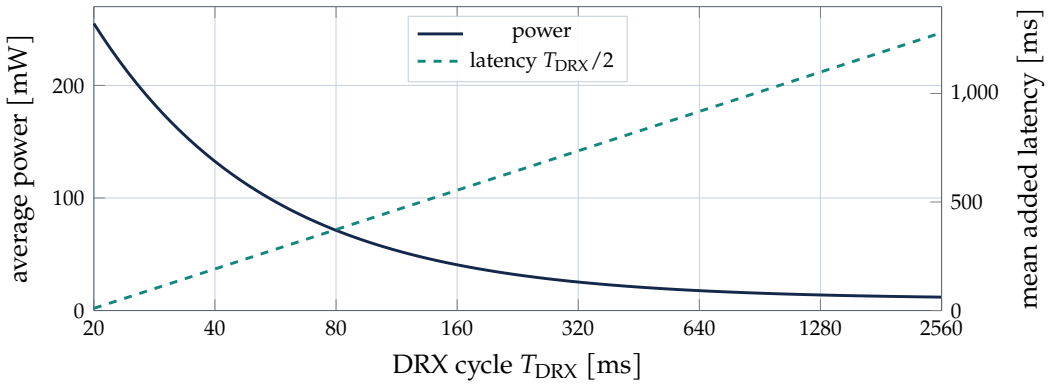


Figure 3. Average power (left axis) and mean added latency (right axis) versus DRX cycle for $T_{\mathrm{on}} = 10\,\mathrm{ms}$, from Eq. (4).

7 Network architecture and diagrams

Figure 4 summarises the radio stacks. On the control plane, RRC terminates in the eNB while NAS passes through it transparently to the MME (carried in RRC towards the UE and in S1AP towards the MME). On the user plane, PDCP, RLC and MAC terminate in the eNB, which relays the IP packets into GTP-U tunnels on S1-U.

8 Step-by-step procedures

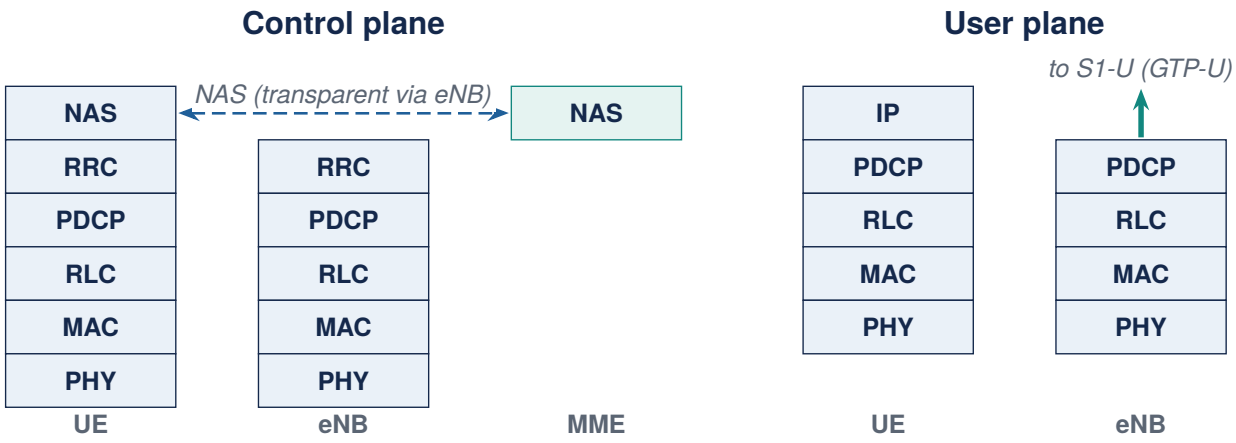


Figure 4. LTE radio protocol stacks: control plane (left) and user plane (right).

Procedure 1 • From RRC_IDLE to data transfer (mobile-originated)

1. **Random access.** The UE sends a preamble, receives a random access response with a timing advance and grant (Lecture 8).
2. **RRC connection request.** On SRB0 it sends RRC Connection Request (identity and cause); the eNB answers RRC Connection Setup, creating SRB1.
3. **NAS transfer.** The UE sends RRC Connection Setup Complete containing the NAS Service Request; the eNB forwards it in S1AP Initial UE Message to the MME selected from the UE's identity.
4. **Context set-up.** The MME answers S1AP Initial Context Setup Request with the security key K_{eNB}, the E-RABs and the S-GW tunnel endpoints.
5. **AS security.** The eNB sends Security Mode Command; the UE derives the AS keys and answers Security Mode Complete. PDCP now ciphers and integrity-protects.
6. **Bearer configuration.** The eNB sends RRC Connection Reconfiguration creating SRB2 and the DRBs; after Reconfiguration Complete it answers Initial Context Setup Response with its own S1-U tunnel endpoints. Uplink and downlink data flow.

9 Worked examples

Worked example 1 • Protocol overhead of a TCP segment

Problem. A 1460-byte TCP payload is sent over LTE with IPv4. Count the header bytes added on the radio by PDCP (2 bytes, 12-bit SN), RLC AM (2 bytes, no segmentation) and MAC (2 bytes subheader), and on S1-U.
Solution. IP packet: $1460 + 20 + 20 = 1500$ bytes. Radio: $1500 + 2 + 2 + 2 = 1506$ bytes (0.4% overhead; ROHC is usually not used for TCP). S1-U: $1500 + 36 = 1536$ bytes (2.4%).
Interpretation. For large packets layer-2 overhead is negligible; the real costs are the reference signals and control channels of the PHY (Lecture 5) and the retransmissions.

Worked example 2 • PDCP window during handover

Problem. A UE receives 150 Mbit/s in 1500-byte packets. During handover, forwarded packets may arrive up to 60 ms late. Is a 12-bit PDCP SN sufficient?
Solution. $\lambda_p = 150 \times 10^6 / 12\,000 = 12\,500$ PDU/s. Equation (2): $T_{win} = 2048/12\,500 = 164$ ms > 60 ms. Yes, with margin; at 300 Mbit/s (82 ms) the margin would be thin.

Worked example 3 • Choosing a DRX cycle

Problem. A messaging application tolerates 200 ms extra latency on average. Using the parameters of Section 6.4, choose the longest standard DRX cycle among 40, 80, 160, 320 ms and compute the power saving relative to no DRX (P_{on} continuously).
Solution. $T_{DRX}/2 \leq 200$ ms allows 320 ms (160 ms latency). $\bar{P} = 25.3$ mW versus 500 mW: a 95% reduction.
Interpretation. Most of the saving is already obtained at 160 ms (40.6 mW); the operator can choose per application through the inactivity timer and cycle settings.

10 Technology comparison

Table 1. Radio access network functions: UTRAN versus E-UTRAN versus NG-RAN.

	UTRAN (3G)	E-UTRAN (4G)	NG-RAN (5G)
Nodes	NodeB + RNC	eNB	gNB (optionally CU + DU)
RLC/PDCP location	RNC	eNB	DU (RLC), CU (PDCP)
Scheduler	RNC (R99), NodeB (HSPA)	eNB, 1 ms	gNB-DU, slot/mini-slot
Inter-node link	Iur	X2	Xn; F1 between CU and DU
Core interface	Iu (RANAP)	S1 (S1AP)	NG (NGAP)
RRC states	Idle, Cell_DCH/FACH/PCH, URA_PCH	Idle, Connected	Idle, Inactive, Connected
New L2 sublayer	–	–	SDAP (QoS-flow mapping)

Trade-off. Putting everything in the eNB minimised latency but made coordination between cells (interference, joint transmission) harder, since it now requires X2 signalling between peers. 5G partially re-centralises the RAN with the CU/DU split (Lecture 14): the time-critical lower layers stay at the antenna site, PDCP and RRC can be pooled – a deliberate compromise between UTRAN's centralisation and E-UTRAN's flatness.

11 Key takeaways

Key takeaways

- The eNB terminates PHY, MAC, RLC, PDCP and RRC: radio decisions stay in one node.
- PDCP: ROHC, ciphering, integrity (SRBs), in-order delivery; COUNT = HFN ‖ SN must never repeat.
- RLC: TM, UM (voice), AM (TCP, signalling); MAC: multiplexing, scheduling, HARQ.
- Logical → transport → physical channels; almost everything uses the shared channels.
- S1-MME (S1AP/SCTP) and S1-U (GTP-U/UDP) towards the core; X2 between eNBs.
- RRC_IDLE vs RRC_CONNECTED (+ RRC_INACTIVE in NR); DRX trades power ($\propto 1/T_{\mathrm{DRX}}$) for latency ($T_{\mathrm{DRX}}/2$).
- SN space must cover the reordering time: $2^{n-1}/\lambda_{\mathrm{p}} > T_{\mathrm{reorder}}$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which layer?

For each function name the LTE layer that performs it: HARQ, ARQ, ciphering of user data, segmentation, header compression, integrity protection of RRC messages, choice of MCS, measurement configuration.

Exercise 2 • Level 1 – conceptual • Why SCTP?

Explain why S1AP and X2AP use SCTP rather than TCP or UDP. Give one failure scenario that multi-homing handles.

Exercise 3 • Level 2 – numerical • RLC window

Using Eq. (3), find the RLC AM window needed for 1 Gbit/s with 1500-byte PDUs and a 15 ms status round trip, and the minimum number of SN bits.

Exercise 4 • Level 2 – numerical • ROHC and IPv6

Repeat Section 6.1 for AMR-WB at 23.85 kbit/s (61-byte frames every 20 ms) with IPv6. What is the bit rate on the radio with and without ROHC, and the capacity gain?

Exercise 5 • Level 3 – analytical • DRX with an inactivity timer

After each received packet the UE stays awake for an inactivity timer T_{in}. Packets arrive as a Poisson process with rate λ. Derive an approximate expression for the fraction of time awake, assuming $T_{\text{in}} \ll 1/\lambda$, and evaluate it for $\lambda = 0.2\,\text{s}^{-1}$, $T_{\text{in}} = 100\,\text{ms}$, $T_{\text{on}} = 10\,\text{ms}$, $T_{\text{DRX}} = 320\,\text{ms}$.

Exercise 6 • Level 3 – analytical • Idle versus connected

Releasing a UE to RRC_IDLE after inactivity saves eNB memory and UE power, but each new data burst then costs about 12 signalling messages. If the eNB can handle 10^4 such transitions per second and each UE generates a burst every 30 s, how many UEs can be served purely from the signalling point of view? Discuss how RRC_INACTIVE changes the answer.

Solutions

Solution to Exercise 1

HARQ: MAC. ARQ: RLC (AM). Ciphering of user data: PDCP. Segmentation: RLC. Header compression: PDCP. Integrity of RRC: PDCP. Choice of MCS: MAC scheduler (with link adaptation input from PHY). Measurement configuration: RRC.

Solution to Exercise 2

SCTP is message-oriented (signalling messages keep their boundaries), reliable and congestion controlled like TCP, supports multiple streams so that a lost message of one UE does not block messages of others (no head-of-line blocking across streams), and supports multi-homing: an association can use several IP paths. If one transport link between eNB and MME fails, the association continues over the other address without tearing down the S1 interface and every UE context with it.

Solution to Exercise 3

$W \geq R\,T_{\text{RTT}}/(8L) = 10^9 \times 0.015/12\,000 = 1250$ PDUs. The SN space must be at least $2W = 2500$, so 12 bits (4096) are needed; LTE Release 8's 10 bits are insufficient (hence the longer RLC sequence numbers of later releases; NR RLC AM uses 12 or 18 bits).

Solution to Exercise 4

IPv6/UDP/RTP: $40 + 8 + 12 = 60$ bytes. Without ROHC: $61 + 60 = 121$ bytes per 20 ms = 48.4 kbit/s; with ROHC (3 bytes): 64 bytes = 25.6 kbit/s. Gain $121/64 = 1.89$. The gain is smaller than for 32-byte frames because the payload is larger.

Solution to Exercise 5

Each packet keeps the UE awake for about T_{in}; with $\lambda T_{\text{in}} \ll 1$ these periods rarely overlap. During the remaining time DRX applies. Fraction awake $f \approx \lambda T_{\text{in}} + (1 - \lambda T_{\text{in}})\,T_{\text{on}}/T_{\text{DRX}} = 0.02 + 0.98 \times 0.03125 = 0.0506$, i.e. 5.1%. With the powers of Section 6.4, $\bar{P} \approx 0.0506 \times 500 + 0.9494 \times 10 = 34.8\,\text{mW}$. The inactivity timer adds about 40% to the energy of the pure DRX case for this traffic.

Solution to Exercise

Each UE causes $1/30$ transitions per second, so $N = 10^4 \times 30 = 3 \times 10^5$ UEs per eNB – more than any cell carries, but the MME sees the sum over many eNBs, and paging, NAS and S1 messages load the core too. With RRC_INACTIVE a resumption needs only about 3 radio messages and no core signalling (the core sees the UE as connected), so the core load disappears and the radio load drops by about $4\times$, at the cost of storing contexts in the RAN.

13 Further reading

The E-UTRAN overall description [1] is the best entry point; the layer specifications are MAC [2], RLC [4], PDCP [3] and RRC [5]; the network interfaces are S1AP [6] and X2AP [7]; SCTP is specified in [10]. Sesia et al. [9] devote several chapters to the layer-2 protocols.

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[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA) and evolved universal terrestrial radio access network (E-UTRAN); overall description; stage 2*, TS 36.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36300.htm.

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07

Lecture 07

LTE Scheduling

زمان‌بندی در LTE

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate to advanced

ESTIMATED STUDY TIME
8–10 hours (3.5 h reading, 4–6 h exercises incl. simulation)

PREREQUISITES
Lectures 5–6; Shannon capacity; Rayleigh fading; basic optimisation

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
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11 Key takeaways . . . 11
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13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-07-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the LTE resource grid – frame, subframe, slot, resource element, resource block – and **count** the resources of any LTE bandwidth.
- **LO2** **Explain** downlink and uplink scheduling signalling: DCI, grants, BSR, PHR, CQI.
- **LO3** **Apply** link adaptation: map CQI to modulation and coding, and **analyse** outer-loop link adaptation (OLLA).
- **LO4** **Analyse** HARQ with eight stop-and-wait processes and compute its throughput effect.
- **LO5** **Compare** round robin, maximum C/I and proportional fair scheduling and **derive** why PF maximises the sum of log-throughputs.
- **LO6** **Evaluate** the throughput–fairness–QoS trade-off with Jain's index and a simulation.

2 Prerequisite knowledge

You need the LTE protocol stack (Lecture 6), in particular the MAC layer's scheduling and HARQ roles, the Shannon capacity $C = B\log_2(1+\mathrm{SNR})$ and the fact that, under Rayleigh fading, the instantaneous SNR is exponentially distributed around its mean. Basic calculus (derivatives, Lagrange-style arguments) is used in Section 6.4.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Resource element (RE)	عنصر منبع	One subcarrier during one OFDM symbol.
Resource block (RB)	بلوک منبع	12 subcarriers (180 kHz) × one slot.
TTI	بازهٔ زمانی ارسال	Scheduling interval: one 1-ms subframe in LTE.
DCI	اطلاعات کنترلی فروسو	Scheduling message on the PDCCH: resources, MCS, HARQ.
CQI	نشانگر کیفیت کانال	UE report: highest MCS decodable at 10% BLER.
MCS	طرح مدولاسیون و کدگذاری	Modulation order + code rate chosen by the scheduler.
BSR / PHR	گزارش وضعیت بافر / حاشیهٔ توان	Uplink reports: queued data / remaining power.
HARQ	درخواست تکرار خودکار ترکیبی	Fast retransmission with soft combining.
Proportional fair (PF)	منصفانهٔ تناسبی	Scheduler maximising $\sum \log R_k$.
Multi-user diversity	تنوع چندکاربره	Gain from serving users at their fading peaks.

3 Introduction and motivation

Every millisecond an LTE eNB must answer one question: *who gets which resource blocks, with which modulation and coding?* The answer determines everything users perceive – throughput, delay, fairness – and everything the operator pays for, because spectrum is the most expensive resource of a mobile network.

The problem is hard because the channel of each user fluctuates (fast fading, mobility, interference), traffic is bursty, and the objectives conflict: maximising total throughput favours users close to the base station, while fairness and QoS guarantees require serving users at the cell edge. LTE gives the eNB a fine-grained tool – 1 ms in time, 180 kHz in frequency – and leaves the scheduling *algorithm* to the vendor. 3GPP standardises the signalling (what the UE reports and how grants are sent), not the decision. This lecture covers both: the standard mechanisms and the classic algorithms with their mathematical foundations.

4 Fundamental concepts

4.1 The resource grid

In the time domain an LTE radio frame lasts 10 ms and contains ten 1 ms *subframes*, each made of two 0.5 ms *slots* of 7 OFDM symbols (normal cyclic prefix) [2]. In the frequency domain, subcar-

riers are spaced $\Delta f = 15\,\text{kHz}$. A *resource element* is one subcarrier for one symbol; a *resource block* is 12 subcarriers (180 kHz) for one slot (84 REs). The scheduler normally assigns *RB pairs* (both slots of a subframe, 168 REs) every TTI of 1 ms.

Table 1. LTE channel bandwidths and resource blocks [2].

Channel bandwidth [MHz]	1.4	3	5	10	15	20
Resource blocks N_{RB}	6	15	25	50	75	100
Occupied bandwidth [MHz]	1.08	2.7	4.5	9.0	13.5	18.0

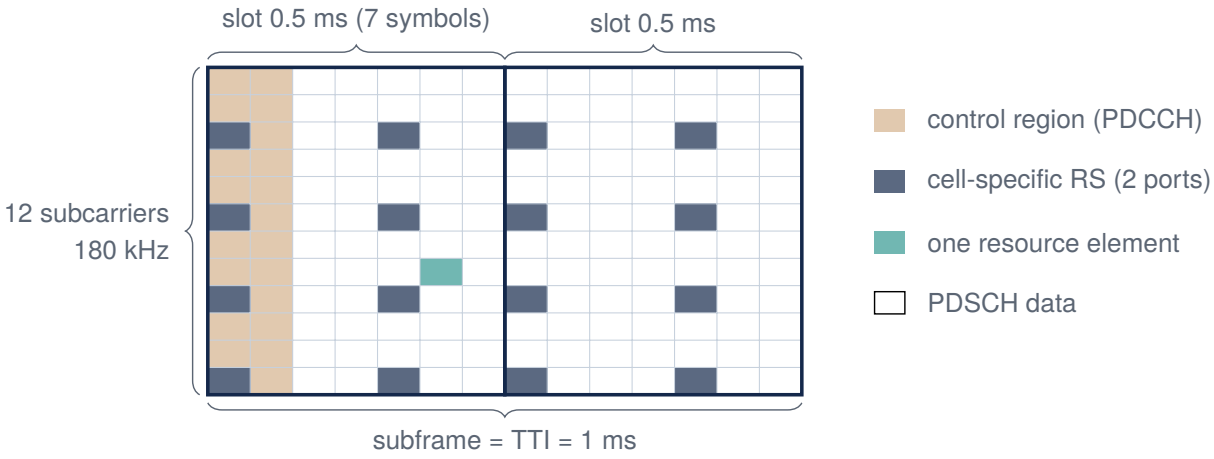


Figure 1. One downlink RB pair (12 subcarriers × 14 symbols) with a two-symbol control region and two-port cell-specific reference signals.

4.2 What the scheduler knows

- **Channel quality**: CQI reports (wideband or per subband), rank and precoder indicators (RI, PMI) in the downlink; sounding reference signals (SRS) in the uplink.
- **Traffic**: downlink buffers are in the eNB; uplink buffers are reported by the UE in *buffer status reports* (BSR), per group of logical channels.
- **QoS**: the QCI, guaranteed bit rate (GBR) and priority of each bearer (Lecture 22).
- **UE power**: *power headroom reports* (PHR) tell how much more power the UE could use, so the uplink scheduler does not grant more RBs than the UE can power.

5 Detailed technical explanation

5.1 Downlink scheduling

For each scheduled UE the eNB sends a *downlink control information* (DCI) message on the PDCCH in the control region of the same subframe [3]. A DCI contains the resource allocation, the MCS index, the HARQ process number, the new-data indicator, the redundancy version and a power-control command for PUCCH. Three resource allocation types trade signalling size against flexibility: type 0 (bitmap of RB groups), type 1 (RBs within a subset) and type 2 (a contiguous

set, compact). The UE decodes the PDSCH in the same subframe and returns ACK/NACK four subframes later.

5.2 Uplink scheduling

The uplink differs in three ways. (i) The eNB does not see the UE's buffer: the UE sends a *scheduling request* (SR) on PUCCH, receives a small grant, and then reports its buffer with a BSR. (ii) Grants are sent on the PDCCH four subframes before the PUSCH transmission (FDD). (iii) SC-FDMA requires the RBs of one UE to be *contiguous* in frequency to keep the low peak-to-average power ratio, which limits frequency-selective scheduling. Uplink HARQ is *synchronous*: retransmissions occur exactly 8 ms later, acknowledged on the PHICH.

5.3 CQI and link adaptation

The UE reports a 4-bit CQI: the highest index whose modulation and code rate would be received with a transport-block error probability not exceeding 10% [3]. Table 2 lists the standard table. The eNB maps the CQI (after correction, see OLLA below) to an MCS index and a transport block size.

Table 2. LTE 4-bit CQI table (TS 36.213, Table 7.2.3-1). Efficiency is in information bits per resource element.

CQI	Modulation	Code rate×1024	Efficiency	CQI	Modulation	Code rate×1024	Efficiency
1	QPSK	78	0.1523	9	16QAM	616	2.4063
2	QPSK	120	0.2344	10	64QAM	466	2.7305
3	QPSK	193	0.3770	11	64QAM	567	3.3223
4	QPSK	308	0.6016	12	64QAM	666	3.9023
5	QPSK	449	0.8770	13	64QAM	772	4.5234
6	QPSK	602	1.1758	14	64QAM	873	5.1152
7	16QAM	378	1.4766	15	64QAM	948	5.5547
8	16QAM	490	1.9141	0	out of range		

Note that the efficiency is simply $Q_m \times (\text{code rate})$: for CQI 15, $6 \times 948/1024 = 5.5547$ bits per RE. Wideband CQI describes the whole carrier; subband CQI (per group of RBs) enables frequency-selective scheduling but costs uplink overhead.

5.4 HARQ

LTE uses eight parallel stop-and-wait HARQ processes in FDD (Figure 2): while process 2 waits for its acknowledgement, processes 3–8 use the following subframes, so the channel never idles. With *Chase combining* the retransmission repeats the same coded bits and the receiver adds the signal energies; with *incremental redundancy* it sends new parity bits, lowering the effective code rate. Typical operation targets a 10% first-transmission BLER and a residual BLER after HARQ of about 10^{-3} or below; RLC ARQ (Lecture 6) catches the rest.

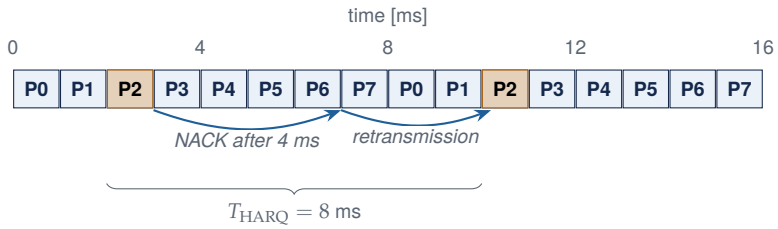


Figure 2. Eight HARQ processes in FDD. A NACK for process P2 arrives four subframes after the transmission; the retransmission occurs eight subframes after the original.

5.5 Scheduling algorithms

Round robin (RR). Users take turns regardless of channel quality. Perfectly fair in *time*, but it ignores the channel and wastes good fading peaks.

Maximum C/I (Max-rate). In each TTI and RB, serve the user with the highest instantaneous rate $r_k(t)$. Maximises cell throughput but starves cell-edge users.

Proportional fair (PF). Serve the user with the largest ratio of instantaneous rate to its own average throughput [5]:

$$k^*(t) = \arg\max_k \frac{r_k(t)}{\bar{R}_k(t)}, \qquad \bar{R}_k(t+1) = \left(1 - \tfrac{1}{t_c}\right)\bar{R}_k(t) + \tfrac{1}{t_c}\, r_k(t)\, \mathbb{1}\{k = k^*(t)\}, \tag{1}$$

where t_c is the averaging window in TTIs. Each user is served when its channel is *good relative to its own average*, so cell-edge users still get their peaks.

QoS-aware scheduling. Real schedulers first satisfy GBR bearers (e.g. VoLTE, often with semi-persistent scheduling), weight metrics by QCI priority and packet delay budget, and use PF-like rules for the remaining best-effort traffic.

6 Mathematical models

6.1 From CQI to data rate

For a UE allocated N_{RB} RB pairs with $N_{\mathrm{RE}}^{\mathrm{data}}$ data REs each and an efficiency η_{CQI} (bits/RE) from Table 2, the rate per layer is

$$R = \eta_{\mathrm{CQI}} \, \frac{N_{\mathrm{RB}}\, N_{\mathrm{RE}}^{\mathrm{data}}}{T_{\mathrm{sf}}}. \tag{2}$$

Example: CQI 9 ($\eta = 2.4063$), 50 RBs, 132 data REs (two control symbols, two CRS ports): $R = 2.4063 \times 50 \times 132/10^{-3} = 15.9\,\mathrm{Mbit/s}$.

6.2 Outer-loop link adaptation

Problem. CQI reports are delayed and biased (estimation errors, interference changes); if the eNB used them directly the actual BLER would drift from the 10% target.

Model. The eNB adds an offset Δ (dB) to the reported SINR before choosing the MCS. After each ACK it raises Δ by δ_{up}; after each NACK it lowers it by δ_{down}.

Derivation. In steady state the expected change is zero. If p is the BLER at the operating point,

$$(1-p)\,\delta_{\text{up}} - p\,\delta_{\text{down}} = 0 \quad\Longrightarrow\quad p = \frac{\delta_{\text{up}}}{\delta_{\text{up}} + \delta_{\text{down}}}. \tag{3}$$

Choosing $\delta_{\text{up}} = \delta_{\text{down}}\, p_{\text{t}}/(1-p_{\text{t}})$ therefore drives the BLER to the target p_{t} regardless of the CQI bias. For $p_{\text{t}} = 0.1$ and $\delta_{\text{down}} = 0.5\,\text{dB}$, $\delta_{\text{up}} = 0.056\,\text{dB}$.

Interpretation. OLLA is a stochastic control loop: small steps give an accurate BLER but slow convergence (about $1/\delta$ steps), large steps react fast but jitter the MCS.

6.3 HARQ throughput

If the first transmission fails with probability p_1 and a retransmission (after combining) fails with probability $p_2 < p_1$, the mean number of transmissions per transport block (neglecting a third attempt's failures) is $\bar{N} \approx 1 + p_1 + p_1 p_2$, and the throughput efficiency is $1/\bar{N}$. With $p_1 = 0.1$ and $p_2 = 0.01$: $\bar{N} = 1.101$, efficiency 90.8%. A 10% BLER target costs only about 9% of throughput but buys a much higher MCS than a 1% target would allow – the reason LTE operates at 10%.

6.4 Why proportional fair is "proportionally fair"

Claim. Among all scheduling policies, PF (asymptotically, for long averaging windows) maximises the utility

$$U = \sum_{k=1}^{K} \log \bar{R}_k. \tag{4}$$

Argument. Consider giving the current TTI to user k. To first order it increases $\bar{R}_k$ by $\epsilon\, r_k(t)$ (with $\epsilon = 1/t_{\text{c}}$), changing the utility by

$$\Delta U \approx \frac{\partial U}{\partial \bar{R}_k}\,\epsilon\, r_k(t) = \epsilon\,\frac{r_k(t)}{\bar{R}_k}. \tag{5}$$

The greedy choice that maximises the utility increase in every TTI is therefore exactly $\arg\max_k r_k(t)/\bar{R}_k$, i.e. Eq. (1). Because U is concave, greedy steps converge to its maximum. A useful property of the optimum is *proportional fairness*: any feasible change of the throughput vector satisfies $\sum_k \Delta\bar{R}_k/\bar{R}_k \le 0$ – no user can gain a percentage of throughput without others losing at least the same total percentage. Max C/I maximises $\sum \bar{R}_k$ instead, and RR equalises time shares.

6.5 Jain's fairness index

To compare schedulers we use [4]

$$J = \frac{\left(\sum_{k=1}^{K} \bar{R}_k\right)^2}{K \sum_{k=1}^{K} \bar{R}_k^2}, \qquad \frac{1}{K} \le J \le 1, \tag{6}$$

which equals 1 when all users get the same throughput and $1/K$ when one user gets everything.

6.6 Simulation: multi-user diversity

Set-up. `scripts/scheduler_sim.py` simulates one 10 MHz carrier (9 MHz occupied) given entirely to one user per TTI; four users with mean SNRs of 20, 12, 5 and 0 dB; independent Rayleigh

block fading per TTI; efficiency $\min(0.75\log_2(1+\mathrm{SNR}), 5.55)$ bit/s/Hz (an attenuated Shannon mapping capped at CQI 15); 20 000 TTIs; $t_c = 100$; fixed random seed.

Results. (Figure 3, Table 3)

Table 3. Simulated per-user and cell throughput [Mbit/s] and Jain's index.

Scheduler	UE1 (20 dB)	UE2 (12 dB)	UE3 (5 dB)	UE4 (0 dB)	Cell	J
Round robin	9.74	5.83	2.88	1.43	19.88	0.710
Prop. fair	12.97	8.34	4.60	2.50	28.40	0.761
Max C/I	35.73	4.18	0.21	0.01	40.13	0.311

Interpretation. PF gives *every* user more throughput than RR: each user still gets about a quarter of the TTIs, but they are its best TTIs. This is *multi-user diversity*, a gain that grows with the number of users and the fading depth. Max C/I raises the cell total by another 41% but essentially switches off UE3 and UE4. The model ignores frequency selectivity (subband scheduling adds further diversity), CQI delay and errors, and traffic dynamics (all users always have data – the "full buffer" assumption).

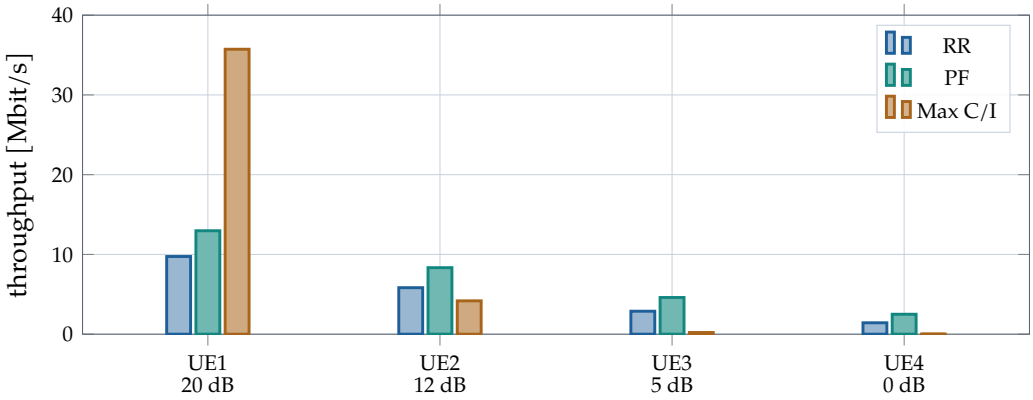


Figure 3. Per-user throughput under round robin, proportional fair and maximum C/I scheduling (simulation of Section 6.6).

7 Network architecture and diagrams

Scheduling is entirely a function of the eNB MAC; the only network-level inputs are the QoS parameters of each bearer, set by the core (QCI, ARP, GBR, MBR, AMBR) through S1AP when bearers are established (Lectures 9 and 22). Between eNBs, X2 *load information* messages allow inter-cell interference coordination: neighbours can agree to avoid scheduling cell-edge users on the same RBs. Figure 4 places the algorithms in the space of objectives.

8 Step-by-step procedures

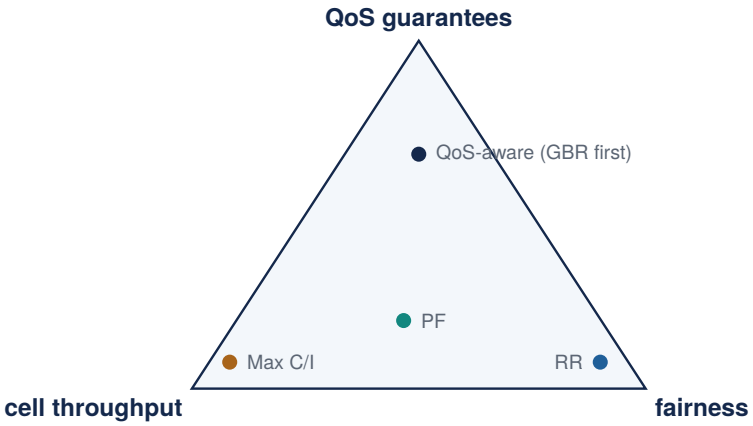


Figure 4. The scheduling trilemma. No algorithm maximises throughput, fairness and QoS at once; practical schedulers combine a QoS stage with a PF-like stage.

Procedure 1 • One TTI of downlink scheduling

1. Collect inputs: latest CQI/RI/PMI per UE, HARQ feedback received for subframe $n-4$, buffer sizes per bearer, QoS parameters.
2. Reserve resources for pending HARQ retransmissions (they have priority to keep delay low).
3. Serve GBR bearers that are below their guaranteed rate (e.g. VoLTE), using the MCS from corrected CQI (OLLA offset).
4. Rank the remaining backlogged UEs with the PF metric per RB group and allocate RBs.
5. For each scheduled UE choose MCS and transport block size; build the MAC PDUs (multiplexing logical channels by priority).
6. Encode the DCIs into PDCCH candidates (if the control region is full, defer UEs to the next TTI) and transmit PDCCH and PDSCH.
7. Update PF averages and OLLA offsets when ACK/NACK arrive four subframes later.

Procedure 2 • Uplink access for a UE with new data

1. New data arrives in an empty buffer; the UE triggers a BSR. Having no grant, it sends a scheduling request on its next PUCCH SR opportunity (or random access if none is configured).
2. The eNB sends a small grant (DCI format 0); four subframes later the UE transmits the BSR (and PHR) on PUSCH.
3. The eNB schedules the UE according to the reported buffer, its uplink SINR (from SRS/DMRS) and its power headroom, with contiguous RBs.
4. Each PUSCH transmission is acknowledged on PHICH after four subframes; non-adaptive retransmissions follow eight subframes after the original.

9 Worked examples

Worked example 1 • Counting resources

Problem. How many REs does a 15 MHz LTE carrier offer per second, and how many of them are PDSCH REs with a three-symbol control region and four CRS ports?
Solution. $75 \times 12 \times 14 \times 1000 = 12.6 \times 10^6$ RE/s. Data REs per RB pair: $168 - 36 - 16 = 116$ (four CRS ports use 24 REs, 8 of which lie in the three control symbols). Total $75 \times 116 \times 1000 = 8.7 \times 10^6$ RE/s, i.e. 69%.

Worked example 2 • OLLA step

Problem. An operator wants a 20% BLER target for a delay-tolerant service and uses $\delta_{\text{down}} = 1$ dB. Find δ_{up} and the offset drift if the actual BLER were 10%.
Solution. $\delta_{\text{up}} = 1 \times 0.2/0.8 = 0.25$ dB. At 10% BLER the expected change per transmission is $0.9 \times 0.25 - 0.1 \times 1 = +0.125$ dB: the offset rises (more aggressive MCS) until the BLER reaches 20%.

Worked example 3 • PF decision

Problem. Two UEs: UE A has average 10 Mbit/s and instantaneous 18 Mbit/s; UE B has average 2 Mbit/s and instantaneous 5 Mbit/s. Whom do Max C/I and PF serve?
Solution. Max C/I: UE A ($18 > 5$). PF: $18/10 = 1.8$ versus $5/2 = 2.5 \Rightarrow$ UE B, which is at a fading peak relative to its own average.

10 Technology comparison

Table 4. Scheduling and link adaptation across generations.

	HSDPA (3.5G)	LTE (4G)	NR (5G)
Scheduling unit	2 ms TTI, codes	1 ms TTI, RB pairs (180 kHz)	slot (1–0.125 ms) or mini-slot, RBs of $12\Delta f$
Frequency domain	none (CDMA)	per RB group (OFDMA)	per RB, bandwidth parts
Channel feedback	CQI	CQI, PMI, RI (subband)	CSI-RS based, beam reports
HARQ	NodeB, 6 processes	8 processes, 8 ms RTT (FDD)	up to 16 processes, flexible timing
Uplink	E-DCH (Rel-6)	SC-FDMA, contiguous RBs	CP-OFDM or DFT-s-OFDM, configured grants

Design trade-offs. Finer granularity enables better adaptation but costs control overhead (more DCIs, more CQI reports); frequent feedback improves accuracy but uses uplink capacity; longer PF windows favour throughput, shorter ones favour short-term fairness and delay. 5G extends the same principles with flexible numerology (Lecture 14).

11 Key takeaways

Key takeaways

- Frame 10 ms → subframe 1 ms → slot 0.5 ms (7 symbols); RB = 12 subcarriers × slot; $N_{\mathrm{RB}} = 6 \ldots 100$ for 1.4–20 MHz.
- 3GPP standardises signalling (DCI, CQI, BSR, PHR, HARQ timing), not the scheduling algorithm.
- CQI → efficiency $Q_m \times$ code rate; OLLA drives the BLER to $\delta_{\mathrm{up}} / (\delta_{\mathrm{up}} + \delta_{\mathrm{down}})$.
- Eight HARQ processes keep the channel busy; a 10% first-transmission BLER costs about 9%.
- Max C/I maximises $\sum \bar{R}_k$, PF maximises $\sum \log \bar{R}_k$, RR equalises time.
- Multi-user diversity: PF can beat RR for *every* user in fading channels.
- Practical schedulers: HARQ retransmissions first, then GBR/QoS, then PF for best effort.

12 Exercises

Exercise 1 • Level 1 – conceptual • Standard versus vendor

Which of the following are standardised by 3GPP and which are vendor implementation? CQI table, proportional fair metric, DCI format, OLLA step sizes, HARQ timing in FDD, priority of GBR bearers.

Exercise 2 • Level 1 – conceptual • Uplink contiguity

Explain why SC-FDMA requires contiguous RBs per UE and what this means for frequency-selective uplink scheduling.

Exercise 3 • Level 2 – numerical • Rate from CQI

A UE reports CQI 12 with rank 2 on a 20 MHz carrier (one control symbol, two CRS ports). It receives 40 RBs. Compute its rate with Eq. (2).

Exercise 4 • Level 2 – numerical • Jain's index

Verify the Jain indices of Table 3 for RR and Max C/I from the per-user values.

Exercise 5 • Level 3 – analytical • Two-user PF analysis

Two users have i.i.d. channels with the same distribution but user 1's rate is always twice user 2's ($r_1 = 2r_2$ in every TTI). Show that PF gives each user half of the TTIs, and compute the resulting throughputs relative to RR. What changes if the channels are independent?

Exercise 6 • Level 3 – analytical • Simulation study

Modify `scheduler_sim.py` to use 8 users with mean SNRs uniformly spaced between 0 and 21 dB and plot cell throughput and Jain's index versus $t_c \in \{10, 100, 1000\}$ for PF. Explain the trend.

Solutions

Solution to Exercise 1

Standardised: CQI table, DCI formats, HARQ timing in FDD (and the meaning of BSR/PHR). Vendor implementation: the PF metric (or any algorithm), OLLA step sizes, the prioritisation policy among bearers (constrained by the QoS parameters but not dictated in detail).

Solution to Exercise 2

SC-FDMA is DFT-spread OFDM: mapping the DFT output to contiguous subcarriers produces a single-carrier-like signal with low peak-to-average power ratio, allowing efficient power amplifiers in the handset. Non-contiguous mapping raises the PAPR. Hence the scheduler can choose *where* a UE's block lies (frequency-selective gain) but not pick scattered good RBs; Release 10 relaxed this with clustered allocation of two clusters.

Solution to Exercise 3

Data REs per RB pair: $168 - 12 - (16 - 4) = 144$. $R = 2 \times 3.9023 \times 40 \times 144/10^{-3} = 45.0\,\text{Mbit/s}$.

Solution to Exercise 4

RR: sum $= 19.88$, sum of squares $= 94.87 + 33.99 + 8.29 + 2.04 = 139.2$; $J = 395.2/(4 \times 139.2) = 0.710$. Max C/I: sum $= 40.13$, squares $\approx 1276.6 + 17.5 + 0.04 = 1294.2$; $J = 1610.4/(4 \times 1294.2) = 0.311$.

Solution to Exercise 5

With $r_1 = 2r_2$ always, $r_1/\bar{R}_1$ versus $r_2/\bar{R}_2$ compares $2r_2/\bar{R}_1$ with $r_2/\bar{R}_2$; equilibrium requires both to be chosen, which happens when $\bar{R}_1 = 2\bar{R}_2$, i.e. when each user gets half the TTIs – exactly RR, because the channels are perfectly correlated and there is no multi-user diversity to exploit. With independent channels PF picks each user at its relative peaks, and both throughputs exceed RR, as in Table 3.

Solution to Exercise 6

With mean SNRs $0, 3, \ldots, 21$ dB and 40 000 TTIs the script gives: $t_c = 10$: cell 29.8 Mbit/s, $J = 0.831$; $t_c = 100$: 32.8 Mbit/s, $J = 0.838$; $t_c = 1000$: 33.2 Mbit/s, $J = 0.839$ (RR: 21.8 Mbit/s, $J = 0.775$). A short window reacts to recent service and sometimes serves a user who is not at a relative peak; longer windows let PF wait for genuine peaks, so throughput rises and saturates once the window is much longer than the fading (here independent from TTI to TTI). The price of long windows in practice is short-term unfairness and larger delay jitter when channels are time-correlated.

13 Further reading

The physical channels and grid are specified in [2], CQI, DCI and HARQ procedures in [3] and MAC procedures (BSR, PHR, SR) in [1]. The proportional fair scheduler originated in CDMA-HDR [5]; its multi-user diversity interpretation is developed in Tse and Viswanath [7]. Jain's index is from [4]. Sesia et al. [6] cover LTE scheduling in depth.

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[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); medium access control (MAC) protocol specification*, TS 36.321, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36321.htm.

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[6] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. Wiley, 2011.

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08

Lecture 08

Random Access Channel (RACH)

کانال دسترسی تصادفی (RACH)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 6–7; slotted ALOHA; complex exponentials and correlation

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-08-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the situations in which an LTE UE must perform random access and **explain** why uplink synchronisation is the core problem.
- **LO2** **Describe** the PRACH: preamble formats, cyclic prefix, guard time and the maximum cell range they imply.
- **LO3** **Explain** Zadoff–Chu sequences, their correlation properties and how 64 preambles are generated per cell with cyclic shifts.
- **LO4** **Trace** contention-based (4-step) and contention-free random access, and the NR 2-step variant.
- **LO5** **Compute** timing advance values and the cell radius supported by a cyclic-shift configuration.
- **LO6** **Model** preamble collisions and RACH overload, and **evaluate** countermeasures such as access class barring and backoff.

2 Prerequisite knowledge

You need the LTE frame structure and uplink scheduling of Lecture 7 (in particular that PUSCH transmissions of different UEs must arrive time-aligned at the eNB), the timing advance concept of GSM (Lecture 1), the slotted ALOHA model of random access from introductory networking, and the definition of the periodic cross-correlation of two complex sequences.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Random access (RA)	دسترسی تصادفی	Uncoordinated first uplink access of a UE.
PRACH	کانال فیزیکی دسترسی تصادفی	Time–frequency resources reserved for preambles.
Preamble	پیش‌نوا	Known sequence sent by the UE in Msg1.
Zadoff–Chu (ZC) sequence	دنبالهٔ زادوف--چو	Constant-amplitude sequence with ideal cyclic autocorrelation.
Cyclic shift (N_{CS})	جابه‌جایی چرخشی	Shift that creates orthogonal preambles from one root.
Timing advance (TA)	پیش‌فرستی زمانی	Uplink advance compensating the round-trip delay.
RAR	پاسخ دسترسی تصادفی	Msg2: TA, uplink grant and temporary identity.
Contention resolution	حل رقابت	Msg4: tells which of the colliding UEs won.
Backoff	عقب‌نشینی	Random wait before a new attempt.
Access class barring (ACB)	مسدودسازی ردهٔ دسترسی	Probabilistic admission control before RA.

3 Introduction and motivation

Scheduled access needs two things the network does not have for a newly arriving UE: an *identity* to address it on the PDCCH and *uplink time alignment*. A UE that has just woken up knows the downlink timing (from the synchronisation signals) but not its distance to the eNB. If it simply transmitted on PUSCH, its symbols would arrive late by the round-trip delay and destroy the orthogonality of SC-FDMA with other users.

Random access is the procedure that solves both problems at once. The UE sends a short, robust signal (a *preamble*) in resources reserved for this purpose; the eNB measures its arrival time, sends back a timing correction and a first grant, and the two sides then resolve the possibility that several UEs chose the same preamble. The procedure is used surprisingly often – at every call set-up from idle, at every handover, after radio link failure and whenever an out-of-sync UE has data – so its capacity, latency and robustness matter. With massive IoT and 5G beam-based access, random access has become one of the most actively engineered parts of the radio interface.

4 Fundamental concepts

4.1 When random access is triggered

In LTE the MAC performs random access [1] for:

- initial access from RRC_IDLE (RRC connection establishment);
- RRC connection re-establishment after radio link failure;
- handover, to synchronise with the target cell;
- downlink data arrival for a connected UE that has lost uplink synchronisation;
- uplink data arrival when the UE is out of sync or has no scheduling-request resources;
- positioning, where the timing advance is needed.

4.2 Contention-based versus contention-free

In **contention-based** random access (CBRA) the UE picks a preamble at random from a common set; two UEs may pick the same one, so a contention-resolution step is needed. In **contention-free** random access (CFRA) the eNB assigns a dedicated preamble in advance (for handover or downlink data arrival); no collision can occur and the procedure ends after the RAR.

4.3 Zadoff–Chu sequences

Definition 1 • Zadoff–Chu sequence

For an odd prime length N_{ZC} and a root index $u \in \{1, \dots, N_{\mathrm{ZC}} - 1\}$,

$$x_u(n) = \exp\!\left(-\mathrm{j}\,\frac{\pi\, u\, n\,(n+1)}{N_{\mathrm{ZC}}}\right), \qquad n = 0, 1, \dots, N_{\mathrm{ZC}} - 1. \tag{1}$$

Its amplitude is constant, its periodic autocorrelation is zero for every non-zero shift, and the cross-correlation of two different roots has constant magnitude $1/\sqrt{N_{\mathrm{ZC}}}$ (normalised).

These properties make ZC sequences ideal preambles: constant amplitude means a low peak-to-average power ratio; zero autocorrelation means that *cyclic shifts* of one root are orthogonal and the delay of a received preamble shows up as a clean correlation peak. LTE uses $N_{\mathrm{ZC}} = 839$ for preamble formats 0–3 [2]; the script `scripts/validate_examples.py` verifies the properties numerically for $u = 25$ and $u = 29$.

5 Detailed technical explanation

5.1 The PRACH

The PRACH occupies six resource blocks (1.08 MHz) in the uplink, in subframes given by the *PRACH configuration index* broadcast in SIB2. Inside this band the preamble uses a subcarrier spacing of 1.25 kHz, so its 839 subcarriers fit in about 1.05 MHz. A preamble consists of a cyclic prefix, the ZC sequence (800 μs, or 1600 μs repeated) and an implicit guard time: Formats with longer CP and guard serve large cells; formats with a repeated sequence collect more energy for coverage-limited cells. Format 4 (TDD only) uses a short sequence of length 139 in the special subframe.

Table 1. LTE FDD preamble formats [2]. The guard time bounds the round-trip delay, hence the cell range $d_{\max} = c\,T_{\mathrm{GT}}/2$.

Format	Subframes	CP [μs]	Sequence [μs]	Guard [μs]	Range [km]
0	1	103.1	800	96.9	14.5
1	2	684.4	800	515.6	77.3
2	2	203.1	1600	196.9	29.5
3	3	684.4	1600	715.6	107.3

5.2 64 preambles per cell

Each cell offers 64 preambles. They are generated from one or more ZC roots by cyclic shifts of N_{CS} samples: one root yields $\lfloor N_{\mathrm{ZC}}/N_{\mathrm{CS}} \rfloor$ preambles, and further roots (consecutive in a broadcast logical order) are used until 64 are obtained. The shift must exceed the maximum round-trip delay plus delay spread; otherwise a distant UE's preamble would be detected as a neighbouring shift (Section 6.2). The 64 preambles are divided into contention-based groups (A and B, the latter signalling a larger Msg3) and a set reserved for contention-free use.

5.3 Contention-based random access (4 steps)

1. **Msg1 – preamble.** The UE selects a preamble at random from its group and transmits it in the next PRACH occasion, with an open-loop power estimate from the downlink path loss.
2. **Msg2 – random access response (RAR).** Within a response window the eNB sends, on PDSCH addressed by an RA-RNTI derived from the PRACH occasion, the detected preamble index (RAPID), a timing advance command (11 bits), an uplink grant for Msg3 and a temporary C-RNTI. If no RAR arrives, the UE *ramps* its power and retries.
3. **Msg3 – scheduled transmission.** The UE applies the TA and sends, on PUSCH with HARQ, its identity and request – for initial access the RRC Connection Request with S-TMSI or a random value.
4. **Msg4 – contention resolution.** The eNB echoes the identity it decoded. A UE whose identity matches has won; the temporary C-RNTI becomes its C-RNTI. Others start the backoff and retry.

Two UEs that chose the same preamble in the same occasion both receive the same RAR and both send Msg3 on the same resources; typically only one Msg3 is decoded (capture effect), which is why Msg4 must echo the identity.

5.4 Contention-free random access

For handover the target cell reserves a preamble and includes it in the handover command; for downlink data arrival the eNB signals it on the PDCCH. The UE sends the dedicated preamble, receives the RAR with TA and grant, and the procedure ends – Lecture 4 relied on this for the access step of X2 handover.

5.5 RACH overload and its control

When many devices access simultaneously – after a power outage, at a stadium, or when thousands of meters report at the same time – collisions multiply, retries multiply further, and the PRACH collapses. LTE offers several countermeasures:

- **Access class barring (ACB)**: SIB2 broadcasts a barring factor a and time; before accessing, a UE draws a uniform number and proceeds only if it is below a, otherwise it waits. Extended access barring (EAB) targets delay-tolerant devices.
- **Backoff indicator** in the RAR: losers wait a random time up to the indicated maximum.
- **More PRACH resources**: additional PRACH occasions per frame, at the cost of uplink capacity.
- **Separate resources for IoT**: NB-IoT and LTE-M define their own random access channels.

5.6 Random access in 5G NR

NR keeps the four-step structure but adapts it to beams and to low latency [5]: each synchronisation signal block (SSB, one per beam) is associated with specific PRACH occasions, so the preamble also tells the gNB which downlink beam the UE sees best; preambles of length 839 or 139 with several subcarrier spacings support everything from long-range rural cells to mmWave; and Release 16 adds **2-step RACH**, in which MsgA combines the preamble and the Msg3 payload and MsgB combines the RAR and contention resolution, halving the round trips.

6 Mathematical models

6.1 Timing advance

Problem. How precisely can LTE align a UE, and how far can it be?

Model. The RAR carries $T_A \in \{0, \ldots, 1282\}$; the UE advances its uplink by $N_{\mathrm{TA}} = 16\, T_A\, T_\mathrm{s}$ with $T_\mathrm{s} = 1/(15000 \times 2048)$ s [3].

Derivation. One step corresponds to a round-trip time of $16 T_\mathrm{s} = 0.521\,\mu$s and therefore to a one-way distance

$$\Delta d = \frac{c \cdot 16\, T_\mathrm{s}}{2} = \frac{2.998 \times 10^8 \times 0.521 \times 10^{-6}}{2} \approx 78.1\,\mathrm{m}, \qquad d_{\max} = 1282\, \Delta d \approx 100\,\mathrm{km}. \tag{2}$$

Interpretation. The step is a fraction of the normal cyclic prefix (4.7 μs), so residual misalignment is absorbed by the CP. Compare with GSM's 553 m step (Lecture 1): the finer step reflects the much shorter LTE symbol and the need to keep all users within the CP. After access, MAC TA commands (6 bits, relative) keep the alignment as the UE moves.

6.2 Cyclic shift and cell radius

Model. The sequence duration $T_{\mathrm{SEQ}} = 800\,\mu$s spans $N_{\mathrm{ZC}} = 839$ samples, so one sample lasts $T_{\mathrm{SEQ}}/N_{\mathrm{ZC}} = 0.954\,\mu$s. A UE at distance r shifts its preamble by the round-trip delay $2r/c$; with multipath delay spread τ_{ds} the received energy occupies $2r/c + \tau_{\mathrm{ds}}$.

Condition. To be unambiguous, this must fit inside one cyclic-shift zone:

$$N_{\mathrm{CS}}\,\frac{T_{\mathrm{SEQ}}}{N_{\mathrm{ZC}}} \;\geq\; \frac{2r}{c} + \tau_{\mathrm{ds}} \quad\Longrightarrow\quad r_{\max} = \frac{c}{2}\left(N_{\mathrm{CS}}\,\frac{T_{\mathrm{SEQ}}}{N_{\mathrm{ZC}}} - \tau_{\mathrm{ds}}\right). \tag{3}$$

Example and interpretation. With $\tau_{\mathrm{ds}} = 5.2\,\mu$s: $N_{\mathrm{CS}} = 13$ gives $r_{\max} \approx 1.1$ km and $\lfloor 839/13 \rfloor = 64$ preambles from a *single* root; $N_{\mathrm{CS}} = 119$ gives $r_{\max} \approx 16.2$ km but only 7 preambles per root, so $\lceil 64/7 \rceil = 10$ roots are needed. Different roots are not orthogonal (cross-correlation $1/\sqrt{839} \approx 0.035$), so large cells have more false alarms and inter-cell preamble interference – a coverage/robustness trade-off planned per cell. The formula ignores a small detection margin used in practice; the guard-time limit of Table 1 must also hold.

6.3 Collision probability

Model. n UEs contend in the same PRACH occasion and choose uniformly among M contention preambles (e.g. $M = 54$ when 10 of the 64 are reserved for CFRA). A UE succeeds if no other UE chose its preamble (ignoring detection errors and Msg3 capture).

Derivation. The probability that a given UE is alone on its preamble is

$$P_{\mathrm{s}}(n) = \left(1 - \frac{1}{M}\right)^{n-1}, \tag{4}$$

and the expected number of successful UEs in the occasion is

$$S(n) = n\left(1 - \frac{1}{M}\right)^{n-1}. \tag{5}$$

Treating n as continuous, $\mathrm{d}S/\mathrm{d}n = 0$ gives $n^{*} = -1/\ln(1 - 1/M) \approx M$, where $S(n^{*}) \approx M/\mathrm{e}$ – exactly the slotted-ALOHA efficiency of $1/\mathrm{e} \approx 37\%$ per preamble.

Interpretation. For $M = 54$: 10 contenders succeed individually with 84.5%; 54 contenders give about 20 successes (37%); 150 contenders give only 9.3 successes – *fewer* than 54 did (Figure 1). Beyond the optimum, more load means less throughput, and the failed UEs retry, pushing the load further: a positive feedback that leads to collapse unless access is throttled (ACB, backoff). The model ignores power ramping, capture and the multiple PRACH occasions per frame.

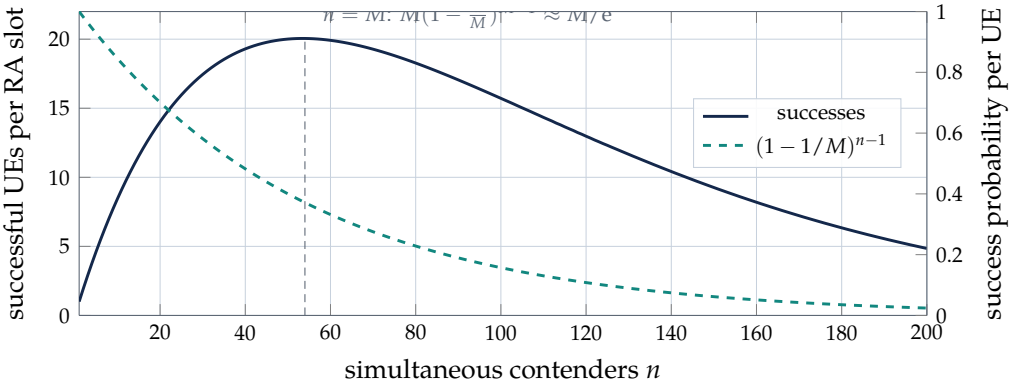


Figure 1. Expected successful UEs per PRACH occasion (left axis, Eq. (5)) and per-UE success probability (right axis, Eq. (4)) for $M = 54$ contention preambles.

6.4 Access delay

If each attempt succeeds independently with probability P_s, the number of attempts is geometric with mean $1/P_s$, and the mean access delay is

$$\bar{D}_{RA} \approx T_{RA} + \left(\frac{1}{P_s} - 1\right)(T_{fail} + \bar{T}_{BO}), \tag{6}$$

where T_{RA} is the duration of a successful 4-step exchange, T_{fail} the time to detect a failure (end of the RAR window or of the contention-resolution timer) and $\bar{T}_{BO}$ the mean backoff (Exercise 4).

7 Network architecture and diagrams

Random access is a purely radio-level procedure between UE and eNB (MAC and PHY), but its outcome feeds the higher layers: Msg3 carries the first RRC message, and Msg4 (in initial access) the RRC Connection Setup, after which the NAS Service Request or Attach Request reaches the MME (Lecture 6). Figure 2 shows the format-0 preamble, Figures 3 to 5 the procedures.

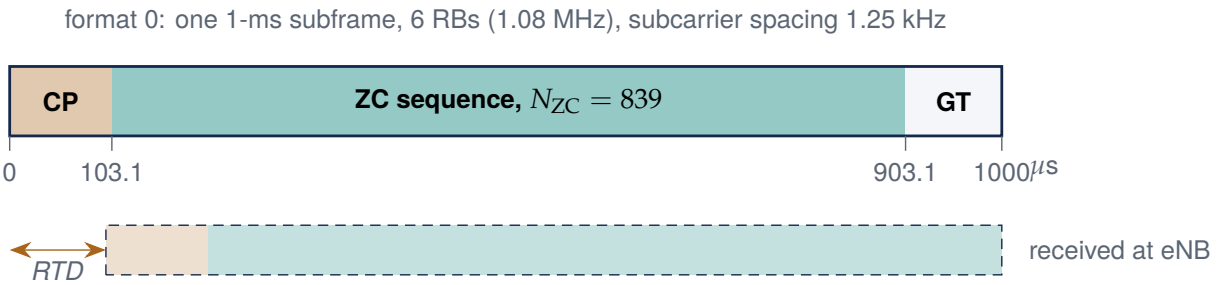


Figure 2. Preamble format 0. The cyclic prefix absorbs the round-trip delay (RTD) and delay spread; the guard time keeps a late preamble from overlapping the next subframe.

8 Step-by-step procedures

Procedure 1 • Contention-based random access for initial access

1. Read SIB2: PRACH configuration index, root sequence, N_{CS}, preamble groups, power parameters, RAR window, contention-resolution timer, ACB parameters.
2. Pass the access-barring check (if configured), select group A or B, pick a preamble at random and transmit it with initial power $P = \min\{P_{max},\ P_{target} + L_p\}$.
3. Monitor the PDCCH for the RA-RNTI during the RAR window. No matching RAR: increase power by the ramping step, apply backoff and go back to step 2 (up to the maximum number of attempts).
4. On RAR: apply TA, store the temporary C-RNTI, send Msg3 on the granted PUSCH, start the contention-resolution timer.
5. On Msg4 with matching identity: success, C-RNTI confirmed, RRC continues. Timer expiry or mismatch: backoff and retry.

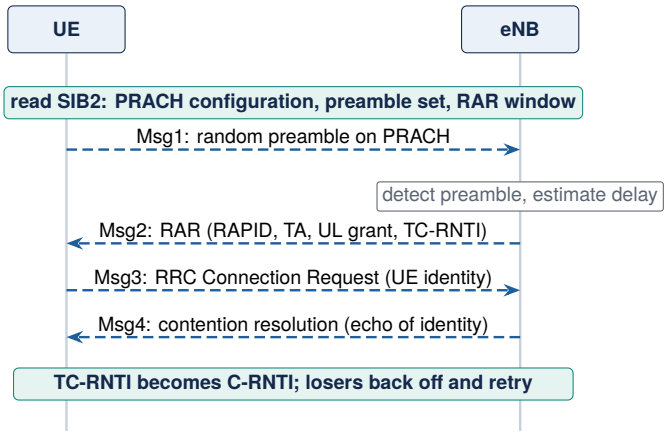


Figure 3. Contention-based (4-step) random access (Procedure 1).

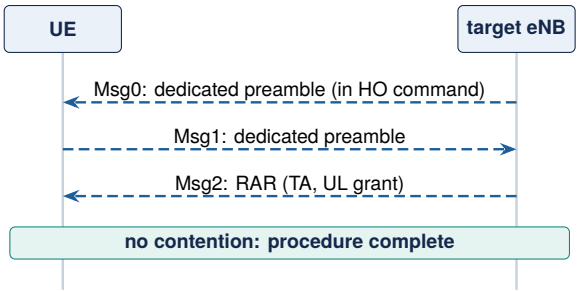


Figure 4. Contention-free random access, e.g. at handover.

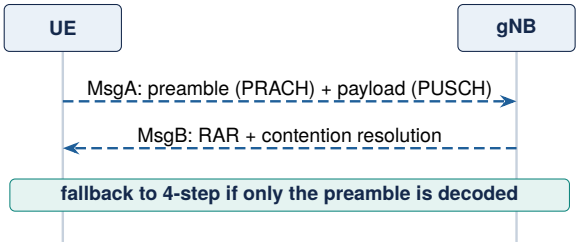


Figure 5. NR 2-step random access (Release 16).

9 Worked examples

Worked example 1 • Timing advance from a measured delay

Problem. The eNB measures a preamble round-trip delay of 21.4 μs. Which T_A is sent and how far is the UE?
Solution. $T_A = \text{round}(21.4/0.521) = 41$; $d \approx 41 \times 78.1 = 3.2\,\text{km}$ (directly: $c\tau/2 = 3.21\,\text{km}$).

Worked example 2 • Planning N_{CS}

Problem. A rural cell must serve UEs up to 12 km with a delay spread of 5.2 μs. Find the smallest N_{CS}, the preambles per root and the number of roots.
Solution. Equation (3): $N_{CS} \geq (2 \times 12\,000/c + 5.2 \times 10^{-6})/0.954 \times 10^{-6} = (80.1 + 5.2)/0.954 = 89.4$. The standard set of N_{CS} values contains 93, which gives $\lfloor 839/93 \rfloor = 9$ preambles per root and $\lceil 64/9 \rceil = 8$ roots.

Worked example 3 • Event-driven IoT burst

Problem. After a power cut, 5000 meters in one cell try to access within 1 s. The cell has one PRACH occasion per 10 ms frame (100 per second) and $M = 54$ contention preambles. Estimate the successes if every meter tries once in a random occasion, and suggest an access-barring factor.
Solution. 50 contenders per occasion on average: $S = 50\,(53/54)^{49} \approx 50 \times 0.40 = 20.0$ successes per occasion, about 2000 in the first second; the 3000 losers retry and add to the next second's load. If the 5000 meters instead all fell into 10 occasions (one per 100 ms), there would be 500 contenders per occasion and $S = 500\,(53/54)^{499} \approx 0.04$: almost nobody succeeds. A barring factor $a \approx 54/500 \approx 0.1$ brings the effective load back to the optimum $n \approx M$, where about $M/\mathrm{e} \approx 20$ UEs succeed per occasion.
Interpretation. In an overloaded random access channel, throttling *increases* the throughput – the central lesson of ALOHA.

10 Technology comparison

Trade-offs. Longer preambles and larger cyclic shifts extend range but reduce the number of orthogonal preambles per root; more PRACH occasions reduce collisions but consume uplink capacity; 2-step RACH halves latency but needs PUSCH resources for payloads that may collide.

11 Key takeaways

Table 2. Random access across generations.

	GSM	LTE	NR
Channel	RACH burst on a timeslot	PRACH, 6 RBs, ZC preambles	PRACH tied to SSB beams
Contention resolution	SDCCH assignment + identity	Msg3/Msg4 with identity	4-step or 2-step (MsgA/MsgB)
TA step / range	553 m / 35 km	78 m / 100 km	depends on numerology (finer at higher SCS)
Contention-free	no	yes (handover, DL data)	yes, plus beam-failure recovery
Overload control	access classes	ACB, EAB, backoff	unified access control

Key takeaways

- Random access provides uplink time alignment and a first identity/grant; it is used for initial access, re-establishment, handover and out-of-sync data.
- Preambles are ZC sequences ($N_{ZC} = 839$): constant amplitude, zero cyclic autocorrelation; 64 per cell from cyclic shifts of one or more roots.
- $r_{max} = \frac{c}{2}(N_{CS}T_{SEQ}/N_{ZC} - \tau_{ds})$; the guard time also bounds the range (format 0: 14.5 km).
- TA step $16T_s = 0.52\,\mu$s $\Rightarrow$ 78 m; 11-bit RAR TA $\Rightarrow \approx 100$ km.
- CBRA: preamble $\rightarrow$ RAR $\rightarrow$ Msg3 $\rightarrow$ contention resolution; CFRA ends after the RAR.
- Successes per occasion $n(1-1/M)^{n-1}$ peak at $\approx M/\mathrm{e}$; overload collapses throughput, so throttle (ACB, backoff).
- NR adds beam-associated PRACH and 2-step RACH.

12 Exercises

Exercise 1 • Level 1 – conceptual • Why not just PUSCH?

Explain what would go wrong if an idle UE transmitted its first message directly on PUSCH without random access. Consider both timing and identity.

Exercise 2 • Level 1 – conceptual • CBRA or CFRA?

For each case state whether CBRA or CFRA is used and why: initial attach; handover; RRC re-establishment after RLF; downlink data arrival for an out-of-sync connected UE.

Exercise 3 • Level 2 – numerical • Format choice

A cell must cover 60 km over flat terrain. Which LTE preamble formats are possible from the guard-time point of view, and what is the smallest N_{CS} (with $\tau_{ds} = 5.2\,\mu s$)? Is it possible with an 800 μs sequence?

Exercise 4 • Level 2 – numerical • Access delay

With $T_{RA} = 20$ ms, $T_{fail} = 15$ ms and a uniform backoff in $[0, 20]$ ms, compute the mean access delay from Eq. (6) for $n = 10$ and $n = 54$ contenders ($M = 54$).

Exercise 5 • Level 3 – analytical • ZC properties

Prove that the periodic autocorrelation of Eq. (1) is zero for every non-zero shift τ when N_{ZC} is prime. (Hint: write $x_u(n+\tau)x_u^*(n)$ and sum a geometric series in n.)

Exercise 6 • Level 3 – analytical • Stability of retries

New UEs arrive at a mean of λ per PRACH occasion; each failed UE retries in the next occasion. Using Eq. (5) with Poisson approximation $S \approx Ge^{-G/M}$ for offered load G per occasion, write the equilibrium condition, find the maximum sustainable λ and discuss what happens beyond it.

Solutions

Solution to Exercise 1

Timing: the UE knows only the downlink timing; its uplink would arrive late by the round-trip delay (up to hundreds of μs), overlapping other users' SC-FDMA symbols and destroying orthogonality. Identity: the eNB has no C-RNTI for the UE and cannot address a grant to it; unscheduled PUSCH resources do not exist. Random access solves both with a robust, delay-tolerant preamble.

Solution to Exercise 2

Initial attach: CBRA (the network does not know the UE). Handover: CFRA (the target reserves a preamble, saving the contention step and time). RRC re-establishment: CBRA (unplanned). Downlink data for an out-of-sync UE: CFRA (the eNB knows the UE and signals a dedicated preamble on the PDCCH), falling back to CBRA if no dedicated preamble is available.

Solution to Exercise 3

The guard time must cover the round trip $2 \times 60\,000/c = 400\,\mu$s: formats 1 (515.6 μs) and 3 (715.6 μs) qualify; 0 and 2 do not. Equation (3) requires $N_{CS} \geq (400 + 5.2)/0.954 = 425$ samples. The largest non-zero value of the standard unrestricted set is 419 ($r_{max} \approx 59$ km), just short; the remaining option is $N_{CS} = 0$ (no cyclic shift), in which each root provides a single preamble and 64 roots are used. This works for any delay, but the 64 preambles are no longer orthogonal (only their low cross-correlation separates them), so detection performance degrades – the price of very large cells. The 800-μs sequence length is not the issue; repetition (format 3) helps the link budget only.

Solution to Exercise 4

$\bar{T}_{\mathrm{BO}} = 10\,\mathrm{ms}$. $n = 10$: $P_{\mathrm{s}} = 0.845$, $1/P_{\mathrm{s}} - 1 = 0.183$, $\bar{D} = 20 + 0.183 \times 25 = 24.6\,\mathrm{ms}$. $n = 54$: $P_{\mathrm{s}} = 0.371$, $1/P_{\mathrm{s}} - 1 = 1.70$, $\bar{D} = 20 + 1.70 \times 25 = 62.4\,\mathrm{ms}$ (optimistic, because retries also add load).

Solution to Exercise 5

$x_u(n+\tau)x_u^*(n) = \exp(-\mathrm{j}\pi u[(n+\tau)(n+\tau+1) - n(n+1)]/N) = \exp(-\mathrm{j}\pi u[\tau^2 + \tau]/N)\exp(-\mathrm{j}2\pi u\tau n/N)$. The first factor is constant in n; summing the second over $n = 0, \ldots, N-1$ gives $\sum_n \mathrm{e}^{-\mathrm{j}2\pi u\tau n/N} = 0$ unless $u\tau \equiv 0 \pmod N$. With N prime and $1 \le u, \tau < N$, $u\tau$ is never a multiple of N, so the autocorrelation is zero for all $\tau \neq 0$. (Periodicity of the quadratic phase modulo N holds for odd N.)

Solution to Exercise 6

In equilibrium the success rate equals the arrival rate: $\lambda = Ge^{-G/M}$. The right-hand side is maximal at $G = M$ with value M/e, so $\lambda_{\max} = M/\mathrm{e} \approx 19.9$ UEs per occasion for $M = 54$. For $\lambda < M/\mathrm{e}$ there are two equilibria: the low-load one ($G < M$) is stable, the high-load one is not. For $\lambda > M/\mathrm{e}$ no equilibrium exists: the backlog grows without bound and successes fall towards zero – the instability that ACB and randomised backoff are designed to prevent.

13 Further reading

The MAC random access procedure is in [1], preamble generation and formats in [2], timing advance in [3], and ACB in the RRC specification [4]. NR random access, including 2-step RACH, is summarised in [5]. Sesia et al. [7] contains a detailed chapter on PRACH design and ZC sequences; Dahlman et al. [6] covers the NR procedures.

References

[1] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); medium access control (MAC) protocol specification*, TS 36.321, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36321.htm.

[2] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); physical channels and modulation*, TS 36.211, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36211.htm.

[3] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); physical layer procedures*, TS 36.213, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36213.htm.

[4] 3GPP, *Evolved universal terrestrial radio access (E-UTRA); radio resource control (RRC); protocol specification*, TS 36.331, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/36331.htm.

[5] 3GPP, *NR; NR and NG-RAN overall description; stage-2*, TS 38.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38300.htm.

[6] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic Press, 2020.

[7] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. Wiley, 2011.

09

Lecture 09

Evolved Packet Core (EPC)

هستهٔ بستهٔ تکامل‌یافته (EPC)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5–6; basic cryptography (symmetric keys, MAC); IP routing

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
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11 Key takeaways . . . 11
12 Exercises . . . 12
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-09-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the design goals of the EPC and the division of work among MME, HSS, S-GW, P-GW and PCRF.
- **LO2** **Describe** the EPS bearer model: default and dedicated bearers, TFTs and the QCI table, and **apply** GBR admission control.
- **LO3** **Trace** EPS-AKA and derive the EPS key hierarchy from K to the access-stratum keys.
- **LO4** **Trace** PDN connectivity and dedicated bearer activation, modification and deactivation.
- **LO5** **Describe** EMM/ECM states, tracking area updates and the EPC's role in handover.
- **LO6** **Analyse** the impact of node failures and **estimate** the signalling storm after a gateway restart.

2 Prerequisite knowledge

This lecture assumes the EPS architecture and reference points of Lecture 5 and the RRC/S1 procedures of Lecture 6. From security you need the idea of a shared symmetric key, of a message authentication code (a keyed checksum that proves integrity and origin) and of a key derivation function (KDF) that produces new keys from old ones without revealing them.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
PDN connection	اتصال PDN	Association between a UE IP address and an APN, anchored in a P-GW.
Default bearer	حامل پیش‌فرض	First bearer of each PDN connection; always on.
Dedicated bearer	حامل اختصاصی	Additional bearer with its own QoS for selected flows.
TFT	قالب جریان ترافیک	Packet filters mapping IP flows to bearers.
QCI	شناسهٔ کلاس کیفیت خدمات	Index selecting priority, delay budget and loss rate.
EPS-AKA	توافق احراز هویت و کلید EPS	Mutual authentication producing K_{ASME}.
KDF	تابع استخراج کلید	HMAC-SHA-256-based derivation of new keys.
EMM / ECM	مدیریت تحرک / اتصال EPS	Registration state / signalling-connection state.
TAU	به‌روزرسانی ناحیهٔ ردیابی	UE procedure when entering a TA outside its list or periodically.
PCC	کنترل سیاست و صورت‌حساب	PCRF decisions enforced by the PCEF in the P-GW.

3 Introduction and motivation

The EPC was designed in the System Architecture Evolution (SAE) study with a clear brief: a *single*, *all-IP* core for LTE that could also serve legacy 3GPP access (via SGSN) and non-3GPP access (WLAN, CDMA2000), with lower latency, fewer nodes and a clean split between control and user planes [3]. The resulting core has only five essential node types, yet it must reproduce – with packets – everything the old CS and PS cores provided: subscriber authentication, mobility with session continuity, QoS for voice, charging and lawful interception.

This lecture is the system view of the EPC. It connects the pieces: how a subscriber is authenticated, how bearers are created and mapped to flows, how the UE's states are tracked, and what happens when a node fails. Lectures 10–13 then zoom into the protocols and nodes (Diameter, HSS, MME, gateways).

4 Fundamental concepts

4.1 Division of labour

- **MME** – control plane only: NAS signalling, authentication, security, tracking areas, paging, bearer control, gateway selection.
- **HSS** – subscription database and authentication centre.
- **S-GW** – user-plane mobility anchor and idle-mode buffer.
- **P-GW** – IP anchor towards each PDN, policy and charging enforcement (PCEF).
- **PCRF** – policy and charging decisions (PCC rules).

The MME and the gateways are connected by S11 (GTPv2-C), the gateways by S5/S8; the MME talks to the HSS over S6a and the P-GW to the PCRF over Gx, both Diameter.

4.2 The bearer model

Definition 1 • EPS bearer and TFT

An EPS bearer is the unit of QoS in LTE: all packets on a bearer receive the same forwarding treatment. A *traffic flow template* (TFT) is a set of packet filters (source/destination address and port ranges, protocol, type of service) that decides which bearer a packet uses: the *UL TFT* in the UE for uplink packets, the *DL TFT* in the P-GW for downlink packets.

Each PDN connection starts with a default bearer without a TFT (it catches everything). Dedicated bearers are created when a service needs different QoS – typically triggered by the PCRF when an application function (e.g. IMS) requests it. Every bearer is characterised by

- **QCI** – a standardised class with priority, packet delay budget (PDB) and packet error loss rate (PELR) (Table 1);
- **ARP** – allocation and retention priority, used for admission and pre-emption;
- for GBR bearers, **GBR** and **MBR** (guaranteed and maximum bit rate);
- for non-GBR bearers collectively, the **APN-AMBR** and **UE-AMBR** aggregate limits.

Table 1. Standardised QCI characteristics (TS 23.203, Table 6.1.7-A, QCI 1–9) [5].

QCI	Type	Priority	PDB [ms]	PELR	Example services
1	GBR	2	100	10^{-2}	Conversational voice (VoLTE)
2	GBR	4	150	10^{-3}	Conversational video (live)
3	GBR	3	50	10^{-3}	Real-time gaming
4	GBR	5	300	10^{-6}	Non-conversational (buffered) video
5	non-GBR	1	100	10^{-6}	IMS signalling
6	non-GBR	6	300	10^{-6}	Buffered video, TCP applications (premium)
7	non-GBR	7	100	10^{-3}	Voice, live video, interactive gaming
8	non-GBR	8	300	10^{-6}	TCP applications (e.g. web, e-mail)
9	non-GBR	9	300	10^{-6}	TCP applications, typical default bearer

4.3 EMM and ECM states

Two orthogonal state machines describe a UE in the EPC (Figure 1): *EPS mobility management* (EMM-DEREGISTERED/REGISTERED) says whether the network knows the UE; *EPS connection management* (ECM-IDLE/CONNECTED) says whether a NAS signalling connection (RRC + S1) exists. A registered UE in ECM-IDLE keeps its IP address and bearers in the core (S5/S8 tunnels stay up), but has no radio or S1-U resources; the S-GW buffers downlink data and asks the MME to page.

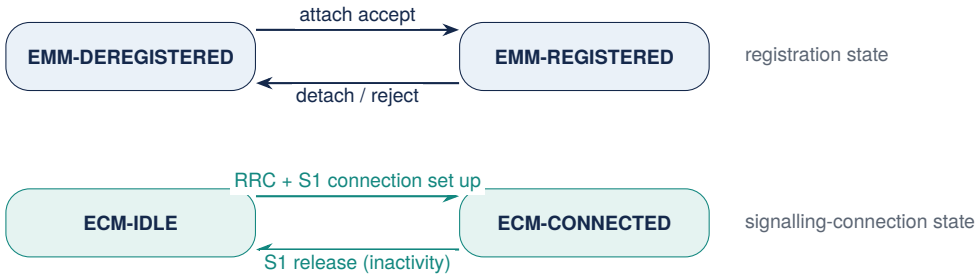


Figure 1. EMM (registration) and ECM (signalling connection) states of a UE in the EPC.

5 Detailed technical explanation

5.1 Authentication: EPS-AKA

EPS-AKA [2] extends UMTS AKA (Lecture 3). The HSS generates an *EPS authentication vector* (RAND, XRES, AUTN, K_{ASME}); the novelty is that K_{ASME} is bound to the serving network identity, so a vector issued for one network is useless in another. The MME sends RAND and AUTN; the USIM checks AUTN (which proves that the challenge comes from the home network and is fresh, via the sequence number SQN), computes RES and the same keys. The MME compares RES with XRES; both sides now share K_{ASME}, which never leaves the MME. The detailed Milenage computations are in Lecture 11.

5.2 NAS and AS security

Security is layered:

- **NAS security** (UE–MME): ciphering and integrity protection of NAS messages with K_{NASenc} and K_{NASint}, activated by the NAS Security Mode Command.
- **AS security** (UE–eNB): integrity and ciphering of RRC (K_{RRCint}, K_{RRCenc}) and ciphering of user data (K_{UPenc}) in PDCP, derived from K_{eNB} which the MME sends to the eNB in Initial Context Setup.

User data is *not* integrity-protected in LTE (only ciphered) – a deliberate trade-off for throughput that 5G revisits. The algorithms are EEA1/EIA1 (SNOW 3G), EEA2/EIA2 (AES) and EEA3/EIA3 (ZUC); EEA0/EIA0 are null algorithms, EIA0 being allowed only for emergency calls without authentication.

5.3 Session management

PDN connectivity. At attach (and later for additional APNs) the UE requests a PDN connection. The MME selects a P-GW from the APN (DNS-based) and an S-GW from the UE's location, and sends Create Session Request; the P-GW allocates the IP address (IPv4, IPv6 prefix or both), asks the PCRF for policy (Gx CCR-I/CCA-I) and creates the default bearer.

Dedicated bearer activation. Normally network-initiated (Figure 5): the PCRF provisions PCC rules, the P-GW sends Create Bearer Request with TFT and QoS through the S-GW to the MME, which sets up the E-RAB in the eNB and sends the NAS Activate Dedicated EPS Bearer Context Request to the UE.

Modification and deactivation. QoS or TFT changes use Update Bearer Request; removal uses Delete Bearer Request. Deleting the default bearer deletes the whole PDN connection.

5.4 Mobility in the EPC

- **Tracking area update (TAU)**: when the UE enters a TA not in its list, or when the periodic timer T3412 expires; the MME may change (context transfer from the old MME) and so may the S-GW.
- **X2 handover**: the core only switches the path (Lecture 4).
- **S1 handover**: used without X2 or when the MME or S-GW changes; the source and target eNBs exchange handover messages via the MME(s).
- **Inter-RAT mobility**: to UTRAN/GERAN through the SGSN (S3/S4 interfaces); the S-GW remains the anchor, the P-GW keeps the IP address.

5.5 Failure scenarios

- **MME failure** – with S1-flex the eNB connects to other MMEs of the pool; UEs whose contexts were lost are recovered when they next contact the network (TAU or service request is rejected with an implicit detach, followed by re-attach).
- **S-GW failure** – detected through GTP echo and the *restart counter* carried in the Recovery information element. The MME and P-GW delete the affected PDN connections; idle UEs are not reachable until they re-attach or the MME restores the sessions on another S-GW (restoration procedures).
- **P-GW failure** – the IP anchor is lost: the UE's IP address and all its sessions on that APN disappear; applications must reconnect.
- **HSS unavailable** – already registered UEs continue (the MME caches subscription and unused vectors); new attaches of UEs unknown to the MME fail.

6 Mathematical models

6.1 GBR admission control

Problem. How many GBR bearers can a cell or gateway admit?

Model. A resource (cell, link) has an average capacity C; the operator reserves a fraction ρ_{GBR} for GBR traffic. Bearer i requests GBR_i. Admission requires

$$\sum_{i \in \mathcal{A}} \mathrm{GBR}_i + \mathrm{GBR}_{\mathrm{new}} \le \rho_{\mathrm{GBR}}\, C, \tag{1}$$

otherwise the request is rejected, or a bearer with lower ARP is pre-empted if the new one may pre-empt and the old one is pre-emptable.

Example. $C = 30\,\mathrm{Mbit/s}$ average cell throughput, $\rho_{\mathrm{GBR}} = 0.5$, VoLTE GBR $= 40\,\mathrm{kbit/s}$: at most $15 \times 10^6 / 40 \times 10^3 = 375$ calls.

Limits. In a radio cell C is not constant: it depends on where the users are (their MCS). Radio admission therefore estimates the *resource* (RBs) a bearer will need at its current channel quality rather than its bit rate (Exercise 3).

6.2 The EPS key hierarchy

Model. Keys are derived with the 3GPP KDF, HMAC-SHA-256 keyed with the parent key over a string S that encodes the purpose (an FC byte and parameters) [2]:

$$K_{\mathrm{ASME}} = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SN\ id},\ \mathrm{SQN} \oplus \mathrm{AK}\big), \tag{2a}$$

$$K_{\mathrm{eNB}} = \mathrm{KDF}\big(K_{\mathrm{ASME}},\ \text{uplink NAS COUNT}\big), \tag{2b}$$

$$K_{\mathrm{NASint}} = \mathrm{KDF}\big(K_{\mathrm{ASME}},\ \text{alg. type},\ \text{alg. id}\big)\ \text{(truncated to 128 bits)}, \tag{2c}$$

and analogously for K_{NASenc}, K_{RRCint}, K_{RRCenc} and K_{UPenc} from K_{eNB} (Figure 3).

Interpretation. Each key is bound to its context: the serving network (Eq. (2a)), the NAS COUNT, hence a fresh K_{eNB} at every transition to connected mode (Eq. (2b)), and the algorithm. Compromise of one eNB's keys reveals nothing about K_{ASME} (one-way KDF). At handover a new K^*_{eNB} is derived so the target does not learn the source's keys.

Brute force. A 128-bit algorithm key has $2^{128} \approx 3.4 \times 10^{38}$ values; at 10^{12} trials per second an exhaustive search takes about 1.1×10^{19} years. Security therefore rests on protecting K and on the algorithms, not on key length.

6.3 Signalling storm after a gateway restart

Problem. An S-GW serving N UEs restarts; all of them must re-establish their sessions. How long does recovery take, and what load does it put on the MME?

Model. Each recovery costs m messages at the MME; the MME processes at most μ messages per second. If all UEs try immediately, the MME is saturated for

$$T_{\mathrm{rec}} = \frac{N\,m}{\mu}. \tag{3}$$

If recovery attempts are spread uniformly over a window $T_{\mathrm{w}} > T_{\mathrm{rec}}$, the offered signalling rate is Nm/T_{w} and the MME stays below saturation.

Example. $N = 10^6$, $m = 10$, $\mu = 2 \times 10^4$ msg/s: $T_{\mathrm{rec}} = 500\,\mathrm{s}$. Spreading over $T_{\mathrm{w}} = 900\,\mathrm{s}$ gives 1.1×10^4 msg/s (56% of capacity) instead of an instantaneous burst of 10^7 messages, which would overflow queues, time out, and trigger retries that make the storm worse.

Lesson. Restoration procedures and randomised timers exist to turn a burst into a spread load – the same positive-feedback hazard as RACH overload (Lecture 8), one level higher.

7 Network architecture and diagrams

Figure 2 shows the bearer as seen end to end. The radio bearer, S1 bearer and S5/S8 bearer are bound one-to-one; the S-GW and eNB simply relay packets between tunnels, while the TFTs at the two ends decide the bearer. The E-RAB (radio + S1) is the part that is released in ECM-IDLE; the S5/S8 part persists.

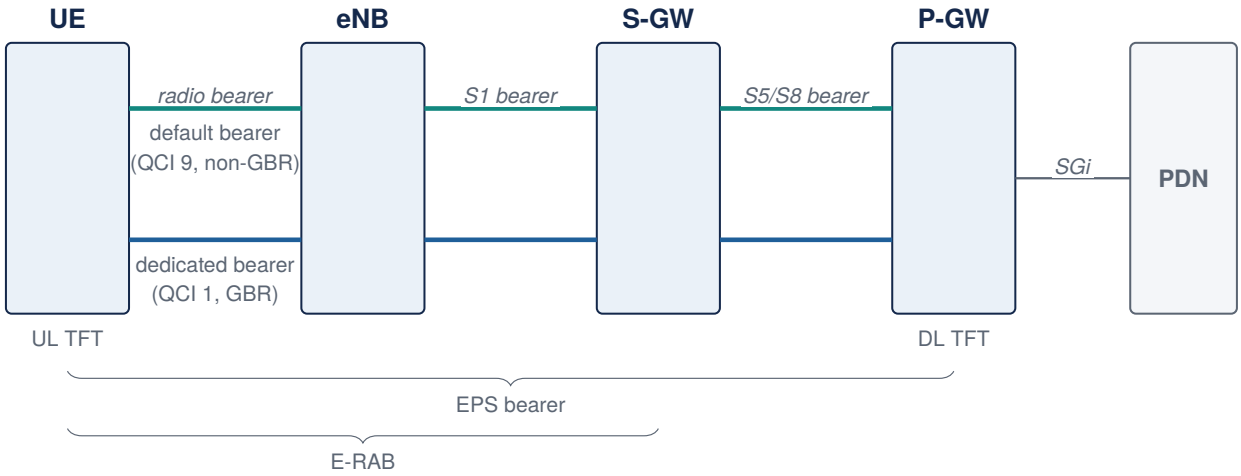


Figure 2. EPS bearer structure with a default and a dedicated bearer. The UL TFT in the UE and the DL TFT in the P-GW map IP flows to bearers.

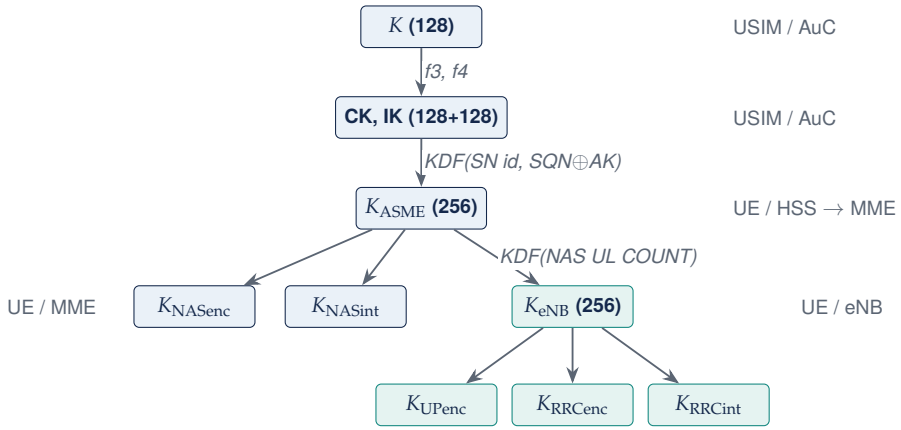


Figure 3. EPS key hierarchy. Key lengths in bits; algorithm keys are truncated to 128 bits.

8 Step-by-step procedures

Procedure 1 • EPS-AKA and NAS security activation

1. The UE sends Attach Request with its GUTI (or IMSI if it has none).
2. If no vector is available, the MME sends S6a Authentication Information Request with the IMSI and the serving network identity; the HSS returns one or more EPS vectors.
3. The MME sends Authentication Request (RAND, AUTN, key set identifier).
4. The USIM verifies AUTN (MAC and SQN freshness), computes RES, CK, IK; the UE derives K_{ASME} and returns Authentication Response (RES).
5. The MME checks RES = XRES and sends Security Mode Command (selected algorithms, integrity protected with K_{NASint}); the UE verifies it and answers Security Mode Complete, ciphered and integrity-protected.

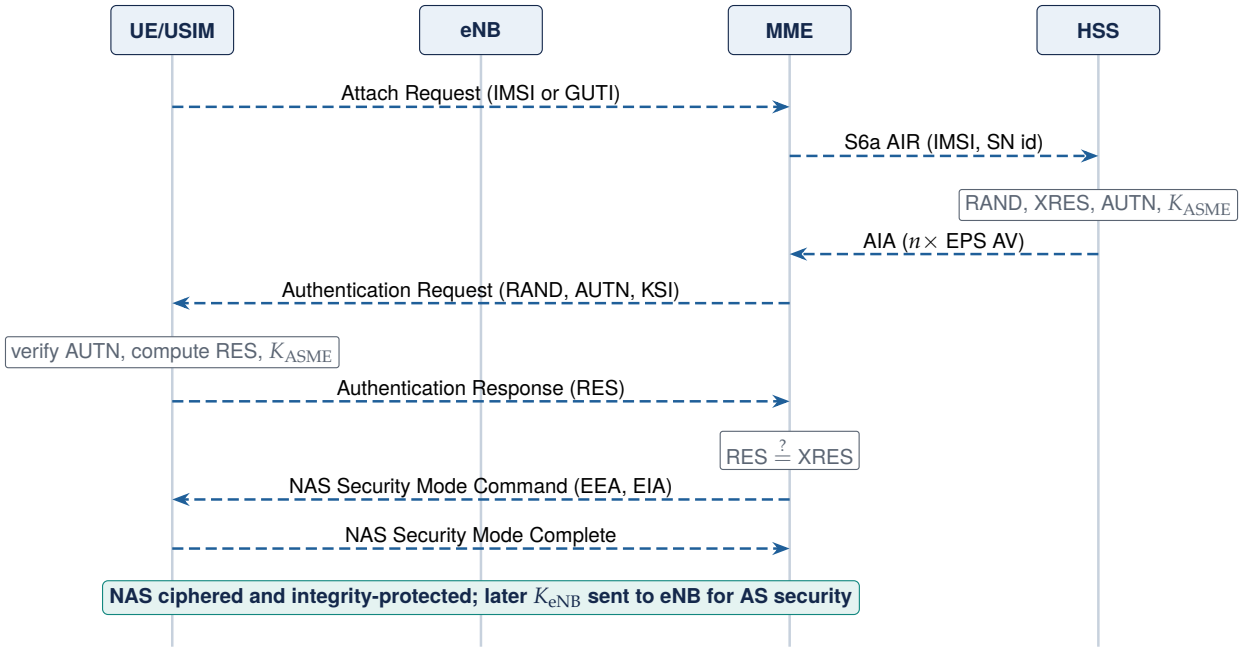


Figure 4. EPS-AKA and NAS security mode (Procedure 1).

Procedure 2 • Network-initiated dedicated bearer activation

1. An application function (e.g. the IMS P-CSCF) requests resources over Rx; the PCRF decides PCC rules and pushes them to the P-GW over Gx.
2. The P-GW binds the rule to a new bearer (QCI, ARP, GBR/MBR, DL TFT) and sends GTPv2-C Create Bearer Request to the S-GW, which forwards it to the MME.
3. The MME sends S1AP E-RAB Setup Request carrying the NAS Activate Dedicated EPS Bearer Context Request (UL TFT, QoS).
4. The eNB admits the bearer, configures a radio bearer with RRC Connection Reconfiguration and answers E-RAB Setup Response; the UE answers the NAS request with *Accept*.
5. Create Bearer Response travels back to the P-GW, which confirms to the PCRF (Gx RAA). Packets matching the TFTs now use the new bearer.

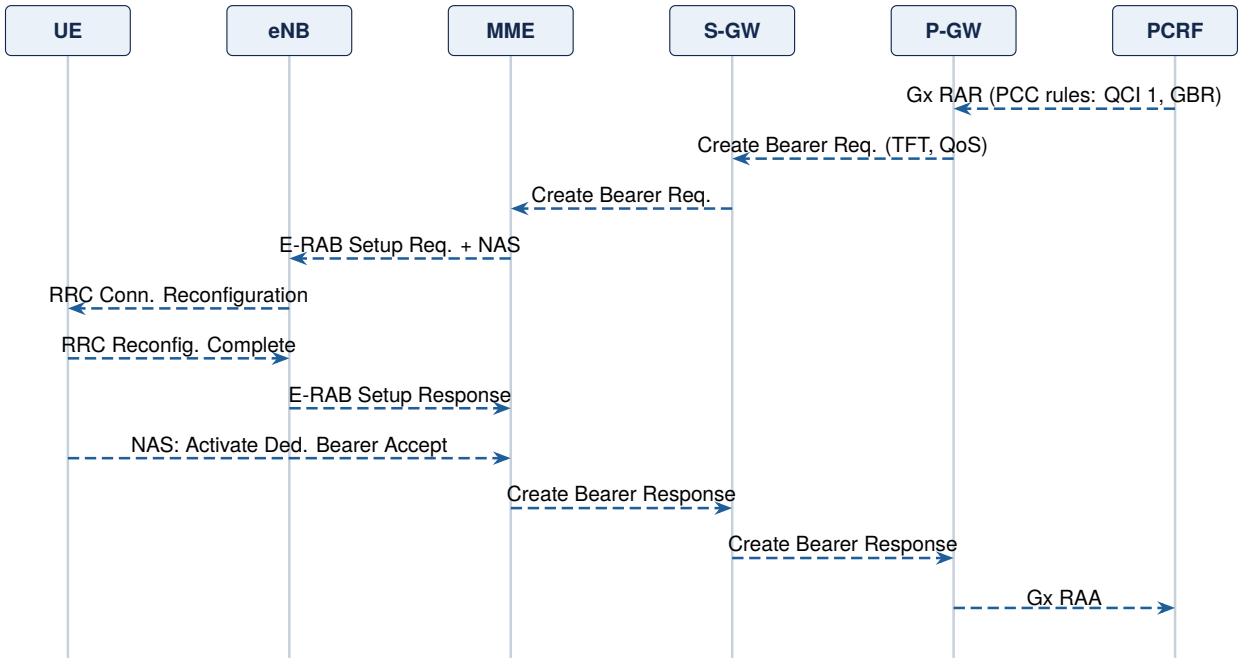


Figure 5. Dedicated bearer activation triggered by policy (Procedure 2).

9 Worked examples

Worked example 1 • Which bearer carries the packet?

Problem. A UE has a default bearer (QCI 9) and a dedicated bearer (QCI 1) whose DL TFT matches UDP packets from 10.1.2.3, ports 40000–40100. Which bearer carries (a) an RTP packet from 10.1.2.3:40050, (b) a TCP packet from 10.1.2.3:443, (c) a DNS response?
Solution. (a) Dedicated bearer (all filter components match). (b) Default bearer (protocol and port do not match). (c) Default bearer (no filter matches; the default bearer has no TFT and receives the rest).
Interpretation. Bearer selection is pure packet classification in the P-GW (downlink) and UE (uplink); the radio then applies the QCI's treatment to the bearer.

Worked example 2 • Admission by radio resources

Problem. A VoLTE bearer needs 40 kbit/s. At its MCS a cell-centre UE carries 800 bits per RB pair (one RB pair for one 1-ms subframe); a cell-edge UE only 160 bits. If 25 RB pairs per subframe are reserved for GBR traffic, how many calls of each kind can be admitted?
Solution. RB pairs per subframe needed $=$ GBR$/$(bits per RB pair $\times$ 1000 s^{-1}): centre $40\,000/(800 \times 1000) = 0.05$, edge $40\,000/(160 \times 1000) = 0.25$. Admissible: $25/0.05 = 500$ centre users or $25/0.25 = 100$ edge users.
Interpretation. The same GBR costs five times more radio resource at the cell edge – admission must be resource-based (Section 6.1).

Worked example 3 • Key freshness

Problem. A UE goes from idle to connected 200 times per day with the same K_{ASME}. How many different K_{eNB} values does it use, and why does that matter?
Solution. 200, because each transition uses a new uplink NAS COUNT in Eq. (2b). Even if the same eNB serves all connections, PDCP COUNT values restart with each new key, so keystreams are never reused under one key.

10 Technology comparison

Table 2. Packet core functions: GPRS/UMTS PS core versus EPC versus 5GC.

	GPRS/UMTS PS	EPC	5GC
Mobility/session control	SGSN (both)	MME (mobility), S-/P-GW-C (session)	AMF (mobility), SMF (session)
User-plane anchor	GGSN	P-GW (S-GW as mobility anchor)	UPF (one or more)
QoS unit	PDP context	EPS bearer (QCI)	QoS flow (5QI) within a PDU session
IP at registration	No (separate activation)	Yes (default bearer)	Optional (PDU session separate)
Authentication	UMTS AKA (quintets)	EPS-AKA (K_{ASME})	5G-AKA / EAP-AKA′, home control
Subscriber database	HLR (MAP)	HSS (Diameter)	UDM/UDR (HTTP/2)

Trade-offs. The bearer model gives precise, per-bearer QoS but every QoS change needs a signalling procedure through four nodes; 5G replaces it with QoS flows that can be added inside an existing session with less signalling (Lecture 22). The MME's concentration of NAS, security and bearer control is efficient but makes it a large stateful node; 5G splits it into AMF and SMF.

11 Key takeaways

Key takeaways

- EPC: MME (control), HSS (subscribers), S-GW (mobility anchor), P-GW (IP anchor, PCEF), PCRF (policy).
- QoS unit = EPS bearer (QCI, ARP, GBR/MBR); TFTs map flows to bearers; default bearer always on.
- EPS-AKA binds K_{ASME} to the serving network; KDF chain $K \rightarrow$ CK,IK $\rightarrow K_{\text{ASME}} \rightarrow K_{\text{eNB}} \rightarrow$ AS keys.
- NAS security UE–MME, AS security UE–eNB; user data ciphered but not integrity-protected in LTE.

- EMM (registered?) and ECM (signalling connection?) are independent state machines.
- GBR admission must be resource-based; a gateway restart can cause a signalling storm of Nm/μ seconds unless recovery is spread.

12 Exercises

Exercise 1 • Level 1 – conceptual • Idle UE

List which resources exist for a UE in EMM-REGISTERED + ECM-IDLE: IP address, S5/S8 tunnel, S1-U tunnel, radio bearer, eNB context, MME context. What triggers their re-creation?

Exercise 2 • Level 1 – conceptual • Serving-network binding

Explain what attack is prevented by including the serving network identity in the derivation of K_{ASME}.

Exercise 3 • Level 2 – numerical • Admission

Repeat Example 2 for a mix of 70% centre and 30% edge users. How many VoLTE calls can be admitted?

Exercise 4 • Level 2 – numerical • Periodic TAU load

An MME serves 5×10^6 registered UEs with T3412 $= 54\,\text{min}$. Each periodic TAU costs 6 messages. Compute the average message rate. What happens to the load if T3412 is reduced to 12 min?

Exercise 5 • Level 3 – analytical • Storm with retries

In Section 6.3, assume that requests waiting longer than 30 s time out and are retried once. Model the MME as a single queue with service rate μ and estimate the total number of messages processed when all N UEs try at once. Compare with spreading over $T_{\text{w}} = 900\,\text{s}$.

Exercise 6 • Level 3 – analytical • Bearer versus flow

A UE runs a VoLTE call, a video call and background web traffic. Count the GTP tunnels on S1-U and S5 in LTE (one PDN for IMS, one for the Internet) and discuss the signalling needed to add a second video stream. How would a flow-based model reduce it?

Solutions

Solution to Exercise 1

Kept: IP address, S5/S8 tunnels, MME context (and S-GW/P-GW contexts). Released: S1-U tunnel, radio bearers, eNB context. They are re-created by a Service Request – triggered by the UE (uplink data or signalling) or by paging after the S-GW receives downlink data and sends a Downlink Data Notification to the MME.

Solution to Exercise 2

Without binding, a vector (or key) obtained by one serving network could be used by another, e.g. a compromised or rogue visited network could reuse vectors to impersonate a different network to the UE, or a network could claim to be a cheaper one for billing. Because both the HSS and the UE derive K_{ASME} with the SN id, a mismatch between the claimed network and the network that requested the vector produces different keys and security mode fails.

Solution to Exercise 3

Mean cost per call $= 0.7 \times 0.05 + 0.3 \times 0.25 = 0.11$ RB pairs per ms; calls $= 25/0.11 \approx 227$.

Solution to Exercise 4

TAU rate $= 5 \times 10^6/(54 \times 60) = 1543$ per s; messages $= 6 \times 1543 = 9259$ msg/s. With 12 min the rate increases by $54/12 = 4.5\times$ to about 41 700 msg/s – shorter timers detect lost UEs faster but cost signalling and UE battery.

Solution to Exercise 5

Total work $Nm = 10^7$ messages at $\mu = 2 \times 10^4$ msg/s needs 500 s. Requests queued longer than 30 s time out: only the first $30\mu/m = 6 \times 10^4$ UEs complete in the first 30 s; the others retry, adding up to N new attempts. A crude bound: with one retry each, offered work approaches $2Nm$ and recovery time about 1000 s, with many UEs failing twice. Spreading over 900 s keeps the queue short (load 56%), no timeouts, total work exactly Nm – faster overall despite the deliberate delay.

Solution to Exercise 6

IMS PDN: default bearer (QCI 5), dedicated QCI 1 (voice) and QCI 2 (video): 3 bearers. Internet PDN: default bearer: 1. Total 4 EPS bearers, i.e. 4 S1-U tunnels and 4 S5 tunnels. A second video stream with the same QCI can be added to the existing QCI 2 bearer by updating its TFT (Update Bearer procedure, still through PCRF, P-GW, S-GW, MME, eNB and UE); a new QCI would need a new bearer. In 5G all flows of a PDU session share one N3 tunnel; adding a QoS flow is a rule update in SMF, UPF and gNB with less per-flow tunnel state (Lecture 22).

13 Further reading

EPC procedures are specified in [3], policy and charging control and the QCI table in [5], security in [2], NAS in [4] and GTPv2-C in [1]. Olsson et al. [6] give a highly readable system description of the EPC.

References

[1] 3GPP, *3GPP evolved packet system (EPS); evolved general packet radio service (GPRS) tunnelling protocol for control plane (GTPv2-C); stage 3*, TS 29.274, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29274.htm.

[2] 3GPP, *3GPP system architecture evolution (SAE); security architecture*, TS 33.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33401.htm.

[3] 3GPP, *General packet radio service (GPRS) enhancements for evolved universal terrestrial radio access network (E-UTRAN) access*, TS 23.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23401.htm.

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[5] 3GPP, *Policy and charging control architecture*, TS 23.203, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23203.htm.

[6] M. Olsson, S. Sultana, S. Rommer, L. Frid, and C. Mulligan, *EPC and 4G Packet Networks: Driving the Mobile Broadband Revolution*, 2nd ed. Academic Press, 2013.

10

Lecture 10

Diameter Protocol

پروتکل Diameter

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5 and 9; TCP/SCTP; binary protocol encoding (TLV)

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
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12 Exercises . . . 11
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-10-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the limitations of RADIUS that Diameter was designed to remove.
- **LO2** **Describe** Diameter nodes and agents, peer connections and realm-based routing.
- **LO3** **Decode** a Diameter header and AVP, including flags, vendor IDs and padding, and **compute** message lengths.
- **LO4** **List** the base commands and result-code classes, and **trace** the S6a, Gx, Rx and Gy applications used in LTE.
- **LO5** **Size** a credit-control quota and **estimate** the transaction load of an HSS.
- **LO6** **Explain** why 5G replaced Diameter with HTTP/2 service-based interfaces, and how the two interwork.

2 Prerequisite knowledge

You need the EPC reference points of Lectures 5 and 9 (S6a, Gx, Rx, Gy) and a working knowledge of transport protocols: TCP's reliable byte stream and SCTP's message-oriented, multi-homed associations (Lecture 6). Binary protocols are usually built from *type–length–value* (TLV) elements; Diameter's AVP is one.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
AAA	احراز هویت، مجوزدهی، حساب‌داری	Authentication, authorisation and accounting framework.
AVP	جفت ویژگی--مقدار	Diameter's TLV information element.
Command code	کد فرمان	Identifies a request/answer pair (e.g. 316 = Update-Location).
Application-ID	شناسهٔ کاربرد	Identifies the interface (e.g. 16777251 = S6a/S6d).
Realm	قلمرو	Administrative domain used for routing (e.g. an operator's domain).
Relay / proxy agent	عامل رله / پراکسی	Agents that forward messages (proxy may modify them).
DRA / DEA	عامل مسیریابی / لبهٔ Diameter	Routing agent inside, edge agent at the border of a network.
CER / DWR / DPR	تبادل قابلیت، پاسبان، قطع	Base peer-management commands.
Result-Code	کد نتیجه	Outcome of a request (2001 = success).
Credit control	کنترل اعتبار	Online charging with quota grants (Gy).

3 Introduction and motivation

Every time an LTE phone attaches, an MME asks the HSS for authentication vectors and the subscriber's profile; every time a VoLTE call starts, the IMS asks the PCRF for a guaranteed bearer; every few megabytes, a prepaid subscriber's P-GW asks the charging system for more credit. All these dialogues – between the control nodes of the core, never touching user data – use one protocol: Diameter [6].

Diameter descends from RADIUS [8], the dial-up era's AAA protocol (the name is a pun: a diameter is twice a radius). RADIUS could not meet operator-grade requirements – reliability, large attributes, server-initiated messages, roaming between realms – so the IETF designed Diameter as an extensible base protocol plus *applications*. 3GPP adopted it for IMS (Release 5) and made it the signalling backbone of the EPC. In 5G it is being replaced by HTTP/2 APIs, but Diameter networks will carry 4G traffic for many years, and the design concepts – realms, agents, capability exchange – recur in the 5G service-based architecture.

4 Fundamental concepts

4.1 From RADIUS to Diameter

Table 1. RADIUS versus Diameter.

	RADIUS	Diameter
Transport	UDP (application-level retransmission)	TCP or SCTP (reliable), port 3868
Attribute length	8-bit length field, at most 253 bytes of data	24-bit length field
Outstanding requests	8-bit identifier: 256 per client–server pair	32-bit hop-by-hop and end-to-end identifiers
Server-initiated messages	Not supported (only extensions)	Supported (e.g. Cancel-Location, Re-Auth)
Peer liveness / failover	No standard mechanism	Watchdog (DWR/DWA) and failover to alternative peers
Capability negotiation	None	CER/CEA exchange supported applications
Agents	Proxies, ad hoc	Relay, proxy, redirect and translation agents
Routing	Static, by server address	By Destination-Realm and Destination-Host
Security	Shared secret, MD5-based hiding	TLS/DTLS or IPsec

4.2 Nodes and agents

A Diameter *client* (e.g. the MME) sends requests to a *server* (e.g. the HSS). Between them, *agents* may forward messages:

- a **relay** forwards based on routing information without inspecting the application;
- a **proxy** forwards and may modify messages to enforce policy;
- a **redirect** agent answers with the address of the right server instead of forwarding;
- a **translation** agent converts to another protocol (e.g. RADIUS).

Operators deploy **Diameter Routing Agents** (DRAs) as central signalling routers, so that each client needs connections to two DRAs instead of a full mesh, and **Diameter Edge Agents** (DEAs) at the border towards roaming partners (via an IP exchange, IPX), hiding the internal topology.

4.3 Peers and realms

Two directly connected nodes are *peers*. Before exchanging application messages they perform a *capabilities exchange* (CER/CEA) announcing their identity (Origin-Host, Origin-Realm) and supported applications. Each node keeps a *peer table* and a *realm routing table*. A request carries Destination-Realm (mandatory for routed requests) and optionally Destination-Host; nodes forward it to a peer that serves that realm and application. In 3GPP networks realms follow the pattern `epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org` [2], so a visited MME can route an S6a request to a subscriber's home HSS using only the IMSI.

5 Detailed technical explanation

5.1 Message structure

A Diameter message is a 20-byte header followed by AVPs (Figure 1) [6].

- **Version** = 1; **Message Length** includes the header and is a multiple of 4.
- **Command flags**: R (request, cleared in answers), P (proxiable), E (error answer), T (potentially retransmitted after a failover).
- **Command Code** (24 bits) identifies the command; request and answer share the code.
- **Application-ID** (32 bits) identifies the application (0 for the base protocol).
- **Hop-by-Hop Identifier**: matches answers to requests on one hop; each agent replaces it.
- **End-to-End Identifier**: unique per request end to end; used to detect duplicates.

Each AVP has a 32-bit code, flags V (vendor-specific: a Vendor-ID follows; 10415 for 3GPP), M (mandatory: the receiver must understand it or reject the message) and P (end-to-end security, deprecated), a 24-bit length that covers header and data but not padding, and data padded to a 4-byte boundary. Data types include Integer32/64, Unsigned32/64, OctetString, UTF8String, DiameterIdentity, Address, Enumerated and *Grouped* – an AVP whose data is itself a sequence of AVPs, which lets Diameter express nested structures such as a subscription profile.

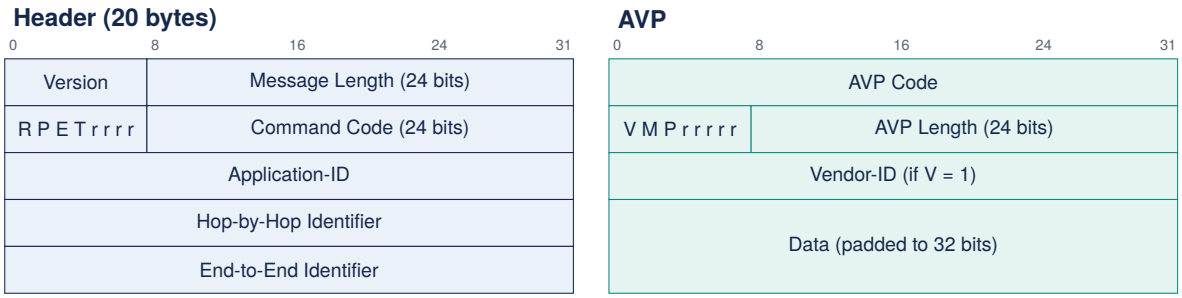


Figure 1. Diameter header (left) and AVP (right). Each row is 32 bits.

5.2 Commands and result codes

Commands are named with a pair of abbreviations ending in R (request) and A (answer). The base protocol defines peer management – **CER/CEA** (Capabilities-Exchange, code 257), **DWR/DWA** (Device-Watchdog, 280), **DPR/DPA** (Disconnect-Peer, 282) – and session commands such as **RAR/RAA** (Re-Auth, 258), **STR/STA** (Session-Termination, 275) and **ASR/ASA** (Abort-Session, 274). Answers carry a `Result-Code` AVP whose thousands digit gives the class (Table 2). Applications define their own codes in an `Experimental-Result` AVP together with

Table 2. Diameter result-code classes with examples [6].

Class	Meaning	Examples
1xxx	Informational	1001 multi-round authentication
2xxx	Success	2001 DIAMETER_SUCCESS
3xxx	Protocol errors	3002 UNABLE_TO_DELIVER, 3004 TOO_BUSY (sent with the E flag)
4xxx	Transient failures	4001 AUTHENTICATION_REJECTED – may succeed later
5xxx	Permanent failures	5001 AVP_UNSUPPORTED, 5012 UNABLE_TO_COMPLY

a Vendor-ID; for example 3GPP's S6a uses 5001 for *user unknown* and 5420 for *unknown EPS subscription* [1]. The same number therefore means different things in Result-Code and in a 3GPP Experimental-Result – a frequent source of confusion when reading traces.

5.3 Diameter applications in LTE

Table 3. Main 3GPP Diameter applications in the EPC.

Interface	Between	Application-ID	Main commands
S6a/S6d	MME/SGSN – HSS	16777251	AIR/AIA (318), ULR/ULA (316), CLR (317), IDR (319), DSR (320), PUR (321), NOR (323)
S13	MME – EIR	16777252	ECR/ECA (324): ME identity check
Gx	PCEF – PCRF	16777238	CCR/CCA (272), RAR/RAA (258)
Rx	AF – PCRF	16777236	AAR/AAA (265), STR/STA, RAR/RAA, ASR/ASA
Gy	PCEF – OCS	4 (credit control)	CCR/CCA (272): INITIAL, UPDATE, TERMINATION
Cx/Dx	CSCF – HSS	16777216	UAR, SAR, MAR, LIR (IMS registration)

S6a. (Figure 3) [1]: the MME obtains vectors with AIR/AIA and registers with ULR/ULA, which returns the subscription data (allowed APNs with QoS profiles, AMBR, access restrictions). The HSS uses CLR to remove the subscriber from a previous MME and IDR/DSR to change the stored profile; the MME uses PUR when it purges an inactive subscriber.

Gx. [3]: at PDN connection set-up the P-GW sends CCR-Initial and receives PCC rules in CCA-Initial; the PCRF later pushes changes with RAR; CCR-Update reports events (e.g. usage thresholds); CCR-Termination ends the session.

Rx. [4]: the P-CSCF describes a call's media in AAR; the PCRF derives QoS and installs rules over Gx (Figure 4); STR ends the media session.

Gy. credit control [5]: the PCEF requests a quota (time or volume) for a rating group in CCR-Initial, reports usage and requests more with CCR-Update, and returns unused quota with CCR-Termination; if credit is exhausted the OCS answers with a final-unit indication.

5.4 Diameter in 5G

The 5G core replaces Diameter by HTTP/2 service-based interfaces (Lectures 18–19): S6a $\rightarrow$ N8 (Nudm), Gx $\rightarrow$ N7 (Npcf_SMPolicyControl), Rx $\rightarrow$ N5 (Npcf_PolicyAuthorization), Gy/Gz $\rightarrow$ N40 (Nchf). Reasons: (i) web technology and tooling (REST, JSON, OpenAPI) are ubiquitous; (ii) network functions register and discover each other dynamically (NRF) instead of relying on static peer tables; (iii) stateless, cloud-native implementations scale more easily with HTTP load balancers. During the 4G/5G coexistence, interworking functions translate between the two worlds, and the HSS and UDM are often a single combined product.

6 Mathematical models

6.1 AVP and message length

Rule. For an AVP with ℓ bytes of data, header length $h = 8$ (or 12 with Vendor-ID):

$$L_{\text{AVP}} = h + \ell, \qquad L_{\text{on wire}} = 4 \left\lceil \frac{h + \ell}{4} \right\rceil, \tag{1}$$

and the message length is $20 + \sum L_{\text{on wire}}$ over all top-level AVPs.

Example. Origin-Host `mme01.epc.mnc010.mcc432.3gppnetwork.org` has 39 characters: $L_{\text{AVP}} = 8 + 39 = 47$, padded to 48 bytes on the wire. Origin-Realm `epc.mnc010.mcc432.3gppnetwork.org` (33 characters): 41, padded to 44. Identity AVPs alone cost about 100 bytes per message, which is why Diameter messages are typically several hundred bytes even for simple requests.

6.2 Failure detection with the watchdog

A peer sends DWR when no traffic has been received for T_{w} (recommended 30 s, with random jitter of ± 2 s to avoid synchronisation). If no answer arrives within a further T_{w}, the connection is considered suspect and traffic is failed over to an alternative peer. The worst-case detection time

on an idle connection is therefore about

$$T_{\text{det}} \approx 2\,T_{\text{w}}, \tag{2}$$

i.e. about a minute with defaults; on a busy connection, unanswered application requests reveal the failure much sooner. Operators tune T_{w} down (e.g. 6 s) on critical links, trading watchdog traffic for faster failover (Exercise 4).

6.3 HSS transaction load

Model. N subscribers each attach a times per day and perform u MME-changing TAUs per day, each needing one AIR and one ULR (AIR only if the MME lacks vectors); other procedures are neglected. The mean S6a transaction rate is

$$\lambda_{\text{S6a}} \approx \frac{N\,(a+u)\,\bar{n}}{86\,400}\ \text{transactions/s}, \tag{3}$$

with $\bar{n}$ the mean number of transactions per procedure.

Example. $N = 10^7$, $a + u = 3$, $\bar{n} = 2$: $\lambda \approx 694$ transactions/s on average. The busy hour is several times higher, and a restart of an MME pool can create bursts far above the mean (Lecture 9), so HSS front-ends are dimensioned for peaks, not averages.

6.4 Credit-control quota sizing

Problem. How large should a Gy volume quota be?

Model. A user consumes data at rate r (bytes/s); each grant gives Q bytes; the PCEF reports and requests a new quota when the current one is used.

Results. The number of CCR-Updates per user per unit time and the maximum overspend (consumption the operator cannot charge if the user's credit ran out just after the grant) are

$$\lambda_{\text{CCR}} = \frac{r}{Q}, \qquad O_{\max} \approx Q. \tag{4}$$

Example and interpretation. $r = 2\,\text{Mbit/s} = 250\,\text{kB/s}$, $Q = 10\,\text{MB}$: one update every 40 s (90 per hour) and up to 10 MB of revenue at risk per user. Doubling Q halves the signalling but doubles the risk – operators therefore give small quotas to low-balance users and large ones to high-balance users. A validity time limits how long a quota may be held.

7 Network architecture and diagrams

In a typical operator network, every MME, P-GW, PCRF, HSS and OCS connects to a redundant pair of DRAs; the DRAs connect to each other and to a pair of DEAs facing the roaming exchange. Routing is by realm and application, with load balancing and session binding (all messages of a Gx/Rx session must reach the same PCRF instance, which DRAs ensure by binding the UE's IP address to the PCRF). Diameter is purely control plane: none of these nodes ever forwards user packets.

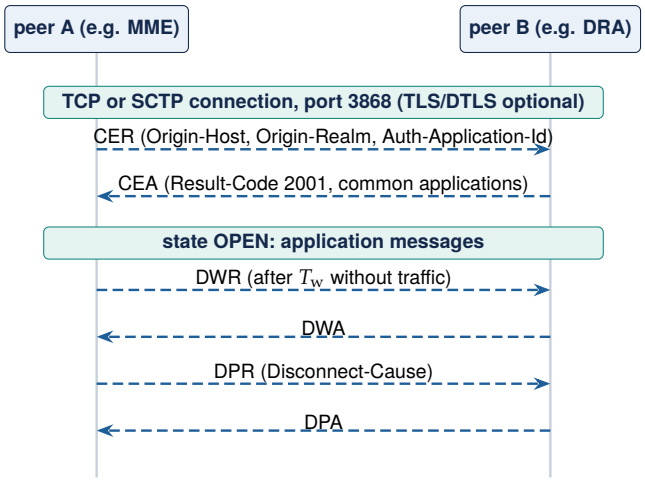


Figure 2. Diameter peer lifecycle: transport set-up, capabilities exchange, watchdog, disconnection.

8 Step-by-step procedures

Procedure 1 • S6a during attach

1. The MME needs vectors for an unknown IMSI. It builds AIR (code 318, Application-ID 16777251, R flag set) with Session-Id, Auth-Session-State, Origin-Host/Realm, Destination-Realm derived from the IMSI, User-Name = IMSI, Visited-PLMN-Id and Requested-EUTRAN-Authentication-Info.
2. The DRA routes on Destination-Realm and Application-ID to an HSS front-end; the HSS returns AIA with Result-Code 2001 and Authentication-Info (grouped E-UTRAN-Vector AVPs).
3. After authentication, the MME sends ULR (316) with RAT-Type and ULR-Flags; the HSS stores the MME identity, sends CLR to the previous MME if there was one, and returns ULA with Subscription-Data.
4. Errors appear as Experimental-Result, e.g. 5001 (user unknown): the MME rejects the attach with an appropriate NAS cause.

Procedure 2 • Policy for a VoLTE call (Rx + Gx)

1. The P-CSCF, having seen the SDP offer/answer of the call, sends Rx AAR to the PCRF with Media-Component-Description AVPs (codec, bandwidth, flow descriptions).
2. The PCRF authorises the media, derives QCI 1 and GBR, answers AAA and sends Gx RAR with Charging-Rule-Install to the P-GW serving the UE's IMS PDN connection.
3. The P-GW creates a dedicated bearer (Lecture 9) and answers RAA.
4. At call end, the P-CSCF sends STR; the PCRF removes the rules (RAR with Charging-Rule-Remove), and the bearer is deleted.

9 Worked examples

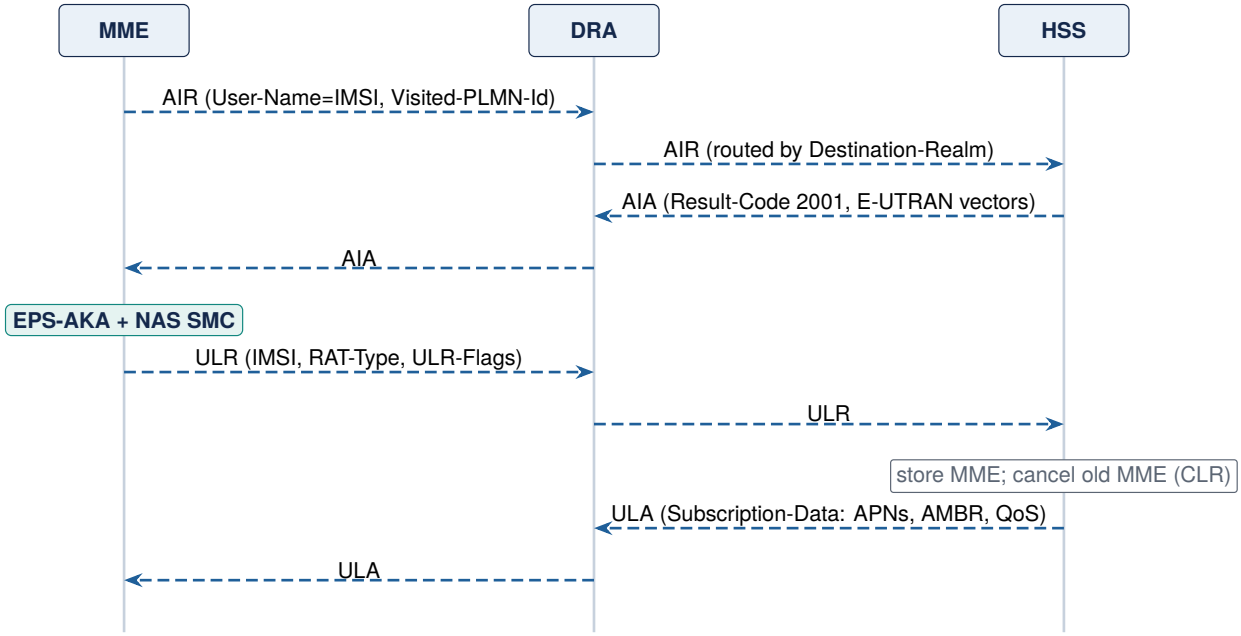


Figure 3. S6a exchanges during attach via a Diameter Routing Agent (Procedure 1).

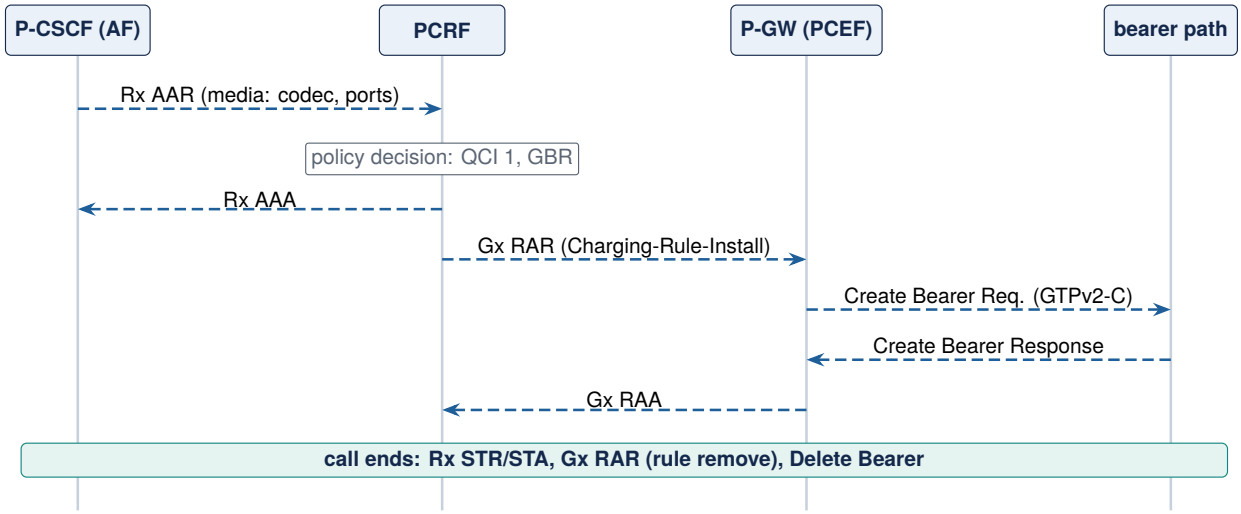


Figure 4. Rx and Gx dialogues for a VoLTE call (Procedure 2).

Worked example 1 • Decoding a header

Problem. The first 20 bytes of a message are (hex) `01 00 01 2C 80 00 01 3C 01 00 00 23` …. Decode version, length, flags, command and application.
Solution. Version 1; length `0x00012C` = 300 bytes; flags `0x80` = R only (a request, not proxiable); command `0x00013C` = 316 = Update-Location; Application-ID `0x01000023` = 16777251 = S6a. So: a ULR from an MME. (Real S6a requests usually also set P.)

Worked example 2 • Message size

Problem. An AIR contains: Session-Id (60 bytes data), Auth-Session-State (Unsigned32), Origin-Host and Origin-Realm as in Section 6.1, Destination-Realm (33 bytes), User-Name (15-digit IMSI), Visited-PLMN-Id (3-byte OctetString, vendor-specific) and Requested-EUTRAN-Authentication-Info (grouped, vendor-specific, containing one Number-Of-Requested-Vectors Unsigned32 AVP, vendor-specific). Compute its length.
Solution. Session-Id $4\lceil 68/4\rceil = 68$; Auth-Session-State $8+4 = 12$; Origin-Host 48; Origin-Realm 44; Destination-Realm 44; User-Name $4\lceil 23/4\rceil = 24$; Visited-PLMN-Id $4\lceil 15/4\rceil = 16$; inner AVP $12+4 = 16$, so the grouped AVP is $12+16 = 28$. Total $20+68+12+48+44+44+24+16+28 = 304$ bytes.

Worked example 3 • Quota policy

Problem. An OCS must handle at most 2000 CCR-Updates per second for 10^6 active data users with a mean rate of 50 kB/s. What is the smallest uniform quota?
Solution. $\lambda_{\text{total}} = Nr/Q \le 2000 \Rightarrow Q \ge 10^6 \times 5 \times 10^4/2000 = 25$ MB. The operator would use 25 MB (or more) and limit risk with validity timers and smaller quotas for low-balance users.

10 Technology comparison

Table 4. Core signalling protocols across generations.

	SS7 MAP (2G/3G)	Diameter (4G, IMS)	HTTP/2 SBI (5G)
Transport	MTP/SCCP (or SIGTRAN over IP)	TCP/SCTP, port 3868	TCP + TLS, HTTP/2
Encoding	ASN.1 BER	Binary AVPs (TLV)	JSON (OpenAPI schemas)
Addressing	Global titles, point codes	Realms, hosts	URIs, NRF discovery
Routing	STPs	DRAs	Direct or via SCP
Roaming edge	International STP	DEA	SEPP
Interaction	TCAP dialogues	Request/answer sessions	RESTful resources, subscribe/notify

Diameter is control plane only. Diameter carries decisions about sessions – who is allowed, with which QoS, how much credit – but no session data. The packets of the subscriber flow through

GTP-U tunnels (Lecture 13) that Diameter never touches; a failed PCRF blocks new policy decisions but not the packets of existing sessions.

11 Key takeaways

Key takeaways

- Diameter = reliable (TCP/SCTP), extensible AAA: base protocol + applications identified by Application-ID.
- 20-byte header (version, length, R/P/E/T flags, command code, application ID, hop-by-hop and end-to-end IDs) + AVPs padded to 32 bits.
- Peers: CER/CEA, DWR/DWA, DPR/DPA; routing by Destination-Realm via relays, proxies and DRAs/DEAs.
- LTE applications: S6a (AIR, ULR, CLR, IDR), Gx (CCR/CCA, RAR), Rx (AAR), Gy (credit control).
- Result-Code classes 1xxx–5xxx; 3GPP-specific errors in Experimental-Result.
- Quota sizing trades signalling r/Q against revenue risk $\approx Q$.
- 5G replaces Diameter by HTTP/2 SBIs; interworking during 4G/5G coexistence.

12 Exercises

Exercise 1 • Level 1 – conceptual • Server-initiated messages

Give two EPC situations in which the server must initiate a Diameter request, and explain why RADIUS could not support them natively.

Exercise 2 • Level 1 – conceptual • Two identifiers

Explain why Diameter needs both a hop-by-hop and an end-to-end identifier. What happens to each when a relay forwards a request?

Exercise 3 • Level 2 – numerical • AVP length

Compute the on-wire length of: (a) a Result-Code AVP; (b) a vendor-specific UTF8String AVP containing the APN `internet`; (c) a grouped AVP containing (a) and (b).

Exercise 4 • Level 2 – numerical • Watchdog tuning

A DRA has 400 idle peer connections. Compute the watchdog message rate for $T_{\mathrm{w}} = 30\,\mathrm{s}$ and 6 s and the corresponding worst-case failure-detection times.

Exercise 5 • Level 3 – analytical • Agent topology

An operator has 20 MMEs, 10 HSS front-ends, 30 P-GWs, 8 PCRFs and 4 OCS nodes. Compare the number of Diameter peer connections for a full mesh of the interfaces S6a, Gx and Gy and for a design with a redundant pair of DRAs (every node connects to both DRAs). Discuss the reliability implications.

Exercise 6 • Level 3 – analytical • Quota with risk budget

A user with balance B is granted quotas $Q = \min(Q_{\max}, \alpha B)$. Show that the expected number of CCR-Updates to consume the whole balance at constant rate is about $\log(B/Q_{\min})/\log(1/(1-\alpha))$ when $Q_{\max}$ is never reached and quotas stop at $Q_{\min}$, and evaluate it for $\alpha = 0.2$, $B = 1\,\text{GB}$, $Q_{\min} = 1\,\text{MB}$.

Solutions

Solution to Exercise

(1) HSS → MME Cancel-Location (CLR) when the subscriber registers in another MME or is withdrawn; (2) PCRF → P-GW Re-Auth (RAR) when a VoLTE call needs a dedicated bearer or a usage limit is reached; also HSS Insert-Subscriber-Data when a profile changes. RADIUS clients send requests and servers only answer; there is no standard way for a server to start a transaction (only later extensions such as Dynamic Authorization), and NAT/firewalls in front of UDP clients make it unreliable.

Solution to Exercise

The hop-by-hop identifier lets a node match an answer to the request it sent on a given connection; each relay replaces it with its own value and restores the original when returning the answer. The end-to-end identifier is set by the originator and never changed; together with Origin-Host it lets the server detect duplicates (e.g. a request retransmitted with the T flag after a failover) for a period of time.

Solution to Exercise

(a) $8 + 4 = 12$ bytes. (b) $12 + 8 = 20$ bytes (already aligned). (c) 8 (or 12 if vendor-specific) $+12 + 20 = 40$ bytes (44 with a Vendor-ID).

Solution to Exercise

Each idle connection produces one DWR and one DWA per T_w from each side that initiates; counting one DWR and one DWA per connection per T_w: $400 \times 2/30 = 26.7$ messages/s for 30 s, 133 messages/s for 6 s – negligible either way. Detection time $\approx 2T_w$: 60 s versus 12 s. The shorter timer is clearly worthwhile on critical links; the risk is false failover on congested links.

Solution to Exercise

Full mesh: S6a $20 \times 10 = 200$, Gx $30 \times 8 = 240$, Gy $30 \times 4 = 120$: 560 connections, each to be configured and monitored. With two DRAs: $(20 + 10 + 30 + 8 + 4) \times 2 = 144$ connections (plus one between the DRAs). DRAs centralise routing, load balancing and session binding and simplify adding nodes, but they become critical single points: they must be deployed in

geo-redundant pairs with enough capacity to carry all traffic if one fails.

Solution to Exercise

Each grant uses a fraction α of the remaining balance, so after k grants the balance is $B(1-\alpha)^k$; quotas fall below $Q_{\min}$ when $\alpha B(1-\alpha)^k = Q_{\min}$, i.e. $k \approx \log(\alpha B/Q_{\min})/\log(1/(1-\alpha))$, essentially the stated expression. For $\alpha = 0.2$, $B = 1000\,\text{MB}$, $Q_{\min} = 1\,\text{MB}$: $k \approx \ln(200)/\ln(1.25) = 5.30/0.223 = 23.7$ grants, after which a few minimum-size grants finish the balance. Geometric quotas keep both signalling (logarithmic in B) and risk (at most α of the balance) small.

13 Further reading

The base protocol is RFC 6733 [6], credit control RFC 8506 [5]; RADIUS is RFC 2865 [8]. The 3GPP applications are specified in [1] (S6a/S6d/S13), [3] (Gx) and [4] (Rx); realm naming follows [2]. Olsson et al. [7] discuss Diameter deployment in the EPC.

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11

Lecture 11

Home Subscriber Server (HSS)

کارساز مشترکین خانگی (HSS)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 9 and 10; basic probability; symmetric cryptography

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 7
8 Step-by-step procedures . . . 8
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10 Technology comparison . . . 9
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-11-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the HSS as the union of HLR, AuC and IMS user-database functions, and **list** its interfaces.
- **LO2** **Explain** the subscription data model of the EPS and how the MME uses it.
- **LO3** **Trace** authentication-vector generation in the AuC and **explain** sequence-number management and re-synchronisation.
- **LO4** **Size** HSS storage and vector-request load, and **evaluate** the trade-off of requesting several vectors at once.
- **LO5** **Compute** the availability of geo-redundant HSS deployments.
- **LO6** **Relate** the HSS to the 5G UDM/AUSF/UDR and to data-layered (UDC) architectures.

2 Prerequisite knowledge

This lecture builds on the EPS-AKA procedure and key hierarchy of Lecture 9 and on the S6a application of Lecture 10. You should know what a block cipher and a message authentication code are, and be able to compute probabilities of independent events (for the availability models).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
HSS	کارساز مشترکین خانگی	Master subscriber database of a 3GPP network (HLR + AuC + IMS data).
HLR	ثبت‌کنندهٔ موقعیت خانگی	2G/3G subscriber database reached over MAP.
AuC	مرکز احراز هویت	Stores the secret key K and generates authentication vectors.
IMSI	شناسهٔ بین‌المللی مشترک موبایل	Permanent subscriber identity (MCC + MNC + MSIN).
Authentication vector	بردار احراز هویت	(RAND, XRES, AUTN, K_{ASME}) for one EPS-AKA run.
SQN	شمارهٔ ترتیب	48-bit counter that proves the freshness of a vector.
AUTS	نشانهٔ همگام‌سازی مجدد	Token the USIM returns to re-synchronise SQN.
OPc	کلید پیکربندی اپراتور	Operator-variant value derived from OP and K for MILENAGE.
UDC / UDR	همگرایی دادهٔ کاربر / مخزن	Architecture separating application logic from a shared data repository.
Geo-redundancy	افزونگی جغرافیایی	Replicating a function across distant sites.

3 Introduction and motivation

A mobile network is, among other things, a very large and very available database. Every subscriber has a permanent identity, a secret key shared only with the SIM, a profile of allowed services and a current location; every attach, many handovers and every incoming call read or write some of these. In 2G and 3G these data lived in the Home Location Register (HLR) and the Authentication Centre (AuC); with IMS (Release 5) 3GPP introduced the *Home Subscriber Server* (HSS), which contains the HLR and AuC functions and adds the IMS user data [6]. In the EPS the HSS is the single subscriber authority for the MME, reached over Diameter S6a (Lecture 10).

The HSS is rarely discussed in textbooks because it does not move packets, yet an HSS outage stops every new attach in a country. Its design questions – how to keep 10^7–10^8 profiles consistent across sites, how to generate fresh authentication vectors at thousands per second, how to recover sequence numbers after a failure – are typical of carrier-grade distributed systems.

4 Fundamental concepts

4.1 Functions of the HSS

- **Identification and addressing**: IMSI, MSISDN, IMS public and private identities [7].
- **Authentication (AuC)**: stores K (and OPc) and generates authentication vectors [1], [2].
- **Subscription management**: stores the profile – allowed APNs, QoS, roaming and access restrictions – and pushes changes to the serving node.
- **Mobility management**: records the serving MME/SGSN/MSC so that the network can reach the subscriber, and cancels the old registration on a change of serving node.
- **IMS support**: stores the serving S-CSCF, IMS service profiles and initial filter criteria, and answers application servers [4].

4.2 Interfaces

Table 1. Main interfaces of the HSS.

Interface	Peer	Protocol	Use
S6a / S6d	MME / SGSN (EPS)	Diameter	Vectors, location update, subscription data [3]
Gr	SGSN (2G/3G)	MAP	Same functions for GPRS/UMTS [5]
C / D	GMSC / MSC-VLR	MAP	Call routing information, CS location update [5]
Cx	I-CSCF / S-CSCF	Diameter	IMS registration, S-CSCF assignment
Sh	Application server	Diameter	Transparent service data, notifications
SWx	3GPP AAA server	Diameter	Authentication for non-3GPP access (e.g. Wi-Fi)

4.3 Subscription data

Definition 1 • EPS subscription profile

The profile that the HSS sends in ULA (and later in IDR) contains, among others, the subscriber status, MSISDN, network-access mode, operator-determined barring, the subscribed UE-AMBR, access-restriction data (e.g. "no E-UTRAN"), the periodic TAU timer and an *APN-configuration profile*: a list of APN configurations, each with a context identifier, the APN name (Service-Selection), PDN type, the default bearer's QCI and ARP and the APN-AMBR [3].

The MME caches the profile while the subscriber is registered with it. The HSS remains the master: it changes the cached copy with Insert-Subscriber-Data (IDR) or Delete-Subscriber-Data (DSR), and withdraws it with Cancel-Location (CLR).

5 Detailed technical explanation

5.1 Authentication-vector generation

For each vector the AuC chooses a random RAND, takes the next sequence number SQN and evaluates five functions of K, RAND, SQN and AMF (Figure 1): f_1 gives the network authentication code MAC, f_2 the expected response XRES, f_3 and f_4 the cipher and integrity keys CK and IK,

and f_5 the anonymity key AK that conceals SQN [1]. The recommended algorithm set, MILENAGE, builds all five from one 128-bit block cipher (AES) and the operator constant OPc [8]. For E-UTRAN the HSS sets the *AMF separation bit* to 1 and derives K_{ASME} from CK and IK, binding the vector to the serving network [2]. The vector is

$$\text{AV} = \big(\text{RAND},\ \text{XRES},\ \text{AUTN},\ K_{\text{ASME}}\big), \qquad \text{AUTN} = (\text{SQN} \oplus \text{AK}) \,\|\, \text{AMF} \,\|\, \text{MAC}. \tag{1}$$

The secret K never leaves the AuC; the MME receives only vectors.

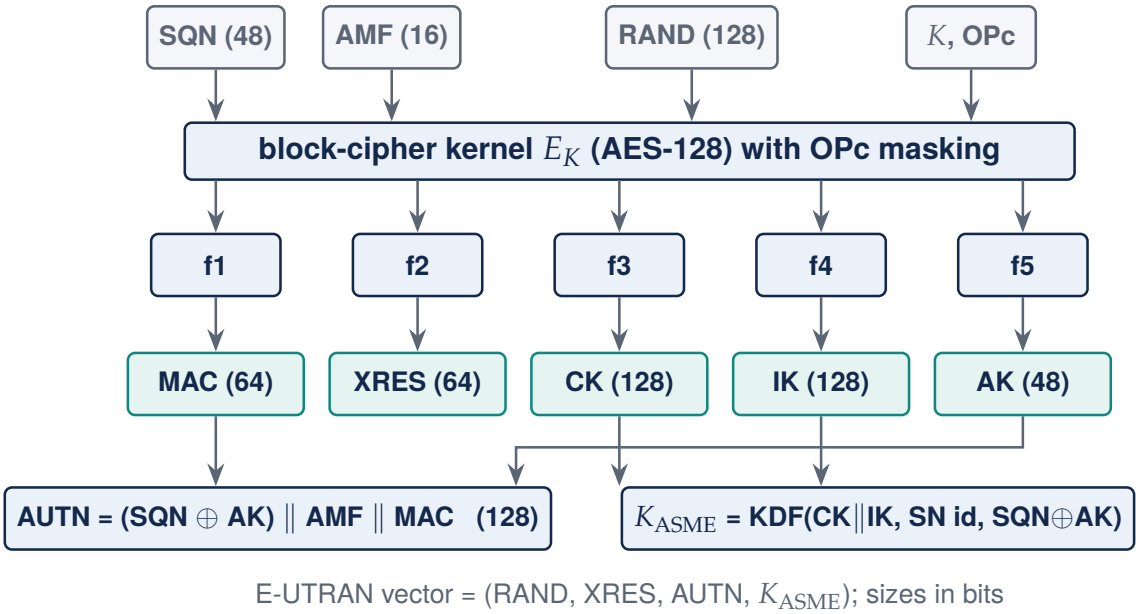


Figure 1. Authentication-vector generation in the AuC with the functions f_1–f_5.

5.2 Sequence numbers and freshness

The USIM accepts AUTN only if MAC is correct *and* SQN is fresh. TS 33.102 describes, in an informative annex, a scheme in which SQN is split into a 43-bit counter SEQ and a 5-bit index IND ($\text{SQN} = \text{SEQ} \,\|\, \text{IND}$) [1]. The USIM keeps an array of the highest SEQ accepted for each of the $2^5 = 32$ values of IND, and accepts a new SQN if its SEQ exceeds the stored value in the slot IND. The array lets vectors that were generated in one order be used in another – for example when two MMEs hold batches of vectors for the same subscriber – without spurious failures. An optional age check also rejects SQN values that are too far behind the highest accepted one.

If the SQN is not fresh, the USIM answers with a *synchronisation failure* carrying AUTS, which contains its own SQN concealed with a key derived like AK and protected by a MAC (MAC-S). The MME sends RAND and AUTS to the HSS in a new AIR (Re-Synchronization-Info); the AuC verifies MAC-S, sets its counter to the USIM's value and returns fresh vectors (Figure 4).

5.3 Mobility management in the HSS

On each ULR the HSS records the serving MME (Origin-Host and realm). If another MME was registered, the HSS sends it CLR so that it deletes its context; this guarantees one serving node per subscriber. An MME that deletes a long-inactive subscriber sends Purge-UE (PUR), after which the HSS marks the subscriber as purged. After a restart, an HSS that may have lost registrations can send Reset (RSR) to the MMEs, which then refresh the affected subscribers at their next contact

[3].

5.4 Implementation: front-ends, data store and UDC

Commercial HSSs separate stateless *front-ends* that terminate MAP and Diameter and run the application logic from a replicated, mostly in-memory *data store*. 3GPP generalised this as *User Data Convergence* (UDC): HLR/HSS, AuC and other front-ends share one User Data Repository (UDR) over the Ud interface, so that a subscriber has one record instead of a copy per node (Figure 2). The 5G core keeps the same split: UDM and AUSF are front-ends, UDR is the repository (Lectures 18–19). Because the HSS is a single point of control, the data store is replicated across geographically separated sites.

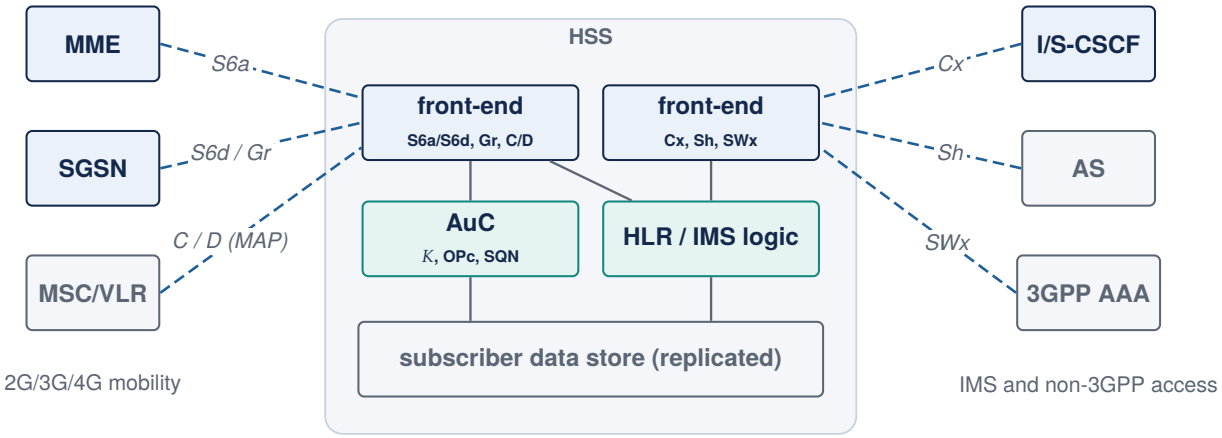


Figure 2. Functional view of the HSS with its interfaces and the front-end/data-store split.

6 Mathematical models

6.1 Sequence-number lifetime

Model. With a b-bit SEQ and one authentication every τ seconds, the counter lasts

$$T_{\text{life}} = 2^b \, \tau. \tag{2}$$

Example. $b = 43$, one authentication per minute: $T_{\text{life}} = 8.8 \times 10^{12}$ min $\approx 1.7 \times 10^7$ years. Exhaustion is not a concern; the counter is large so that schemes that skip values (for example a SEQ derived from a clock) remain possible.

6.2 Storage

Model. N subscribers with a mean profile size s, replicated R times, with a fill factor η (fraction of memory that may be used):

$$M = \frac{N \, s \, R}{\eta}. \tag{3}$$

Example. $N = 10^7$, $s = 4$ kB, $R = 3$, $\eta = 1$: $M = 120$ GB – small enough to keep in memory on a few servers. Storage is therefore not the hard problem; consistency and availability are.

6.3 Vector requests

Model. If authentications occur at rate λ_{auth} and the MME requests n vectors per AIR and uses all of them, the AIR rate is

$$\lambda_{\text{AIR}} = \frac{\lambda_{\text{auth}}}{n}. \tag{4}$$

Example. 10^7 subscribers authenticated 10 times a day: $\lambda_{\text{auth}} = 1157\,\text{s}^{-1}$; $n = 5$ gives $\lambda_{\text{AIR}} = 231\,\text{s}^{-1}$.

Limitation. Vectors left unused when the subscriber moves to another MME are wasted, and stored vectors consume MME memory, so n is usually small (Exercise 4).

6.4 Availability of redundant sites

Model. Each site is available with probability a (unavailability $u = 1 - a$), independently of the others. If service needs at least k of N sites,

$$A_{k\text{-of-}N} = \sum_{i=k}^{N} \binom{N}{i} a^i (1-a)^{N-i}. \tag{5}$$

Special cases: $A_{1\text{-of-}2} = 1 - u^2$; $A_{2\text{-of-}3} = 1 - (3u^2 - 2u^3)$; $A_{2\text{-of-}2} = a^2$.

Example. $a = 0.999$ (8.76 h downtime per year): two sites each able to carry the full load give $u^2 = 10^{-6}$, i.e. 31.5 s per year; two sites that are both needed give 17.5 h per year (Figure 3).

Assumptions and interpretation. Independence is the critical assumption: software bugs, configuration errors and replication faults hit all sites at once, and in practice dominate HSS outages. The model gives an upper bound on what redundancy can achieve, not a prediction.

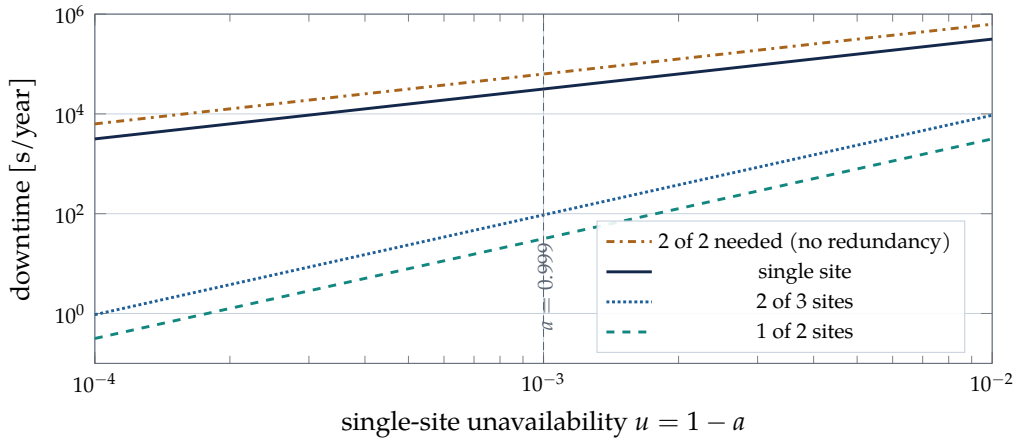


Figure 3. Annual downtime as a function of single-site unavailability for four deployment options (independent failures assumed).

7 Network architecture and diagrams

Figure 2 shows the HSS between the circuit and packet core of 2G/3G, the EPS and the IMS. The same subscriber record serves all of them: a VoLTE subscriber is authenticated for E-UTRAN over

S6a, registered in the IMS over Cx, and reached by an MSC over MAP if the call falls back to the circuit domain. In the EPS, the HSS has exactly one control-plane peer type for mobility, the MME, and is usually reached via DRAs (Lecture 10).

8 Step-by-step procedures

Procedure 1 • Generating an E-UTRAN vector in the AuC

1. Receive AIR with the IMSI and the serving-network identity (Visited-PLMN-Id).
2. Look up K and OPc; generate a fresh 128-bit RAND; take the next SQN (SEQ and IND).
3. Compute MAC $= f_1$, XRES $= f_2$, CK $= f_3$, IK $= f_4$, AK $= f_5$ with AMF separation bit $= 1$.
4. Form AUTN with Eq. (1); derive K_{ASME} from CK, IK, the serving-network identity and SQN$\oplus$AK.
5. Return the vectors in AIA; the HSS discards CK and IK.

Procedure 2 • Sequence-number re-synchronisation

1. The USIM verifies MAC, recovers SQN and finds it not fresh; it computes AUTS and the UE sends Authentication Failure (synch failure).
2. The MME discards its remaining vectors for this subscriber and sends AIR with Re-Synchronization-Info = RAND $\|$ AUTS.
3. The AuC verifies MAC-S, sets its SQN to the USIM's value and generates new vectors.
4. The MME restarts authentication with a new vector; it now succeeds.

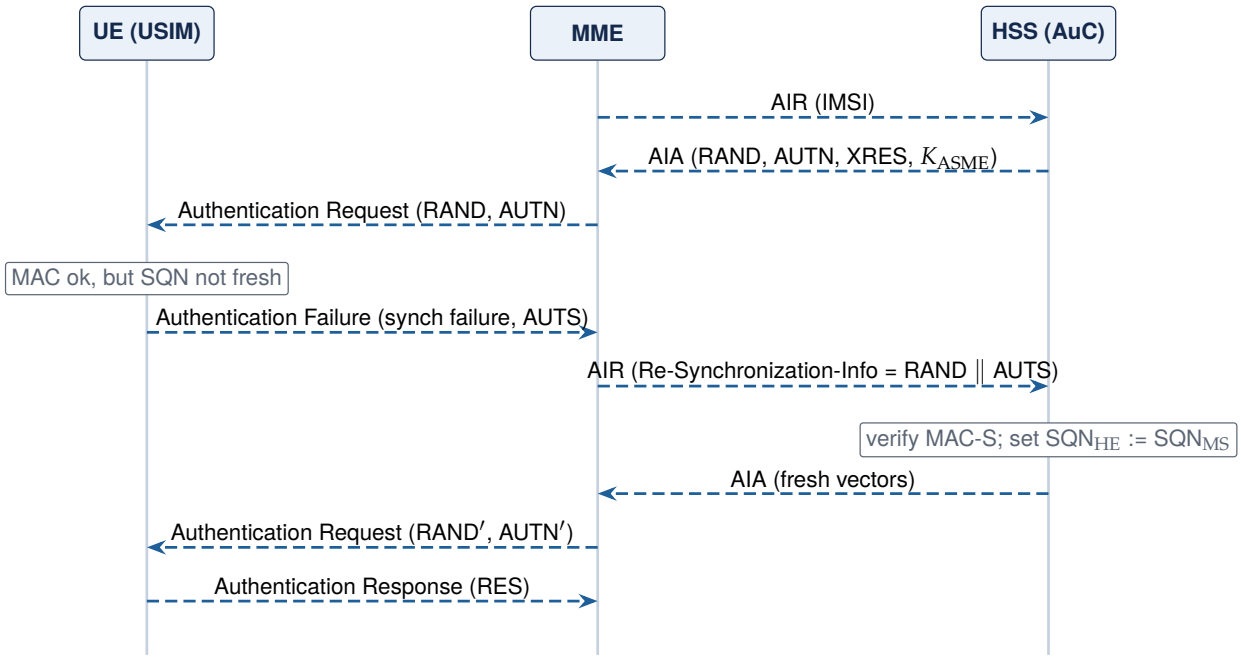


Figure 4. Sequence-number re-synchronisation (Procedure 2).

9 Worked examples

Worked example 1 • Size of an AIA

Problem. Compute the size of the Authentication-Info AVP carrying five E-UTRAN vectors (all AVPs vendor-specific, optional Item-Number omitted).
Solution. Using the AVP length rule of Lecture 10 with a 12-byte header: RAND $12+16=28$, XRES (64 bits) $12+8=20$, AUTN $12+16=28$, K_{ASME} $12+32=44$. One E-UTRAN-Vector: $12+120=132$ bytes; five vectors in the grouped Authentication-Info: $12+5\times 132=672$ bytes.

Worked example 2 • Restore from backup

Problem. An HSS of 10^7 subscribers is restored from a one-day-old backup, so its SQN counters are behind the USIMs'. If all subscribers authenticate within the next hour, how much extra S6a load is created?
Solution. Each subscriber's first authentication fails with a synchronisation failure and needs one extra AIR: $10^7/3600=2778$ extra AIR per second for an hour – on top of normal traffic and comparable to it. Operators avoid this by replicating SQN updates continuously rather than relying on backups, or by advancing restored counters by a safe margin.

Worked example 3 • Choosing a redundancy scheme

Problem. Sites have $a=0.999$. Compare one site, two sites each able to carry the full load, and three sites of which any two can carry the load.
Solution. From Eq. (5): one site $u=10^{-3}$ (8.76 h/year); 1-of-2 $u^2=10^{-6}$ (31.5 s/year); 2-of-3 $3u^2-2u^3\approx 3.0\times 10^{-6}$ (94.5 s/year). The 2-of-3 design is three times less available than 1-of-2 but needs only 50 % capacity per site instead of 100 %, i.e. 150 % total instead of 200 %.

10 Technology comparison

Table 2. Subscriber-data functions across generations.

	HLR/AuC (2G/3G)	HSS (4G, IMS)	UDM/AUSF/UDR (5G)
Protocol	MAP over SS7/SIGTRAN	Diameter (S6a, Cx, Sh)	HTTP/2 SBI (Nudm, Nausf, Nudr)
Authentication	GSM triplets, UMTS quintets	EPS vectors with K_{ASME}	5G-AKA, EAP-AKA′
Identity privacy	TMSI only	TMSI/GUTI only	Concealed SUPI (SUCI)
Data model	Per-node records	Profile per access + IMS	Common UDR, NF front-ends
Discovery	Global titles	Realms, DRAs	NRF

HSS versus visitor databases. The HSS is the master copy; the MME (like the VLR in 2G/3G) caches what it needs for its subscribers. Caching keeps the HSS out of most procedures – a TAU within the same MME needs no S6a message – at the price of keeping copies consistent with IDR/DSR/CLR.

11 Key takeaways

Key takeaways

- HSS = HLR + AuC + IMS user data; the single subscriber authority, reached over S6a, Cx, Sh, SWx and MAP.
- The AuC keeps K secret and issues vectors (RAND, XRES, AUTN, K_{ASME}) built with f_1–f_5 (MILENAGE).
- SQN = SEQ $\|$ IND proves freshness; failures are repaired with AUTS re-synchronisation.
- The HSS records the serving MME, cancels old registrations and pushes profile changes.
- Storage is small ($\approx$100 GB for 10^7 users); availability and consistency are the hard parts, modelled by k-of-N redundancy with independence as the key assumption.
- The front-end/repository split (UDC) leads directly to UDM/AUSF/UDR in 5G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Master and cache

Which subscriber data are stored permanently in the HSS and which are cached in the MME? Which messages keep the cache consistent?

Exercise 2 • Level 1 – conceptual • Realm from IMSI

For IMSI 432101234567890 with a two-digit MNC, write the EPC home-network realm that a visited MME uses as Destination-Realm.

Exercise 3 • Level 2 – numerical • Storage dimensioning

Compute the memory needed for 5×10^7 subscribers with a 6 kB profile, three replicas and a fill factor of 0.7.

Exercise 4 • Level 2 – numerical • How many vectors per request?

With $\lambda_{\mathrm{auth}} = 1157\,\mathrm{s}^{-1}$, compute the AIR rate for $n = 1, 3, 5$. If a subscriber uses on average two vectors per MME visit before moving to another MME, what fraction of generated vectors is wasted for $n = 5$?

Exercise 5 • Level 3 – analytical • Redundancy design

Derive the expressions for $A_{2\text{-of-}3}$ and $A_{1\text{-of-}2}$ from Eq. (5). Then suppose that, besides independent failures, a common-mode failure (e.g. a faulty software release) disables all sites with probability $p_c = 10^{-5}$. Recompute the 1-of-2 downtime and comment.

Exercise 6 • Level 3 – analytical • The IND array

MME A holds an unused vector with IND = 7. The HSS keeps issuing vectors to MME B, incrementing IND cyclically. After how many newly issued vectors can the vector held by MME A become unacceptable, and under which condition does it actually fail?

Solutions

Solution to Exercise 1

Permanent in the HSS: IMSI, MSISDN, K/OPc and SQN (AuC), the subscription profile and the identity of the serving MME. Cached in the MME: the profile (from ULA), unused vectors and the security context. Consistency: IDR/DSR (profile changes), CLR (withdrawal), PUR (MME deletes the subscriber), RSR (HSS restart).

Solution to Exercise 2

MCC = 432, MNC = 10 (padded to three digits as 010), so the realm is

`epc.mnc010.mcc432.3gppnetwork.org`

[7]. The MNC length is not encoded in the IMSI; the MME needs it from configuration.

Solution to Exercise 3

$M = 5 \times 10^7 \times 6 \times 10^3 \times 3/0.7 = 1.29 \times 10^{12}$ bytes ≈ 1.3 TB – distributed over a cluster, still feasible in memory.

Solution to Exercise 4

$\lambda_{\text{AIR}} = 1157,\,386,\,231\ \text{s}^{-1}$. With $n = 5$ each visit triggers one AIR and uses two of five vectors: three of five (60 %) are wasted, so the AuC computes $5/2 = 2.5$ times as many vectors as needed. With $n = 3$, one AIR per visit and 33 % waste. The best n matches the typical number of authentications per visit.

Solution to Exercise 5

$A_{1\text{-of-}2} = 2a(1-a) + a^2 = 1 - (1-a)^2$. $A_{2\text{-of-}3} = 3a^2(1-a) + a^3$, so $1 - A = 1 - 3a^2 + 2a^3 = 3u^2 - 2u^3$ with $u = 1 - a$. With a common mode: $1 - A \approx p_c + u^2 = 10^{-5} + 10^{-6} = 1.1 \times 10^{-5}$, i.e. 347 s per year instead of 31.5 s. The common-mode term dominates by a factor of ten: beyond two sites, staged software roll-outs and configuration discipline improve availability more than extra hardware.

Solution to Exercise

IND takes 32 values, so the next vector with IND $= 7$ (and a higher SEQ) is issued after 32 new vectors. The vector held by MME A fails only if that newer vector is *used* by the USIM before MME A uses its own: the USIM then stores the higher SEQ in slot 7 and rejects the older one. Up to 31 newer vectors can therefore be used in any order without affecting it.

13 Further reading

The HSS is defined in TS 23.002 [6]; identities and realms in TS 23.003 [7]. The authentication framework and the SQN management annex are in TS 33.102 [1], the EPS-specific parts in TS 33.401 [2], and MILENAGE in TS 35.206 [8]. S6a is specified in TS 29.272 [3], MAP in TS 29.002 [5]. Olsson et al. [9] describe the HSS from a deployment perspective.

References

[1] 3GPP, *3G security; security architecture*, TS 33.102, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33102.htm.

[2] 3GPP, *3GPP system architecture evolution (SAE); security architecture*, TS 33.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33401.htm.

[3] 3GPP, *Evolved packet system (EPS); mobility management entity (MME) and serving GPRS support node (SGSN) related interfaces based on Diameter protocol*, TS 29.272, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29272.htm.

[4] 3GPP, *IP multimedia subsystem (IMS); stage 2*, TS 23.228, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23228.htm.

[5] 3GPP, *Mobile application part (MAP) specification*, TS 29.002, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29002.htm.

[6] 3GPP, *Network architecture*, TS 23.002, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23002.htm.

[7] 3GPP, *Numbering, addressing and identification*, TS 23.003, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23003.htm.

[8] 3GPP, *Specification of the MILENAGE algorithm set: An example algorithm set for the 3GPP authentication and key generation functions; document 2: Algorithm specification*, TS 35.206, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/35206.htm.

[9] M. Olsson, S. Sultana, S. Rommer, L. Frid, and C. Mulligan, *EPC and 4G Packet Networks: Driving the Mobile Broadband Revolution*, 2nd ed. Academic Press, 2013.

12

Lecture 12

Mobility Management Entity (MME)

موجودیت مدیریت تحرک (MME)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 4, 6, 9 and 11; basic queueing (M/M/1)

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-12-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the functions and reference points of the MME and **explain** why it is a pure control-plane node.
- **LO2** **Decode** a GUTI and **explain** how temporary identities protect privacy and route signalling inside an MME pool.
- **LO3** **Trace** the initial attach procedure end to end.
- **LO4** **Evaluate** paging strategies by their expected signalling cost and delay.
- **LO5** **Dimension** an MME pool with weight factors and failure margins, and **analyse** MME overload with a queueing model.
- **LO6** **Compare** the MME with the 2G/3G SGSN/VLR and the 5G AMF.

2 Prerequisite knowledge

You need the EPC architecture and EMM/ECM states of Lecture 9, the S1AP and RRC procedures of Lecture 6, location-area design from Lecture 4 and the S6a dialogue with the HSS (Lectures 10–11). The overload model uses the M/M/1 queue: Poisson arrivals at rate λ, exponential service at rate μ, one server.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
MME	موجودیت مدیریت تحرک	Control-plane node of the EPC that terminates NAS signalling.
NAS	لایهٔ غیردسترسی	Signalling between UE and MME (EMM and ESM) [7].
GUTI	شناسهٔ موقت یکتای سراسری	Temporary UE identity allocated by the MME.
S-TMSI	TMSI کوتاه‌شده	MMEC + M-TMSI; used in paging and RRC set-up.
TAI list	فهرست شناسه‌های ناحیهٔ ردیابی	Set of tracking areas in which the UE need not report.
Paging	فراخوانی	Locating an idle UE for mobile-terminated traffic.
MME pool	MME مخزن	Group of MMEs serving the same pool area (S1-flex).
Relative MME capacity	MME ظرفیت نسبی	Weight used by eNBs to share new UEs among pool members.
Overload control	کنترل اضافه‌بار	Mechanisms that reject or delay signalling when the MME is saturated.
T3412	زمان‌سنج TAU دوره‌ای	Periodic tracking-area update timer.

3 Introduction and motivation

In GPRS and UMTS the SGSN did everything for packet subscribers: it authenticated them, tracked them, held their sessions and forwarded their packets. The EPS split this node into two. The *Mobility Management Entity* (MME) keeps the control functions; the serving gateway keeps the packets [6]. The split lets operators scale signalling capacity (which grows with the number of devices and their mobility) independently of user-plane capacity (which grows with traffic volume), and it is the first step of the control/user-plane separation that 5G completes.

The MME is the UE's counterpart in the core: every NAS message – attach, tracking-area update, service request, bearer request – terminates there. This lecture looks at what the MME stores, how it locates idle devices, how MMEs share load in a pool, and what happens when signalling exceeds capacity.

4 Fundamental concepts

4.1 Functions of the MME

- **NAS signalling and security**: EMM and ESM procedures, authentication with vectors from the HSS, NAS ciphering and integrity [2], [7].
- **Reachability and mobility**: tracking-area lists, paging of idle UEs, inter-eNB handover with MME relocation (S10), inter-RAT mobility to 2G/3G (S3).
- **Session management**: default and dedicated bearer set-up via the S-GW (S11), selection of S-GW and P-GW.
- **Subscription handling**: registration with the HSS (S6a), enforcement of access restrictions, equipment-identity check (S13).
- **Interworking**: CS fallback and SMS over SGs [3], SRVCC over Sv [9], location services (SLs).

The MME never touches user packets; its load is measured in procedures per second and in stored contexts, not in bits per second.

Table 1. MME reference points.

Interface	Peer	Protocol	Purpose
S1-MME	eNB	S1AP over SCTP	UE-associated and non-UE signalling, NAS transport [5]
S11	S-GW	GTPv2-C	Session and bearer management [1]
S6a	HSS	Diameter	Authentication data, location update [4]
S10	MME	GTPv2-C	Context transfer on MME relocation
S3	SGSN	GTPv2-C	Inter-RAT mobility with 2G/3G
S13	EIR	Diameter	Equipment identity (IMEI) check
SGs	MSC/VLR	SGsAP over SCTP	CS fallback, SMS [3]
Sv	MSC	GTPv2-C	SRVCC handover of voice to CS [9]

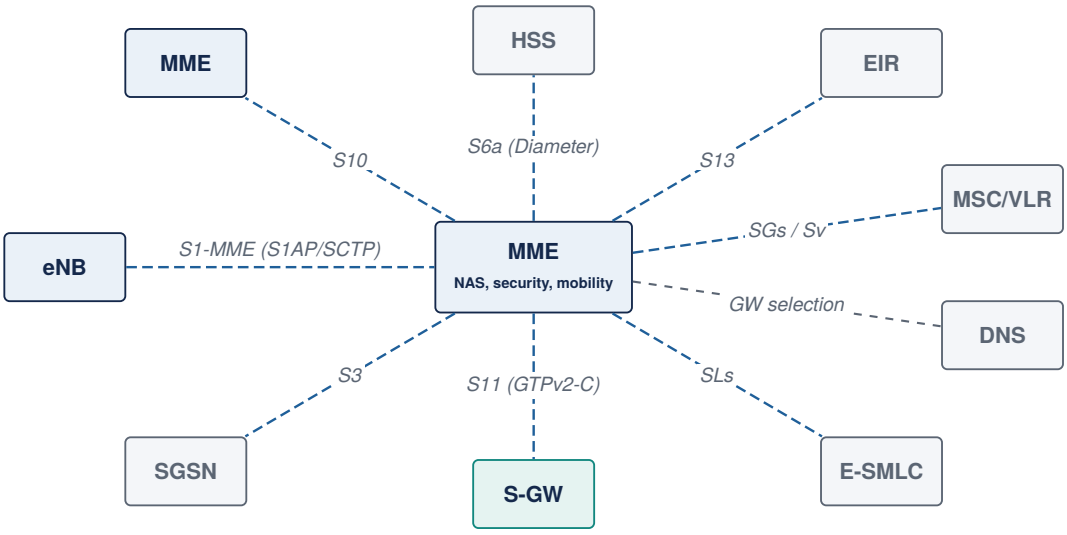


Figure 1. The MME and its reference points. Dashed lines are control-plane interfaces.

4.2 Identities

The permanent IMSI is sent over the air as rarely as possible. After the first attach the MME allocates a *Globally Unique Temporary Identity* (GUTI) [8] (Figure 2). Its first part, the GUMMEI,

names the MME that allocated it (PLMN, 16-bit MME group ID, 8-bit MME code); its last part, the 32-bit M-TMSI, names the UE within that MME. The 40-bit S-TMSI (MME code + M-TMSI) is enough inside one pool and is used in paging and RRC connection set-up. Reallocating the GUTI regularly prevents an observer from linking a UE's transactions.

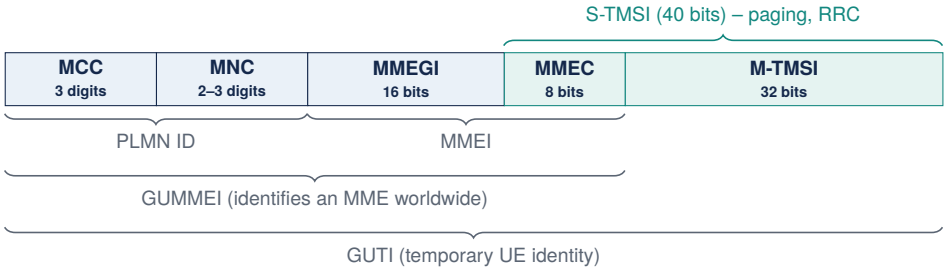


Figure 2. Structure of the GUTI and the identities derived from it.

4.3 MME pools

An eNB can connect to several MMEs (S1-flex). The MMEs that serve the same set of tracking areas form a *pool*. When a UE first connects, the eNB's NAS node selection function chooses an MME, weighted by the *relative MME capacity* each MME announced in S1 Setup; afterwards the MME code in the S-TMSI routes every new connection of that UE to the same MME. Pools provide load sharing, resilience (a failed MME's UEs re-register with the others) and the ability to take an MME out of service gracefully.

5 Detailed technical explanation

5.1 UE context

For each registered UE the MME stores the IMSI and GUTI, the security context (keys, NAS counts, algorithms), the subscription profile from the HSS, the TAI list and last known TAI and eNB, the EMM/ECM state, the selected S-GW and the bearer contexts (EBI, QoS, S-GW TEIDs). In ECM-CONNECTED the context also holds the S1AP identifiers of the UE's eNB connection. This state is what makes the MME the *anchor* for the UE's signalling: losing it (e.g. after an MME failure) forces the UE to attach again.

5.2 Tracking areas and paging

The MME gives each UE a list of tracking areas (TAI list); the UE performs a TAU only when it enters a TA outside the list or when the periodic timer T3412 expires (default 54 min) [7]. When downlink data arrive for an idle UE, the S-GW sends Downlink Data Notification and the MME pages the UE in the cells of its TAI list with its S-TMSI. Operators reduce paging load with *staged* strategies – first the last-seen eNB, then the last TA, then the whole list – at the price of extra delay (Section 6.1).

If no periodic TAU arrives, the MME starts the *mobile reachable timer* (slightly longer than T3412); when it expires, the MME stops paging the UE, and after a further implicit-detach timer it detaches it.

5.3 Gateway selection

At attach, the MME selects an S-GW that serves the UE's tracking area and a P-GW that serves the requested APN, normally by DNS queries on names built from the TAI and the APN, which return candidate gateways with priorities and weights. The P-GW must stay the same for the life of the PDN connection (it owns the IP address); the S-GW may change with mobility (Lecture 13).

5.4 Load balancing, rebalancing and overload control

Under normal conditions the eNBs distribute new UEs in proportion to the relative capacities. To empty an MME (for maintenance) or to rebalance, the MME releases UEs with the cause *load-balancing TAU required*: the UE performs a TAU, and because its S-TMSI is withheld, the eNB selects a new MME. When an MME is overloaded it sends S1AP Overload Start to its eNBs, which then reject new RRC connections of selected types (e.g. all except emergency and mobile-terminated) until Overload Stop [5]. NAS back-off timers and access-class barring (Lecture 8) complement this at the UE.

6 Mathematical models

6.1 Staged paging

Model. The MME pages in stages $s = 1, 2, \ldots$; stage s covers n_s cells and finds the UE with (unconditional) probability p_s, with $\sum_s p_s = 1$. Each stage takes one paging response time T_{p}. The expected number of cells paged and the expected delay are

$$\mathbb{E}[M] = \sum_{s} n_s \Big(1 - \sum_{j<s} p_j\Big), \qquad \mathbb{E}[D] = T_{\mathrm{p}} \sum_{s} \Big(1 - \sum_{j<s} p_j\Big). \tag{1}$$

Example. Three stages: last eNB ($n_1 = 1$, $p_1 = 0.6$), last TA ($n_2 = 20$, $p_2 = 0.3$), TAI list ($n_3 = 60$, $p_3 = 0.1$): $\mathbb{E}[M] = 1 + 0.4 \times 20 + 0.1 \times 60 = 15$ cells, compared with 60 for blanket paging; $\mathbb{E}[D] = 1.5\, T_{\mathrm{p}}$ instead of T_{p}.

Interpretation. Staging pays off when the probabilities are concentrated in the early stages (static users, small cells), and is unattractive for delay-sensitive services such as incoming voice calls, for which operators often page the whole list at once.

6.2 Pool weights and failure margin

Model. The share of new UEs that MME i receives is

$$s_i = \frac{w_i}{\sum_j w_j}, \tag{2}$$

with w_i its relative capacity. If a pool of m equal MMEs must survive the loss of one without overload, each MME may be loaded at most to

$$\rho_{\max} = \frac{m-1}{m}. \tag{3}$$

Example. Weights 100, 100, 50 give shares 0.4, 0.4, 0.2. A pool of four equal MMEs may run at 75 % in normal operation; a pool of two at only 50 %.

6.3 Processing delay and overload

Model. Treat the MME's signalling processor as an M/M/1 queue with capacity μ procedures per second and offered load λ ($\rho = \lambda/\mu < 1$). The mean response time is

$$W = \frac{1}{\mu - \lambda} = \frac{1}{\mu(1-\rho)}. \tag{4}$$

Example. $\mu = 5000\,\mathrm{s}^{-1}$: $W = 1$ ms at $\lambda = 4000$, 10 ms at 4900 and 100 ms at 4990 (Figure 3).

Interpretation. The delay grows without bound as $\rho \to 1$. Once response times approach NAS timers (T3410 = 15 s for attach), UEs retransmit, the offered load rises further and the system can collapse; overload control must therefore act well before $\rho = 1$. The M/M/1 model is optimistic for the bursty arrivals of a signalling storm.

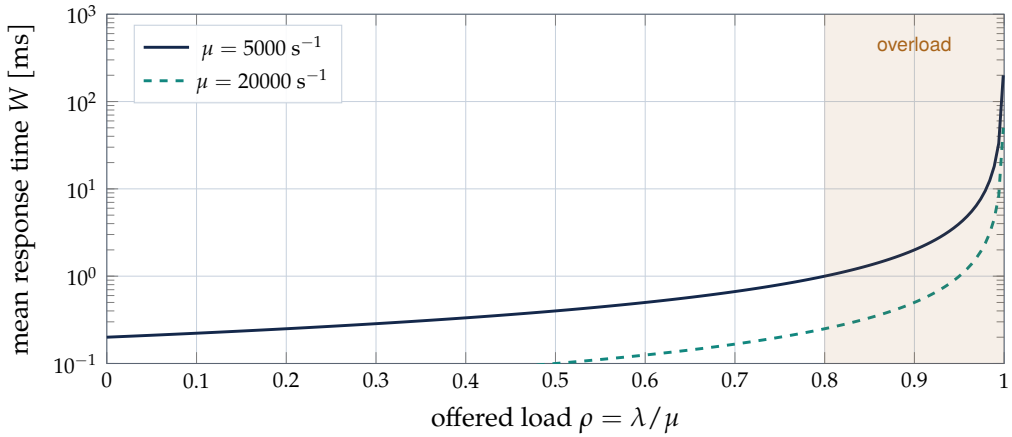


Figure 3. Mean response time of an M/M/1 signalling processor versus load (Eq. (4)).

7 Network architecture and diagrams

In a typical network, the eNBs of a region connect to all MMEs of a pool of two to eight MMEs; pools of neighbouring regions overlap so that a UE moving between them changes MME only at the pool boundary. The MMEs reach the HSS via DRAs, the S-GWs and P-GWs over S11 (often in the same data centre), and the MSC/VLR over SGs for CS fallback and SMS (Figure 1).

8 Step-by-step procedures

Procedure 1 • Initial attach (simplified)

1. The UE sends Attach Request with a PDN Connectivity Request inside RRC set-up; the eNB selects an MME and forwards it in S1AP Initial UE Message.
2. The MME identifies the UE (GUTI or IMSI), runs EPS-AKA with a vector from the HSS and activates NAS security.

3. The MME registers with the HSS (ULR/ULA) and receives the subscription profile.
4. The MME selects S-GW and P-GW and sends Create Session Request; the P-GW allocates the IP address and, with a PCRF, obtains the policy; responses return the S-GW's uplink tunnel endpoint.
5. The MME sends Initial Context Setup Request with Attach Accept and the default bearer to the eNB, which configures the radio bearer with RRC Connection Reconfiguration.
6. The UE answers Attach Complete; the MME sends Modify Bearer Request with the eNB's downlink tunnel endpoint to the S-GW. The default bearer now carries traffic in both directions.

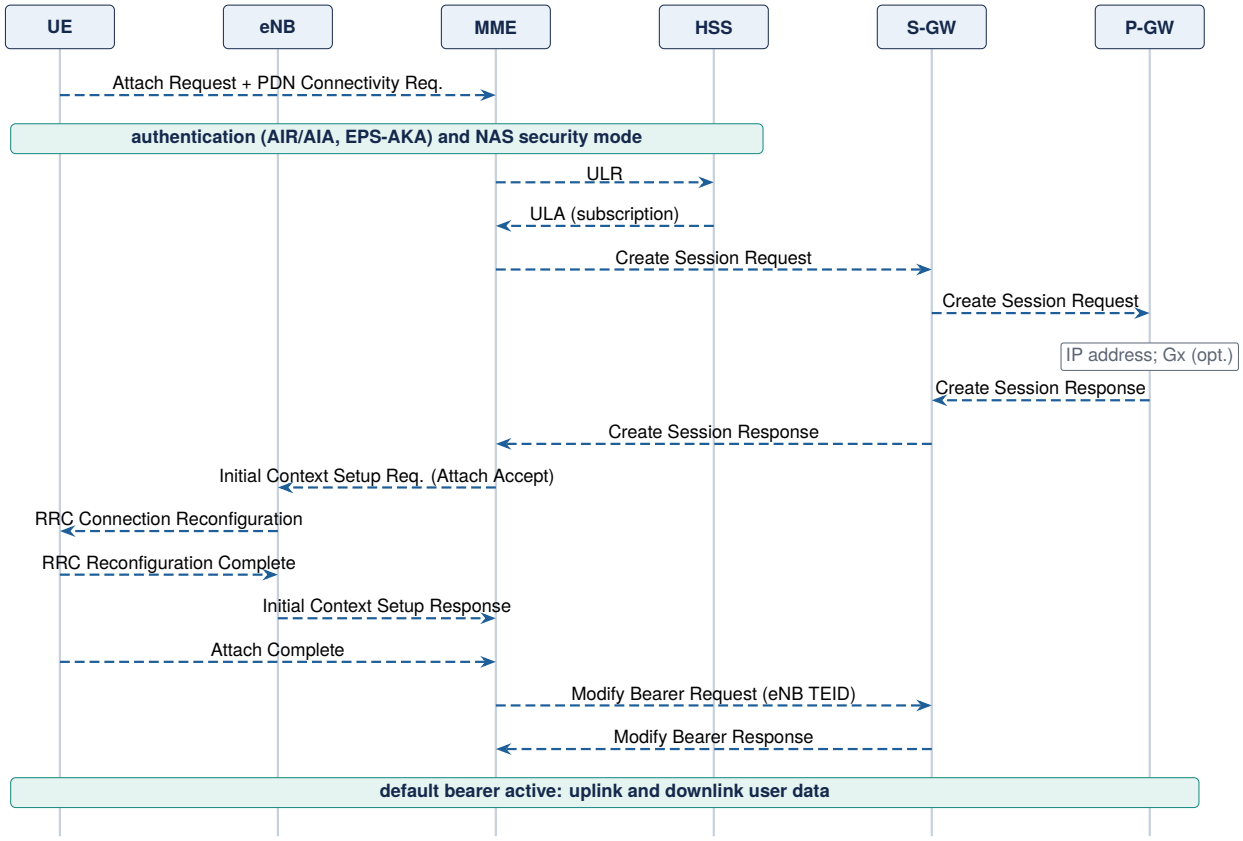


Figure 4. Simplified E-UTRAN initial attach (Procedure 1) [6].

Procedure 2 • Emptying an MME for maintenance

1. Set the MME's relative capacity to 0 (S1AP MME Configuration Update) so that eNBs stop selecting it for new UEs.
2. Release connected UEs with cause *load-balancing TAU required*; page idle UEs or wait for their next contact and release them in the same way.
3. Each UE performs a TAU without S-TMSI routing; the eNB selects another MME, which fetches the context over S10 and registers with the HSS.
4. When no contexts remain, the MME can be taken out of service.

9 Worked examples

Worked example 1 • Decoding a GUTI

Problem. A GUTI has MCC 432, MNC 35, MMEGI `0x8001`, MMEC `0x02` and M-TMSI `0xC0A80001`. Which identities does the eNB use, and which MME serves the UE?
Solution. GUMMEI = (432, 35, 32769, 2): MME code 2 in group 32769 of PLMN 432-35. The S-TMSI is `0x02C0A80001` (40 bits); paging and RRC set-up carry only this. The eNB's NAS node selection function maps MMEC 2 to the corresponding S1 connection in its pool.

Worked example 2 • Paging cost

Problem. A city has 5×10^5 mobile-terminated events per busy hour. Compare the paging messages per busy hour for blanket paging over 60 cells with the three-stage strategy of Section 6.1.
Solution. Blanket: $5 \times 10^5 \times 60 = 3.0 \times 10^7$ cell pagings per hour. Staged: $5 \times 10^5 \times 15 = 7.5 \times 10^6$, a factor 4 less, at a mean delay of $1.5\,T_{\mathrm{p}}$.

Worked example 3 • Pool sizing

Problem. A region has 6×10^6 registered UEs. Each MME supports 2×10^6 contexts. How many MMEs are needed if the pool must survive one failure?
Solution. Without redundancy $\lceil 6/2 \rceil = 3$. With one failure tolerated, the remaining $m-1$ MMEs must hold all contexts: $(m-1) \times 2 \times 10^6 \geq 6 \times 10^6 \Rightarrow m = 4$, each loaded to $6/8 = 75\,\%$ in normal operation, consistent with Eq. (3).

10 Technology comparison

Table 2. Mobility-management nodes across generations.

	SGSN + MSC/VLR (2G/3G)	MME (4G)	AMF (+ SMF) (5G)
User plane	SGSN forwards packets (3G: optional direct tunnel)	None	None
Session control	SGSN	MME (via S-GW)	SMF (separate)
Location unit	Location / routing area	TAI list	Registration area (TAI list)
Temporary ID	TMSI / P-TMSI	GUTI, S-TMSI	5G-GUTI, 5G-S-TMSI
Pooling	Iu-flex (optional)	S1-flex	AMF sets, NRF
Interfaces	MAP, Gn/Gp, Iu	S1AP, GTPv2-C, Diameter	NGAP, HTTP/2 SBI

From MME to AMF. The 5G AMF keeps the MME's access and mobility functions but hands session management to a separate SMF, and replaces GTPv2-C and Diameter with service-based interfaces (Lecture 19). The concepts – temporary identities, registration areas, pools, overload control – carry over almost unchanged.

11 Key takeaways

Key takeaways

- The MME is the EPC's control-plane anchor: NAS, security, mobility, bearer control, gateway selection; no user data.
- GUTI = GUMMEI + M-TMSI; the S-TMSI routes signalling to the right MME in a pool and protects the IMSI.
- Attach combines authentication, HSS registration, session creation and radio-bearer set-up.
- Staged paging trades delay for signalling: $\mathbb{E}[M] = \sum n_s(1 - \sum_{j<s} p_j)$.
- Pools share load by weights; $N+1$ redundancy limits normal load to $(m-1)/m$.
- Response time explodes as $\rho \to 1$; overload control must act early.

12 Exercises

Exercise 1 • Level 1 – conceptual • Control or user plane?

For each function, state whether the MME, the S-GW or the P-GW performs it: (a) paging; (b) buffering downlink packets for an idle UE; (c) IP address allocation; (d) NAS ciphering; (e) choosing the S-GW.

Exercise 2 • Level 1 – conceptual • Why temporary identities?

Give two reasons why the network uses the GUTI/S-TMSI instead of the IMSI in paging and RRC set-up.

Exercise 3 • Level 2 – numerical • Paging strategy

Evaluate Eq. (1) for $p = (0.4, 0.4, 0.2)$ and $n = (1, 30, 90)$. Compare with blanket paging over 90 cells. Which strategy would you use for incoming voice calls?

Exercise 4 • Level 2 – numerical • Weights after a failure

A pool has five MMEs with relative capacities $2, 2, 1, 1, 1$ (capacity proportional to weight). Compute the shares of new UEs before and after the failure of one weight-2 MME, and the maximum normal-operation utilisation that still allows the pool to absorb this failure.

Exercise 5 • Level 3 – analytical • Overload

An MME has $\mu = 5000$ procedures/s. (a) What load keeps $W \leq 10$ ms in the M/M/1 model? (b) After a regional power outage, 2×10^6 UEs try to attach within 60 s. Compare the offered load with μ and explain which mechanisms keep the system stable.

Exercise 6 • Level 3 – analytical • Choosing T3412

An MME holds 5×10^6 idle UEs. Compute the periodic-TAU rate for T3412 = 54 min and 12 min, and discuss the trade-off between signalling, UE battery and the time needed to detect unreachable UEs. Why do IoT devices use much longer values?

Solutions

Solution to Exercise 1

(a) MME (triggered by the S-GW's Downlink Data Notification); (b) S-GW; (c) P-GW; (d) MME; (e) MME.

Solution to Exercise 2

Privacy: the IMSI is permanent, so sending it in the clear would let an eavesdropper track a subscriber; the GUTI is reallocated. Efficiency and routing: the 40-bit S-TMSI is shorter than the IMSI and its MME code tells the eNB which pool member holds the context.

Solution to Exercise 3

$\mathbb{E}[M] = 1 + 0.6 \times 30 + 0.2 \times 90 = 37$ cells versus 90; $\mathbb{E}[D] = (1 + 0.6 + 0.2)\, T_\mathrm{p} = 1.8\, T_\mathrm{p}$. For voice, call set-up delay is critical and the UE must be found at once: page the whole TAI list (or at least skip the single-cell stage).

Solution to Exercise 4

Before: $2/7 = 28.6\,\%$ for each weight-2 MME and $1/7 = 14.3\,\%$ for each weight-1 MME. After: total weight 5, so 40 % for the remaining weight-2 MME and 20 % for each weight-1 MME. Capacity falls from 7 to 5 units, so the normal utilisation must not exceed $5/7 = 71\,\%$.

Solution to Exercise 5

(a) $1/(\mu - \lambda) \le 0.01 \Rightarrow \lambda \le 4900\,\mathrm{s}^{-1}$ ($\rho \le 0.98$); with bursty traffic the real limit is lower. (b) $2 \times 10^6/60 = 3.3 \times 10^4$ attaches/s, almost seven times μ (and each attach involves several messages). Stability requires S1AP overload control (eNBs reject or delay connections), access-class barring and RACH back-off at the UE, randomised NAS retry timers, and a pool so that several MMEs share the load.

Solution to Exercise 6

$5 \times 10^6/3240 = 1543\,\mathrm{TAU/s}$ for 54 min; $5 \times 10^6/720 = 6944\,\mathrm{TAU/s}$ for 12 min. A short timer detects lost UEs sooner (detection takes about T3412 plus the reachability margin) but multiplies signalling and drains batteries. IoT devices send rarely and must last years on a battery; extended timers of many hours, combined with power saving mode, reduce their signalling and energy use, and the network accepts that they are unreachable most of the time.

13 Further reading

The MME's role and procedures are specified in TS 23.401 [6]; NAS in TS 24.301 [7]; S1AP in TS 36.413 [5]; GTPv2-C in TS 29.274 [1]; identities in TS 23.003 [8]. CS fallback and SRVCC are in TS 23.272 [3] and TS 23.216 [9]. Olsson et al. [11] and Sesia et al. [12] give textbook treatments; Kleinrock [10] covers the M/M/1 queue.

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[12] S. Sesia, I. Toufik, and M. Baker, *LTE – The UMTS Long Term Evolution: From Theory to Practice*, 2nd ed. Wiley, 2011.

13

Lecture 13

Serving and PDN Gateways (S-GW and P-GW)

دروازه‌های خدمت‌دهنده و (S-GW PDN) و P-GW)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 9 and 12; IP addressing, UDP, tunnelling

In this lecture

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2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-13-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Distinguish** the functions of the serving gateway and the PDN gateway, and **explain** why the user plane has two anchors.
- **LO2** **Describe** GTP-U tunnelling and **compute** its overhead and its effect on the MTU.
- **LO3** **Trace** downlink delivery to an idle UE and the path switch after a handover.
- **LO4** **Model** APN-AMBR policing with a token bucket and **size** gateway buffers and packet rates.
- **LO5** **Explain** control/user-plane separation (CUPS) with PFCP rules, and **compare** home-routed and local-breakout roaming.

2 Prerequisite knowledge

This lecture uses the bearer model of Lecture 9 (default and dedicated bearers, TFTs, QCI, AMBR) and the MME procedures of Lecture 12 (attach, paging, service request). From networking you need IPv4/IPv6 addressing, UDP and the idea of tunnelling: carrying one packet as the payload of another.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
S-GW	دروازهٔ خدمت‌دهنده	User-plane anchor for mobility between eNBs; buffers for idle UEs.
P-GW	دروازهٔ PDN	Anchor towards a packet data network; allocates the UE's IP address.
GTP-U	پروتکل تونل‌زنی GPRS، کاربر صفحهٔ	UDP-based tunnelling protocol for user packets [4].
TEID	شناسهٔ نقطهٔ پایانی تونل	32-bit identifier of a tunnel at the receiving node.
PCEF	تابع اعمال سیاست و صورت‌حساب	Enforcement point of PCC rules in the P-GW.
DDN	اعلان دادهٔ فروسو	S-GW's request to the MME to page an idle UE.
End marker	نشانگر پایان	GTP-U packet that marks the last packet on the old path after a switch.
Token bucket	سطل نشانه	Policing model with rate r and depth b.
CUPS	جداسازی صفحهٔ کنترل و کاربر	Split of each gateway into a control and a user-plane function [2].
PFCP	پروتکل کنترل پیش‌برد بسته	Protocol by which the control plane installs packet rules [5].

3 Introduction and motivation

Every byte that an LTE subscriber sends or receives crosses two gateways. The *serving gateway* (S-GW) faces the radio network: it stays fixed while the UE moves between eNBs and holds packets while the UE sleeps. The *PDN gateway* (P-GW) faces the Internet or the IMS: it gives the UE its IP address and enforces the operator's policy and charging. The P-GW is the UE's point of presence in the IP world, so it must not change during a session; the S-GW is a regional anchor and may change with mobility [3].

Gateways are where the core meets traffic volume. An MME is dimensioned in procedures per second; a gateway is dimensioned in bits and packets per second, and its design is dominated by forwarding performance, buffering and per-packet rule processing. This lecture follows a packet through the gateways, quantifies the cost of tunnelling and policing, and ends with the separation of control and user planes that leads to the 5G UPF.

4 Fundamental concepts

Table 1. Functions of the S-GW and the P-GW [3].

S-GW	P-GW
Local mobility anchor for inter-eNB handover; sends end markers	UE IP address allocation (IPv4, IPv6 prefix)
Downlink buffering in ECM-IDLE; triggers paging via DDN	Policy enforcement (PCEF): gating, QoS, APN-AMBR
Mobility anchor towards 2G/3G (S4, S12)	Downlink packet filtering with TFTs to select the bearer
Lawful interception	Lawful interception, deep packet inspection
Per-bearer accounting (e.g. for roaming settlement)	Online (Gy) and offline (Gz) charging
DSCP marking on the transport network	DSCP marking, NAT and firewall functions (operator choice)

4.1 Division of labour

A UE has exactly one S-GW at a time, which serves all its PDN connections; it may have several P-GWs, one per PDN connection (for example one for the Internet APN and one for IMS). Operators often deploy a *combined S/P-GW* for non-roaming traffic, in which S5 becomes internal.

4.2 Tunnels instead of routes

The UE's IP address belongs to the P-GW's address pool. If routers between eNB and P-GW forwarded the UE's packets by their IP header, every handover would require routing updates. Instead each bearer is carried in GTP-U tunnels: S1-U between eNB and S-GW, S5/S8 between S-GW and P-GW (Figure 1). A handover only changes the tunnel endpoint (address and TEID) at the S-GW; the UE's address and the P-GW never change. Each bearer has its own pair of tunnels in each direction, identified by the receiver's TEID.

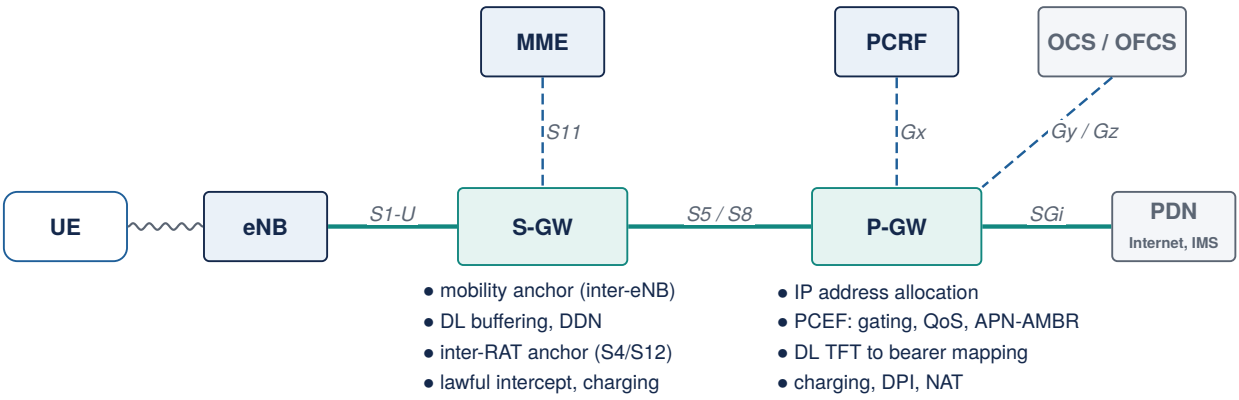


Figure 1. S-GW and P-GW on the user-plane path with their main functions.

5 Detailed technical explanation

5.1 GTP-U

GTP-U runs over UDP port 2152 [4]. Its mandatory header is 8 bytes: flags (version 1, protocol type, and the E, S and PN bits that announce 4 optional bytes for an extension header, a sequence number and an N-PDU number), the message type (255 for G-PDU, i.e. user data), the length of what follows the mandatory header, and the TEID (Figure 2). Signalling messages of GTP-U itself include Echo Request/Response for path supervision, Error Indication (the receiver has no context for a TEID) and End Marker (type 254). The inner packet (T-PDU) is not modified: the gateways see and police the UE's own IP header.

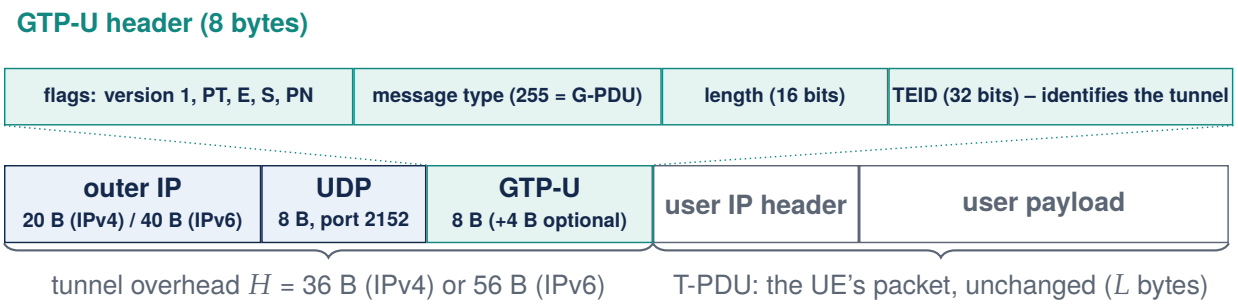


Figure 2. GTP-U encapsulation of a user packet and the mandatory GTP-U header.

5.2 IP address allocation

At attach (or PDN connection set-up) the UE requests PDN type IPv4, IPv6 or IPv4v6. For IPv4 the P-GW returns an address in the Create Session Response, which reaches the UE in NAS; for IPv6 it allocates a /64 prefix and announces it in Router Advertisements over the default bearer, from which the UE builds its address. Private IPv4 addresses with carrier-grade NAT at or behind the P-GW are common, which is one motivation for IPv6-only APNs.

5.3 Policy enforcement

The PCEF in the P-GW applies the PCC rules received over Gx (Lecture 10): it maps downlink packets to bearers with DL TFTs, blocks or allows service data flows (gating), polices GBR/MBR per bearer and APN-AMBR across the non-GBR bearers of a PDN connection (Section 6.3), counts volume and time per rating group for charging, and marks packets with DSCP values derived from the QCI so that the transport network can prioritise them. UE-AMBR is enforced by the eNB.

5.4 Idle mode and handover

When a UE goes to ECM-IDLE, the eNB releases its context and the MME asks the S-GW to release the S1-U tunnels; the S5/S8 part stays. A downlink packet then finds no S1-U tunnel: the S-GW buffers it and sends DDN to the MME, which pages the UE (Figure 4). After an X2 handover the target eNB asks the MME for a path switch; the MME sends Modify Bearer Request to the S-GW, which switches the downlink to the target eNB and sends one or more *end marker* packets on the old path, so that the source eNB knows when forwarding of in-flight packets is complete and the target can deliver packets in order.

5.5 Control/user-plane separation (CUPS)

Release 14 split each gateway into a control function (SGW-C, PGW-C) and one or more user-plane functions (SGW-U, PGW-U), connected by the Sx interfaces [2]. The control function keeps S11, S5-C, Gx and Gy; the user-plane function only forwards packets according to rules installed with PFCP [5]:

- **PDR** (packet detection rule): which packets (TEID, UE address, SDF filter) the rule applies to;
- **FAR** (forwarding action rule): forward, drop, buffer or duplicate, and where to;
- **QER** (QoS enforcement rule): gating and rate limits;
- **URR** (usage reporting rule): what to count and when to report;
- **BAR** (buffering action rule): how to buffer for idle UEs.

User-plane functions can then be placed close to the radio network (for low latency and local traffic) and scaled independently of the control plane. The 5G UPF and its N4 interface reuse this model (Lecture 23).

6 Mathematical models

6.1 Tunnelling overhead and MTU

Model. A user packet of L bytes is carried with tunnel overhead H (outer IP + UDP + GTP-U): $H = 20 + 8 + 8 = 36$ bytes with IPv4 transport, $H = 56$ bytes with IPv6. The transport efficiency and the largest user packet that fits a transport MTU M without fragmentation are

$$\eta = \frac{L}{L+H}, \qquad L_{\max} = M - H. \tag{1}$$

Example. $L = 1400$, IPv4 transport: $\eta = 97.5\,\%$. With $M = 1500$, $L_{\max} = 1464$ bytes (1444 with IPv6 transport), which is why operators configure the UE's MTU below 1500 or enlarge the transport MTU.

Interpretation. Overhead matters for small packets: for voice it is a large fraction of the packet (Example 1). Header compression (ROHC) on the radio does not help in the core, where the full headers and the tunnel are carried.

6.2 Downlink buffering

Model. A source sends at rate R to an idle UE; until the S1-U tunnel is restored after a paging and service-request delay D, the S-GW must buffer

$$B = R\,D. \tag{2}$$

Example. $R = 2\,\text{Mbit/s}$, $D = 1.5\,\text{s}$: $B = 3\,\text{Mbit} = 375\,\text{kB}$. In practice TCP sources send only a few packets before waiting for acknowledgements, so actual buffers are much smaller, and gateways limit the buffer per bearer and drop beyond it.

6.3 Policing with a token bucket

Model. A token bucket with rate r and depth b admits a traffic stream if, for every interval of length T, the data sent $A(T)$ satisfy

$$A(T) \le b + r\,T. \tag{3}$$

A source with peak rate $R > r$ starting with a full bucket can send at its peak for $T^{*} = b/(R-r)$.

Example. APN-AMBR $r = 10\,\text{Mbit/s}$, $b = 8\,\text{Mbit}$ (1 MB): in 3 s at most $8 + 30 = 38\,\text{Mbit}$ (4.75 MB) pass; a 40 Mbit/s source can burst for $8/30 = 0.27\,\text{s}$ before being limited to 10 Mbit/s (Figure 3).

Interpretation. The rate fixes the long-term share; the depth determines how bursty traffic may be. Too small a bucket hurts TCP (losses at every burst), too large a bucket lets short bursts exceed the subscribed rate noticeably.

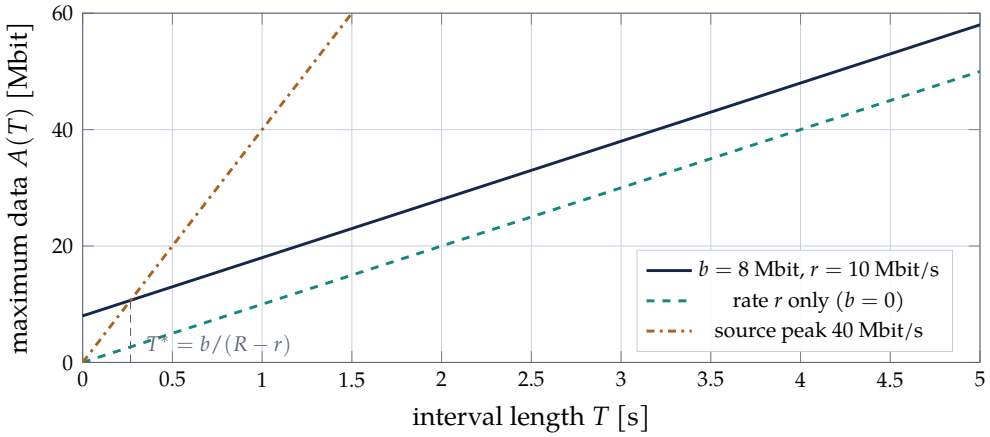


Figure 3. Token-bucket envelope $b + rT$ (Eq. (3)) and a peak-rate source.

6.4 Packet rate

Model. Forwarding cost is per packet, not per byte. A gateway carrying throughput C with mean packet size $\bar{L}$ handles

$$P = \frac{C}{8\,\bar{L}}\ \text{packets/s}. \tag{4}$$

Example. $C = 40\,\text{Gbit/s}$, $\bar{L} = 800\,\text{bytes}$: $P = 6.25\,\text{Mpps}$ – each packet must be classified against the PDRs of its session, counted and encapsulated within about 160 ns on average per packet for a single processing pipeline, which is why user planes use kernel-bypass packet I/O and hardware offload.

7 Network architecture and diagrams

In a non-roaming network the S-GW and P-GW are usually co-located in regional data centres; with CUPS, user-plane functions move towards the edge while the control functions stay central. For roaming, the visited network provides the S-GW; the P-GW is either in the home network (*home-routed*, over S8, the usual case for data) or in the visited network (*local breakout*, used e.g. for

IMS voice in some agreements). Home routing keeps charging and policy in the home network but lengthens the path (Exercise 5).

8 Step-by-step procedures

Procedure 1 • Downlink data for an idle UE

1. A downlink packet reaches the P-GW, which maps it to a bearer with the DL TFT and forwards it over S5 to the S-GW.
2. The S-GW has no S1-U tunnel for the bearer: it buffers the packet and sends Downlink Data Notification to the MME (further packets are buffered without a new DDN).
3. The MME pages the UE in its TAI list; the UE answers with a Service Request.
4. The MME sets up the context in the eNB (Initial Context Setup), learns the eNB's TEID and sends Modify Bearer Request to the S-GW.
5. The S-GW forwards the buffered packets to the eNB, followed by new traffic.

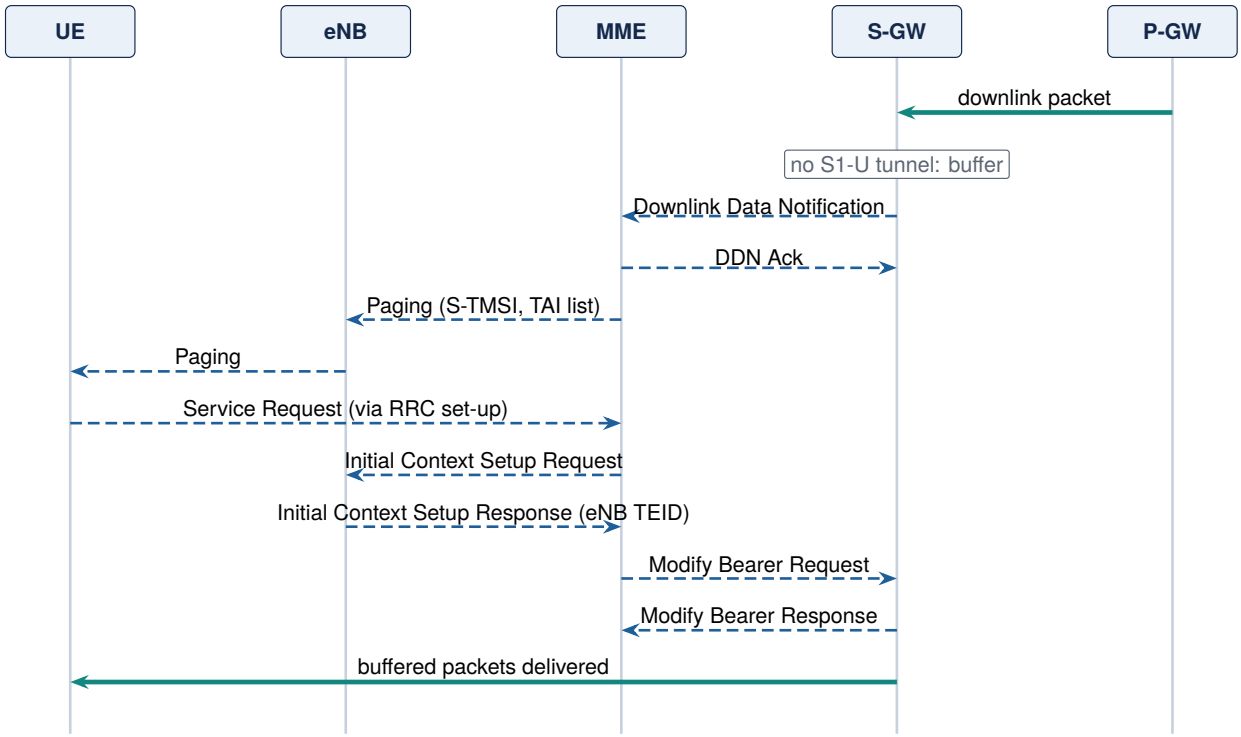


Figure 4. Network-triggered service request with S-GW buffering (Procedure 1).

Procedure 2 • Path switch after an X2 handover

1. During the handover the source eNB forwards in-flight downlink packets to the target eNB over X2.
2. The target eNB sends Path Switch Request to the MME with its downlink TEIDs.
3. The MME sends Modify Bearer Request to the S-GW, which switches the downlink tunnels to the target.

4. The S-GW sends end-marker packets to the source eNB on the old tunnels; the source relays them to the target, which then delivers the forwarded packets before the new ones.
5. The source eNB releases the UE context.

9 Worked examples

Worked example 1 • Overhead of a voice packet

Problem. An AMR-WB 12.65 kbit/s frame produces about 33 bytes of RTP payload every 20 ms; the UE uses IPv6. Compute the size of the packet on S1-U (IPv4 transport) and the fraction that is headers.
Solution. Inner packet: $33 + 12\,(\text{RTP}) + 8\,(\text{UDP}) + 40\,(\text{IPv6}) = 93$ bytes. On S1-U: $93 + 36 = 129$ bytes. Tunnel overhead alone is $36/129 = 28\,\%$; all headers together are $(129 - 33)/129 = 74\,\%$. At 50 packets/s the call uses $129 \times 8 \times 50 = 51.6$ kbit/s on S1-U for 12.65 kbit/s of speech.

Worked example 2 • Token-bucket admission

Problem. With $r = 10$ Mbit/s and $b = 8$ Mbit, a user downloads a 5 MB file at 40 Mbit/s from a server, starting with a full bucket. How long does the download take?
Solution. 5 MB = 40 Mbit. During the initial burst of $T^* = 0.27$ s, $40 \times 0.27 = 10.7$ Mbit pass; the remaining 29.3 Mbit pass at 10 Mbit/s in 2.93 s. Total ≈ 3.2 s, versus 1 s unpoliced and 4 s with $b = 0$.

Worked example 3 • Gateway packet rate

Problem. A P-GW serves 2×10^5 simultaneously active users with a mean of 200 kbit/s each. Compute throughput and packet rate for a mean packet of 800 bytes.
Solution. $C = 2 \times 10^5 \times 2 \times 10^5 = 40$ Gbit/s; $P = 40 \times 10^9/6400 = 6.25$ Mpps (Eq. (4)). Uplink packets (many small TCP acknowledgements) add to the packet rate while adding little throughput.

10 Technology comparison

One anchor or two?. Two anchors separate concerns: the S-GW hides radio mobility from the P-GW, and the P-GW hides the PDN from the radio side. The cost is an extra hop and extra state. 5G keeps the idea but makes it flexible: a session may have one UPF or a chain of them, chosen per session by the SMF.

11 Key takeaways

Table 2. Packet gateways across generations.

	SGSN + GGSN (2G/3G)	S-GW + P-GW (4G)	UPF (+ SMF) (5G)
Anchors	GGSN (IP), SGSN (mobility)	P-GW (IP), S-GW (mobility)	PDU-session anchor UPF, intermediate UPFs
Tunnelling	GTP-U (Gn, Iu-PS)	GTP-U (S1-U, S5/S8)	GTP-U (N3, N9)
Control of forwarding	Integrated	Integrated; CUPS from Rel-14	Always separated: PFCP over N4
QoS granularity	PDP context	EPS bearer	QoS flow (QFI)
Local breakout / edge	Rare	Possible with CUPS	Native (UL classifier, branching point)

Key takeaways

- S-GW: mobility anchor between eNBs, buffering and DDN for idle UEs, end markers.
- P-GW: IP anchor (address allocation), PCEF (TFT mapping, gating, AMBR, charging), SGi to the PDN.
- GTP-U tunnels carry bearers; overhead $H = 36$ (IPv4) or 56 (IPv6) bytes, $L_{\max} = M - H$.
- Buffering $B = RD$; policing $A(T) \leq b + rT$; forwarding cost $P = C/(8\bar{L})$ packets/s.
- CUPS splits gateways into control and user planes connected by PFCP rules (PDR, FAR, QER, URR, BAR) – the direct ancestor of the 5G UPF.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which gateway?

Which gateway (S-GW or P-GW) changes in each case: (a) X2 handover between two eNBs of the same S-GW area; (b) the UE moves to a region served by another S-GW; (c) the UE opens a second PDN connection to the IMS APN?

Exercise 2 • Level 1 – conceptual • End markers

Explain what could go wrong at the target eNB after a handover if the S-GW did not send end-marker packets.

Exercise 3 • Level 2 – numerical • MTU

The transport network has an MTU of 1500 bytes and uses IPv6. What is the largest UE packet that avoids fragmentation? What MTU would the transport network need to carry 1500-byte UE packets?

Exercise 4 • Level 2 – numerical • Buffer sizing

An S-GW limits the buffer per bearer to 256 kB. For a paging and service-request delay of 1.2 s, up to which source rate is no packet dropped? If 10^4 UEs are paged simultaneously and each buffer fills, how much memory is needed?

Exercise 5 • Level 3 – analytical • Home-routed roaming

A subscriber roams 4000 km from the home network and accesses a server located in the visited country. Estimate the extra round-trip time of home routing compared with local breakout (propagation at 2×10^8 m/s, ignoring processing), and discuss why operators still prefer home routing for data.

Exercise 6 • Level 3 – analytical • Scaling a user plane

A PGW-U core forwards 1.5 Mpps. How many cores are needed for 100 Gbit/s with a mean packet of 700 bytes if cores may be loaded to at most 70 %? What does CUPS allow the operator to do when traffic doubles but the number of subscribers does not?

Solutions

Solution to Exercise 1

(a) Neither: only the S1-U endpoint (eNB) changes. (b) The S-GW changes (S-GW relocation); the P-GW and IP address stay. (c) A new P-GW (possibly) is added for the new PDN connection; the S-GW is shared.

Solution to Exercise 2

The target receives packets on two paths: forwarded over X2 from the source and directly from the S-GW. Without an end marker it cannot know when the forwarded stream has ended; delivering new packets first would reorder the flow (hurting TCP), and waiting by timer would add delay. The end marker closes the old path explicitly.

Solution to Exercise 3

$L_{\max} = 1500 - 56 = 1444$ bytes. To carry 1500-byte UE packets the transport MTU must be at least $1500 + 56 = 1556$ bytes (or 1540 with IPv4 transport); operators often run jumbo frames in the backhaul.

Solution to Exercise 4

$R \le B/D = 256 \times 8$ kbit$/1.2$ s $= 1.71$ Mbit/s. Worst-case memory $10^4 \times 256$ kB $= 2.56$ GB; in practice far less because TCP sources stall after one window and most bursts are a few packets.

Solution to Exercise 5

Home routing sends each packet visited $\rightarrow$ home $\rightarrow$ server (in the visited country) and back the same way: an extra 2×4000 km each direction, i.e. 4×4000 km$/(2 \times 10^8$ m/s$) = 80$ ms of extra RTT. Operators accept it because the home network keeps control of policy, charging, lawful interception and service access (e.g. home APNs), and because roaming settlement is

simpler; local breakout needs agreements on policy and charging in the visited network.

Solution to Exercise 6

$P = 100 \times 10^9/(8 \times 700) = 17.9$ Mpps; $17.9/1.5 = 11.9$ cores at full load, $11.9/0.7 = 17.0 \rightarrow$ 18 cores. With CUPS the operator adds user-plane capacity (more PGW-U instances or cores, possibly at new edge sites) without touching the control plane, whose load depends on subscribers and procedures, not on traffic volume.

13 Further reading

Gateway functions are specified in TS 23.401 [3], GTPv2-C in TS 29.274 [1] and GTP-U in TS 29.281 [4]; CUPS and PFCP in TS 23.214 [2] and TS 29.244 [5]; policy and charging control in TS 23.203 [6]. Olsson et al. [7] discuss gateway deployment, and Rommer et al. [8] the evolution towards the 5G user plane.

References

[1] 3GPP, *3GPP evolved packet system (EPS); evolved general packet radio service (GPRS) tunnelling protocol for control plane (GTPv2-C); stage 3*, TS 29.274, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29274.htm.

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[3] 3GPP, *General packet radio service (GPRS) enhancements for evolved universal terrestrial radio access network (E-UTRAN) access*, TS 23.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23401.htm.

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[7] M. Olsson, S. Sultana, S. Rommer, L. Frid, and C. Mulligan, *EPC and 4G Packet Networks: Driving the Mobile Broadband Revolution*, 2nd ed. Academic Press, 2013.

[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.

14

Lecture 14

5G Networks: Introduction

مقدمه‌ای بر شبکه‌های نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 5–7 and 9; OFDM basics; Shannon capacity

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 6
6 Mathematical models . . . 7
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
9 Worked examples . . . 9
10 Technology comparison . . . 9
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-14-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **State** the IMT-2020 usage scenarios and key minimum requirements, and **relate** each to a design choice of 5G.
- **LO2** **Explain** NR's scalable numerology and **compute** slot and symbol durations for each subcarrier spacing.
- **LO3** **Compute** the approximate peak data rate of an NR carrier with the 3GPP formula.
- **LO4** **Compare** non-standalone (EN-DC) and standalone deployments.
- **LO5** **Estimate** latency, connection density and area capacity for given deployments, and **judge** whether a requirement is met.
- **LO6** **Outline** the 5G core and the topics of the following lectures.

2 Prerequisite knowledge

You need the LTE architecture and radio protocols (Lectures 5–7), the EPC (Lecture 9), the structure of an OFDM resource grid (subcarriers, symbols, cyclic prefix, resource blocks) and Shannon's capacity formula $C = B\log_2(1+\mathrm{SNR})$ [11].

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
IMT-2020	IMT-2020	ITU-R's name for the 5G family of radio interfaces and its requirements.
eMBB / URLLC / mMTC	سه سناریوی استفاده	Enhanced broadband, ultra-reliable low latency, massive machine type.
NR	رادیوی جدید	3GPP's 5G radio interface (Release 15 onward).
Numerology μ	عددشناسی	Subcarrier spacing $15 \cdot 2^{\mu}$ kHz and the matching slot length.
FR1 / FR2	بازه‌های بسامدی ۱ و ۲	Sub-7 GHz and millimetre-wave frequency ranges.
Bandwidth part	بخش پهنای باند	Subset of a carrier configured for a UE.
EN-DC	اتصال دوگانهٔ E-UTRA--NR	LTE master and NR secondary node with the EPC (NSA).
SA / NSA	مستقل / غیرمستقل	NR with the 5G core / NR anchored on LTE.
5GC	هستهٔ نسل پنجم	Service-based core network (AMF, SMF, UPF, ...).
Network slice	برش شبکه	Logical end-to-end network for a service type.

3 Introduction and motivation

LTE was designed for one dominant use: mobile broadband to smartphones. By the mid-2010s operators and industries expected three different kinds of demand: far higher capacity for video and fixed wireless access, wireless control of machines with reliability and latency comparable to wired fieldbuses, and connectivity for very large numbers of cheap, battery-powered sensors. ITU-R captured these as the three IMT-2020 usage scenarios (Figure 1) [9] and set minimum technical requirements for any candidate radio interface [10]. 3GPP's answer is the 5G system: the NR radio interface [2], [8] and the service-based 5G core [7].

This lecture opens Part IV. It surveys the requirements, the main ideas of NR that address them, and the migration from LTE; the core network, procedures, slicing and QoS follow in Lectures 18–22, while software-defined networking and virtualisation, the technologies on which the 5G core is built, are the topic of Part III.

4 Fundamental concepts

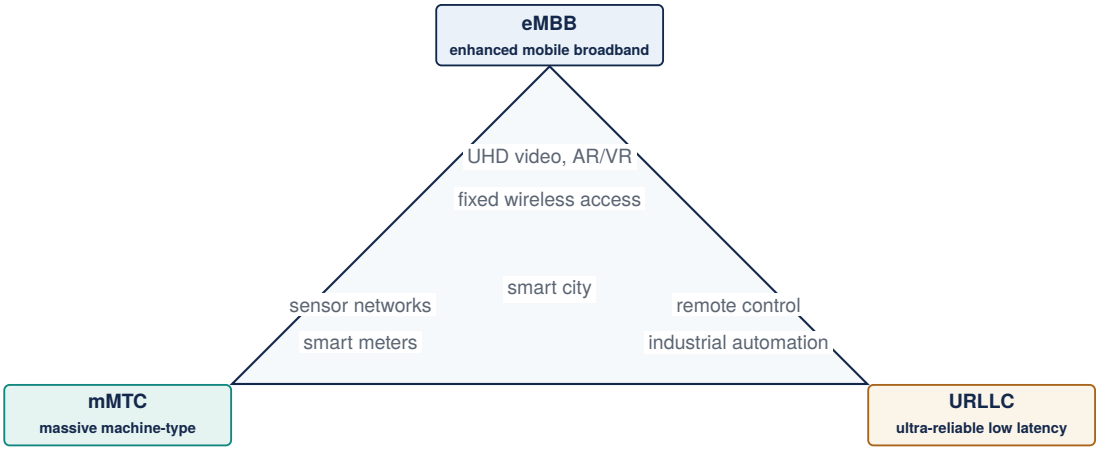


Figure 1. The three IMT-2020 usage scenarios with example applications [9].

Table 1. Selected IMT-2020 minimum technical requirements (ITU-R M.2410) [10].

Requirement	Value	Scenario / remark
Peak data rate	DL 20 Gbit/s, UL 10 Gbit/s	eMBB
Peak spectral efficiency	DL 30, UL 15 bit/s/Hz	eMBB
User-experienced data rate	DL 100 Mbit/s, UL 50 Mbit/s	eMBB, dense urban (5th percentile)
Area traffic capacity	10 Mbit/s/m^2	eMBB, indoor hotspot, DL
User-plane latency	4 ms (eMBB), 1 ms (URLLC)	One way, unloaded
Control-plane latency	20 ms	Idle to active
Connection density	10^6 devices/km^2	mMTC
Reliability	$1 - 10^{-5}$ for 32 bytes in 1 ms	URLLC
Mobility	up to 500 km/h	High-speed trains
Bandwidth	at least 100 MHz; up to 1 GHz above 6 GHz	–

4.1 Requirements

No single deployment has to meet all values at once: each requirement is evaluated in its own test environment. Figure 2 shows the improvement over IMT-Advanced (4G) that ITU-R envisaged for each key capability.

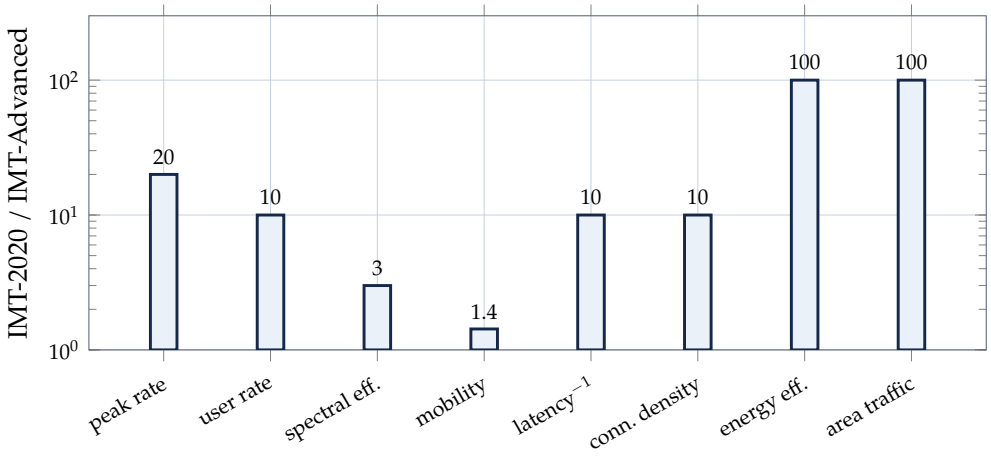


Figure 2. Improvement factors of IMT-2020 key capabilities over IMT-Advanced, as envisaged in ITU-R M.2083 [9]. Latency is shown as its inverse.

4.2 Design principles of NR

- **Wide spectrum range**: FR1 (410 MHz–7.125 GHz) and FR2 (24.25–52.6 GHz, extended in later releases), with carriers up to 100 MHz in FR1 and 400 MHz in FR2 [2].
- **Scalable numerology**: subcarrier spacing $15 \cdot 2^{\mu}$ kHz (Section 6.1).
- **Ultra-lean design**: no always-on cell-specific reference signals; signals are sent only when needed, which saves energy and reduces interference.
- **Beam-centric operation**: synchronisation signal blocks are swept in beams; beam management is built in, which is essential at millimetre waves.
- **Flexible scheduling**: slots and mini-slots (2, 4 or 7 symbols), dynamic TDD, bandwidth parts.
- **New channel coding**: LDPC for data, polar codes for control [3].
- **Forward compatibility**: reserved resources so that future features do not break legacy UEs.

4.3 Deployment options

3GPP defined several ways to combine LTE, NR, the EPC and the 5GC. The two that matter in practice (Figure 3):

- **Non-standalone (NSA), option 3 family (EN-DC)**: the UE is connected to an LTE eNB (master node, control plane to the MME) and an NR en-gNB (secondary node, adding capacity) [1]. It reuses the EPC and allowed the first 5G launches in 2019 without a new core, but provides none of the new core features (slicing, new QoS, service-based architecture).
- **Standalone (SA), option 2**: the gNB connects to the 5GC over N2 (control) and N3 (user plane). All 5G features, including low latency and slicing, require SA.

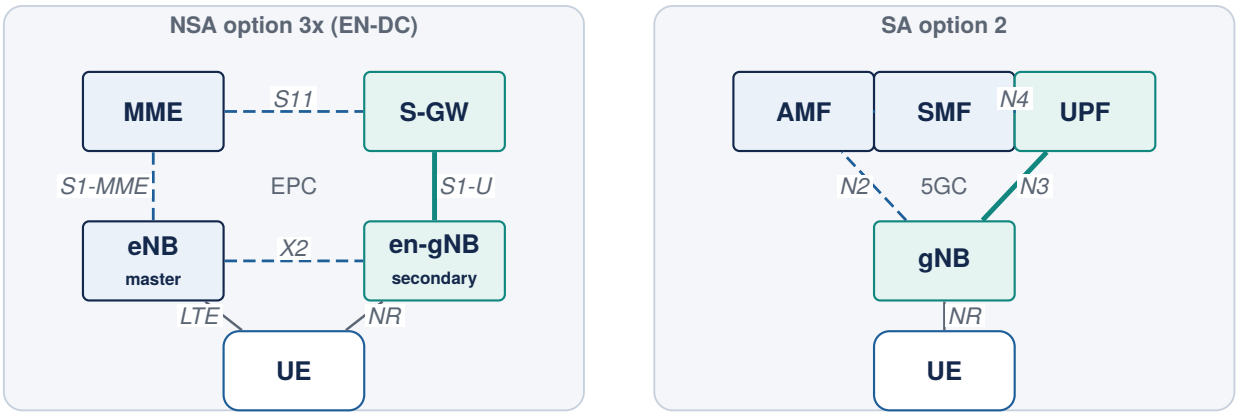


Figure 3. Non-standalone EN-DC (option 3x) with the EPC, and standalone NR (option 2) with the 5GC.

5 Detailed technical explanation

5.1 Frame structure

NR keeps LTE's 10 ms frame and 1 ms subframe. A slot has 14 OFDM symbols (normal cyclic prefix); with numerology μ a subframe contains 2^{μ} slots (Figure 4). A resource block is 12 subcarriers in frequency; a carrier has at most 275 resource blocks. Because 1 ms is a fixed grid for all numerologies, symbols of different numerologies align at subframe boundaries, which allows mixing them on one carrier [3].

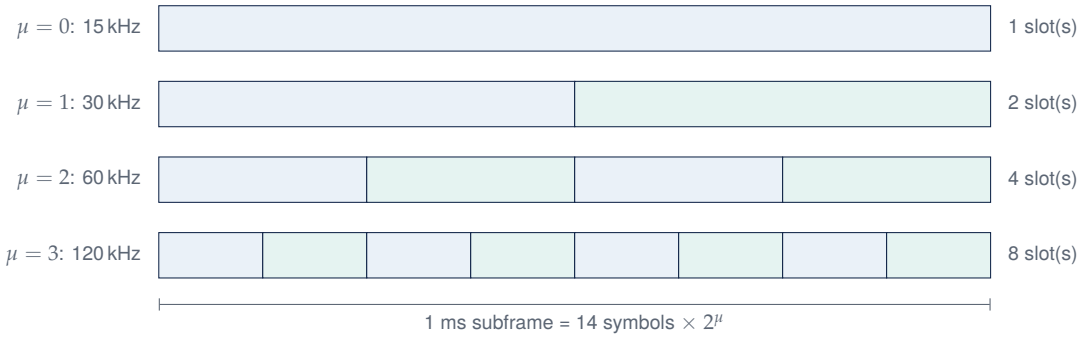


Figure 4. Slots per 1 ms subframe for numerologies $\mu = 0$–3.

5.2 Initial access and beams

A gNB transmits synchronisation signal blocks (SSBs: PSS, SSS and PBCH) in a burst, each in a different beam. The UE detects the best SSB, reads the system information and performs random access on the RACH occasion associated with that SSB; the gNB thereby learns the best beam. During a connection, beams are refined with CSI-RS measurements and reports. In FR1 a few wide beams suffice; in FR2 up to 64 narrow beams compensate the higher path loss with array gain.

5.3 From EPC to 5GC

The 5G core separates access and mobility management (AMF) from session management (SMF) and the user plane (UPF), stores subscription data in the UDM/UDR, authenticates through the AUSF and controls policy in the PCF. Network functions expose services over HTTP/2 and discover each other through the NRF [7]. The QoS model uses QoS flows within PDU sessions instead of EPS bearers, and network slices allow several logical networks on one infrastructure. These topics are developed in Lectures 18–22.

6 Mathematical models

6.1 Numerology

Definitions. For $\mu = 0, 1, 2, 3, 4$:

$$\Delta f = 15 \cdot 2^{\mu} \text{ kHz}, \qquad T_{\text{slot}} = \frac{1 \text{ ms}}{2^{\mu}}, \qquad \bar{T}_{\text{sym}} = \frac{T_{\text{slot}}}{14}, \tag{1}$$

where $\bar{T}_{\text{sym}}$ is the mean symbol duration including the cyclic prefix. The normal cyclic prefix is about $4.7/2^{\mu}$ µs.

Interpretation. Wider spacing shortens symbols and slots (lower latency) and tolerates more phase noise and Doppler (needed in FR2), but its shorter cyclic prefix tolerates less delay spread, which suits small cells. FR1 macro cells typically use 15 or 30 kHz; FR2 uses 120 kHz.

6.2 Peak data rate

Formula. 3GPP approximates the maximum data rate of a carrier as [4]

$$R \approx 10^{-6}\, v\, Q_m\, f\, R_{\max} \frac{N_{\text{PRB}} \cdot 12}{T_s^{\mu}} \left(1 - \text{OH}\right) \text{Mbit/s}, \qquad T_s^{\mu} = \frac{10^{-3}}{14 \cdot 2^{\mu}} \text{ s}, \tag{2}$$

with v MIMO layers, Q_m bits per modulation symbol, scaling factor $f \leq 1$, maximum code rate $R_{\max} = 948/1024$, N_{PRB} resource blocks and overhead OH (0.14 in FR1 DL, 0.18 in FR2 DL, 0.08 in FR1 UL, 0.10 in FR2 UL). For carrier aggregation the rates of the carriers are added.

Example. FR1, 100 MHz at 30 kHz ($N_{\text{PRB}} = 273$), 4 layers, 256-QAM ($Q_m = 8$), $f = 1$: $R \approx 2.34$ Gbit/s (Example 1).

6.3 User-plane latency

Model. A simple estimate of one-way radio latency is

$$T_{\text{UP}} \approx T_{\text{align}} + T_{\text{tx}} + T_{\text{proc}} \approx \tfrac{1}{2} T_{\text{slot}} + T_{\text{slot}} + 2\, T_{\text{slot}}, \tag{3}$$

counting on average half a slot to wait for the next transmission opportunity, one slot of transmission and about two slots of processing at transmitter and receiver.

Example. $\mu = 0$: 3.5 ms; $\mu = 1$: 1.75 ms; $\mu = 3$: 0.44 ms. Mini-slots and pre-configured (grant-free) uplink shorten T_{align} and T_{tx} further.

Limitations. The model ignores scheduling requests, HARQ retransmissions, queueing and the transport network and core; it shows why the URLLC target of 1 ms requires larger numerologies or mini-slots.

6.4 Connection density and area capacity

Models. With device density ρ (devices/km^2) and cell radius r, a cell serves $N = \rho\pi r^2$ devices; if each accesses the network once per period T, the access rate is N/T. For area capacity, a cell with bandwidth B and mean spectral efficiency $\bar{\eta}$ covering area A_{cell} offers

$$C_{\text{area}} = \frac{B\,\bar{\eta}}{A_{\text{cell}}}. \tag{4}$$

Examples. $\rho = 10^6$/km^2, $r = 500$ m: $N = 7.9 \times 10^5$ devices; one report every 2 h gives 109 accesses/s per cell (Example 3). For $C_{\text{area}} = 10$ Mbit/s/m^2 with $B = 400$ MHz and $\bar{\eta} = 5$ bit/s/Hz, each cell may cover only $A_{\text{cell}} = 200$ m^2: area capacity is achieved by densification, not by spectral efficiency alone.

7 Network architecture and diagrams

Figure 3 summarises the two main architectures. In NSA the EPC remains the core, and the NR node is an extension of the LTE radio network (it has no control-plane connection to the core). In SA, the gNB (split into central and distributed units, Lecture 23) connects to the 5GC; LTE and NR then interwork through the core (N26 between AMF and MME) for mobility and voice fallback. Most operators started with NSA and introduced SA later, often first for specific segments such as fixed wireless access or enterprise slices.

8 Step-by-step procedures

Procedure 1 • Adding an NR secondary node in EN-DC

1. The UE is connected to the LTE eNB (master node) and reports NR measurements configured by it.
2. The eNB decides to add an NR leg and sends SgNB Addition Request over X2 to the en-gNB.
3. The en-gNB allocates resources and answers with the NR radio configuration; the eNB forwards it in RRC Connection Reconfiguration.
4. The UE performs random access on the NR cell and confirms; the eNB informs the MME (E-RAB modification) so that the S-GW can send the user plane directly to the en-gNB (option 3x).
5. Data now flow over NR (and LTE, if the bearer is split); mobility and control remain with LTE.

Procedure 2 • Initial access in standalone NR

1. Search the SSB raster, detect PSS/SSS, determine the cell ID and the best SSB beam.
2. Read the MIB on PBCH and SIB1 (scheduled by it) to obtain RACH configuration and cell parameters.
3. Transmit a preamble in the RACH occasion of the best SSB; receive the random-access response (Lecture 8).
4. Complete RRC set-up and send the Registration Request to the AMF (Lecture 20).

9 Worked examples

Worked example 1 • Peak rate of an FR1 carrier

Problem. Evaluate Eq. (2) for 100 MHz at 30 kHz, 4 layers, 256-QAM, $f = 1$, DL.
Solution. $T_s^1 = 10^{-3}/28 = 35.7\,\mu\text{s}$. $N_{\text{PRB}} \cdot 12/T_s = 273 \times 12/35.7 \times 10^{-6} = 9.17 \times 10^7$ REs/s. $R = 4 \times 8 \times 0.926 \times 9.17 \times 10^7 \times 0.86 = 2.34 \times 10^9$ bit/s. To reach 20 Gbit/s, NR needs FR2 bandwidths, more layers and carrier aggregation.

Worked example 2 • Latency and numerology

Problem. Which numerologies satisfy the 1 ms URLLC target with the model of Eq. (3)?
Solution. $T_{\text{UP}} = 3.5\,T_{\text{slot}} \leq 1\,\text{ms}$ requires $T_{\text{slot}} \leq 0.286\,\text{ms}$, i.e. $\mu \geq 2$ (0.875 ms at 60 kHz). With 30 kHz, mini-slots of 2 symbols reduce the transmission and alignment terms and can meet the target as well.

Worked example 3 • mMTC access load

Problem. A cell of radius 500 m in an area with 10^6 devices/km^2; each device sends one report every two hours. Compute the access rate and compare with LTE's typical RACH capacity of the order of a few hundred attempts per second per cell.
Solution. $N = 10^6 \times \pi \times 0.25 = 7.85 \times 10^5$; rate $= 7.85 \times 10^5/7200 = 109\,\text{s}^{-1}$ on average – manageable, but synchronised reporting (all devices at the full hour) would be catastrophic (Lecture 8). mMTC therefore relies on randomised schedules, early data transmission and NB-IoT/LTE-M features, not only on raw capacity.

10 Technology comparison

Evolution, not replacement. LTE continues to evolve (LTE-M, NB-IoT, and dynamic spectrum sharing with NR on the same carrier), and NR reuses many LTE concepts: OFDM, HARQ, the layer-2 stack, and the attach/registration and bearer/flow logic of the core.

11 Key takeaways

Table 2. LTE and NR compared.

	LTE (4G)	NR (5G)
Spectrum	Below 6 GHz	FR1 and FR2 (mmWave)
Max carrier bandwidth	20 MHz (aggregation up to 32 carriers)	100 MHz (FR1), 400 MHz (FR2)
Subcarrier spacing	15 kHz (fixed)	$15 \cdot 2^{\mu}$ kHz
Scheduling unit	1 ms subframe	Slot ($1/2^{\mu}$ ms), mini-slot
Reference signals	Always-on CRS	On demand (ultra-lean)
Channel coding	Turbo (data), convolutional (control)	LDPC (data), polar (control)
Beamforming	Added in later releases	Native, including initial access
Core	EPC	EPC (NSA) or 5GC (SA)
QoS unit	EPS bearer	QoS flow

Key takeaways

- IMT-2020 defines three scenarios (eMBB, URLLC, mMTC) and minimum requirements such as 20 Gbit/s peak, 1 ms URLLC latency and 10^6 devices/km^2.
- NR: wide spectrum (FR1, FR2), numerology $15 \cdot 2^{\mu}$ kHz, slots of $1/2^{\mu}$ ms, lean and beam-based.
- Peak rate $\propto v\, Q_m\, R_{\max}\, N_{\mathrm{PRB}} \cdot 12/T_s\,(1-\mathrm{OH})$; 100 MHz FR1 gives about 2.3 Gbit/s with 4 layers.
- NSA (EN-DC) reuses the EPC; SA with the 5GC unlocks slicing, new QoS and low latency.
- Area capacity comes from densification; latency from numerology and mini-slots; density from access design.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which scenario?

Assign each service to eMBB, URLLC or mMTC and name its most demanding requirement: (a) 8K video streaming; (b) motion control in a factory; (c) water meters; (d) remote surgery; (e) fixed wireless access to homes.

Exercise 2 • Level 1 – conceptual • Why NSA first?

Give two reasons why most operators launched 5G as NSA, and one capability they could not offer until SA.

Exercise 3 • Level 2 – numerical • FR2 peak rate

Compute the peak DL rate of a 400 MHz FR2 carrier at 120 kHz ($N_{\mathrm{PRB}} = 264$) with 2 layers and 64-QAM. How many such carriers would be needed for 20 Gbit/s?

Exercise 4 • Level 2 – numerical • Cyclic prefix and cell size

The normal cyclic prefix is about $4.7/2^{\mu}$ µs. Compute it for $\mu = 0$–3 and the corresponding path-length difference it can absorb. Which numerologies suit a rural macro cell with a delay spread of 2 µs?

Exercise 5 • Level 3 – analytical • Peak spectral efficiency

Using Eq. (2), compute the peak spectral efficiency of a 100 MHz FR1 carrier at 30 kHz with 8 DL layers and 256-QAM. Does it meet the IMT-2020 requirement of 30 bit/s/Hz? Which assumptions make this a theoretical value?

Exercise 6 • Level 3 – analytical • Densification

An indoor hotspot must provide 10 Mbit/s/m^2. Compare the cell area needed with (a) 400 MHz at 5 bit/s/Hz and (b) 100 MHz at 7 bit/s/Hz. Discuss the implications for backhaul and for the core network.

Solutions

Solution to Exercise 1

(a) eMBB – user data rate; (b) URLLC – latency and reliability; (c) mMTC – connection density and battery life; (d) URLLC (with eMBB video) – reliability and latency; (e) eMBB – capacity per area and per user.

Solution to Exercise 2

NSA reused the existing EPC (no new core, faster time to market) and LTE coverage for control and mobility, while NR added capacity; early devices and spectrum were also tuned for it. Features needing the 5GC – network slicing, QoS flows, service-based exposure, low-latency edge user planes, and NR-only coverage – required SA.

Solution to Exercise 3

$T_s^3 = 10^{-3}/112 = 8.93$ µs; $R = 2 \times 6 \times 0.926 \times 264 \times 12/8.93 \times 10^{-6} \times 0.82 = 3.23$ Gbit/s. For 20 Gbit/s about $20/3.23 = 6.2$, i.e. 7 carriers – or more layers and higher-order modulation.

Solution to Exercise 4

$\mu = 0$: 4.69 µs (1406 m); $\mu = 1$: 2.34 µs (703 m); $\mu = 2$: 1.17 µs (352 m); $\mu = 3$: 0.59 µs (176 m), using $d = c\,T_{\mathrm{CP}}$. With 2 µs delay spread, $\mu = 0$ (and marginally $\mu = 1$) are suitable; 60 kHz needs the extended CP, and 120 kHz is for small FR2 cells.

Solution to Exercise

With 8 layers the rate doubles to 4.67 Gbit/s, i.e. 46.7 bit/s/Hz over 100 MHz – above 30. The value assumes 8 spatially separable layers, 256-QAM at the highest code rate (very high SINR), the fixed overhead factor and full allocation to one UE; real systems reach a fraction of it.

Solution to Exercise

(a) $400 \times 10^6 \times 5 = 2\,\text{Gbit/s}$ per cell, so $A = 2 \times 10^9 / 10^7 = 200\,\text{m}^2$; (b) 0.7 Gbit/s, so $70\,\text{m}^2$. Either way cells are rooms, not streets. Each cell needs multi-gigabit backhaul (fibre), and the core must handle traffic of 10 Tbit/s per km^2 of such hotspots locally – an argument for edge user planes (Lecture 23).

13 Further reading

ITU-R M.2083 [9] sets out the vision and M.2410 [10] the requirements; 3GPP's own scenarios are in TR 38.913 [6] and service requirements in TS 22.261 [5]. NR is described in TS 38.300 [2] and TS 38.211 [3], EN-DC in TS 37.340 [1] and the 5G system in TS 23.501 [7]. Dahlman et al. [8] is the standard textbook on NR.

References

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15

Lecture 15

Software-Defined Networking (SDN)

شبکه‌سازی نرم‌افزارمحور (SDN)

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
IP routing and switching; Lecture 13 (CUPS); M/M/1 queue (Lecture 12)

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-15-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the separation of control and data planes and the motivation for logically centralised control.
- **LO2** **Describe** the SDN layers and interfaces (northbound, southbound) and the OpenFlow switch model: flow tables, pipeline, groups and meters.
- **LO3** **Trace** reactive and proactive flow installation and **compute** their latency and table requirements.
- **LO4** **Explain** data-plane programmability with P4.
- **LO5** **Analyse** controller placement and controller load.
- **LO6** **Identify** SDN principles in mobile networks (CUPS, PFCP, transport and slicing).

2 Prerequisite knowledge

You should know how IP routers and Ethernet switches forward packets (longest-prefix match, MAC learning) and that routing protocols such as OSPF and BGP compute forwarding tables in a distributed way. From the course, you need CUPS and PFCP (Lecture 13) and the M/M/1 delay formula (Lecture 12).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Control plane	صفحهٔ کنترل	Logic that decides how traffic is forwarded.
Data plane	صفحهٔ داده	Per-packet forwarding according to installed state.
SDN controller	کنترل‌گر SDN	Logically centralised software that programs the data plane.
Southbound API	API جنوبی	Controller–device interface (OpenFlow, P4Runtime, NETCONF).
Northbound API	API شمالی	Controller–application interface (e.g. REST).
Flow table	جدول جریان	Prioritised match–action rules in a switch.
Packet-In / Flow-Mod	پیام‌های OpenFlow	Switch-to-controller packet report / rule installation.
Reactive / proactive	واکنشی / پیش‌کنشی	Rules installed on first packet / in advance.
P4	زبان P4	Language for programming protocol-independent packet processors.
TCAM	حافظهٔ آدرس‌پذیر محتوایی سه‌حالته	Hardware memory for wildcard matching.

3 Introduction and motivation

In a traditional network every router runs its own control software: it exchanges routing messages with its neighbours, computes paths and fills its forwarding table. Control and forwarding are bundled in each box, and each vendor implements the protocols in its own closed software. This design is robust, but it makes the network hard to change: a new forwarding behaviour needs a new protocol, a standard and years of implementation, and network-wide policies (traffic engineering, isolation, access control) must be expressed indirectly through many per-box configurations.

Software-defined networking (SDN) separates the two planes [5], [6]. Forwarding devices become simple, programmable match–action engines; a logically centralised controller with a global view computes their state and installs it over an open interface. Network behaviour becomes software that can be written, tested and deployed like other software. Mobile networks adopted the same idea: CUPS and PFCP (Lecture 13) are SDN applied to packet gateways, and the 5G core with its UPF is built on it. This lecture opens Part III, which covers the software technologies – SDN, NFV, service-based architecture – on which 5G networks rest.

4 Fundamental concepts

4.1 Three principles

1. **Separation of control and data planes**: devices forward; controllers decide.
2. **Logically centralised control**: one (possibly distributed and replicated) controller has a global view of topology and state.
3. **Programmability**: open interfaces let software applications define network behaviour.

4.2 Layers and interfaces

RFC 7426 describes SDN in terms of planes – forwarding, operational, control, management and application – and the abstraction layers between them [4]. In the common simplified view (Figure 1) applications use a *northbound* API (often REST) of the controller, and the controller programs devices through a *southbound* API: OpenFlow [7] for flow tables, P4Runtime for P4-programmed devices, or NETCONF/YANG for configuration.

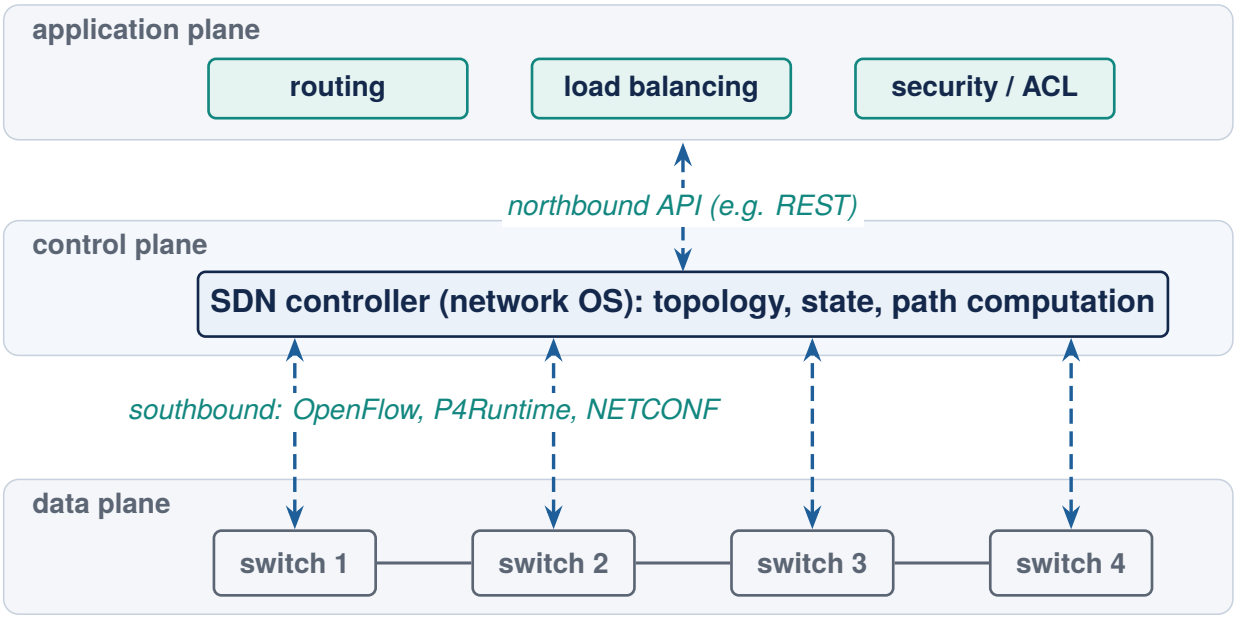


Figure 1. SDN layers: applications, controller and data plane with northbound and southbound APIs.

4.3 The OpenFlow switch

Definition 1 • Flow entry

A flow entry consists of match fields (ingress port and packet header fields from Ethernet to transport layer), a priority, counters, instructions (apply or write actions, go to another table), idle and hard timeouts, and an opaque cookie [7]. A packet is matched against the entries of a table; the highest-priority match wins.

A switch contains a pipeline of flow tables (Figure 2), a *group table* (for multicast, load balancing with *select* groups and fast failover) and a *meter table* (rate limiting). If no entry matches (*table miss*), the table-miss entry decides: drop, continue, or send the packet to the controller. The switch talks to the controller over a TCP connection (port 6653, optionally TLS).

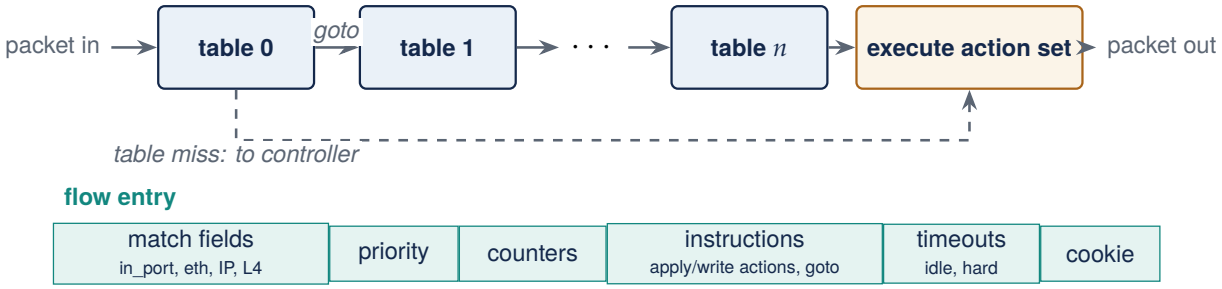


Figure 2. OpenFlow pipeline with multiple flow tables, and the fields of a flow entry.

5 Detailed technical explanation

5.1 OpenFlow messages

- **Controller-to-switch**: Features (capabilities), Flow-Mod (add, modify, delete entries), Group-Mod, Meter-Mod, Packet-Out (send a packet from a port), Multipart/Stats (read counters), Barrier (ensure earlier messages are processed).
- **Asynchronous**: Packet-In (a packet sent to the controller, by default with its first 128 bytes and a buffer identifier), Flow-Removed (an entry expired), Port-Status (a port changed).
- **Symmetric**: Hello, Echo (liveness), Error.

5.2 Reactive and proactive control

In *reactive* mode the first packet of a new flow misses, the switch sends a Packet-In, and the controller computes a path and installs entries along it before releasing the packet with Packet-Out (Figure 3). Fine-grained control is possible, but every new flow pays a controller round trip, and the controller load grows with the flow arrival rate. In *proactive* mode the controller installs (usually wildcard) rules in advance, from its knowledge of topology and policy; no packet waits, the controller is off the critical path, but rules must be aggregated to fit the tables. Practical deployments mix both: proactive rules for the bulk of traffic, reactive handling for exceptions.

5.3 Data-plane programmability: P4

OpenFlow fixes the header fields a switch can match; adding a new protocol requires a new version of the specification and of the hardware. P4 [3] goes one step further: the program defines the headers, the parser state machine, the match–action tables and the control flow, and a compiler maps it to a programmable target (switch ASIC, smart NIC, software switch) organised as a parser, ingress and egress match–action pipelines and a deparser (Figure 4). The controller then fills the tables at run time through P4Runtime. P4 has been used, for example, to implement GTP-U encapsulation and UPF functions directly in switch hardware.

5.4 Distributed controllers and consistency

A single controller is a single point of failure and a bottleneck. Production controllers run as clusters that replicate state with consensus protocols and partition switches among instances; a switch connects to several controllers with master/slave (later primary/secondary) roles. Updat-

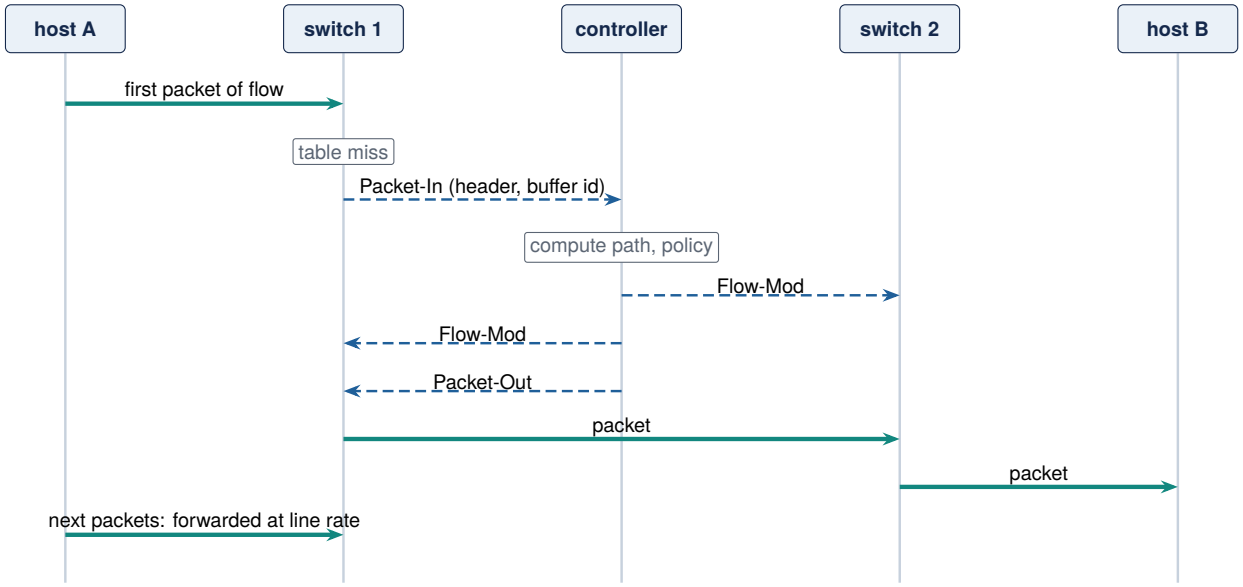


Figure 3. Reactive flow set-up: table miss, Packet-In, Flow-Mod along the path, Packet-Out.

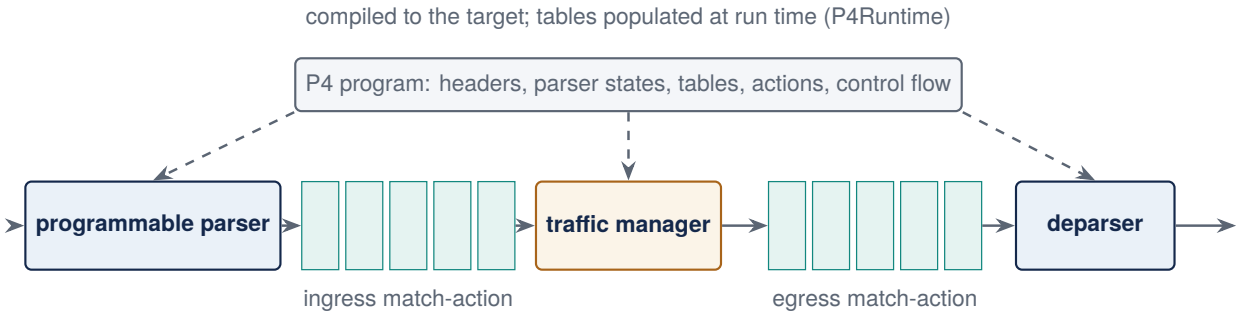


Figure 4. Protocol-independent switch architecture programmed with P4.

ing many switches is not atomic: during an update some packets may see old rules on one switch and new ones on the next, causing loops or policy violations. *Consistent update* techniques tag packets with a configuration version at the ingress and install the new version on all switches before flipping the tags, so that each packet sees either the old or the new configuration, never a mix.

5.5 SDN in mobile networks

- **Gateways**: CUPS (Lecture 13) separates SGW/PGW control from user-plane functions; PFCP plays the role of a southbound protocol with rules (PDR, FAR, QER, URR) that resemble OpenFlow entries.
- **5G core**: the SMF controls UPFs over N4 with PFCP (Lecture 23).
- **Transport**: SDN controllers configure backhaul and fronthaul paths, segment routing and transport slices.
- **RAN**: the O-RAN architecture opens RAN control to software applications through a RAN intelligent controller – the SDN idea applied to radio resource management.

6 Mathematical models

6.1 Reactive flow-set-up latency

Model. The first packet of a flow waits for the switch–controller round trip τ and the controller's processing. With S switches each generating λ_s new flows per second and a controller modelled as an M/M/1 server of rate μ,

$$T_{\text{setup}} \approx \tau + \frac{1}{\mu - S\lambda_s}, \qquad S\lambda_s < \mu. \tag{1}$$

Example. $\tau = 2\,\text{ms}$, $S = 100$, $\lambda_s = 1000\,\text{s}^{-1}$, $\mu = 1.5 \times 10^5\,\text{s}^{-1}$: $T_{\text{setup}} = 2 + 0.02 = 2.02\,\text{ms}$. At $S\lambda_s = 1.49 \times 10^5$ the queueing term rises to 1 ms (Figure 5).

Interpretation. For moderate load the propagation delay to the controller dominates, which argues for controllers close to the switches; near saturation the queueing term explodes and Packet-In storms (e.g. during a scan or a failure) can overload the controller – one reason to prefer proactive rules.

6.2 Flow-table occupancy

Model. With exact-match (per-flow) rules, new flows arrive at rate λ and each entry stays for the flow duration plus the idle timeout, $\bar{T}$ on average. By Little's law the mean number of entries is

$$\bar{N} = \lambda\,\bar{T}. \tag{2}$$

Example. $\lambda = 2000\,\text{s}^{-1}$, mean flow duration 10 s, idle timeout 10 s: $\bar{N} = 40\,000$ entries – far more than the few thousand TCAM entries of many switches. Per-flow reactive control does not scale to such switches; wildcard rules, aggregation, or larger exact-match tables (hash-based) are needed.

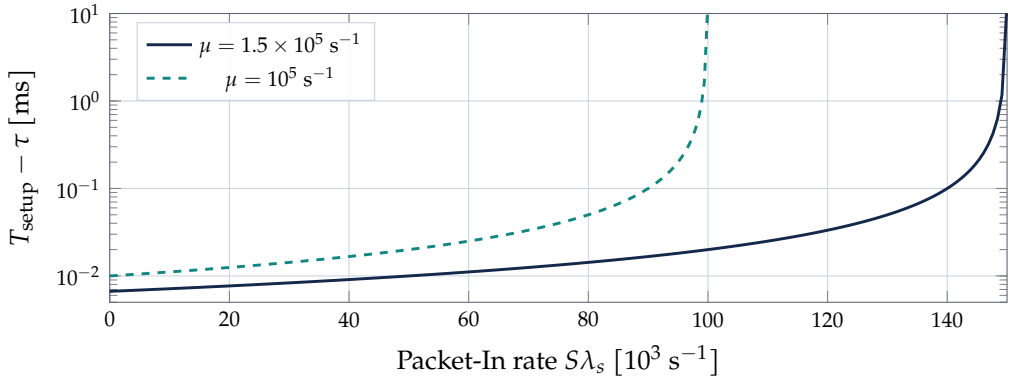


Figure 5. Controller queueing delay versus Packet-In rate (Eq. (1)).

6.3 Controller placement

Model. For switches $v \in V$ and controller locations C, with $d(v, c)$ the latency between them, the average and worst-case switch–controller latencies are

$$\bar{L}(C) = \frac{1}{|V|} \sum_{v \in V} \min_{c \in C} d(v, c), \qquad L_{\max}(C) = \max_{v \in V} \min_{c \in C} d(v, c). \tag{3}$$

Minimising $\bar{L}$ for $|C| = k$ is the k-median problem; minimising $L_{\max}$ is the k-centre problem.

Example. Five switches on a line, 1 ms apart: one controller at the centre gives $\bar{L} = 1.2$ ms ($L_{\max} = 2$ ms); at an end, $\bar{L} = 2$ ms. Two controllers at the second and fourth switches give $\bar{L} = 0.6$ ms and $L_{\max} = 1$ ms.

7 Network architecture and diagrams

In a data centre, the SDN controller programs the leaf–spine fabric and the virtual switches in the servers; in a WAN, it computes traffic-engineered paths between sites; in a mobile operator, SDN controllers manage transport networks and, through the core's control functions, the user-plane nodes. The controller is itself a distributed application deployed in redundant instances, often on the virtualised infrastructure of Lecture 16.

8 Step-by-step procedures

Procedure 1 • Reactive flow installation

1. The first packet of a new flow arrives at switch 1 and misses all entries; the table-miss entry sends it to the controller as Packet-In (header and buffer identifier).
2. The controller identifies source and destination, checks policy and computes a path with its topology view.
3. It sends Flow-Mod messages to all switches on the path (destination side first, to avoid further misses), optionally followed by a Barrier.

4. It sends Packet-Out to switch 1 to release the buffered packet along the new path.
5. Subsequent packets match the new entries and are forwarded at line rate; entries expire after the idle timeout and Flow-Removed reports their counters.

Procedure 2 • Topology discovery

1. On connection, each switch reports its ports (Features reply).
2. The controller sends, via Packet-Out, a link-discovery frame (LLDP) on every port of every switch.
3. A neighbouring switch receives the frame and, by a pre-installed rule, returns it as Packet-In with its ingress port.
4. From (sending switch, port) and (receiving switch, port) the controller learns each link; it repeats periodically and updates the topology on Port-Status events.

9 Worked examples

Worked example 1 • Set-up delay budget

Problem. A campus network has 100 switches; each sees 1000 new flows per second. The controller handles 1.5×10^5 Packet-In/s and is 1 ms away (one way). What delay does the first packet of a flow see, and what fraction of short web flows (10 packets, 20 ms) is it?
Solution. $\tau = 2$ ms, queueing 0.02 ms (Eq. (1)): about 2 ms, i.e. 10 % of the flow's duration. The controller runs at 67 % load; a 50 % increase in flow rate would saturate it.

Worked example 2 • Table size

Problem. A top-of-rack switch with 4000 TCAM entries sees 2000 new flows/s, each lasting 10 s on average, idle timeout 10 s. Can reactive per-flow rules be used? What if the idle timeout is 1 s?
Solution. $\bar{N} = 2000 \times 20 = 40\,000 > 4000$: no. With 1 s idle timeout, $\bar{N} = 22\,000$: still no. Rules must be aggregated, e.g. per destination prefix (proactive), which needs entries per destination rather than per flow.

Worked example 3 • Placement

Problem. Verify the values of the line example in Section 6.3.
Solution. Centre (switch 3): distances 2, 1, 0, 1, 2; mean 1.2 ms, max 2 ms. End (switch 1): 0, 1, 2, 3, 4; mean 2 ms, max 4 ms. Controllers at 2 and 4: 1, 0, 1, 0, 1; mean 0.6 ms, max 1 ms.

10 Technology comparison

Not a replacement for routing. SDN does not abolish distributed protocols: many deployments keep BGP or IS-IS for basic reachability and fast local failover, and use the controller for policy

Table 1. Traditional distributed control versus SDN.

	Distributed control (OSPF, BGP)	SDN
Control location	In every device	Logically centralised controller
View	Local, built by protocol exchange	Global topology and state
New behaviour	New protocol, standard, firmware	New controller application
Failure handling	Local, fast, autonomous	Needs controller or pre-installed backups
Scalability limit	Protocol convergence, table size	Controller capacity, control-channel latency
Policy expression	Indirect (metrics, filters per box)	Direct, network-wide
Vendor coupling	Hardware and software bundled	Open interfaces, merchant silicon

and optimisation on top (hybrid SDN).

11 Key takeaways

Key takeaways

- SDN separates control from forwarding, centralises control logically and makes the network programmable.
- OpenFlow switches: prioritised flow tables in a pipeline, groups, meters; Packet-In, Flow-Mod, Packet-Out.
- Reactive control costs $T_{\text{setup}} \approx \tau + 1/(\mu - S\lambda_s)$ per new flow; proactive control needs aggregated rules.
- Table occupancy follows Little's law, $\bar{N} = \lambda \bar{T}$; per-flow rules rarely fit TCAMs.
- P4 makes the parser and pipeline programmable; controller placement is a k-median/k-centre problem.
- CUPS/PFCP and the 5G UPF apply SDN to the mobile core.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which plane?

Classify as control or data plane: (a) OSPF shortest-path computation; (b) TTL decrement; (c) installing a Flow-Mod; (d) matching a packet against a TCAM; (e) LLDP-based topology discovery.

Exercise 2 • Level 1 – conceptual • Reactive or proactive?

For each case choose reactive or proactive control and justify: (a) a data-centre fabric with destination-based forwarding; (b) a campus network where guests must authenticate before their first flow is allowed; (c) a mobile backhaul network.

Exercise 3 • Level 2 – numerical • Control-channel bandwidth

A controller receives 10^5 Packet-In messages per second, each carrying 128 bytes of packet plus 32 bytes of OpenFlow and TCP/IP overhead (amortised). Compute the control-channel bandwidth. How does it change if the switch sends whole 1500-byte packets instead of buffering them?

Exercise 4 • Level 2 – numerical • Timeouts

Using Eq. (2), find the idle timeout that keeps the mean occupancy at 15 000 entries for 1000 new flows/s with mean duration 8 s. What is the price of a very short idle timeout?

Exercise 5 • Level 3 – analytical • Placement on a ring

Six switches form a ring with 1 ms links. Compute $\bar{L}$ and $L_{\max}$ for one controller, and for two controllers placed optimally. What happens to the worst case if one of the two controllers fails?

Exercise 6 • Level 3 – analytical • Consistent update

A path change moves traffic from route A–B–D to A–C–D. Show how updating switch A before installing rules on C can drop packets, and describe a version-tagging update that avoids both drops and mixing. How many rules does it need temporarily?

Solutions

Solution to Exercise 1

(a) control; (b) data; (c) control (the act of installing; the entry is then used by the data plane); (d) data; (e) control.

Solution to Exercise 2

(a) Proactive: destinations are known, traffic volume is huge, and per-flow set-up delay is unacceptable. (b) Reactive for the first flow of each guest (the controller must see it to apply authentication), then proactive rules for authorised hosts. (c) Proactive: paths are planned and must survive controller outages.

Solution to Exercise 3

$10^5 \times 160\,\text{bytes} = 16\,\text{MB/s} = 128\,\text{Mbit/s}$. With whole packets: $10^5 \times 1532 \times 8 = 1.23\,\text{Gbit/s}$ – ten times more, and the controller must also store or forward them. Buffering in the switch saves control-channel capacity.

Solution to Exercise 4

$15\,000 = 1000\,(8 + T_{\text{idle}}) \Rightarrow T_{\text{idle}} = 7\,\text{s}$. A very short idle timeout evicts entries of flows that pause briefly; their next packet misses again and causes another Packet-In, increasing controller load and latency.

Solution to Exercise

One controller: distances 0, 1, 1, 2, 2, 3; $\bar{L} = 9/6 = 1.5\,\text{ms}$, $L_{\max} = 3\,\text{ms}$. Two opposite controllers: 0, 1, 1, 0, 1, 1; $\bar{L} = 4/6 = 0.67\,\text{ms}$, $L_{\max} = 1\,\text{ms}$. If one fails, the survivor serves all with $L_{\max} = 3\,\text{ms}$ – the placement must be evaluated for failure cases too.

Solution to Exercise

If A forwards to C before C has a rule for the flow, C misses and drops (or sends to the controller). Order matters: install on C first, then switch A ("make before break"). For general updates: install the new rules on C and D matching version tag 2 (while version-1 rules remain), then make A stamp tag 2 on new packets; once old packets have drained, delete version-1 rules. Temporarily the affected switches hold both versions – up to twice the rules.

13 Further reading

McKeown et al. [6] introduced OpenFlow, whose current switch specification is published by the ONF [7]; Kreutz et al. [5] survey SDN comprehensively; RFC 7426 [4] defines SDN layers and terminology; Bosshart et al. [3] present P4. For the mobile context see TS 23.214 [1] and TS 29.244 [2].

References

[1] 3GPP, *Architecture enhancements for control and user plane separation of EPC nodes*, TS 23.214, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23214.htm.

[2] 3GPP, *Interface between the control plane and the user plane nodes*, TS 29.244, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29244.htm.

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[4] E. Haleplidis, K. Pentikousis, S. Denazis, J. Hadi Salim, D. Meyer, and O. Koufopavlou, "Software-defined networking (SDN): Layers and architecture terminology", IRTF, RFC 7426, 2015. [Online]. Available: https://www.rfc-editor.org/rfc/rfc7426.

[5] D. Kreutz, F. M. V. Ramos, P. E. Veríssimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey", *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14–76, 2015.

[6] N. McKeown, T. Anderson, H. Balakrishnan, *et al.*, "OpenFlow: Enabling innovation in campus networks", *ACM SIGCOMM Computer Communication Review*, vol. 38, no. 2, pp. 69–74, 2008.

[7] Open Networking Foundation, *OpenFlow switch specification, version 1.5.1*, Open Networking Foundation, 2015.

16

Lecture 16

Network Functions Virtualisation (NFV)

مجازی‌سازی کارکردهای شبکه (NFV)

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 11 and 15; operating systems (processes, virtual memory); basic probability

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 4
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 7
8 Step-by-step procedures . . . 8
9 Worked examples . . . 8
10 Technology comparison . . . 9
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-16-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** why operators moved network functions from dedicated appliances to software on shared infrastructure.
- **LO2** **Describe** the ETSI NFV architectural framework: VNFs, NFVI and the MANO functions with their reference points.
- **LO3** **Compare** virtual machines and containers, and **explain** cloud-native network functions.
- **LO4** **Trace** the VNF lifecycle, including scaling and healing.
- **LO5** **Compute** availability of virtualised stacks, scaling needs and per-packet processing budgets.
- **LO6** **Relate** NFV to SDN and to the virtualised EPC and the cloud-native 5G core.

2 Prerequisite knowledge

You should know what an operating system kernel, a process and virtual memory are, and how series and parallel availability are combined (Lecture 11). SDN (Lecture 15) is the natural companion of NFV.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
NFV	مجازی‌سازی کارکردهای شبکه	Running network functions as software on general-purpose servers.
VNF / VNFC	کارکرد شبکهٔ مجازی / مؤلفه	A virtualised network function / one of its components.
NFVI	زیرساخت NFV	Compute, storage, network hardware and the virtualisation layer.
MANO	مدیریت و هماهنگ‌سازی	NFV orchestrator, VNF manager and virtualised infrastructure manager.
VNFD / NSD	توصیفگر VNF / خدمت شبکه	Templates describing a VNF / a network service.
Hypervisor	ابرناظر	Software that runs virtual machines on a host.
Container	کانتینر	Isolated process group sharing the host kernel.
CNF	کارکرد شبکهٔ ابرزاد	Cloud-native network function built from containers/microservices.
SR-IOV / DPDK	فناوری‌های شتاب I/O	Direct NIC access for VMs / user-space packet processing.
Service function chain	زنجیرهٔ کارکرد خدمت	Ordered sequence of functions a flow traverses.

3 Introduction and motivation

Until the 2010s every network function – MME, HSS, gateway, firewall, load balancer – was a dedicated box: specialised hardware, a proprietary operating system and the function software, sold together. Adding capacity meant buying and installing more boxes; introducing a service meant integrating yet another appliance; each type had its own spares, tools and lifecycle. In 2012 a group of operators proposed, in a white paper that launched the ETSI NFV Industry Specification Group, to run these functions as software on standard servers in data centres, managed by common orchestration [1].

The promised benefits are those of cloud computing: shared hardware and lower cost, capacity that follows demand (elastic scaling), faster introduction of services, and automation. The costs are equally real: packet processing on general-purpose CPUs is harder than on ASICs, and availability now depends on a stack of hardware, virtualisation and software layers operated by different teams. The virtualised EPC was the first large mobile application; the 5G core was designed from the start as cloud-native software (Lectures 18–19).

4 Fundamental concepts

4.1 The ETSI NFV framework

The framework (Figure 1) [1] has three domains:

- **VNFs**, each with an element manager (EM) for its application-level configuration.
- **NFVI**: physical compute, storage and network resources, abstracted by a virtualisation layer into virtual resources.
- **MANO** [2]: the *NFV orchestrator* (NFVO) manages network services and global resources; the *VNF manager* (VNFM) manages the lifecycle of VNF instances; the *virtualised infrastructure manager* (VIM, e.g. an OpenStack or Kubernetes control plane) manages the NFVI resources.

Named reference points (Os-Ma, Or-Vnfm, Vi-Vnfm, Or-Vi, Ve-Vnfm, Nf-Vi, Vn-Nf, Vl-Ha) define who talks to whom, so that components from different vendors can be combined.

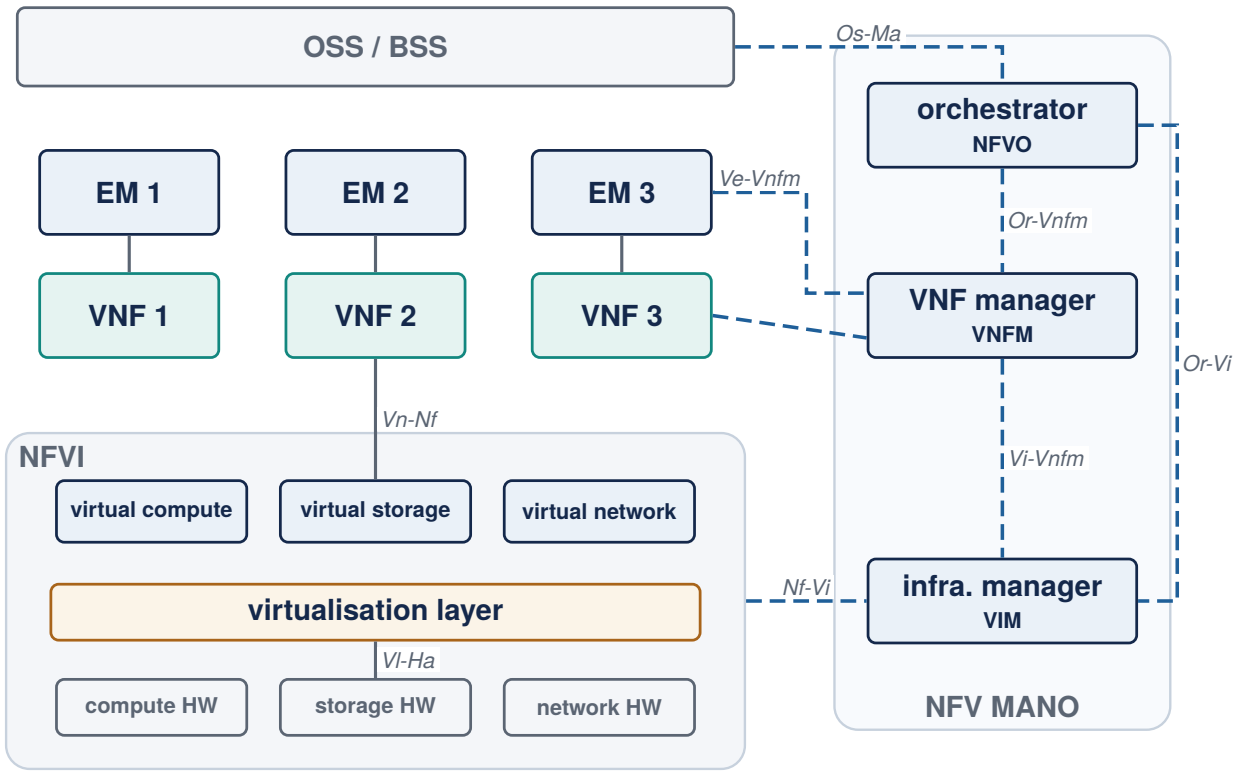


Figure 1. ETSI NFV architectural framework with its main reference points [1].

4.2 Descriptors and network services

A VNF is delivered as a package containing software images and a *VNF descriptor* (VNFD): its components (VNFCs), their resource needs, internal virtual links, connection points, scaling aspects and lifecycle scripts. A *network service* combines VNFs and the virtual links between them, described by a network service descriptor (NSD) and, where traffic must traverse functions in order, a forwarding graph. Descriptors make deployment declarative: the orchestrator turns a template into running instances.

4.3 Virtual machines and containers

A *virtual machine* runs a complete guest operating system on a hypervisor; it gives strong isolation and supports any OS, at the cost of memory and boot time. A *container* isolates a group of processes with kernel namespaces and control groups while sharing the host kernel; it starts in seconds and has little overhead (Figure 2). *Cloud-native network functions* (CNFs) are built as sets of containerised microservices, keep session state in separate data stores so that instances are disposable, and are deployed, scaled and healed by a container orchestrator such as Kubernetes.

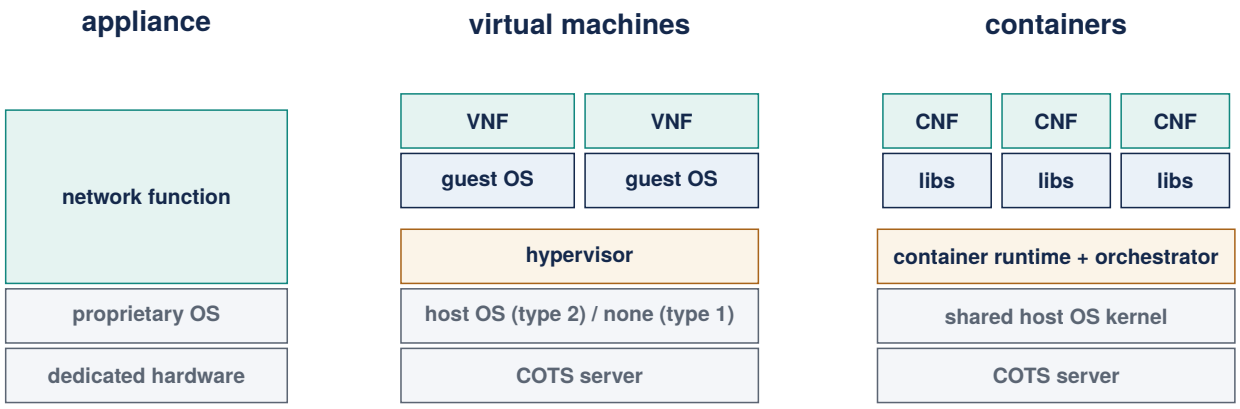


Figure 2. From appliances to virtual machines and containers.

5 Detailed technical explanation

5.1 Lifecycle management

The VNFM drives a VNF through the lifecycle operations of the MANO specifications [2]: *onboarding* (the package is validated and its images stored), *instantiation*, *scaling* out/in (adding or removing VNFC instances) or up/down (changing their resources), *healing* (replacing failed instances), *update/upgrade*, and *termination*. Scaling and healing can be triggered by the VNF or EM, by the VNFM from performance and fault data, or by the NFVO for a whole network service (Figure 4).

5.2 High-performance data planes

Gateways and UPFs must forward millions of packets per second. Plain virtual switching through the kernel and hypervisor cannot keep up, so NFVI nodes for user-plane functions use:

- **DPDK**: polling-mode drivers and packet processing in user space, avoiding interrupts and kernel copies;
- **SR-IOV**: the NIC presents virtual functions that are passed directly to VMs or containers;
- **CPU pinning and NUMA awareness**: dedicated cores on the socket attached to the NIC, huge pages for packet buffers;
- **Hardware offload**: smart NICs or programmable switches (Lecture 15) for encapsulation, classification or encryption.

These techniques trade some of NFV's flexibility (portability, overcommitment) for performance; the per-packet budget of Section 6.3 explains why they are needed.

5.3 Resilience in a virtualised world

In an appliance, redundancy was built into the box (redundant boards, synchronised state). In NFV, resilience is a property of the software design: instances are spread across hosts and sites (anti-affinity rules), state is replicated or externalised, and failed instances are healed by the orchestrator. Common-mode failures move up the stack: a faulty hypervisor update or orchestrator bug can affect every function at once.

5.4 NFV in mobile networks

The virtualised EPC (vEPC) packaged MME, S/P-GW, HSS and PCRF as VNFs, initially as "lift and shift" of appliance software into VMs. The 5G core takes the next step: network functions are defined as services (Lecture 18), implemented as CNFs and deployed on Kubernetes-based platforms, with the user plane at edge sites. In the RAN, virtualised and Cloud RAN run the baseband (CU and DU) on servers with acceleration hardware.

6 Mathematical models

6.1 Availability of a virtualised stack

Model. A VNF instance needs its host hardware, the virtualisation layer and its own software, with availabilities a_{hw}, a_{v} and a_{sw} (independent). In series, and with k independent redundant instances of which one suffices,

$$A_1 = a_{\mathrm{hw}}\, a_{\mathrm{v}}\, a_{\mathrm{sw}}, \qquad A_k = 1 - (1 - A_1)^k. \tag{1}$$

Example. $a_{\mathrm{hw}} = 0.999$, $a_{\mathrm{v}} = 0.9995$, $a_{\mathrm{sw}} = 0.999$: $A_1 = 0.9975$, i.e. 21.9 h of downtime per year – worse than each component. Two instances on independent hosts: $A_2 = 1 - (0.0025)^2 = 0.999994$, about 197 s per year, provided failover is instantaneous.

Interpretation. Redundancy at the instance level more than compensates for the less reliable layers, but only if failures are independent (different hosts, racks, sites) and state survives the failover.

6.2 Elastic scaling

Model. An instance serves μ requests/s; the operator keeps utilisation at most ρ_{t}. For offered load λ the number of instances is

$$n = \left\lceil \frac{\lambda}{\rho_{\mathrm{t}}\, \mu} \right\rceil. \tag{2}$$

If load can grow at rate g (requests/s per second) and a new instance takes T_{i} to become ready, the spare capacity must cover the growth during instantiation:

$$n\mu - \lambda \geq g\, T_{\mathrm{i}} \quad \Longleftrightarrow \quad \rho \leq 1 - \frac{g\, T_{\mathrm{i}}}{n\mu}. \tag{3}$$

Example. $\lambda = 40\,000\,\mathrm{s}^{-1}$, $\mu = 5000\,\mathrm{s}^{-1}$, $\rho_\mathrm{t} = 0.7$: $n = \lceil 11.4 \rceil = 12$. With $g = 100\,\mathrm{s}^{-2}$ and $T_\mathrm{i} = 60\,\mathrm{s}$ the headroom must be $6000\,\mathrm{s}^{-1}$, so $\rho \le 1 - 6000/60\,000 = 0.9$: the 70 % target is safe. Containers (T_i of seconds) allow much higher utilisation than VMs (minutes).

6.3 Per-packet processing budget

Model. At line rate R with packets of L bytes (plus 20 bytes of Ethernet preamble and inter-frame gap), one core at clock f has

$$c = \frac{f}{R/\big(8(L+20)\big)} \tag{4}$$

cycles per packet.

Example. $f = 2.5\,\mathrm{GHz}$, $R = 10\,\mathrm{Gbit/s}$, $L = 64$: 14.88 Mpps and $c = 168$ cycles – less than the cost of a single cache miss to main memory in the worst case. At 100 Gbit/s only 17 cycles remain (Figure 3).

Interpretation. Line-rate small-packet forwarding needs many cores in parallel, batching, cache-friendly data structures and offload; real traffic with larger average packets relaxes the budget considerably.

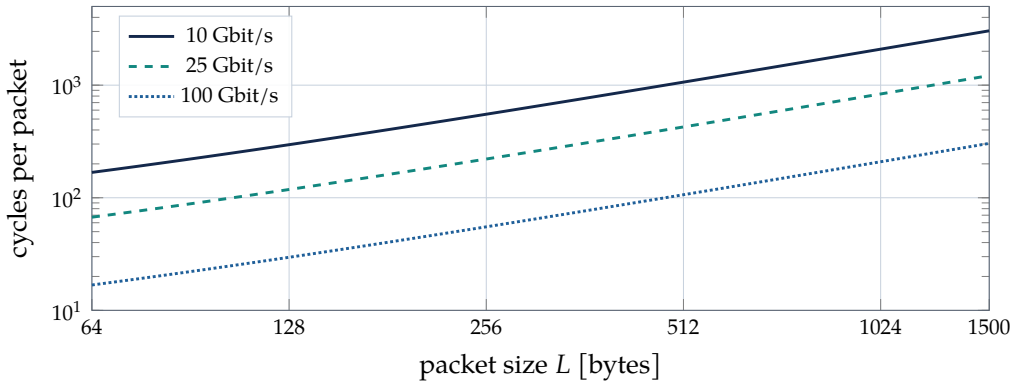


Figure 3. Cycles available per packet on one 2.5 GHz core at line rate (Eq. (4)).

6.4 Placement as bin packing

Model. VNFCs with CPU demands d_j must be placed on servers of capacity C. The number of servers is at least

$$m \ge \left\lceil \frac{\sum_j d_j}{C} \right\rceil, \tag{5}$$

and finding the minimum is the NP-hard bin-packing problem; heuristics such as first-fit decreasing come close. Anti-affinity (replicas on different servers) and latency constraints (user plane at the edge) add constraints that can increase m.

7 Network architecture and diagrams

An operator's NFV deployment typically consists of a few central data centres (control-plane functions, subscriber databases, OSS) and many edge sites (user-plane functions, RAN baseband), all

managed by a common orchestration layer. SDN controllers (Lecture 15) configure the data-centre fabrics and the transport between sites, so that the virtual links of network services are realised end to end. Figure 1 shows the functional view; physically, MANO components themselves run as virtualised applications on the same infrastructure.

8 Step-by-step procedures

Procedure 1 • Instantiating a network service

1. The operator (OSS) requests instantiation of a network service from the NFVO, referencing an onboarded NSD.
2. The NFVO checks resources, selects VIMs/sites and asks the VNFM of each VNF to instantiate it.
3. The VNFM requests a resource grant from the NFVO, then asks the VIM to create VMs or containers, networks and storage according to the VNFD.
4. The VIM allocates the resources on the NFVI and boots the images; the VNFM configures the instances and informs the EM.
5. The NFVO creates the virtual links between the VNFs (possibly through an SDN controller) and reports the service as instantiated.

Procedure 2 • Automatic scale-out

1. The VNF or EM reports load above a threshold defined in the VNFD (or the VNFM detects it from metrics).
2. The VNFM decides the scaling step and obtains a grant from the NFVO.
3. The VIM allocates resources and starts a new VNFC instance.
4. The VNFM configures it and adds it to the VNF's internal load distribution; traffic shifts to it.

9 Worked examples

Worked example 1 • Redundancy design

Problem. With the component availabilities of Section 6.1, compare one instance, two instances on the same host, and two instances on different hosts.
Solution. One: $A_1 = 0.9975$ (21.9 h/year). Same host: hardware and hypervisor are shared, so $A = a_{\text{hw}} a_{\text{v}} \left[1 - (1 - a_{\text{sw}})^2\right] = 0.9985$ (13.1 h/year): only software failures are covered. Different hosts: $A_2 = 0.999994$ (197 s/year). Anti-affinity is essential.

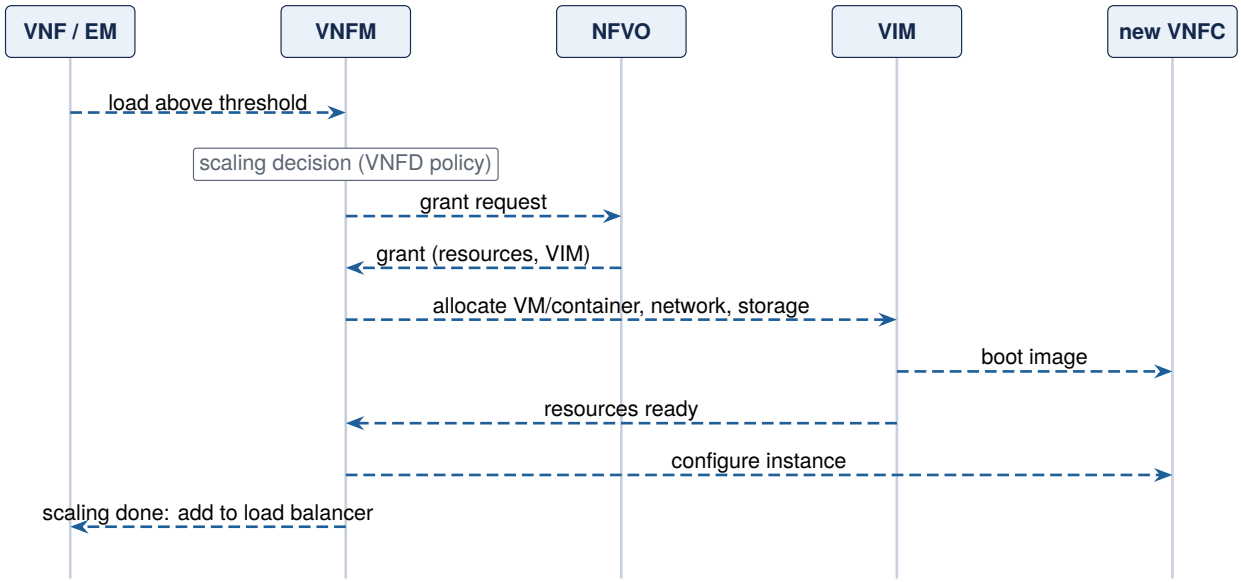


Figure 4. Simplified VNF scale-out (Procedure 2).

Worked example 2 • Scaling a signalling function

Problem. An AMF-like function must handle a busy hour of 40 000 procedures/s; each instance handles 5000/s. How many instances are needed at 70 % target utilisation, and what is the utilisation after one instance fails?
Solution. $n = 12$ (Eq. (2)); utilisation $40\,000/60\,000 = 67\,\%$. After a failure: $40\,000/55\,000 = 73\,\%$ – still below saturation, and the orchestrator heals the instance in the meantime.

Worked example 3 • Packing VNFCs

Problem. VNFCs need 6, 5, 5, 4, 4, 3, 3 and 2 cores; servers have 16 cores. Find the lower bound and a first-fit decreasing placement.
Solution. $\sum d_j = 32$, so $m \geq 2$. First-fit decreasing: server 1 gets 6, 5, 5 (16 cores); server 2 gets 4, 4, 3, 3, 2 (16 cores). Two servers, which is optimal – but with no spare capacity for healing.

10 Technology comparison

NFV and SDN. The two are complementary and independent: NFV changes *where* functions run (software on shared servers), SDN changes *how* forwarding is controlled (programmable devices, central control). Together they allow a network service – functions and the links between them – to be created by software.

11 Key takeaways

Table 1. Physical appliances, VM-based VNFs and cloud-native CNFs.

	Appliance	VNF (VMs)	CNF (containers)
Hardware	Dedicated, vendor-specific	Shared COTS servers	Shared COTS servers
Isolation	Physical	Hypervisor	Kernel namespaces, cgroups
Start-up time	Hours to days (install)	Minutes	Seconds
Scaling	Buy and install boxes	Add VMs (scale out/up)	Add pods automatically
State handling	Inside the box	Often inside the VM	Externalised data stores
Lifecycle	Vendor-specific tools	MANO (NFVO, VNFM, VIM)	Kubernetes + operators, CI/CD
Data-plane speed	ASIC line rate	Needs DPDK, SR-IOV	Needs DPDK, SR-IOV, offload

Key takeaways

- NFV runs network functions as software on shared infrastructure, managed by MANO (NFVO, VNFM, VIM).
- Descriptors (VNFD, NSD) make deployment declarative; lifecycle operations include scale and heal.
- Containers and cloud-native design (stateless instances, external state) underpin the 5G core.
- Series availability drops with each layer; independent redundant instances restore it: $A_k = 1 - (1 - A_1)^k$.
- Scaling: $n = \lceil \lambda / (\rho_t \mu) \rceil$ with headroom gT_i; packet budget $c = 8f(L + 20)/R$ cycles.

12 Exercises

Exercise 1 • Level 1 – conceptual • Who does what?

Which MANO component (NFVO, VNFM, VIM) is responsible for: (a) creating a virtual machine; (b) deciding to add a VNFC instance; (c) connecting several VNFs into a network service; (d) keeping an inventory of NFVI resources?

Exercise 2 • Level 1 – conceptual • VM or container?

Give one argument for running a virtualised HSS in VMs and one for running it as containers.

Exercise 3 • Level 2 – numerical • Same-host redundancy

Using Eq. (1), compute the annual downtime of three instances placed on three different hosts that share one power and network domain with availability 0.9999. Which term dominates?

Exercise 4 • Level 2 – numerical • Packet budget

A UPF core runs at 3 GHz and needs 400 cycles per packet. What throughput can one core sustain with 64-byte and with 800-byte packets? How many cores for 100 Gbit/s at 800 bytes?

Exercise 5 • Level 3 – analytical • Scaling lag

Load ramps at $g = 200\,\mathrm{s}^{-2}$. Instances serve $\mu = 4000\,\mathrm{s}^{-1}$. Compare the maximum safe utilisation for $n = 10$ instances when a new instance takes 120 s (VM) or 10 s (container) to become ready. Discuss what else limits the speed of scaling.

Exercise 6 • Level 3 – analytical • Placement with anti-affinity

The VNFCs of Example 3 consist of two replicas of a 6-core component (A), two of a 5-core component (B) and the rest (4, 4, 3, 3, 2) without constraints; replicas of the same component must be on different servers, and each server must keep 2 cores free for healing. Find a placement with the fewest 16-core servers.

Solutions

Solution to Exercise 1

(a) VIM; (b) VNFM (possibly at the request of the EM/VNF or NFVO); (c) NFVO; (d) VIM (with a global view kept by the NFVO).

Solution to Exercise 2

VMs: strong isolation for a security-critical database holding subscriber keys, and support for an existing vendor image. Containers: fast scaling and healing, smaller footprint and integration with cloud-native tooling, if the HSS software is designed with external, replicated state.

Solution to Exercise 3

Three independent instances: $1 - (0.0025)^3 \approx 1 - 1.6 \times 10^{-8}$. Shared domain in series: $A = 0.9999 \times (1 - 1.6 \times 10^{-8}) \approx 0.99990$, i.e. about 53 min/year, almost all due to the shared domain. Independence of failure domains matters more than the number of instances.

Solution to Exercise 4

One core: $3 \times 10^9 / 400 = 7.5$ Mpps. 64-byte packets: $7.5 \times 10^6 \times 84 \times 8 = 5.0$ Gbit/s; 800-byte packets: $7.5 \times 10^6 \times 820 \times 8 = 49$ Gbit/s (on the wire). For 100 Gbit/s at 800 bytes: $100/49 = 2.03$, i.e. 3 cores (more with headroom).

Solution to Exercise 5

Capacity $n\mu = 40\,000\,\mathrm{s}^{-1}$. VM: headroom $200 \times 120 = 24\,000$, so $\rho \leq 0.4$; container: $200 \times 10 = 2000$, so $\rho \leq 0.95$. Other limits: detection delay of the monitoring loop, image download, warm-up (caches, state synchronisation), load-balancer reconfiguration and the availability of free hardware at the site.

Solution to Exercise

Usable capacity is 14 cores per server; total demand 32, so at least 3 servers. Placement: server 1: A (6) + B (5) + 3 = 14; server 2: A (6) + B (5) + 3 = 14; server 3: 4 + 4 + 2 = 10. Anti-affinity holds (each server has one replica of A and B at most) and 2 cores remain free on each: three servers.

13 Further reading

The NFV architectural framework is ETSI GS NFV 002 [1] and management and orchestration ETSI GS NFV-MAN 001 [2]. Kreutz et al. [4] discuss the relationship between SDN and NFV; Rommer et al. [5] describe the cloud-native 5G core; Kleinrock [3] covers the queueing models used for dimensioning.

References

[1] ETSI, "Network functions virtualisation (NFV); architectural framework", European Telecommunications Standards Institute, Group Specification GS NFV 002, 2014.

[2] ETSI, "Network functions virtualisation (NFV); management and orchestration", European Telecommunications Standards Institute, Group Specification GS NFV-MAN 001, 2014.

[3] L. Kleinrock, *Queueing Systems, Volume 1: Theory*. Wiley, 1975.

[4] D. Kreutz, F. M. V. Ramos, P. E. Veríssimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey", *Proceedings of the IEEE*, vol. 103, no. 1, pp. 14–76, 2015.

[5] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.

17

Lecture 17

3GPP Standardisation

استانداردسازی 3GPP

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
5–7 hours (3 h reading, 2–4 h exercises)

DIFFICULTY
Introductory to intermediate

PREREQUISITES
Lectures 1–16 (as examples); basic probability

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 7
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 9
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-17-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** how 3GPP is organised: organisational partners, TSGs and working groups, and its relation to ITU-R and the IETF.
- **LO2** **Explain** the three-stage methodology, specification numbering and the release model.
- **LO3** **Decode** a specification version number and **classify** change requests.
- **LO4** **Trace** how a feature moves from a study item to a specification and to products.
- **LO5** **Locate and read** a specification efficiently: scope, normative text, annexes.
- **LO6** **Reason quantitatively** about interoperability, work backlogs and time to market.

2 Prerequisite knowledge

This lecture is about the process behind everything in this course. The earlier lectures provide the examples: the specifications cited there (TS 23.401, TS 29.272, TS 38.300, …) are the products of the process described here. Exercise 5 uses Little's law (Lecture 15).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
3GPP	پروژهٔ مشارکت نسل سوم	Partnership of standards organisations that specifies mobile systems.
Organisational partner	شریک سازمانی	Regional/national standards body that is a member of 3GPP.
TSG / WG	گروه مشخصات فنی / گروه کاری	Decision-making groups and their technical sub-groups.
TS / TR	مشخصات فنی / گزارش فنی	Normative specification / informative study report.
Stage 1, 2, 3	مراحل ۱، ۲، ۳	Requirements, architecture, detailed protocols.
Release	انتشار	Consistent set of features and specifications frozen together.
Freeze	انجماد	Point after which only corrections are accepted.
Change request (CR)	درخواست تغییر	Formal proposal to modify a specification under change control.
Work item / study item	مورد کاری / مطالعه	Approved task to specify / to study a feature.
SEP / FRAND	پتنت ضروری استاندارد	Standard-essential patent / fair, reasonable and non-discriminatory licensing.

3 Introduction and motivation

A phone bought in one country works on a network built from equipment of several vendors in another country. That this works at all is the result of standards: thousands of documents that specify every message, every timer and every bit, written by competitors who cooperate in the same rooms. For mobile networks the dominant body is the *3rd Generation Partnership Project* (3GPP), created in 1998 to specify the third generation and since responsible for GSM's evolution, LTE, 5G and, now, 6G.

Engineers in the mobile industry spend much of their time with 3GPP specifications: implementing them, testing against them, or contributing to them. Understanding how the documents are organised, how they evolve and how decisions are made is therefore a practical skill, not trivia. It also explains why the systems look the way they do: many design choices in this course are compromises reached in standardisation meetings.

4 Fundamental concepts

4.1 Organisation

3GPP is not a legal entity of its own: it is a partnership of seven *organisational partners* – ARIB and TTC (Japan), ATIS (USA), CCSA (China), ETSI (Europe), TSDSI (India) and TTA (Korea) – which transpose its specifications into their own standards. Companies participate as members of one of these partners; *market representation partners* (industry associations) contribute requirements. The technical work is done in three *Technical Specification Groups* and their working groups (Figure 1) [7]:

- **TSG RAN**: radio access (physical layer, radio protocols, RAN architecture, RF, UE conformance tests);
- **TSG SA**: service and system aspects (requirements, architecture, security, codecs, management) and overall coordination;
- **TSG CT**: core network and terminals (NAS, core protocols, policy, USIM).

Working groups prepare documents; TSG plenaries, meeting quarterly, approve them. Decisions are made by consensus; formal voting is possible but rare.

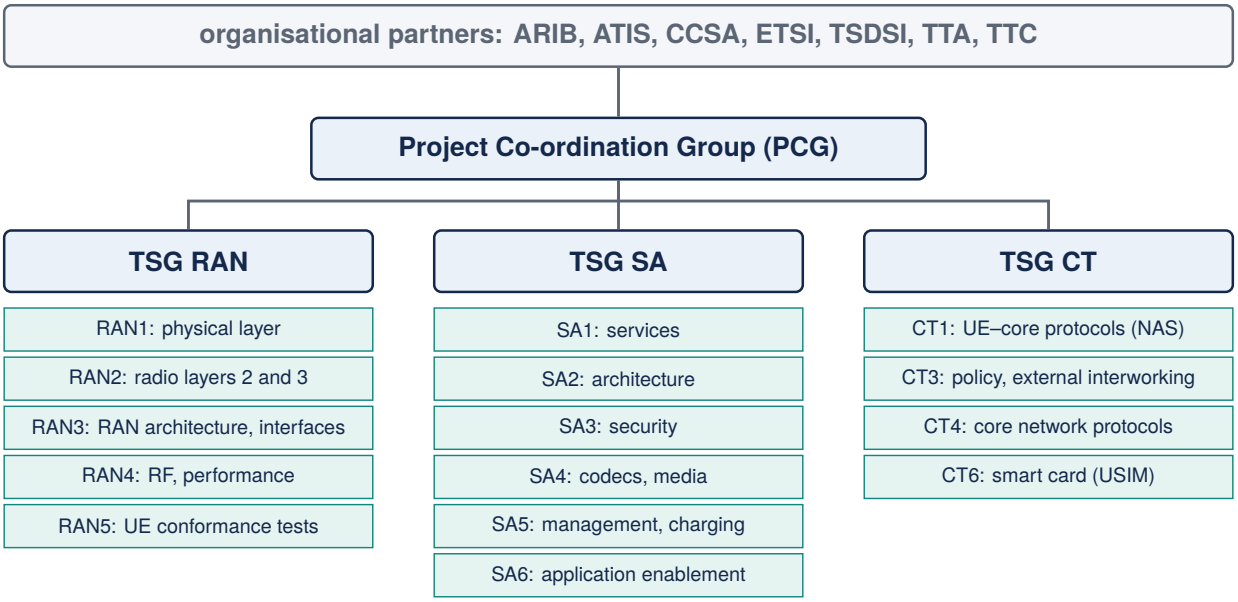


Figure 1. Organisation of 3GPP: organisational partners, project co-ordination, TSGs and their working groups (selection).

4.2 Relation to other bodies

ITU-R defines the requirements of each IMT generation (Lecture 14) [12], [13]; 3GPP's technologies are submitted, through the organisational partners, as candidate IMT radio interfaces and evaluated. The IETF specifies the Internet protocols that 3GPP reuses – SCTP, Diameter, HTTP/2, SIP, IPsec, TLS [10], [14]–[16] – and 3GPP requests extensions when needed. Industry bodies such as the GSMA define operational guidelines (e.g. roaming, VoLTE profiles), and certification organisations test devices against 3GPP conformance specifications.

4.3 Three stages

Each feature is specified at three levels (Figure 2):

- **Stage 1** describes the service from the user's point of view: requirements, without solutions (e.g. TS 22.261 [3]).
- **Stage 2** describes the architecture: functional entities, reference points and information flows (e.g. TS 23.501 [6]).
- **Stage 3** specifies protocols bit by bit: messages, information elements, encodings, state machines (e.g. TS 24.501 [1], TS 38.331 [2]).

Conformance-test specifications then define how devices are tested against stage 3.

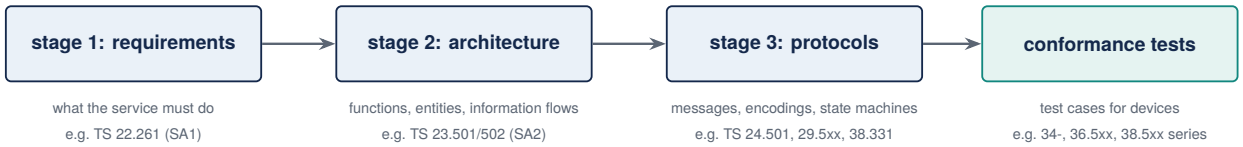


Figure 2. Three-stage methodology with typical specifications.

4.4 Specification numbering

Every specification has a number *ss.nnn* whose series *ss* indicates the subject [7]. A *technical specification* (TS) is normative: products must comply with it. A *technical report* (TR) is informative: it records studies, evaluations and options, often in the 8xx and 9xx ranges (e.g. TR 38.913 [5], TR 38.901 [4]).

Table 1. Selected 3GPP specification series.

Series	Subject	Example
21	Requirements, vocabulary, working methods	TR 21.905 [8]
22	Service aspects (stage 1)	TS 22.261
23	Technical realisation (stage 2)	TS 23.401, 23.501
24	Signalling protocols UE–core (stage 3)	TS 24.301, 24.501
26	Codecs	–
28, 32	Management and charging	TS 28.530
29	Core network protocols (stage 3)	TS 29.274, 29.500
31	SIM/USIM	–
33	Security	TS 33.401, 33.501
35	Security algorithms	TS 35.206
36	LTE (E-UTRA) radio	TS 36.300, 36.331
37	Multiple radio technologies	TS 37.340
38	NR radio	TS 38.300, 38.331

5 Detailed technical explanation

5.1 Releases

3GPP publishes its work in *releases*: consistent sets of features specified together (Figure 3). Each release has freeze dates: after the *stage 3 freeze* no new functions are added, only corrections; after the *ASN.1 freeze* the encoding of radio protocol messages is stable so that implementations become interoperable. Releases overlap: while one is being corrected, the next is under development and the one after is being studied. Vendors and operators implement a subset of features of a release, and devices declare their capabilities, so "supporting Release 17" means little without a feature list.

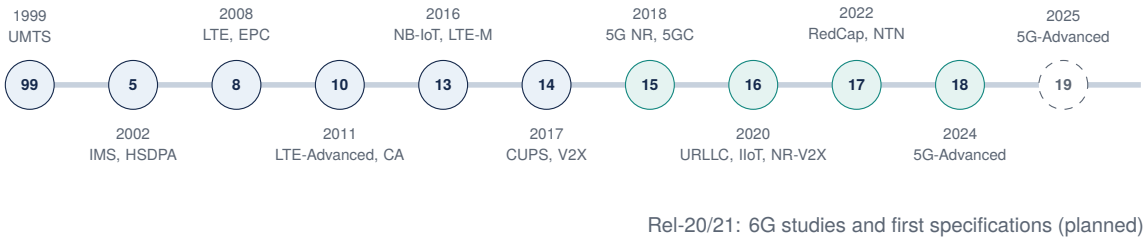


Figure 3. Selected 3GPP releases with headline features. Years indicate approximate completion; teal marks 5G releases.

5.2 Versions

Each specification has a version *x.y.z* [7]. Before approval, $x = 1$ (presented for information) or $x = 2$ (presented for approval). Once approved and under change control, x equals the release (e.g. 17 for Release 17); y increases with every set of approved technical changes; z increases for purely editorial changes. A specification thus exists in parallel versions, one per release, and corrections are often mirrored into several releases.

5.3 Change requests

A specification under change control changes only through *change requests* (CRs), each with a category [7]: **F** (correction of an error), **A** (mirror of a category-F correction in a later release), **B** (addition of a feature), **C** (functional modification of a feature) and **D** (editorial). After freeze only F, A and D are normally accepted. CRs are agreed in the working group and approved by the TSG; the specification is then republished with an incremented version (Figure 4).

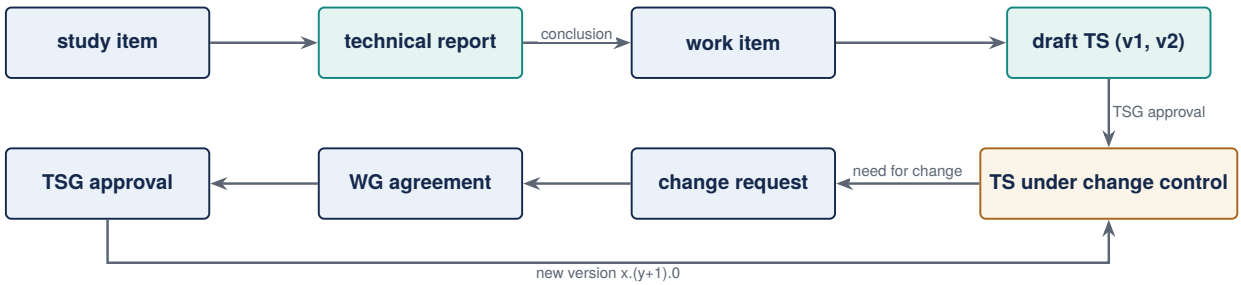


Figure 4. From study item to specification, and the change-request cycle.

5.4 How work is done

Companies submit written contributions (temporary documents, "Tdocs") to working-group meetings, held several times per release cycle in different regions, with electronic meetings in between. Proposals are discussed, merged and revised until consensus emerges; chairs record agreements. Typical contributions are proposals for solutions in a study, text proposals for a TR or TS, and CRs. All documents are public on the 3GPP website, which makes the history of every design decision traceable.

5.5 Reading a specification

A 3GPP specification follows a fixed template: scope, references, definitions and abbreviations, then the technical clauses and annexes. Normative text uses "shall" (mandatory), "should" (recommended) and "may" (permitted); informative annexes give background and examples. Practical advice: read the stage-2 description first, then the stage-3 procedure, and only then the message and information-element definitions; check the release and version; and search across specifications, because a procedure is usually split over several of them (e.g. an attach involves TS 23.401, 24.301, 36.413, 29.274 and 29.272).

5.6 Intellectual property

Companies that contribute technology often hold patents essential to implementing a standard (*standard-essential patents*, SEPs). Under the IPR policies of the organisational partners, they declare such patents and commit to license them on fair, reasonable and non-discriminatory (FRAND) terms. Licensing is negotiated bilaterally, outside 3GPP, and the level of aggregate royalties is a recurring debate (Section 6.4).

6 Mathematical models

The models below are deliberately simple; their numerical examples use illustrative assumptions, not measured 3GPP statistics.

6.1 Interoperability across interfaces

Model. An end-to-end service crosses k multi-vendor interfaces. If each is implemented correctly with probability p, independently, the service works with probability

$$P_{\text{e2e}} = p^k. \tag{1}$$

Example. $p = 0.98$, $k = 20$: $P_{\text{e2e}} = 0.67$. To reach 0.9 with $k = 20$ requires $p \geq 0.9^{1/20} = 0.995$.

Interpretation. The more interfaces a system has, the more precise the specifications and the more thorough the testing must be – the reason for stage-3 detail, conformance tests and interoperability test events. It is also an argument for fewer, simpler interfaces.

6.2 Work in progress

Model. If CRs on a specification group arrive at rate λ (per month) and each spends on average T months from first submission to approval, Little's law gives the number of open CRs:

$$N = \lambda\, T. \tag{2}$$

Example. With $\lambda = 400$ CRs/month and $T = 3$ months, about 1200 CRs are open at any time. Halving T (e.g. with electronic meetings between plenaries) halves the backlog for the same throughput.

6.3 Time to market

Model. A feature passes through study, normative work, implementation and testing; phases can partly overlap. The time from study start to commercial availability is

$$T_{\text{market}} = \sum_{i} T_i - \sum_{j} \Delta_j, \tag{3}$$

with phase durations T_i and overlaps Δ_j.

Example. Study 12 months, stage 2 12, stage 3 12, implementation 18, testing and certification 6, with 6 months of overlap between implementation and late stage 3: $T_{\text{market}} = 54$ months, i.e. about 4.5 years – consistent with the gap of several years between the start of a generation's studies and its first commercial networks.

6.4 Royalty stacking

Model. A device price P carries royalties to N SEP holders. If each independently asks a rate r_i, the total is $R = \sum_i r_i$; under an aggregate cap $R_{\max}$ allocated by each holder's share s_i of essential patents,

$$R = \sum_{i=1}^{N} r_i, \qquad r_i^{\text{cap}} = R_{\max}\, s_i, \quad \sum_i s_i = 1. \tag{4}$$

Example. Twenty holders each asking 1 % stack to 20 % of the device price; with a 5 % cap, a holder with 10 % of the essential patents receives 0.5 %. The economics of SEPs are contested; the model only shows why the question arises.

7 Network architecture and diagrams

The organisation of 3GPP mirrors the architecture of the systems it specifies: RAN working groups own the radio interface and RAN interfaces (Uu, X2/Xn, S1/NG, F1), SA2 owns the system architecture and reference points, CT groups own the core protocols and the UE–core NAS. A new feature that crosses these boundaries – network slicing, for example – needs coordinated work items in several groups, with SA2's stage 2 as the common reference.

8 Step-by-step procedures

Procedure 1 • Introducing a new feature

1. Companies propose a study item (SI) with objectives; the TSG approves it.
2. The working groups study use cases and candidate solutions and document them in a TR, with evaluations and a conclusion.
3. A work item (WI) is approved with a scope, target release and responsible groups.
4. Stage 2 and stage 3 specifications are written or extended (new TS or CRs to existing ones) until the stage 3 and ASN.1 freezes.
5. After freeze, corrections continue through category-F CRs; conformance tests are specified; products implement and certify the feature.

Procedure 2 • Finding the right specification

1. Identify the topic and stage: requirement (22), architecture (23), protocol (24, 29, 36, 38), security (33).
2. Start from the stage-2 overview (e.g. TS 23.501 or TS 38.300) and follow its references.
3. Choose the release that matches the implementation and download the latest version of that release.
4. Read scope and definitions, then the procedure clause, then the message definitions; check the CR history for recent changes.

9 Worked examples

Worked example 1 • Decoding a version

Problem. What do "TS 38.331 V17.5.0" and "TS 23.501 V2.0.0" tell you?
Solution. TS 38.331 is in series 38 (NR radio): the NR RRC protocol. V17.5.0: Release 17, under change control, fifth set of technical changes after its first Release-17 version, no editorial-only update since. TS 23.501 V2.0.0: a draft presented to the TSG for approval (before it became V15.0.0 in Release 15).

Worked example 2 • Interoperability budget

Problem. A VoLTE call involves about 20 interfaces between UE, eNB, MME, gateways, PCRF and IMS nodes from different vendors. What per-interface correctness is needed for 99 % end-to-end success?
Solution. $p \geq 0.99^{1/20} = 0.9995$. Field experience with early VoLTE, which needed extensive interoperability testing and an industry profile to narrow the options, illustrates the point.

Worked example 3 • Classifying CRs

Problem. Classify: (a) a wrong timer value in Release 16, fixed in Release 16; (b) the same fix copied into Release 17; (c) a new optional information element enabling a feature; (d) a

typo.
Solution. (a) F; (b) A; (c) B (only before the release's freeze); (d) D.

10 Technology comparison

Table 2. Standards bodies relevant to mobile networks.

Body	Role	Typical output
ITU-R	Spectrum regulation and IMT requirements	Recommendations and reports (e.g. M.2083, M.2410, M.2160)
3GPP	Specification of mobile systems	TS and TR (series 21–38)
IETF	Internet protocols	RFCs (e.g. SCTP, Diameter, HTTP/2)
ETSI	European standards; NFV, MEC industry groups	ENs, group specifications (e.g. NFV)
ONF, O-RAN Alliance	Open interfaces for SDN and RAN	Specifications (e.g. OpenFlow, O-RAN interfaces)
GSMA	Operator association	Permanent reference documents, profiles

Consensus versus speed. Open, consensus-based standardisation produces specifications that many companies can implement and that last for decades, at the cost of time and complexity: options are added to satisfy many parties. Industry alliances and open-source projects move faster but with less formal commitment; modern networks combine all of them.

11 Key takeaways

Key takeaways

- 3GPP = seven organisational partners; work in TSG RAN, SA and CT and their WGs; consensus decisions.
- Stage 1 (requirements), stage 2 (architecture), stage 3 (protocols); TS normative, TR informative.
- Series numbers identify the topic (23 architecture, 24/29 protocols, 33 security, 36 LTE, 38 NR).
- Releases freeze in stages; versions $x.y.z$ = release, technical changes, editorial changes.
- Change requests (F, A, B, C, D) are the only way to change a specification under change control.
- Interoperability decays as p^k; specifications and testing must be precise.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which series?

In which series would you look for: (a) the NAS protocol of 5G; (b) the 5G security architecture; (c) the LTE MAC protocol; (d) the service requirements of 5G; (e) the GTP-U protocol?

Exercise 2 • Level 1 – conceptual • TS or TR?

Is a channel model used to evaluate candidate technologies normative? Where is it published, and why does it matter for comparing simulation results?

Exercise 3 • Level 2 – numerical • Versions

A vendor's documentation states compliance with "TS 24.501 V16.5.0". The operator's network implements V16.9.0. What kinds of differences can exist between the two, and how would you find them?

Exercise 4 • Level 2 – numerical • Interfaces and correctness

Using Eq. (1), compute P_{e2e} for $p = 0.995$ and $k = 20$ and 50. How many interfaces can a service cross at $p = 0.999$ if $P_{\text{e2e}} \geq 0.95$ is required?

Exercise 5 • Level 3 – analytical • Backlog

A working group handles 250 CRs per month with an average time to approval of 4 months. A new release adds a burst so that arrivals rise to 350 per month while approval capacity stays at 250 per month. Using Eq. (2) and a simple fluid argument, estimate the backlog after 6 months of the burst. What can the group do?

Exercise 6 • Level 3 – analytical • Overlapping phases

With the phase durations of Section 6.3, how much earlier could commercial products appear if implementation started when stage 2 is complete (overlapping stage 3 fully)? What risk does this create, and how do the freeze dates mitigate it?

Solutions

Solution to Exercise 1

(a) 24 (TS 24.501); (b) 33 (TS 33.501); (c) 36 (TS 36.321); (d) 22 (TS 22.261); (e) 29 (TS 29.281).

Solution to Exercise 2

No: channel models are informative; they are published in TRs (e.g. TR 38.901). They matter because simulation results from different companies are comparable only if they use the same channel model and assumptions, which the TR fixes.

Solution to Exercise 6

Both are Release 16; V16.9.0 includes four more sets of technical changes – mostly corrections (category F), possibly clarifications that change behaviour in corner cases. Compare the CR lists between the versions (the change history annex and the CR database) and check whether any affect implemented procedures.

Solution to Exercise 7

$0.995^{20} = 0.905$; $0.995^{50} = 0.778$. With $p = 0.999$: $k \leq \ln 0.95 / \ln 0.999 = 51.3$, i.e. at most 51 interfaces.

Solution to Exercise 8

Initially $N = 250 \times 4 = 1000$ open CRs. During the burst the backlog grows by $350 - 250 = 100$ per month: after 6 months about 1600, and by Little's law the time to approval rises to about $1600/250 = 6.4$ months. Options: more meeting time (additional electronic sessions), prioritisation of essential CRs, postponing optional features to the next release.

Solution to Exercise 9

Implementation would overlap the whole 12 months of stage 3 instead of 6: products about 6 months earlier (48 instead of 54 months). Risk: implementing a moving target – later stage-3 changes force rework and early products may not interoperate. Freeze dates (especially the ASN.1 freeze) mark when the specification is stable enough to implement without such risk.

13 Further reading

TR 21.900 [7] describes 3GPP working methods, version numbering and CR categories; TR 21.905 [8] is the vocabulary. The ITU-R documents on IMT-2020 and IMT-2030 [11]–[13] show how 3GPP's work connects to global requirements. Dahlman et al. [9] describe the standardisation history of NR.

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18

Lecture 18

5G Service-Based Architecture (SBA)

معماری سرویس‌محور نسل پنجم (SBA)

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 10, 14 and 16; HTTP, REST and JSON basics

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
9 Worked examples . . . 9
10 Technology comparison . . . 10
11 Key takeaways . . . 10
12 Exercises . . . 10
13 Further reading . . . 12

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-18-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the principles of the service-based architecture: NF services, producers and consumers, request/response and subscribe/notify.
- **LO2** **Describe** the protocol stack of service-based interfaces (HTTP/2, TLS, JSON, OpenAPI) and the naming and URI conventions of NF services.
- **LO3** **Trace** NF registration, discovery, authorisation and service invocation through the NRF.
- **LO4** **Compare** direct and indirect (SCP) communication and **explain** the SEPP's role in roaming.
- **LO5** **Quantify** connection needs, discovery load and latency of service-based signalling.
- **LO6** **Contrast** SBA with the point-to-point Diameter/GTP-C design of the EPC.

2 Prerequisite knowledge

You need the Diameter-based EPC interfaces (Lecture 10) as a contrast, the 5G overview of Lecture 14, the cloud-native principles of Lecture 16, and familiarity with HTTP methods (GET, POST, PUT, PATCH, DELETE), status codes, URIs and JSON.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
SBA / SBI	معماری / واسط سرویس‌محور	Architecture in which NFs offer services over a common interface.
NF service	خدمت کارکرد شبکه	Capability exposed by an NF (e.g. Nsmf_PDUSession).
Producer / consumer	تولیدکننده / مصرف‌کننده	NF offering / using a service.
NRF	تابع مخزن شبکه	Registry for NF registration, discovery and access tokens.
SCP	پراکسی ارتباط خدمت	Proxy for indirect communication and delegated discovery.
SEPP	پراکسی حفاظت لبهٔ امنیت	Edge proxy securing SBI signalling between PLMNs (N32).
OpenAPI	OpenAPI	Machine-readable description of a REST API.
HTTP/2 stream	جریان HTTP/2	One request/response exchange multiplexed on a connection.
OAuth 2.0	2.0 OAuth	Authorisation framework; the NRF issues access tokens [7].
NF profile	نمایهٔ NF	Description of an NF instance stored in the NRF.

3 Introduction and motivation

The EPC was designed as a set of boxes connected by point-to-point interfaces: S6a between MME and HSS, S11 between MME and S-GW, Gx between P-GW and PCRF, each with its own protocol (Diameter, GTPv2-C) and its own peer configuration. Adding a function or a consumer meant defining a new interface and configuring every peer. In a cloud-native core, where instances come and go with load, this rigidity becomes the bottleneck.

The 5G core adopts the patterns of web-scale software: each network function exposes *services* through a common *service-based interface* (SBI) built on HTTP/2 and JSON; any authorised NF can consume them; NFs register with a repository function (NRF) and discover each other at run time [3], [6]. This lecture closes Part III by showing how SDN, NFV and web technology come together in the 5G control plane; Part IV then uses these services to build the 5G procedures.

4 Fundamental concepts

4.1 Network functions and services

The control plane of the 5G core consists of NFs connected to the SBI (Figure 1); the user plane (UPF) and the access network keep classical reference points (N2, N3, N4, N6). Table 1 lists the main NFs.

Table 1. Main 5G core network functions and example services [6].

NF	Role	Example service
AMF	Access and mobility management, NAS termination	Namf_Communication
SMF	Session management, UPF control	Nsmf_PDUSession
UPF	User-plane forwarding (not on SBI)	– (N4/PFCP)
AUSF	Authentication server	Nausf_UEAuthentication
UDM / UDR	Subscription data management / repository	Nudm_SDM, Nudm_UECM
PCF	Policy control	Npcf_SMPolicyControl
NRF	NF repository: registration, discovery, tokens	Nnrf_NFDiscovery
NSSF	Network slice selection	Nnssf_NSSelection
NEF	Exposure of network capabilities to external AFs	Nnef_EventExposure
CHF	Charging	Nchf_ConvergedCharging
NWDAF	Network data analytics	Nnwdaf_AnalyticsInfo
SCP, SEPP	Indirect communication, inter-PLMN security	– (proxies)

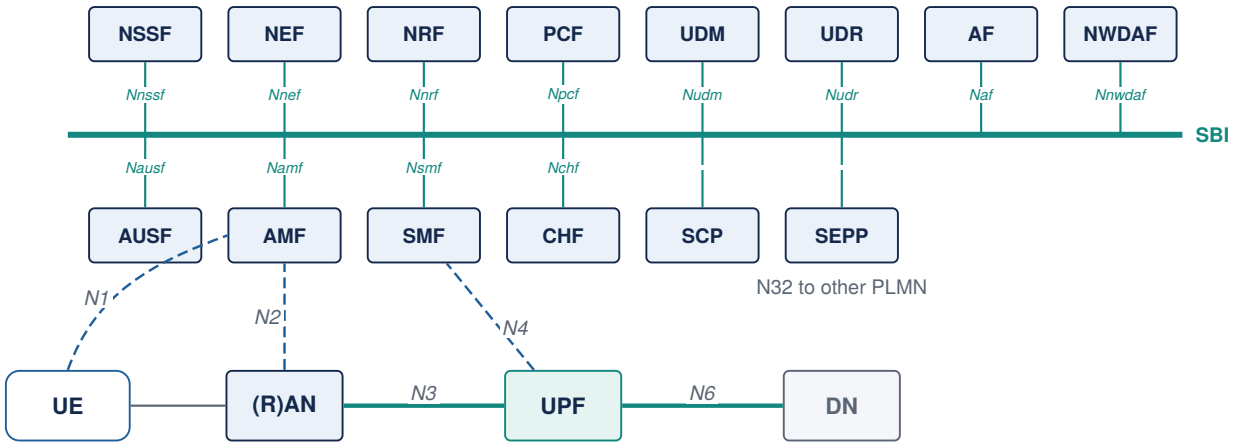


Figure 1. 5G system with service-based control plane (SBI) and reference points to the RAN and UPF.

4.2 Producers, consumers and interaction patterns

A *producer* NF offers a service; a *consumer* NF uses it. Services are named *N<nf>_<Service>_<Operation>*, e.g. Nsmf_PDUSession_CreateSMContext. Two interaction patterns exist:

- **request/response**: the consumer requests an operation and receives the result;
- **subscribe/notify**: the consumer subscribes to events (e.g. AMF subscribes to UDM for subscription changes) and the producer sends notifications to a callback URI given by the consumer.

Services are decoupled from the NFs that use them: a new consumer can use an existing service

without changing the producer – the main architectural gain over point-to-point interfaces.

4.3 Protocol stack

SBIs use HTTP/2 [9] over TCP with TLS, JSON bodies, and APIs described in OpenAPI 3.0 documents published with the specifications [2], [3]. Resources are identified by URIs of the form `{apiRoot}/<api-name>/<version>/<resource>`, e.g. `{apiRoot}/nsmf-pdusession/v1/sm-contexts`. HTTP methods map to operations (Table 2); errors are returned as HTTP status codes with a JSON problem description.

Table 2. Use of HTTP methods on service-based interfaces [2].

Method	Meaning	Typical answer
POST	Create a resource in a collection, or invoke a custom operation	201 Created (Location), 200 OK
GET	Read a resource or query a collection	200 OK
PUT	Create or replace a resource with a client-chosen URI	201 Created, 200/204
PATCH	Partially modify a resource (JSON Patch / merge patch)	200 OK, 204 No Content
DELETE	Remove a resource	204 No Content

5 Detailed technical explanation

5.1 The NRF

The NRF provides three services [1]:

- **Nnrf_NFManagement**: an NF instance registers its *NF profile* (instance ID, type, PLMN, supported slices, services and their endpoints, capacity, priority) with PUT, keeps it alive with periodic heartbeats, updates it with PATCH and deregisters with DELETE; NFs can subscribe to changes of other profiles.
- **Nnrf_NFDiscovery**: a consumer queries for producers by target NF type and criteria (slice, DNN, TAI, SUPI range, ...); the answer contains matching profiles and a validity period during which they may be cached.
- **Nnrf_AccessToken**: the NRF acts as OAuth 2.0 authorisation server and issues access tokens (client credentials grant) that a producer verifies before serving a request [7].

5.2 Direct and indirect communication

In *direct* communication the consumer sends requests straight to the producer, either from configuration (option A) or after NRF discovery (option B). In *indirect* communication (Release 16) requests pass through a *service communication proxy* (SCP): the consumer either discovers the producer itself (option C) or delegates discovery and selection to the SCP (option D) (Figure 2). The SCP plays a role similar to a Diameter routing agent or a service mesh: load balancing, routing, overload handling and observability in one place, at the cost of an extra hop.



Figure 2. Direct communication (options A/B) and indirect communication via an SCP (options C/D).

5.3 Resilience: NF sets and binding

NF instances that share context (e.g. via a common data store) can form an *NF set*; any instance of the set can serve a UE's requests, so a failed instance is replaced without losing the session. *Binding indications* tell consumers and SCPs which instance or set should receive subsequent requests or notifications for a given context. This is the SBA counterpart of the stateless-instance design of cloud-native functions (Lecture 16).

5.4 Roaming and the SEPP

Between PLMNs, NFs never talk directly: each PLMN has a *security edge protection proxy* (SEPP) at its border, and the SEPPs interconnect over N32. The SEPPs negotiate security on N32-c and protect forwarded messages on N32-f, either with TLS end to end between the SEPPs or with application-layer protection when intermediate IPX providers must be able to read or modify selected fields [5]. The SEPP thus plays the role of the Diameter edge agent of the EPC, with stronger security.

5.5 Service operations in practice

A few examples of resource-style operations (simplified) [1]:

- registration: `PUT{apiRoot}/nnrf-nfm/v1/nf-instances/{nfInstanceId}` with the NF profile;
- discovery: `GET{apiRoot}/nnrf-disc/v1/nf-instances?target-nf-type=SMF&requester-nf-type=AMF`;
- session creation: `POST{apiRoot}/nsmf-pdusession/v1/sm-contexts` with a JSON body containing SUPI, DNN, S-NSSAI and the N1 message; the SMF answers 201 with the URI of the new SM context.

6 Mathematical models

6.1 Concurrency on HTTP/2 connections

Model. A consumer sends λ requests per second to a producer; each request is outstanding for a response time R (network round trip plus processing). By Little's law the number of concurrent

requests is

$$N_{\text{inflight}} = \lambda R. \tag{1}$$

With HTTP/1.1 without pipelining, each outstanding request needs its own connection; with HTTP/2 a single connection carries up to the peer's maximum number of concurrent streams (recommended to be at least 100 [9]).

Example. $\lambda = 10^4\,\text{s}^{-1}$, $R = 5\,\text{ms}$: $N_{\text{inflight}} = 50$ – fifty HTTP/1.1 connections or one HTTP/2 connection (in practice a few, for redundancy and load spreading).

6.2 Discovery load and caching

Model. Without caching, every request triggers a discovery and the NRF sees the full request rate λ. If each of C consumers caches discovery results for K target types for a validity period T, the NRF sees approximately

$$\lambda_{\text{NRF}} \approx \frac{C\,K}{T}. \tag{2}$$

Example. $C = 100$, $K = 5$, $T = 60\,\text{s}$: 8.3 discoveries/s instead of 10^4/s (Figure 3).

Interpretation. Caching makes the NRF a light-weight directory, but cached profiles become stale: a consumer may keep sending requests to an overloaded or failed instance until the cache expires or a notification arrives. The validity period trades NRF load against reaction time.

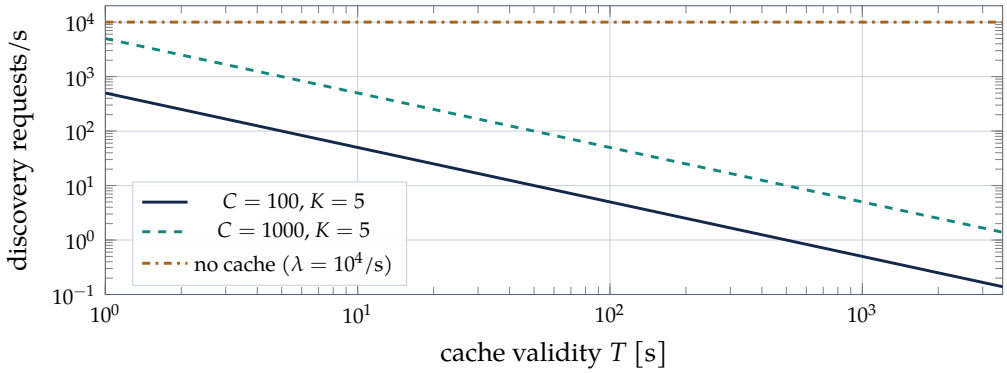


Figure 3. NRF discovery load versus cache validity (Eq. (2)).

6.3 Latency of indirect communication

Model. Let T_{dir} be the response time of a direct request. Routing through an SCP adds two network hops of one-way delay d each direction and SCP processing t_{s}; with delegated discovery the SCP occasionally (probability p_{m} of a cache miss) queries the NRF, costing T_{NRF}:

$$T_{\text{ind}} \approx T_{\text{dir}} + 2d + t_{\text{s}} + p_{\text{m}}\,T_{\text{NRF}}. \tag{3}$$

Example. $T_{\text{dir}} = 3\,\text{ms}$, $d = 0.25\,\text{ms}$, $t_{\text{s}} = 0.5\,\text{ms}$, $p_{\text{m}} = 0.05$, $T_{\text{NRF}} = 2\,\text{ms}$: $T_{\text{ind}} = 4.1\,\text{ms}$. A registration that needs ten sequential SBI exchanges gains about 11 ms.

6.4 Header compression

Model. HTTP/2 compresses headers with HPACK: the first request on a connection sends headers of H_1 bytes; later requests mostly reference table entries and cost h bytes. The mean header size

over n requests is

$$\bar{H}_n = \frac{H_1 + (n-1)\,h}{n} \xrightarrow{n\to\infty} h. \tag{4}$$

Example. $H_1 = 400\,\text{bytes}$, $h = 20\,\text{bytes}$, $n = 100$: $\bar{H} = 23.8\,\text{bytes}$ per request – a reason to keep long-lived connections between NFs.

7 Network architecture and diagrams

In a deployment, NF instances run as containers in one or more data centres; each NF type is replicated and registered in the NRF (itself redundant); SCPs or a service mesh connect them; SEPPs sit at the border; AMFs and UPFs connect to the RAN over N2 and N3. Unlike the EPC, there is no fixed peer table: who talks to whom is decided at run time from the NRF's view, the slice and the location of the UE (Figure 1).

8 Step-by-step procedures

Procedure 1 • NF registration and discovery

1. A new SMF instance starts and sends PUT with its NF profile to the NRF (Nnrf_NFManagement_NFRegister); the NRF stores it and answers 201 with the heartbeat interval.
2. The SMF sends periodic heartbeats; if they stop, the NRF marks it suspended and notifies subscribers.
3. An AMF needing an SMF for a PDU session sends a discovery GET with target type SMF, DNN and S-NSSAI.
4. The NRF returns matching profiles with a validity period; the AMF selects one (priority, capacity, locality) and caches the result.

Procedure 2 • Authorised service invocation

1. The AMF requests an access token from the NRF for scope "nsmf-pdusession" (Nnrf_AccessToken).
2. The NRF authenticates the AMF (TLS client certificate) and returns a signed token.
3. The AMF sends POST to the SMF's sm-contexts resource with the token in the Authorization header.
4. The SMF verifies the token's signature, scope and audience, creates the SM context and answers 201 with its URI; later operations use that URI.

9 Worked examples

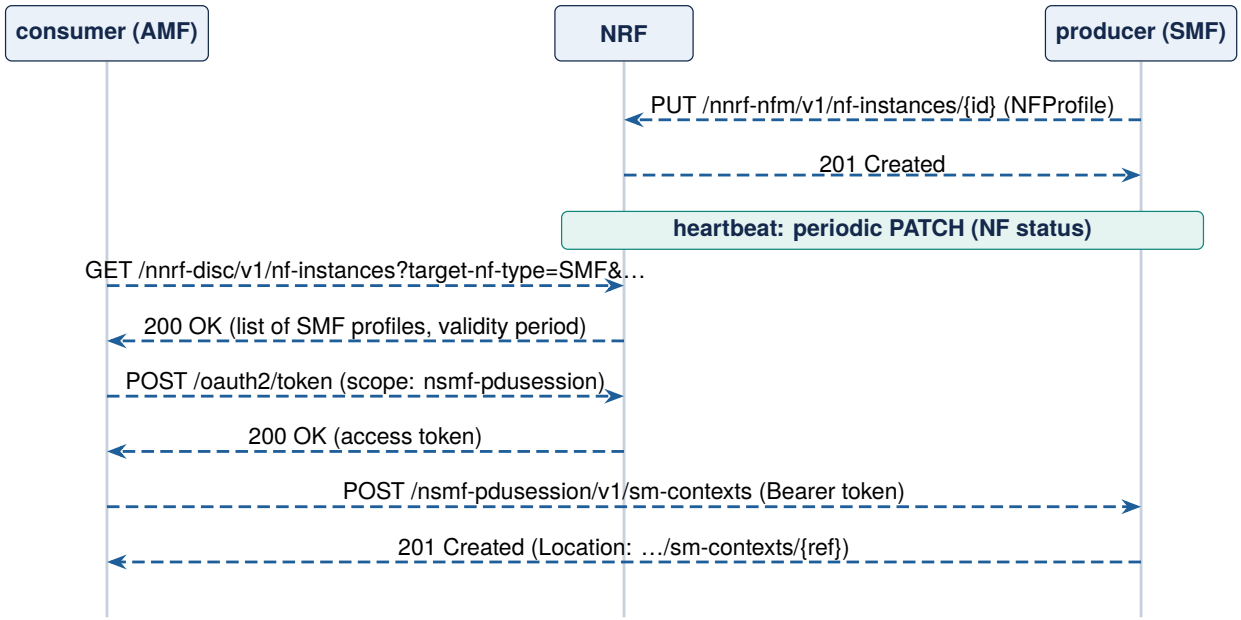


Figure 4. Registration, discovery, token and service invocation (Procedures 1 and 2).

Worked example 1 • Connections between NFs

Problem. An AMF sends 2×10^4 requests/s to a set of SMFs with mean response time 8 ms. How many requests are outstanding, and how many HTTP/2 connections are needed if each allows 100 concurrent streams and connections should be at most 50 % utilised?
Solution. $N = 2 \times 10^4 \times 0.008 = 160$. With 50 streams used per connection, at least $\lceil 160/50 \rceil = 4$ connections, spread over SMF instances for redundancy.

Worked example 2 • NRF sizing

Problem. A core has 1000 consumer instances, each discovering 5 target types with a validity of 300 s, plus 500 NF instances sending a heartbeat every 10 s. What load does the NRF see?
Solution. Discovery $1000 \times 5/300 = 16.7\,\mathrm{s}^{-1}$; heartbeats $500/10 = 50\,\mathrm{s}^{-1}$; total about 67 requests/s – small, but critical: the NRF must be highly available, because new sessions cannot be placed without it once caches expire.

Worked example 3 • SCP or direct?

Problem. With the numbers of Section 6.3, by how much does indirect communication lengthen a procedure with 6 sequential SBI exchanges? What does the operator gain?
Solution. $6 \times 1.1 = 6.6$ ms. The gains: consumers need no discovery logic or per-producer connection management; load balancing, retries, overload control and monitoring are centralised; producers can be scaled or moved transparently.

10 Technology comparison

Table 3. EPC point-to-point signalling versus 5G SBA.

	EPC (Diameter, GTPv2-C)	5GC SBA
Interface model	Point-to-point reference points	Services on a common bus; any authorised consumer
Transport	SCTP/TCP (Diameter), UDP (GTP-C)	TCP + TLS, HTTP/2
Encoding	Binary AVPs / IEs	JSON, OpenAPI-defined
Peer discovery	Static configuration, DNS, DRAs	NRF registration and discovery
Interaction	Request/answer	Request/response, subscribe/notify
Routing proxy	DRA	SCP (optional)
Roaming edge	DEA (Diameter), GTP firewalls	SEPP with N32 security
Authorisation	Implicit (peer configured)	OAuth 2.0 tokens from the NRF
Extensibility	New AVPs/IEs, new applications	New services/operations, API versions

Not everything is service-based. N1 (NAS), N2 (NGAP), N4 (PFCP) and the user plane (N3, N9 with GTP-U) keep point-to-point protocols, because they involve the RAN, the UE or high-rate forwarding where HTTP brings no benefit.

11 Key takeaways

Key takeaways

- 5GC control-plane NFs expose services (Nnf_Service_Operation) on a common SBI: HTTP/2, TLS, JSON, OpenAPI.
- Request/response and subscribe/notify; producers are decoupled from consumers.
- The NRF handles registration, discovery and OAuth 2.0 tokens; SCP enables indirect communication; SEPP secures roaming.
- Little's law sizes concurrency (λR); caching keeps NRF load at CK/T; an SCP adds about $2d + t_s$ per exchange.
- N1, N2, N4 and the user plane remain point-to-point.

12 Exercises

Exercise 1 • Level 1 – conceptual • Producers and consumers

During PDU session establishment the AMF calls Nsmf_PDUSession_CreateSMContext and the SMF calls Nudm_SDM_Get and Npcf_SMPolicyControl_Create. Identify producer and consumer in each case.

Exercise 2 • Level 1 – conceptual • Which method?

Choose the HTTP method for: (a) reading a UE's subscribed slices; (b) creating an SM context; (c) registering an NF profile with its known instance ID; (d) changing the load value in an NF profile; (e) releasing a policy association.

Exercise 3 • Level 2 – numerical • Connections

A UDM receives 5×10^4 requests/s from 40 consumer instances, response time 4 ms. Compute the concurrency and compare the number of connections needed with HTTP/1.1 (no pipelining) and HTTP/2 (100 streams per connection, one connection per consumer instance). Is one HTTP/2 connection per consumer enough?

Exercise 4 • Level 2 – numerical • Cache validity

An operator wants the NRF discovery load below $50\,\mathrm{s}^{-1}$ with 2000 consumers and 6 target types. What is the minimum validity period? What is the drawback of choosing it much larger?

Exercise 5 • Level 3 – analytical • Latency budget

A registration procedure has 12 sequential SBI exchanges with $T_{\mathrm{dir}} = 2.5$ ms. The control-plane budget for registration in the core is 50 ms. Using Eq. (3) with $d = 0.5$ ms, $t_{\mathrm{s}} = 0.8$ ms, $p_{\mathrm{m}} = 0.1$, $T_{\mathrm{NRF}} = 3$ ms, check whether indirect communication fits. What changes if NFs are in two data centres 300 km apart?

Exercise 6 • Level 3 – analytical • Design critique

Give two arguments for and two against using HTTP/2 + JSON instead of a binary protocol such as Diameter for core signalling, and explain why N2 and N4 were not made service-based.

Solutions

Solution to Exercise 1

CreateSMContext: SMF producer, AMF consumer. SDM_Get: UDM producer, SMF consumer. SMPolicyControl_Create: PCF producer, SMF consumer.

Solution to Exercise 2

(a) GET; (b) POST (to the sm-contexts collection); (c) PUT (to nf-instances/{id}); (d) PATCH; (e) DELETE (or POST to a custom "delete" operation where the API defines one).

Solution to Exercise 3

$N = 5 \times 10^4 \times 0.004 = 200$ concurrent requests. HTTP/1.1: 200 connections. HTTP/2: per consumer $200/40 = 5$ concurrent streams – one connection per consumer suffices in capacity, but two are advisable for redundancy and for spreading load over UDM instances.

Solution to Exercise 4

$CK/T \leq 50 \Rightarrow T \geq 2000 \times 6/50 = 240\,\text{s}$. A much larger validity makes consumers slow to notice new or removed producers; they rely on NRF notifications or error responses to update, and load balancing over newly scaled instances is delayed.

Solution to Exercise 5

Per exchange: $2.5 + 1.0 + 0.8 + 0.3 = 4.6\,\text{ms}$; 12 exchanges: 55.2 ms – over the 50 ms budget (direct: 30 ms). With 300 km between data centres (one-way about 1.5 ms in fibre), every cross-site exchange adds about 3 ms of round trip: the procedure must keep the NFs of one UE in one site (locality-aware discovery) or parallelise exchanges.

Solution to Exercise 6

For: widely available tools, libraries and skills; easy extensibility and versioning with OpenAPI; native fit with cloud platforms (load balancers, service meshes, observability). Against: larger messages and parsing cost than binary TLV; TCP and TLS connection management; more freedom (optional fields, versions) that can hurt interoperability. N2 and N4 involve the RAN and the user plane, which need compact, low-latency, often hardware-assisted protocols and long-lived associations; their peers are not generic consumers, so the SBA's decoupling brings little benefit there.

13 Further reading

The SBA is defined in TS 23.501 [6] and its procedures in TS 23.502 [4]; TS 29.500 [3] and TS 29.501 [2] specify the technical realisation and API design rules; TS 29.510 [1] specifies the NRF; TS 33.501 [5] covers SBA security including the SEPP. HTTP/2 is RFC 9113 [9] and OAuth 2.0 RFC 6749 [7]. Rommer et al. [8] give a textbook treatment.

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[1] 3GPP, *5G system; network function repository services; stage 3*, TS 29.510, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29510.htm.

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19

Lecture 19

5G Core Network Architecture

معماری هستهٔ نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate to advanced

PREREQUISITES
Lectures 9, 12, 13, 14 and 18

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 7
7 Network architecture and diagrams . . . 8
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11 Key takeaways . . . 10
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-19-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G system architecture in reference-point form and **map** each NF to its EPC predecessor.
- **LO2** **Explain** the 5G identities (SUPI, SUCI, 5G-GUTI, GPSI, PEI) and how SUCI protects subscriber privacy.
- **LO3** **Describe** PDU sessions, their types and the session and service continuity (SSC) modes.
- **LO4** **Design** UPF topologies with uplink classifiers and local anchors for edge computing, and **estimate** the latency and backhaul gains.
- **LO5** **Explain** RM/CM states, registration areas and interworking with the EPC.

2 Prerequisite knowledge

You need the EPC architecture (Lectures 9, 12, 13), the 5G overview (Lecture 14) and the service-based mechanics of Lecture 18 (NRF, SBI). This lecture describes *what* the 5G core consists of; Lecture 20 describes the procedures that use it, Lecture 21 slicing and Lecture 22 QoS.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
SUPI / SUCI	شناسهٔ دائمی / پنهان‌شدهٔ مشترک	Permanent identity / its encrypted form sent over the air.
5G-GUTI	شناسهٔ موقت سراسری	Temporary identity allocated by the AMF.
PDU session	نشست PDU	Association between UE and a data network through the 5GC.
DNN	نام شبکهٔ داده	Name of a data network (the 5G equivalent of an APN).
SSC mode	حالت پیوستگی نشست و خدمت	Rules for keeping or changing the session anchor during mobility.
PSA	لنگر نشست PDU	UPF terminating N6 towards the data network.
UL CL / BP	طبقه‌بند فراسو / نقطهٔ انشعاب	UPF functions that split traffic between anchors.
RM / CM	مدیریت ثبت / اتصال	Registration and connection management states.
N26	واسط N26	AMF–MME interface for EPC–5GC interworking.
N3IWF	تابع همکاری غیر 3GPP	Gateway for untrusted non-3GPP access (e.g. Wi-Fi).

3 Introduction and motivation

The 5G core (5GC) is the first mobile core designed from the outset for cloud deployment. It keeps the proven concepts of the EPC – a mobility anchor in the control plane, a session anchor in the user plane, a subscriber database, policy control – but reorganises them: access and mobility are separated from session management, the user plane is fully controlled through PFCP, subscription data are split between front-ends and a repository, and every control-plane function is a service producer (Lecture 18) [7]. The new structure supports features the EPC could not offer natively: access-agnostic registration (3GPP and non-3GPP), multiple user-plane anchors per session for edge computing, Ethernet and unstructured sessions, network slicing and a flow-based QoS model.

4 Fundamental concepts

4.1 Network functions and reference points

Figure 1 shows the 5G system in reference-point representation. Each reference point is realised on the SBI by the services of the producer (e.g. N11 by Nsmf services, N8 by Nudm services),

except N1, N2, N3, N4, N6 and N9.

Table 1. 5GC network functions and their EPC counterparts.

5GC	Main responsibilities	EPC counterpart
AMF	Registration, connection, reachability, mobility; NAS termination and security; SMS	MME (mobility part)
SMF	PDU sessions, IP allocation, UPF selection and control, charging trigger	MME (session part), SGW-C, PGW-C
UPF	Packet routing, forwarding, QoS enforcement, usage reporting, anchoring	SGW-U, PGW-U
AUSF	Authentication server	HSS (AuC part)
UDM / UDR	Credentials processing, subscription data / repository	HSS
PCF	Policy for access, mobility, sessions and UE	PCRF
NSSF	Slice selection	–
NEF	Exposure of capabilities to external applications	SCEF
NRF, SCP, SEPP	Discovery, indirect communication, roaming security	DNS, DRA, DEA

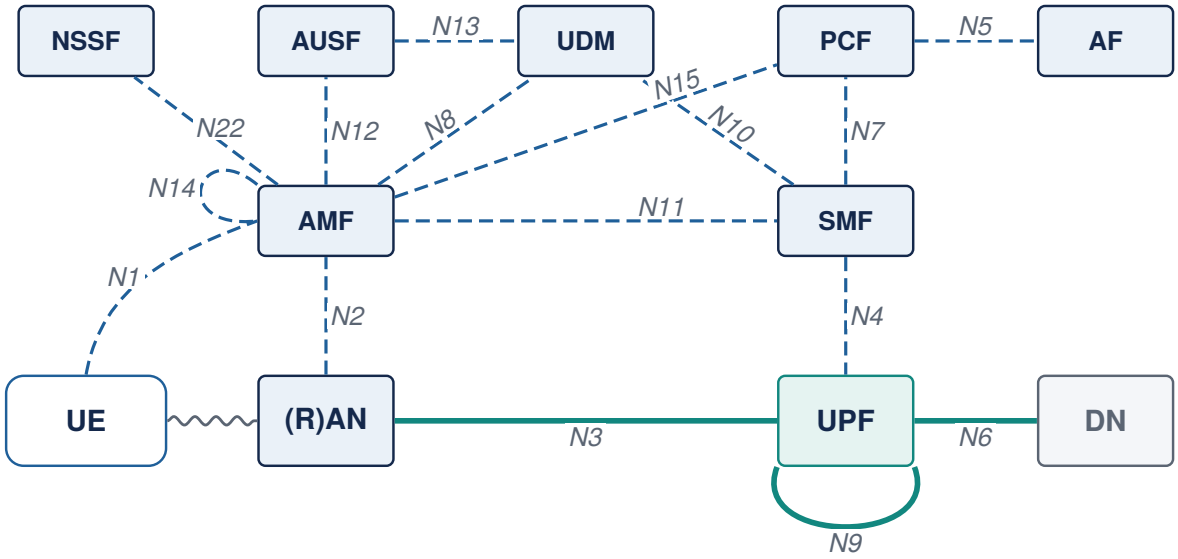


Figure 1. 5G system architecture in reference-point representation (non-roaming, selected NFs) [7].

4.2 Identities

- **SUPI**: the permanent subscription identifier, usually an IMSI (or a network access identifier).
- **SUCI**: the SUPI concealed with the home network's public key; only the routing part is in clear (Figure 2). The UDM's de-concealment function recovers the SUPI [6].
- **5G-GUTI**: temporary identity = GUAMI (PLMN, AMF region, AMF set, AMF pointer) + 5G-TMSI; the 48-bit 5G-S-TMSI is used in paging and RRC.
- **PEI**: permanent equipment identifier (IMEI); **GPSI**: generic public identifier (MSISDN or

external ID) used towards external applications.

Unlike LTE, a 5G UE never sends its permanent identity in clear over the radio interface: even the first registration uses a SUCI, which closes the "IMSI catcher" weakness of earlier generations.

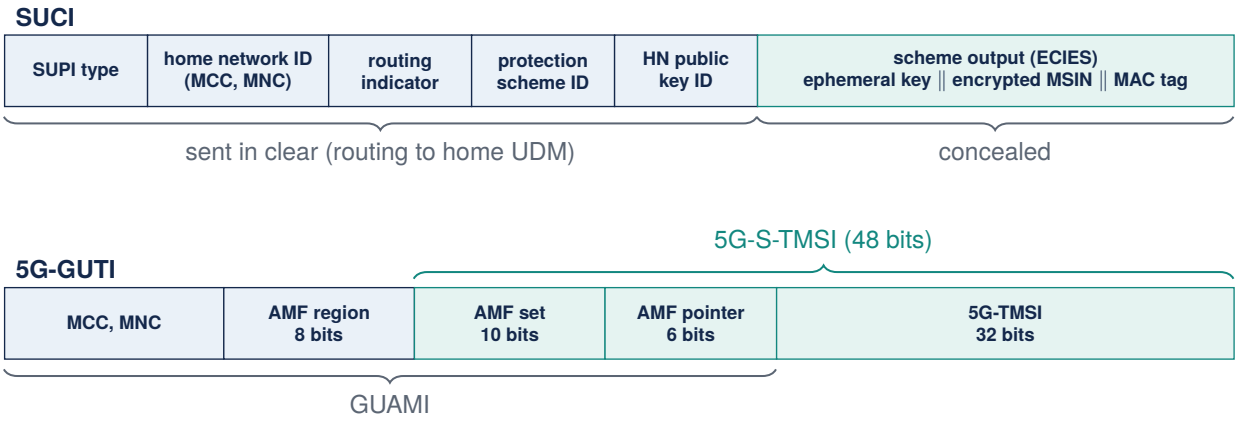


Figure 2. Structure of the SUCI and of the 5G-GUTI.

4.3 PDU sessions

A PDU session connects the UE to a data network identified by a DNN and a slice (S-NSSAI). Session types are IPv4, IPv6, IPv4v6, *Ethernet* (the UPF switches Ethernet frames, e.g. for industrial LANs) and *Unstructured* (opaque data). Within a session, traffic is carried in QoS flows (Lecture 22); a UE may have several sessions to different DNNs or slices at the same time.

4.4 Session and service continuity

The *SSC mode* of a session defines what happens to its anchor when the UE moves [7]:

- **SSC mode 1**: the PSA (and the IP address) is kept for the life of the session – like the EPC's P-GW.
- **SSC mode 2**: the network may release the session and ask the UE to establish a new one with a closer anchor ("break before make").
- **SSC mode 3**: the network establishes a new session with a new anchor before releasing the old one ("make before break"); the application can migrate gracefully.

5 Detailed technical explanation

5.1 The AMF

The AMF terminates N1 (NAS) and N2 (NGAP) [2], [3]. It handles registration, reachability and paging, connection management, mobility and access authentication with the AUSF. Unlike the MME it does not process session signalling: session NAS messages are relayed transparently to the SMF. One AMF serves a UE across 3GPP and non-3GPP access, so registration is access-agnostic. AMFs are organised in *AMF sets*: any AMF of a set can take over a UE's context from

shared storage.

5.2 The SMF and UPF

The SMF manages PDU sessions: it selects UPFs, allocates IP addresses, installs PFCP rules (PDR, FAR, QER, URR, BAR – Lecture 13) over N4, interacts with the PCF for policy and with the CHF for charging. The UPF is the only user-plane function: an *intermediate UPF* (I-UPF) may connect the RAN, an *uplink classifier* (UL CL) or *branching point* (for IPv6 multi-homing) may divert part of the traffic, and one or more *PDU session anchors* (PSAs) terminate N6 (Figure 3).

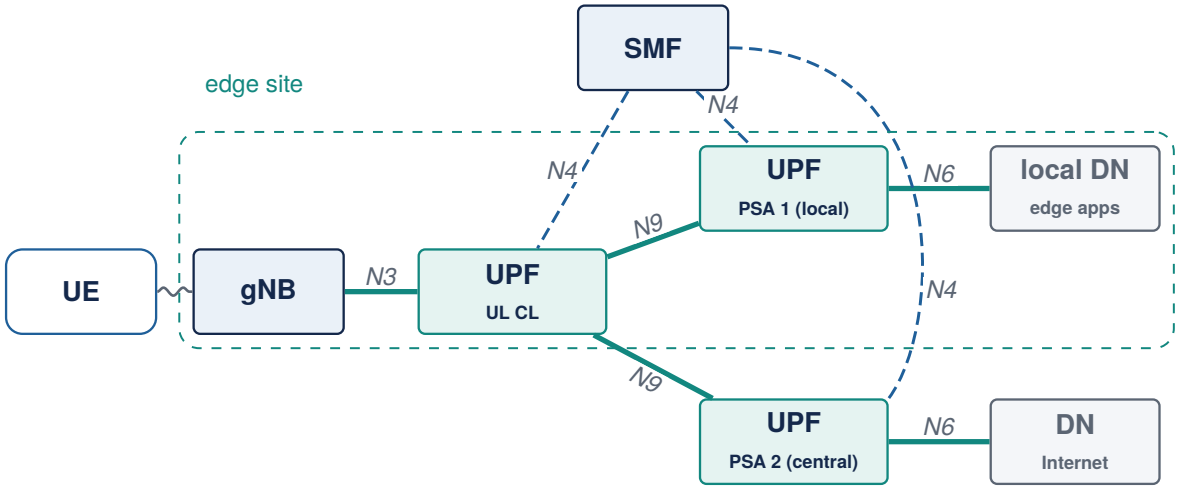


Figure 3. A PDU session with an uplink classifier and two anchors: local edge breakout and central Internet access.

5.3 Edge computing

To serve low-latency applications, the SMF can insert a UL CL and a local PSA at an edge site, so that traffic for edge application addresses is routed locally while the rest goes to the central anchor. Application functions can influence this routing through the NEF or PCF, and TS 23.548 adds discovery of edge application servers [1]. The result is a single PDU session with both local and central breakout – impossible in the EPC without special arrangements.

5.4 States and registration areas

RM states: RM-DEREGISTERED and RM-REGISTERED; CM states: CM-IDLE and CM-CONNECTED. A UE in CM-CONNECTED may additionally be in RRC_INACTIVE: the N2 connection and context stay in place while the RAN pages the UE within a RAN notification area, which saves signalling for frequent small transmissions. The AMF allocates a *registration area* (a TAI list) and performs mobility and periodic registration updates, as the MME did with TAUs.

5.5 Interworking with the EPC

During the long coexistence of 4G and 5G, a UE moves between LTE (EPC) and NR (5GC). With *N26* the MME and AMF exchange contexts directly and IP sessions survive the transition; this requires combined nodes: SMF+PGW-C, UPF+PGW-U, HSS+UDM and PCF+PCRF serve both systems (Figure 4). Without N26 the UE re-registers and the session continuity depends on the

combined anchor. Voice is supported by EPS fallback (a 5G voice call is handed over to LTE for VoLTE) where VoNR is not available.

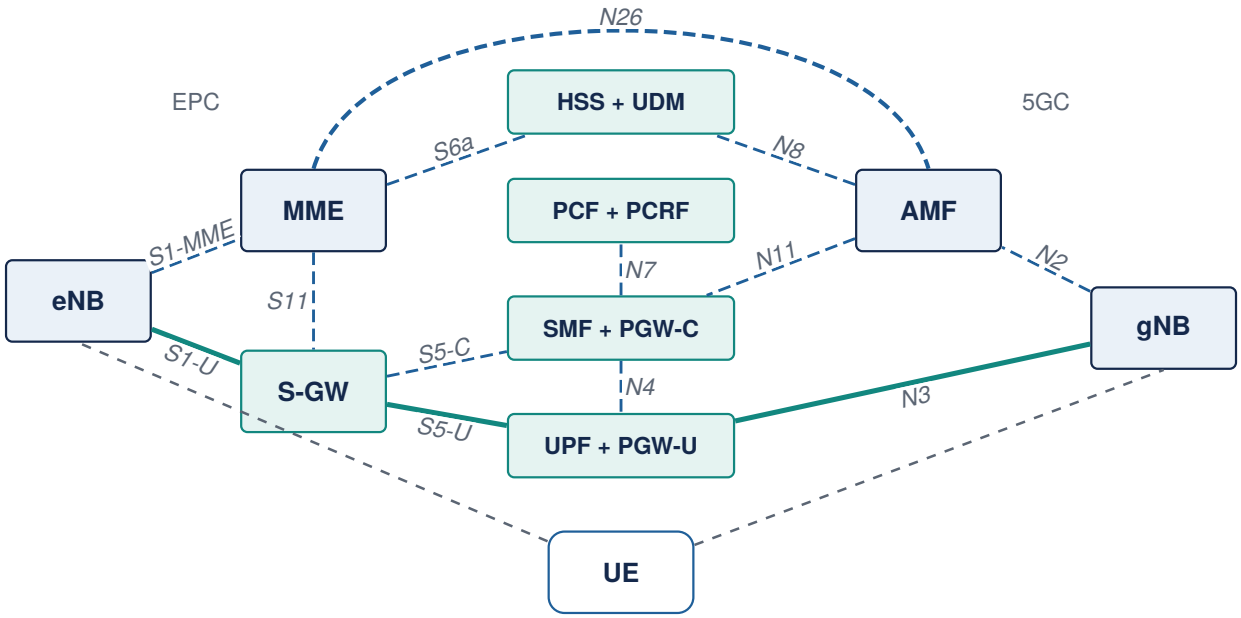


Figure 4. EPC–5GC interworking with N26 and combined nodes.

5.6 Non-3GPP access

Untrusted non-3GPP access (e.g. public Wi-Fi) connects through an N3IWF, which terminates an IPsec tunnel from the UE and presents N2/N3 towards the core; trusted non-3GPP access uses a TNGF. In both cases the UE registers with the same AMF and uses the same NAS and PDU session procedures – the access-agnostic design of the 5GC.

6 Mathematical models

6.1 SUCI size

Model. With ECIES, the scheme output consists of the ephemeral public key (k bytes), the encrypted MSIN (the same length m as the plaintext, BCD encoded) and a MAC tag (t bytes):

$$L_{\text{out}} = k + m + t. \tag{1}$$

Example. A 10-digit MSIN ($m = 5$ bytes): profile A (X25519, $k = 32$, $t = 8$) gives 45 bytes; profile B (P-256 compressed point, $k = 33$) gives 46 bytes [6]. The overhead is small compared with the privacy gained, and the computation is done once per registration attempt that needs a permanent identity.

6.2 Latency and UPF placement

Model. Round-trip latency between UE and an application server behind a PSA at fibre distance D is

$$\mathrm{RTT} \approx \mathrm{RTT}_{\mathrm{radio}} + 2\,\frac{D}{v} + T_{\mathrm{proc}}, \tag{2}$$

with $v \approx 2 \times 10^8$ m/s in fibre (about 5 µs per km).

Example. $D = 500$ km (central data centre) adds 5 ms; $D = 20$ km (edge site) adds 0.2 ms. With a 4 ms radio RTT and 1 ms processing, the RTT falls from 10 ms to 5.2 ms – the edge halves it.

Interpretation. Propagation is only one term; for URLLC targets of a few milliseconds every component – radio numerology, scheduling, transport and processing – must be engineered together.

6.3 Backhaul offload

Model. If a fraction α of a site's traffic T is served locally (UL CL to a local PSA and edge cache or application), the traffic carried towards the central core is

$$T_{\mathrm{bh}} = (1 - \alpha)\,T. \tag{3}$$

Example. With 60 % of video served from an edge cache, a site with 10 Gbit/s busy-hour traffic needs 4 Gbit/s of backhaul towards the central core instead of 10.

6.4 Interruption in SSC modes

Model. When the anchor changes, the service interruption is approximately

$$T_{\mathrm{int}} \approx \begin{cases} 0 & \text{SSC 1 (no anchor change) and SSC 3 (make before break),} \\ T_{\mathrm{rel}} + T_{\mathrm{est}} + T_{\mathrm{app}} & \text{SSC 2 (break before make),} \end{cases} \tag{4}$$

where T_{rel} and T_{est} are the release and establishment times and T_{app} the time for the application to reconnect with the new address.

Example. $T_{\mathrm{rel}} = 50$ ms, $T_{\mathrm{est}} = 150$ ms, $T_{\mathrm{app}} = 300$ ms: half a second without service in SSC 2 – acceptable for a background download, not for a video call, which should use SSC 1 or 3.

7 Network architecture and diagrams

A national 5GC typically has a few central sites hosting AMF sets, SMFs, UDM/UDR, PCF, NRF and the central UPFs, and many edge sites with UPFs (and possibly edge application platforms) close to the RAN. The SMF chooses, per session, which UPFs to chain: for an Internet session a central PSA, for an industrial Ethernet session a local PSA on the factory premises, for a session with edge services a UL CL with both (Figure 3). Slices (Lecture 21) may use dedicated SMFs and UPFs.

8 Step-by-step procedures

Procedure 1 • SUCI calculation and de-concealment

1. The USIM (or ME) holds the home network public key and its identifier, and the protection scheme to use.
2. For each registration that needs a permanent identity, the UE generates an ephemeral key pair, derives a shared secret with the home network public key and encrypts the MSIN; it computes the MAC tag.
3. The UE sends the SUCI (routing part in clear plus scheme output) in the Registration Request.
4. The AMF forwards it via the AUSF to the home UDM, whose SIDF uses the private key to recover the SUPI.

Procedure 2 • Adding a local anchor for edge traffic

1. An edge application is deployed and its address range is configured in (or reported through the NEF/PCF to) the SMF.
2. When the UE is in the edge site's area, the SMF selects a UPF there as UL CL and a local PSA, and installs PFCP rules: traffic matching the edge addresses is forwarded to the local PSA, the rest over N9 to the central PSA.
3. The SMF updates the RAN's N3 tunnel to point to the UL CL (via the AMF and N2).
4. When the UE leaves the area, the SMF removes the UL CL and local PSA; the session continues through the central anchor.

9 Worked examples

Worked example 1 • Mapping EPC to 5GC

Problem. Which 5GC functions take over (a) the MME's tracking-area list management, (b) the MME's bearer set-up via S11, (c) the P-GW's IP address allocation, (d) the PCRF's Gx rules?
Solution. (a) AMF (registration areas); (b) SMF (PDU session management, N4 to the UPF); (c) SMF (allocation) with the UPF as anchor; (d) PCF over N7 (Npcf_SMPolicyControl) to the SMF.

Worked example 2 • Edge latency budget

Problem. An AR application needs an RTT below 10 ms. Radio RTT is 4 ms, processing 1 ms. Up to which fibre distance can the application server be placed?
Solution. $4 + 1 + 2D/v \leq 10 \Rightarrow D \leq 5\text{ ms} \times v/2 = 500\,\text{km}$ of fibre path. Fibre routes are typically 1.5 times the straight-line distance, so about 330 km in a straight line – a regional, not necessarily a cell-site, edge.

Worked example 3 • Choosing an SSC mode

Problem. Choose SSC modes for (a) IMS voice, (b) a vehicle downloading map updates, (c) a cloud game that can migrate between edge servers.
Solution. (a) SSC 1 – the IMS registration is bound to the IP address; (b) SSC 2 – a short interruption is tolerable and a local anchor reduces backhaul; (c) SSC 3 – the application opens a connection via the new anchor before the old one is released.

10 Technology comparison

Table 2. EPC and 5GC architecture compared.

	EPC	5GC
Mobility vs session control	Both in MME	AMF and SMF separated
User plane control	Integrated (CUPS optional)	Always via PFCP (N4)
Session types	IP (non-IP via SCEF)	IP, Ethernet, Unstructured
Anchors per session	One P-GW	One or more PSAs, UL CL/BP
Continuity	Anchor fixed	SSC modes 1, 2, 3
Identity privacy	IMSI may be sent in clear	SUCI (always concealed)
Access	3GPP; non-3GPP via ePDG	Access-agnostic AMF (N3IWF, TNGF)
Signalling	Diameter, GTPv2-C	SBI (HTTP/2), NAS, NGAP, PFCP

Continuity of ideas. Many EPC concepts survive with new names: TAI lists become registration areas, APNs become DNNs, bearers become QoS flows, the HSS becomes UDM/UDR/AUSF. Understanding the EPC therefore remains the best preparation for the 5GC.

11 Key takeaways

Key takeaways

- 5GC separates AMF (access, mobility) from SMF (sessions) and controls UPFs only through PFCP.
- SUPI is never sent in clear; SUCI = routing info + ECIES output ($k + m + t$ bytes).
- PDU sessions (IP, Ethernet, Unstructured) can have several anchors; UL CL and BP enable local breakout.
- SSC modes 1/2/3 trade anchor stability against optimal routing; SSC 2 interrupts service, SSC 3 does not.
- Edge UPFs cut RTT by $2D/v$ and backhaul by a factor $(1 - \alpha)$.
- Interworking with the EPC relies on N26 and combined nodes (SMF+PGW-C, UPF+PGW-U, HSS+UDM, PCF+PCRF).

12 Exercises

Exercise 1 • Level 1 – conceptual • Who does it?

Which NF performs: (a) paging; (b) IP address allocation; (c) slice selection at registration; (d) SUCI de-concealment; (e) exposure of UE location to an authorised external application?

Exercise 2 • Level 1 – conceptual • Ethernet sessions

Give an application of Ethernet PDU sessions and explain why IP sessions would be inconvenient for it.

Exercise 3 • Level 2 – numerical • SUCI size

Compute the scheme-output length for a 9-digit MSIN under profiles A and B. How many bytes does concealment add compared with the null scheme, which sends the MSIN in clear?

Exercise 4 • Level 2 – numerical • Edge offload

A city has 50 edge sites, each with 8 Gbit/s busy-hour traffic. What edge-served fraction is needed to keep total backhaul to the central core below 200 Gbit/s?

Exercise 5 • Level 3 – analytical • Latency design

A factory needs RTT below 5 ms between a robot controller and the UE. Radio RTT is 2 ms with mini-slots and processing 0.5 ms. Where must the PSA and the controller be, and which session type and SSC mode would you choose? What if the radio RTT were 4 ms?

Exercise 6 • Level 3 – analytical • Interworking without N26

Explain what happens to an ongoing IP session when a UE moves from NR to LTE coverage (a) with N26 and combined anchors, (b) without N26 but with combined anchors, (c) without combined anchors. Which case keeps the IP address?

Solutions

Solution to Exercise 1

(a) AMF (and the RAN for RRC_INACTIVE); (b) SMF; (c) AMF with the NSSF; (d) UDM (SIDF); (e) NEF, obtaining the information from the AMF/UDM.

Solution to Exercise 2

Industrial LANs (e.g. time-sensitive networking between controllers and machines) and enterprise Layer-2 VPNs. The devices use Ethernet protocols and MAC addressing, sometimes non-IP; carrying them over IP sessions would need tunnelling or gateways, adding latency and complexity, whereas an Ethernet session lets the UPF switch frames directly.

Solution to Exercise 3

9 digits in BCD occupy 5 bytes (last half-byte padded). Profile A: $32 + 5 + 8 = 45$ bytes; profile B: $33 + 5 + 8 = 46$ bytes. The null scheme carries the 5-byte MSIN, so concealment adds 40 or 41 bytes.

Solution to Exercise 4

Total traffic $50 \times 8 = 400\,\text{Gbit/s}$; $(1 - \alpha) \times 400 \leq 200 \Rightarrow \alpha \geq 0.5$: half of the traffic must be served at the edge.

Solution to Exercise 5

$2 + 0.5 + 2D/v \leq 5 \Rightarrow D \leq 250\,\text{km}$: in principle a regional site, but transport hops, queueing and jitter consume budget, so the PSA and controller are placed on the factory premises (local UPF). An Ethernet session (if the control protocol is Layer-2) with SSC 1 (fixed local anchor). With 4 ms radio RTT only 1 ms remains: $D \leq 100\,\text{km}$, and in practice the same on-premises design, with little margin.

Solution to Exercise 6

(a) The AMF passes the context to the MME over N26; the session is moved to the EPC through the same SMF+PGW-C/UPF+PGW-U: IP address kept, short interruption. (b) The UE attaches/registers in the EPC itself with "handover" indications; the combined anchor can restore the PDN connection with the same address, but with a longer interruption. (c) A new PDN connection with a new IP address: the application must reconnect. Only (a) and (b) keep the IP address.

13 Further reading

TS 23.501 [7] specifies the architecture and TS 23.502 [5] the procedures; TS 23.503 [4] policy; TS 23.548 [1] edge computing; TS 33.501 [6] security including SUCI; TS 24.501 [3] NAS; TS 38.413 [2] NGAP. Rommer et al. [8] give a thorough textbook account of the 5G core.

References

[1] 3GPP, *5G system enhancements for edge computing; stage 2*, TS 23.548, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23548.htm.

[2] 3GPP, *NG-RAN; NG application protocol (NGAP)*, TS 38.413, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38413.htm.

[3] 3GPP, *Non-access-stratum (NAS) protocol for 5G system (5GS); stage 3*, TS 24.501, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/24501.htm.

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[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.

20

Lecture 20

5G System Procedures

رویه‌های سامانهٔ نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 9, 12, 18 and 19

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-20-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Trace** initial registration including 5G-AKA, NAS security and subscription retrieval.
- **LO2** **Derive** the 5G-AKA verification values and key hierarchy at the level of their inputs and purpose.
- **LO3** **Trace** PDU session establishment across AMF, SMF, PCF, UPF and the RAN.
- **LO4** **Explain** UE- and network-triggered service request, RRC_INACTIVE and Xn/N2 handover.
- **LO5** **Estimate** procedure latency, AMF signalling load and handover forwarding buffers.

2 Prerequisite knowledge

This lecture uses the 5GC architecture and identities of Lecture 19, the SBI and NRF of Lecture 18, the EPS-AKA and EPC procedures of Lectures 9 and 12 (as a comparison) and the PFCP rules of Lecture 13. Procedures are presented in simplified form following TS 23.502 [6]; optional steps (roaming, interworking, many error cases) are omitted.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Registration	ثبت‌نام	Procedure that makes the UE known to the 5GC (initial, mobility, periodic).
5G-AKA	احراز هویت و توافق کلید ۵G	Authentication with home-network confirmation of RES*.
SEAF	تابع لنگر امنیتی	Security anchor in the serving network (co-located with the AMF).
NAS SMC	فرمان حالت امنیتی NAS	Activation of NAS ciphering and integrity.
PDU session establishment	برقراری نشست PDU	Creation of a session, its QoS flows and tunnels.
N1N2 message transfer	انتقال پیام N1N2	AMF service carrying SMF messages to UE (N1) and RAN (N2).
Service request	درخواست خدمت	Transition from CM-IDLE to CM-CONNECTED.
RRC_INACTIVE	حالت غیرفعال RRC	RAN-level suspended state with core context kept.
Path switch	تغییر مسیر	Moving the N3 tunnel to the target gNB after Xn handover.
ABBA	پارامتر ضدحمله به پایین‌برداری	Anti-bidding-down parameter bound into K_{AMF}.

3 Introduction and motivation

An architecture becomes concrete only through its procedures: the ordered exchanges that make a UE known to the network, secure its signalling, give it a data session and keep that session alive while the UE sleeps and moves. In the 5GC these procedures combine three kinds of signalling: NAS between UE and AMF (TS 24.501 [4]), NGAP between RAN and AMF (TS 38.413 [3]) and service operations between NFs (Lecture 18), plus PFCP towards the UPF.

Understanding the procedures is essential for troubleshooting (where did an attach fail?), for dimensioning (how many messages does each NF handle?) and for latency engineering (how long until the first packet flows?). This lecture traces the four central procedures and quantifies their cost.

4 Fundamental concepts

4.1 Types of registration

- **Initial registration**: the UE is RM-DEREGISTERED and registers for the first time (or after a long absence).
- **Mobility registration update**: the UE enters a tracking area outside its registration area, or its capabilities or requested slices change.
- **Periodic registration update**: on expiry of the periodic timer, to confirm reachability.
- **Emergency registration**: limited service for emergency calls.

4.2 Security anchors

5G-AKA [7] differs from EPS-AKA in one important respect: the *home network* confirms the authentication. The serving network's SEAF checks a hashed expected response (HXRES*) for a first verification, but the AUSF in the home network verifies RES* itself before releasing the anchor key K_{SEAF} and the SUPI. A visited network therefore cannot claim that a subscriber was present without proof – an answer to fraud concerns in roaming.

4.3 Session establishment roles

The AMF only relays session signalling: it selects an SMF and transports N1 (NAS SM) and N2 (NGAP resource set-up) messages for it. The SMF owns the session: it obtains policy from the PCF, selects and programs the UPF over N4, and asks the AMF to deliver the NAS accept to the UE and the resource request to the RAN in a single *N1N2 message transfer*.

5 Detailed technical explanation

5.1 Initial registration

The UE sends a Registration Request with its SUCI (or a 5G-GUTI if it has one), requested slices (NSSAI) and capabilities (Figure 1) [6]. The RAN selects an AMF (from the GUAMI in the 5G-S-TMSI or by load) and forwards the message in an NGAP Initial UE Message. If the UE gave a 5G-GUTI from another AMF, the new AMF fetches the context from the old AMF (Namf_Communication_UEContextTransfer). The AMF then authenticates the UE (5G-AKA, Section 5.2), activates NAS security, registers with the UDM (Nudm_UECM_Registration), retrieves the access and mobility subscription (Nudm_SDM_Get) and subscribes to changes, creates an AM policy association with the PCF, and, if needed, asks the NSSF to select slices. Registration Accept carries the new 5G-GUTI, the registration area and the allowed NSSAI; the UE acknowledges with Registration Complete.

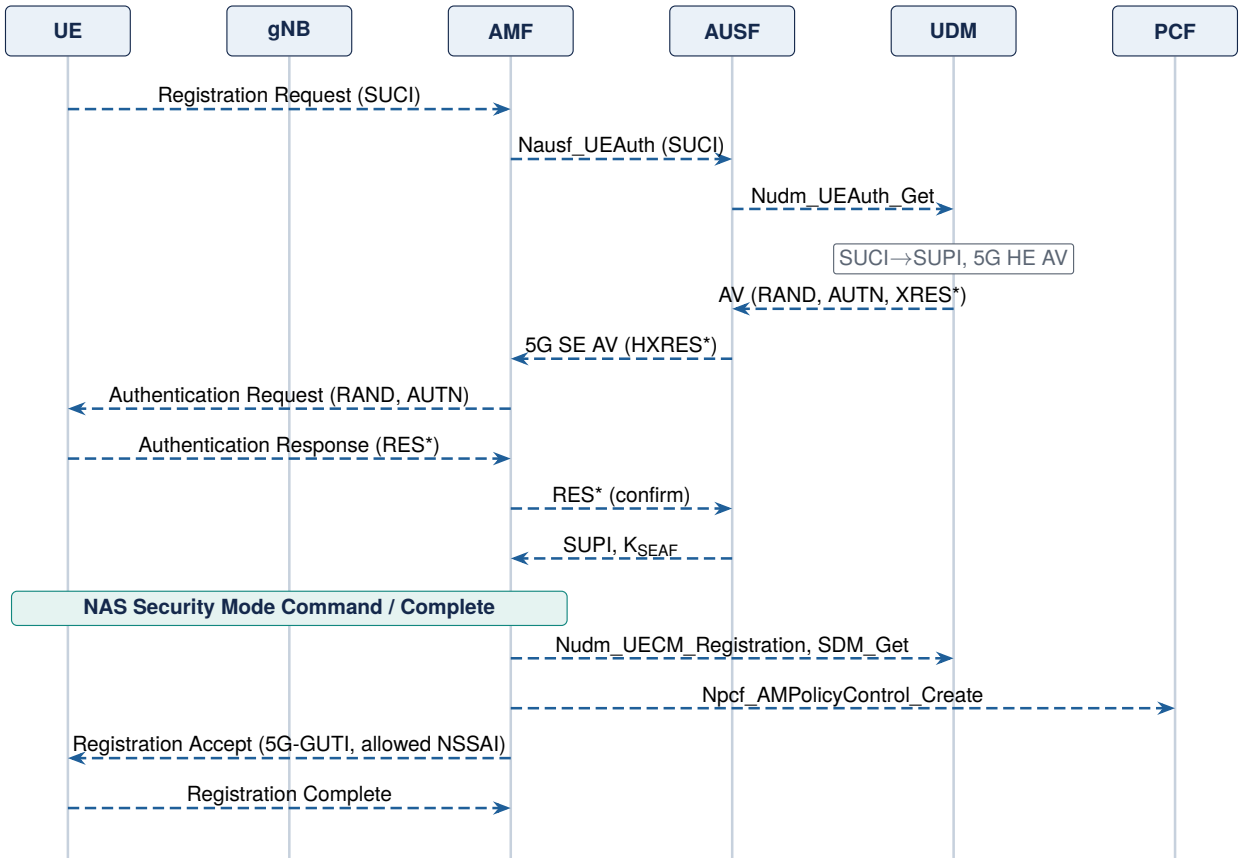


Figure 1. Initial registration with 5G-AKA (simplified) [6], [7].

5.2 5G-AKA

The UDM generates an authentication vector with the same functions f_1–f_5 as in Lecture 11, but derives 5G-specific values bound to the *serving network name* (SNN, of the form `5G:mnc<MNC>.mcc<MCC>.3gppnetwork.org`) [7]:

$$\mathrm{XRES}^* = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SNN},\ \mathrm{RAND},\ \mathrm{XRES}\big)\ \text{(128 bits)}, \tag{1a}$$

$$\mathrm{HXRES}^* = \mathrm{SHA\text{-}256}\big(\mathrm{RAND} \,\|\, \mathrm{XRES}^*\big)\ \text{(truncated to 128 bits)}, \tag{1b}$$

$$K_{\mathrm{AUSF}} = \mathrm{KDF}\big(\mathrm{CK} \,\|\, \mathrm{IK},\ \mathrm{SNN},\ \mathrm{SQN} \oplus \mathrm{AK}\big), \tag{1c}$$

followed by

$$K_{\text{SEAF}} = \text{KDF}(K_{\text{AUSF}}, \text{SNN}), \qquad K_{\text{AMF}} = \text{KDF}(K_{\text{SEAF}}, \text{SUPI}, \text{ABBA}),$$
$$K_{\text{gNB}} = \text{KDF}(K_{\text{AMF}}, \text{UL NAS COUNT}, \ldots).$$

The AUSF stores XRES* and gives the SEAF only HXRES*: the SEAF can check whether the hash of RAND and RES* computed as in Eq. (1b) matches, but cannot compute RES* itself, so it must forward RES* to the home network for the final decision.

5.3 PDU session establishment

The UE sends a PDU Session Establishment Request (with a new PDU session ID, DNN, S-NSSAI, session type and SSC mode) inside an UL NAS Transport message (Figure 2). The AMF selects an SMF (NRF discovery by DNN and slice) and calls Nsmf_PDUSession_CreateSMContext. The SMF retrieves session-management subscription data from the UDM, creates an SM policy association with the PCF (which returns PCC rules), selects a UPF and establishes the N4 session (PDRs, FARs, QERs, URRs), obtaining the uplink tunnel endpoint. It then calls Namf_Communication_N1N2MessageTransfer with the NAS accept (QoS rules, IP address) for the UE and the N2 resource set-up information (QoS flows and profiles, UPF tunnel endpoint) for the gNB. The gNB configures data radio bearers with RRC Reconfiguration and returns its downlink tunnel endpoint, which the SMF installs in the UPF with an N4 modification. QoS flows are now active in both directions.

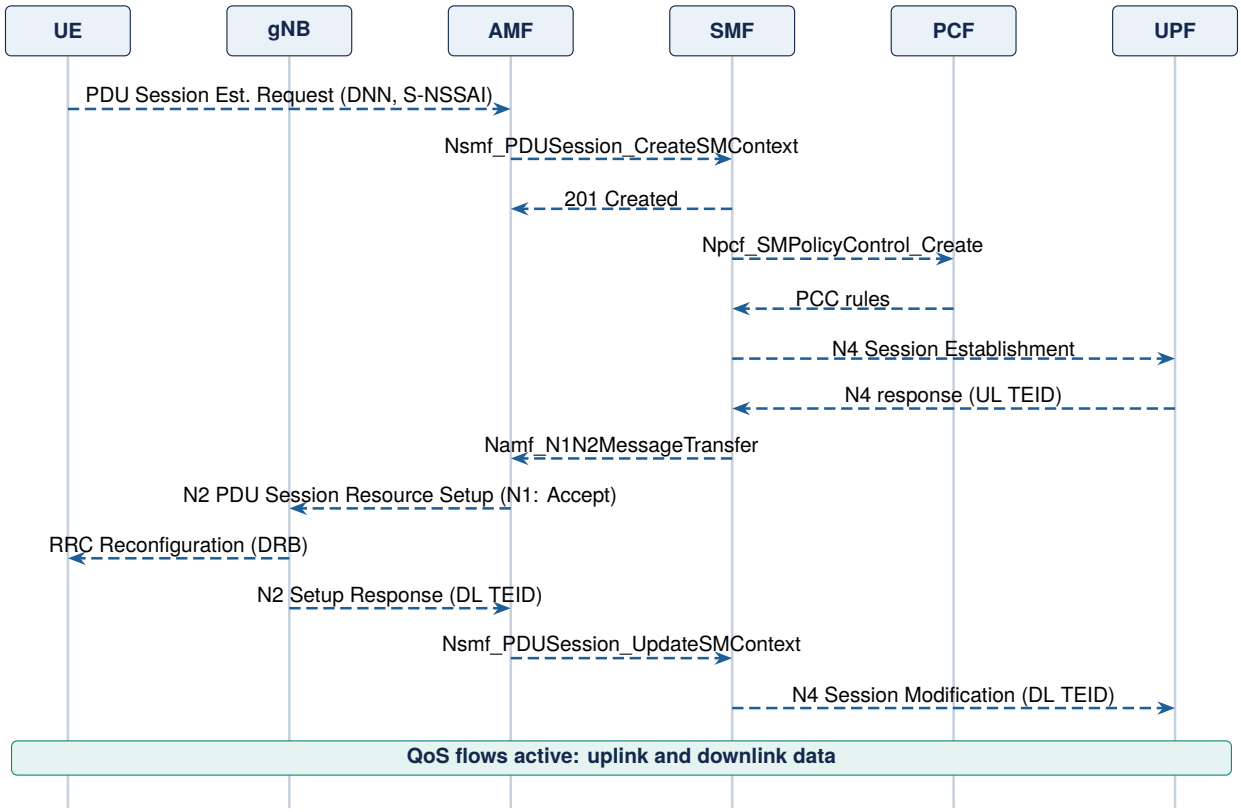


Figure 2. PDU session establishment (simplified) [6].

5.4 Service request and paging

A UE in CM-IDLE with uplink data sends a Service Request (with its 5G-S-TMSI) in RRC set-up; the AMF re-activates the user plane of the selected PDU sessions through the SMFs (UpdateSM-Context, N4 modification) and the RAN (N2 Initial Context Setup). For downlink data, the UPF buffers (or the SMF buffers, by FAR configuration) and notifies the SMF over N4; the SMF calls N1N2MessageTransfer; the AMF pages the UE in its registration area; the UE answers with a Service Request. A UE in RRC_INACTIVE skips most of this: the core context and N3 tunnels are kept, the RAN pages within its notification area and resumes the RRC connection locally, so no core signalling is needed unless the UE has moved out of the area.

5.5 Handover

With an Xn interface between gNBs, the source decides, prepares the target over Xn, commands the UE, forwards buffered data; the target then sends an N2 Path Switch Request so that the AMF and SMF move the N3 tunnel to the target, with end markers on the old path (Figure 3). Without Xn (or when the AMF or UPF must change), an *N2 handover* is prepared through the AMF(s), which also handles AMF relocation and inter-system handover to LTE.

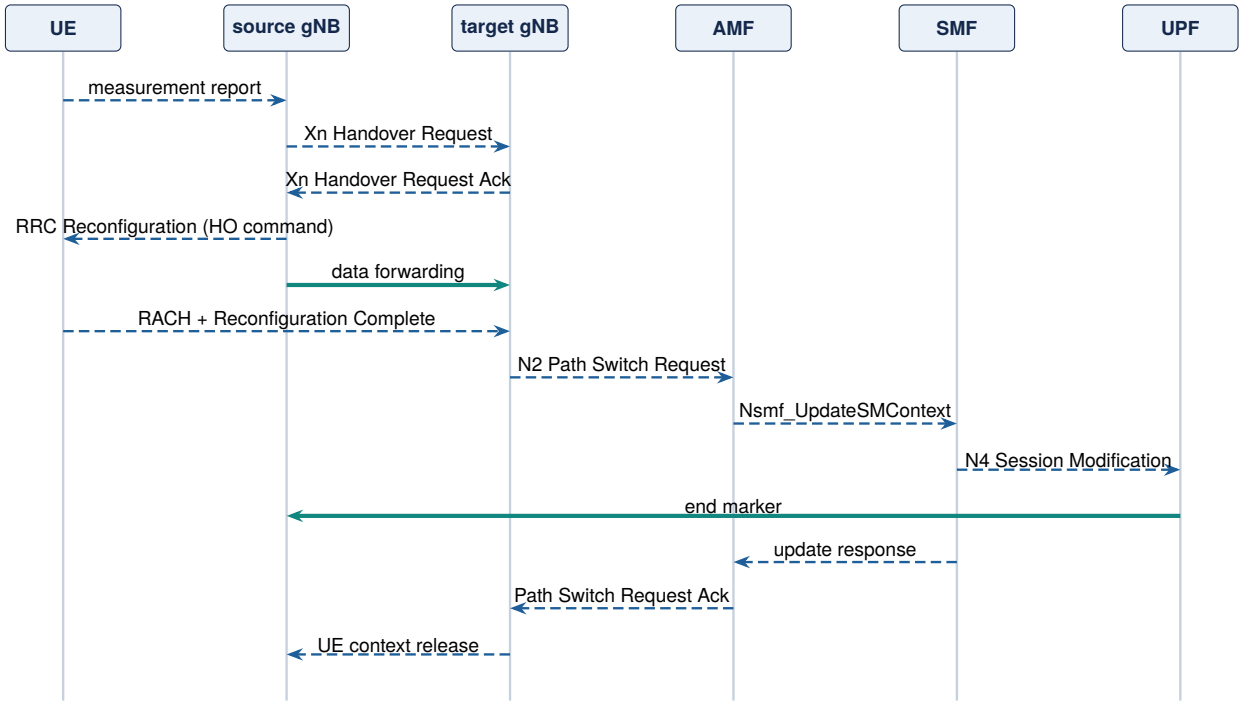


Figure 3. Xn handover with path switch (simplified).

6 Mathematical models

6.1 Procedure latency

Model. A procedure is a chain of sequential exchanges; parallel branches count only once. With n_{N} UE–AMF round trips of duration T_{N} (radio and RAN included) and n_{S} sequential core exchanges

of duration T_S,

$$T_{\text{proc}} \approx n_N\, T_N + n_S\, T_S. \tag{2}$$

Example. Initial registration (Figure 1) has about 6 UE–AMF round trips (registration, authentication, security mode, accept plus RRC) and 8 sequential SBI exchanges. With $T_N = 15\,\text{ms}$ and $T_S = 3\,\text{ms}$: $T_{\text{proc}} \approx 90 + 24 = 114\,\text{ms}$. The radio round trips dominate; core optimisation helps mainly when NFs are far apart or overloaded.

6.2 Signalling load at the AMF

Model. If procedure type k occurs r_k times per UE per hour and costs m_k messages at the AMF (NAS, NGAP and SBI counted), an AMF serving N UEs handles

$$\Lambda = \frac{N}{3600} \sum_k r_k\, m_k \quad \text{messages/s}. \tag{3}$$

Example. Illustrative values for $N = 10^6$: registrations 0.5/h at 14 messages, service requests 20/h at 6, Xn handovers 2/h at 4, PDU session establishments 0.2/h at 10: $\Lambda = 10^6 \times 137/3600 = 3.8 \times 10^4$ messages/s, of which service requests make up 88 % (Figure 4). If 70 % of idle-to-active transitions use RRC_INACTIVE (no core messages), Λ falls to 1.5×10^4 messages/s.

Interpretation. Frequent short sessions – smartphones and IoT devices waking for a few packets – dominate core signalling; RRC_INACTIVE is primarily a signalling-reduction feature.

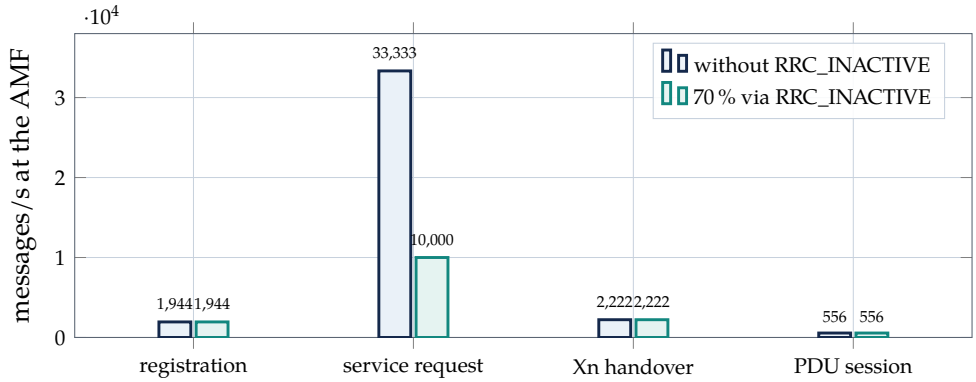


Figure 4. AMF message load per procedure type for the illustrative model of Section 6.2.

6.3 Forwarding buffer during handover

Model. Between the moment the source stops transmitting and the moment the path switch is complete, downlink data at rate R are forwarded over Xn and buffered at the target until the UE is ready. With interruption T_s,

$$B = R\, T_s. \tag{4}$$

Example. $R = 100\,\text{Mbit/s}$, $T_s = 20\,\text{ms}$: $B = 2\,\text{Mbit} = 250\,\text{kB}$ per UE. For an eMBB cell with several fast users the target needs megabytes of buffer; for URLLC, dual active protocol stack handover avoids the interruption altogether.

6.4 Authentication security

Model. An attacker without K who guesses RES* (128 bits) succeeds with probability 2^{-128} per attempt; one who has observed HXRES* must invert a 128-bit truncated hash. The expected number of attempts for a brute-force search over b bits is

$$\mathbb{E}[A] = 2^{b-1}. \tag{5}$$

Example. $b = 128$: 1.7×10^{38} attempts – infeasible; the security of 5G-AKA rests on the secrecy of K and the freshness of SQN, not on the length of the verification values.

7 Network architecture and diagrams

The figures of this lecture are *message sequence charts*: vertical lifelines for UE, RAN and NFs, horizontal arrows for messages. They abstract from transport (SCTP for NGAP, HTTP/2 for SBI, UDP for PFCP), from intermediate SCPs and from parallelism. In a real trace the same procedure appears as dozens of packets on several interfaces; reading traces efficiently means mapping them back to these charts.

8 Step-by-step procedures

Procedure 1 • Initial registration

1. UE → AMF: Registration Request (SUCI, requested NSSAI, capabilities) via RRC set-up and NGAP Initial UE Message.
2. AMF → AUSF → UDM: authentication request; the UDM de-conceals the SUCI and returns a 5G home-environment vector.
3. AMF (SEAF) → UE: Authentication Request; UE verifies AUTN and returns RES*; SEAF checks HXRES*; AUSF checks RES* and returns SUPI and K_{SEAF}.
4. NAS Security Mode Command/Complete; the AMF derives K_{AMF} and NAS keys.
5. AMF registers with the UDM and retrieves subscription data; creates the AM policy association; selects slices.
6. Registration Accept (5G-GUTI, registration area, allowed NSSAI); Registration Complete.

Procedure 2 • PDU session establishment

1. UE → AMF: PDU Session Establishment Request (DNN, S-NSSAI, type, SSC mode) in UL NAS Transport.
2. AMF selects an SMF and calls CreateSMContext; SMF fetches SM subscription, creates the SM policy association.
3. SMF selects a UPF and sends N4 Session Establishment with the rules; receives the UL tunnel endpoint.

4. SMF → AMF: N1N2MessageTransfer (N1: Accept with QoS rules and IP address; N2: QoS flows, UL TEID).
5. AMF → gNB: PDU Session Resource Setup; gNB configures DRBs and returns the DL TEID.
6. AMF → SMF: UpdateSMContext; SMF → UPF: N4 modification with the DL TEID.

9 Worked examples

Worked example 1 • Who checks what in 5G-AKA

Problem. A visited network's AMF receives RES* from the UE. Explain which checks it can perform and why it needs the AUSF.
Solution. The SEAF has HXRES* and RAND; it computes SHA-256(RAND ‖ RES*) (truncated) and compares with HXRES* (Eq. (1b)) – this filters wrong responses locally. It cannot derive XRES* (it lacks CK, IK) and does not receive the SUPI or K_{SEAF} until the AUSF has compared RES* with XRES*. The home network thus confirms the authentication.

Worked example 2 • Time to first packet

Problem. After power-on, a UE registers and then establishes a PDU session. The PDU session has 3 UE–AMF round trips (request, RRC reconfiguration with accept) and 6 sequential SBI/N4 exchanges. With the values of Section 6.1, when can the first packet flow?
Solution. Registration 114 ms; PDU session $3 \times 15 + 6 \times 3 = 63$ ms; total about 180 ms (plus cell search and system information acquisition, typically longer). Running the PDU session request piggybacked on registration (allowed for some cases) or pre-establishing sessions shortens it.

Worked example 3 • Dimensioning with RRC_INACTIVE

Problem. An AMF instance handles 2×10^4 messages/s. How many instances are needed for the load of Section 6.2 with and without RRC_INACTIVE, at 70 % target utilisation?
Solution. Without: $3.8 \times 10^4/(0.7 \times 2 \times 10^4) = 2.7 \rightarrow 3$ instances; with: $1.47 \times 10^4/1.4 \times 10^4 = 1.05 \rightarrow 2$ instances (plus redundancy in both cases).

10 Technology comparison

Always-on versus on-demand. The EPS attach always created a default bearer ("always-on IP"); 5G separates registration from session establishment. A device can register (e.g. for SMS or for a later session) without consuming user-plane resources, and sessions are created per DNN and slice as needed.

11 Key takeaways

Table 1. EPS and 5GS procedures compared.

	EPS	5GS
Attach / registration	Attach always creates a default bearer	Registration without session; PDU sessions separately
Authentication	EPS-AKA, decision in serving network	5G-AKA or EAP-AKA′, home confirmation
Identity	IMSI may be sent in clear	SUCI (concealed)
Session set-up	MME → S-GW → P-GW (GTPv2-C)	AMF relays; SMF drives PCF, UPF (SBI, PFCP)
Idle optimisation	ECM-IDLE only	CM-IDLE and RRC_INACTIVE
Handover path switch	MME → S-GW (S11)	AMF → SMF → UPF (SBI, N4)

Key takeaways

- Registration = identification (SUCI/5G-GUTI) + 5G-AKA + NAS SMC + UDM registration and subscription + policy + accept.
- 5G-AKA binds keys to the serving network name; SEAF checks HXRES*, the home AUSF confirms RES*.
- PDU session: AMF relays, SMF orchestrates PCF and UPF, N1N2 transfer delivers NAS accept and N2 set-up together.
- Service requests dominate AMF load; RRC_INACTIVE removes most of them from the core.
- Latency $\approx n_N T_N + n_S T_S$; load $= (N/3600)\sum r_k m_k$; forwarding buffer $= RT_s$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Order of steps

Why must NAS security be activated before the AMF sends the Registration Accept? What could an attacker do otherwise?

Exercise 2 • Level 1 – conceptual • N1N2 message transfer

Explain why the SMF sends both the NAS accept and the N2 resource information to the AMF in one service operation rather than in two.

Exercise 3 • Level 2 – numerical • Latency

A mobility registration update needs 2 UE–AMF round trips and 4 sequential SBI exchanges (context transfer from the old AMF, UDM update, policy update, subscription retrieval). Compute its latency with $T_N = 15\,\text{ms}$ and $T_S = 3\,\text{ms}$, and with $T_S = 12\,\text{ms}$ if the old AMF is in a distant data centre.

Exercise 4 • Level 2 – numerical • Forwarding buffer

A cell has 5 users each receiving 300 Mbit/s when a train carries them all into a neighbouring cell simultaneously; the interruption is 30 ms. How much buffer does the target gNB need? What would reduce it?

Exercise 5 • Level 3 – analytical • IoT signalling

10^6 IoT devices per AMF send one small report every 15 minutes. Each report needs a service request (6 AMF messages) if the device returns to CM-IDLE, or none if it stays in RRC_INACTIVE (resumed in the RAN). Compute the AMF load in both cases, and discuss what RRC_INACTIVE costs the RAN instead.

Exercise 6 • Level 3 – analytical • Authentication in roaming

A visited network claims to have served a roaming subscriber for 2 hours and bills the home operator. With EPS-AKA and with 5G-AKA, what evidence does the home network have that the subscriber was really authenticated in the visited network? Relate your answer to Eq. (1).

Solutions

Solution to Exercise 1

The Registration Accept carries the new 5G-GUTI, the allowed slices and the registration area. Without integrity protection an attacker could modify these (e.g. restrict the UE's services or force it to reveal itself), and without ciphering the GUTI–SUCI link would be visible, undermining identity privacy. The security mode procedure also confirms the UE's security capabilities, preventing bidding-down attacks.

Solution to Exercise 2

The NAS accept and the RAN resource set-up belong together: the RAN must create radio bearers for the QoS flows the UE is told about, and the AMF can deliver both in one NGAP message (PDU Session Resource Setup with the NAS PDU inside). One operation keeps them consistent and saves a round trip; if the RAN rejects the resources, the AMF can report the failure to the SMF without the UE ever seeing an inconsistent accept.

Solution to Exercise 3

$2 \times 15 + 4 \times 3 = 42$ ms; with $T_S = 12$ ms: $30 + 48 = 78$ ms. Distant core NFs can dominate procedures with many sequential exchanges – an argument for locality-aware NF selection.

Solution to Exercise 4

$B = 5 \times 300 \times 10^6 \times 0.03/8 = 5.6$ MB. Reductions: shorter interruption (RACH-less or conditional handover, prepared target), dual active protocol stack (no interruption), or reducing the forwarded volume by letting TCP slow down; on trains, a moving relay or group handover helps.

Solution to Exercise 5

Reports: $10^6/900 = 1111\,\mathrm{s}^{-1}$. CM-IDLE: $1111 \times 6 = 6667$ AMF messages/s; RRC_INACTIVE: none for the reports (only periodic RAN notification area updates, also handled in the RAN). The RAN must keep the UE contexts and security state for all inactive UEs (memory) and handle RAN paging and resume; when a UE moves out of the RAN notification area, a RAN-based update is needed.

Solution to Exercise 6

With EPS-AKA the serving network compares RES with XRES itself; the home network only knows that it sent authentication vectors – it cannot prove the UE responded. With 5G-AKA the AUSF receives RES* and compares it with XRES* (Eq. (1a)); RES* can only be computed by a USIM holding K in response to that RAND and bound to that serving network name. The home network thus has cryptographic evidence that the subscriber was authenticated in that network at that time.

13 Further reading

The procedures are specified in TS 23.502 [6], with NAS in TS 24.501 [4], NGAP in TS 38.413 [3] and security (5G-AKA, key hierarchy, SUCI) in TS 33.501 [7]; the SBI operations used are in TS 29.500 [2] and the NRF in TS 29.510 [1]. RRC_INACTIVE and handover on the RAN side are in TS 38.300 [5]. Rommer et al. [8] walk through the procedures in detail.

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[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.

21

Lecture 21

Network Slicing

برش‌بندی شبکه

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 2 (Erlang B), 16, 19 and 20; normal distribution

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-21-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Define** a network slice and **decode** the S-NSSAI and the NSSAI sets used in signalling.
- **LO2** **Trace** slice selection at registration and slice-specific PDU session set-up.
- **LO3** **Explain** how slices are realised in the RAN, transport and core, and the isolation options.
- **LO4** **Describe** slice management: lifecycle, templates and the roles of operator and tenant.
- **LO5** **Quantify** the statistical-multiplexing cost of isolation, RAN resource partitions and slice admission control.

2 Prerequisite knowledge

You need the 5GC architecture and procedures (Lectures 19–20), the virtualisation concepts of Lecture 16, and the Erlang-B formula of Lecture 2. The multiplexing model uses the normal distribution: for a Gaussian load with mean μ and standard deviation σ, the capacity exceeded with probability ϵ is $\mu + z_{1-\epsilon}\sigma$ ($z_{0.99} = 2.326$).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Network slice	برش شبکه	Logical network providing specific capabilities and characteristics.
S-NSSAI	شناسهٔ برش	Single network slice selection assistance information (SST + SD).
SST / SD	نوع برش / متمایزکننده	Slice/service type and optional slice differentiator.
NSSAI	مجموعهٔ S-NSSAIها	Configured, requested, allowed and rejected sets.
NSSF	تابع انتخاب برش	NF that selects allowed slices and the AMF set.
NSSAA	احراز هویت ویژهٔ برش	Slice-specific authentication by a third-party AAA server.
NSAC	کنترل پذیرش برش	Limits on UEs or PDU sessions per slice.
Isolation	جداسازی	Degree to which one slice is protected from another.
Tenant	مستأجر	Customer (e.g. enterprise) for whom a slice is operated.
Slice template	الگوی برش	Standardised description of slice attributes (e.g. latency, throughput).

3 Introduction and motivation

The three IMT-2020 scenarios (Lecture 14) make conflicting demands: an eMBB service wants throughput and tolerates delay, a URLLC service wants guaranteed latency and reliability, an mMTC service wants huge numbers of cheap, rarely active connections. Building three separate networks would be uneconomic; running all on one network with best-effort sharing would fail the most demanding customers. *Network slicing* runs several logical networks on one physical infrastructure, each with its own capabilities, policies and, where needed, its own network functions [5], [6].

Slicing is also a business concept: an operator can offer a slice to an enterprise, a public-safety agency or a vertical industry as a product with guaranteed characteristics, managed through standard interfaces [1]. It is made possible by the technologies of Part III: virtualisation creates dedicated instances cheaply, SDN programs the transport, and the SBA lets NFs be selected per slice.

4 Fundamental concepts

4.1 Identifying slices

A slice is identified by an *S-NSSAI* (Figure 1): an 8-bit *slice/service type* (SST) and an optional 24-bit *slice differentiator* (SD) that distinguishes several slices of the same type (e.g. two enterprises' URLLC slices) [2]. Standardised SST values include 1 (eMBB), 2 (URLLC), 3 (massive IoT) and 4 (V2X), with further values added in later releases; operators may also define their own. A UE handles sets of S-NSSAIs:

- **configured NSSAI**: provisioned in the UE per PLMN;
- **requested NSSAI**: sent in the Registration Request;
- **allowed NSSAI**: returned by the network for the registration area (at most 8 S-NSSAIs);
- **rejected NSSAI**: requested slices refused, with a cause (e.g. not available in this area).

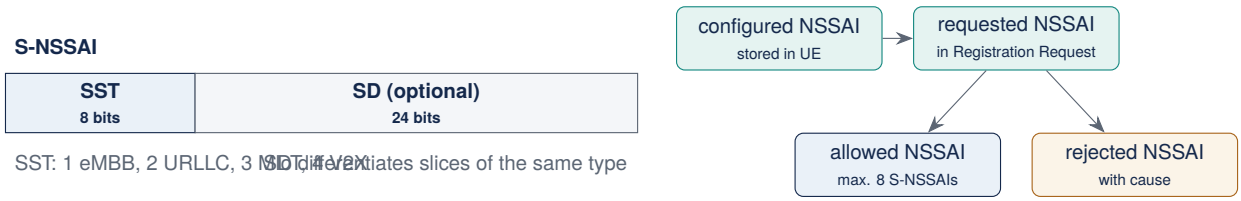


Figure 1. S-NSSAI structure and the NSSAI sets exchanged at registration.

4.2 What a slice consists of

An end-to-end slice spans the RAN, the transport network and the core (Figure 2). Some functions are typically shared (AMF, NSSF, NRF, UDM, the RAN nodes), others may be slice-specific (SMF, UPF, PCF, application functions). A UE can use up to eight slices simultaneously, but all through the same AMF; each PDU session belongs to exactly one slice.

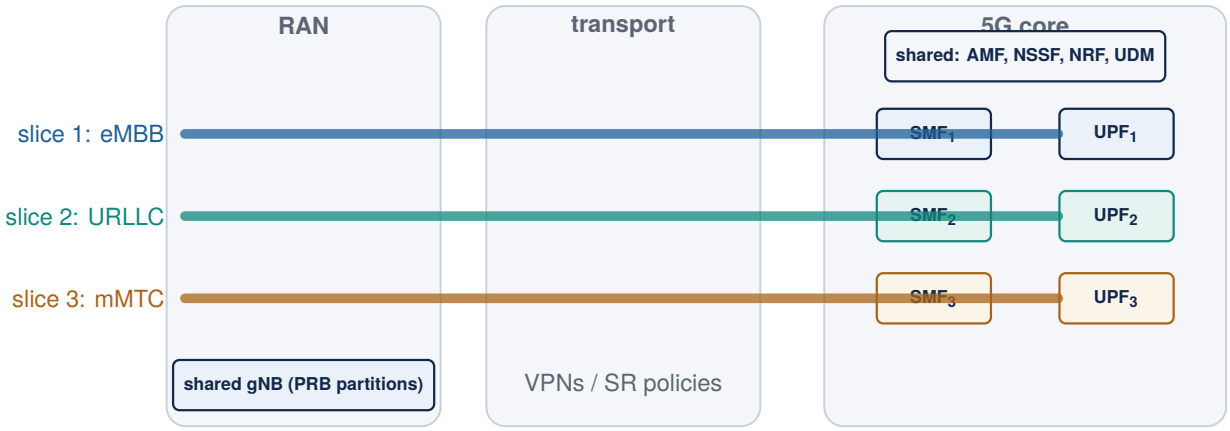


Figure 2. Three end-to-end slices sharing RAN and common control functions, with slice-specific SMFs and UPFs.

4.3 Isolation

Isolation can be achieved at several levels: *logical* (separate policies and QoS on shared NFs), *virtual* (dedicated NF instances on shared infrastructure) or *physical* (dedicated hardware, spectrum or sites). Stronger isolation protects a slice from others' load, faults and security incidents, but forfeits the statistical-multiplexing gain of sharing (Section 6.1). The right level depends on the service-level agreement: a public-safety slice may need dedicated UPFs and reserved radio resources, while a consumer video slice can share almost everything.

5 Detailed technical explanation

5.1 Slice selection

At registration the AMF checks the requested NSSAI against the subscribed S-NSSAIs from the UDM and, if necessary, asks the NSSF, which returns the allowed NSSAI for the UE's location and possibly a different AMF set better suited to the slices (Figure 3). The AMF may then re-route the registration to an AMF of that set. When the UE requests a PDU session for an S-NSSAI and DNN, the AMF discovers an SMF serving that slice through the NRF, and the SMF selects a UPF of the slice. Slice information is also given to the RAN, which may use it for resource allocation and mobility decisions.

5.2 Slice-specific authentication and admission control

Release 16 added *network slice-specific authentication and authorisation* (NSSAA): after primary authentication, access to a slice can additionally depend on credentials verified by an AAA server of the slice tenant (e.g. an enterprise), so that the tenant controls which devices use its slice. Release 17 added *network slice admission control* (NSAC): the network can limit the number of registered UEs and of PDU sessions per slice, enforcing the capacity sold to a tenant [5].

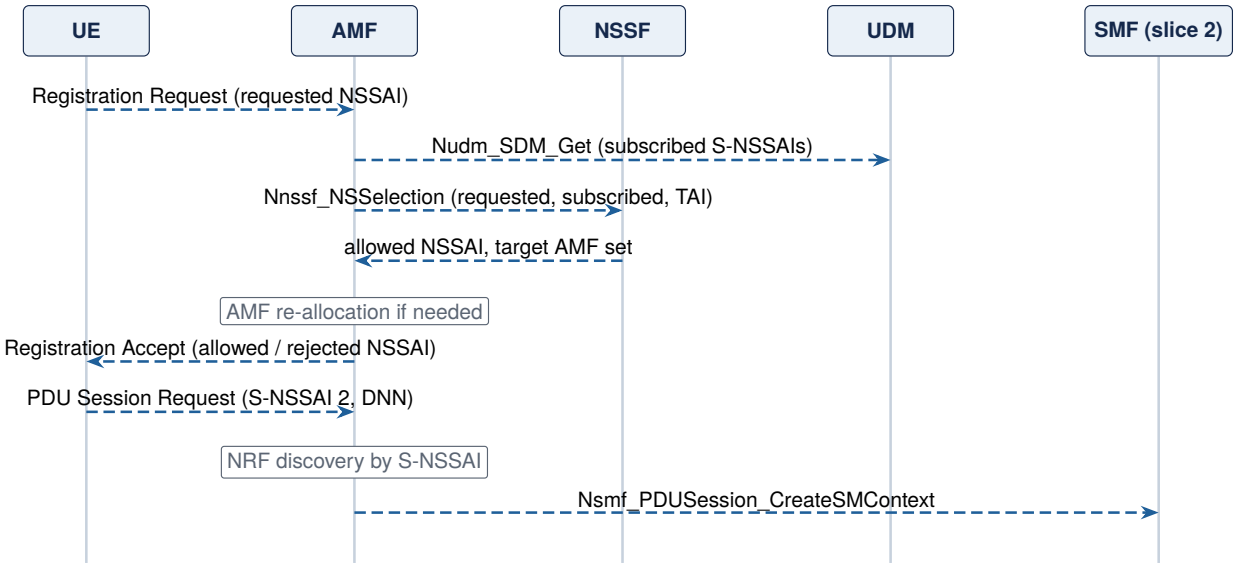


Figure 3. Slice selection at registration and slice-specific PDU session set-up (simplified).

5.3 Slicing in the RAN and transport

The RAN is shared; slices are realised by scheduling policies: minimum, maximum and dedicated shares of physical resource blocks per slice, slice-aware admission control and priority, and possibly separate bandwidth parts or carriers. The transport network maps slices to virtual private networks, segment-routing policies or hard-isolated channels with guaranteed bandwidth and latency. Without these, a core slice alone guarantees nothing end to end.

5.4 Slice management

3GPP management specifications describe slices as managed objects: a *network slice* composed of *network slice subnets* (RAN, transport, core), with a lifecycle of preparation, commissioning, operation and decommissioning [1]. Customers' requirements are expressed with slice templates – for example the GSMA's generic network slice template lists attributes such as throughput, latency, availability, number of devices and isolation – which the operator's orchestration translates into NF instances, policies and RAN and transport configuration, using NFV MANO (Lecture 16) and SDN controllers (Lecture 15).

6 Mathematical models

6.1 The cost of isolation

Model. Slice i has a Gaussian load with mean μ_i and standard deviation σ_i, independent of the others. To keep the probability of overload below ϵ, dedicated capacity per slice and shared capacity for all slices are

$$C_i = \mu_i + z\,\sigma_i, \qquad C_{\text{shared}} = \sum_i \mu_i + z\sqrt{\sum_i \sigma_i^2}, \tag{1}$$

with $z = z_{1-\epsilon}$. The *multiplexing gain* of sharing is

$$G = \frac{\sum_i C_i}{C_{\text{shared}}} \geq 1. \tag{2}$$

Example. Three slices with $\mu = 100$, $\sigma = 30$ (in any capacity unit), $\epsilon = 1\,\%$ ($z = 2.326$): dedicated $3 \times 169.8 = 509.3$, shared $300 + 2.326\sqrt{3} \times 30 = 420.9$; $G = 1.21$ – isolation costs 21 % more capacity (Figure 4).

Interpretation. The gain grows with the number of slices and with the burstiness σ/μ; it vanishes for constant loads. Isolation is therefore expensive exactly for bursty services – the economic argument for sharing what can be shared and isolating only what the SLA requires. Correlated loads (e.g. all slices peaking in the same hour) reduce the gain.

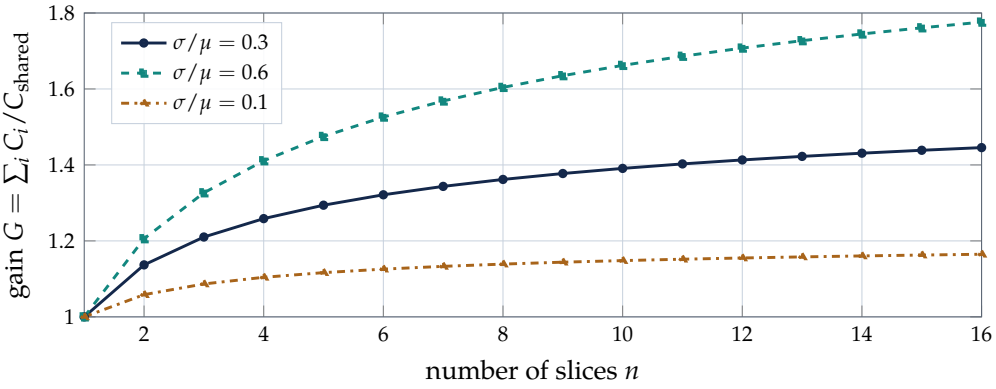


Figure 4. Multiplexing gain of a shared pool versus dedicated slice capacity (Eq. (2), $\epsilon = 1\,\%$).

6.2 RAN resource partitions

Model. A cell has N resource blocks; slice i is guaranteed a minimum share ρ_i and, with a robust modulation and coding scheme, a rate r_i per resource block. Its guaranteed throughput is

$$R_i^{\min} = \lfloor \rho_i N \rfloor \, r_i, \tag{3}$$

while unused guaranteed resources may be lent to other slices unless they are *dedicated*.

Example. $N = 273$ (100 MHz at 30 kHz), URLLC minimum 20 % (54 resource blocks), $r = 0.5$ Mbit/s per block: $R^{\min} = 27$ Mbit/s, sufficient for many control flows, with the rest of the cell available to eMBB when URLLC is idle.

6.3 Slice admission control

Model. If a tenant's slice is limited to c simultaneous PDU sessions and sessions arrive as a Poisson stream with offered load A Erlang, the probability that a new session is rejected is the Erlang-B formula of Lecture 2:

$$B(A, c) = \frac{A^c / c!}{\sum_{k=0}^{c} A^k / k!}. \tag{4}$$

Example. $A = 40$ E: with $c = 50$, $B = 1.9\,\%$; with $c = 60$, $B = 0.07\,\%$. A quota of $c = 53$ is the smallest giving $B \leq 1\,\%$. The quota is part of the contract: it protects other slices and defines what the tenant pays for.

7 Network architecture and diagrams

Figure 2 shows a common deployment pattern: shared RAN with per-slice scheduling policies, transport VPNs per slice, shared AMF and subscriber data, and per-slice SMF and UPF – the URLLC slice's UPF placed at the edge, the eMBB slice's UPFs centrally. An enterprise slice may additionally have a UPF on the enterprise premises (Lecture 19) and NSSAA towards the enterprise's AAA server. The management system keeps the mapping from each slice to its subnets and resources.

8 Step-by-step procedures

Procedure 1 • Registration with slice selection

1. The UE sends a Registration Request with the requested NSSAI (from its configured NSSAI).
2. The AMF retrieves the subscribed S-NSSAIs from the UDM.
3. If it cannot decide alone, the AMF queries the NSSF with requested and subscribed NSSAI and the UE's TAI; the NSSF returns the allowed NSSAI and possibly a target AMF set.
4. If another AMF set is needed, the registration is re-routed; slices requiring NSSAA are authorised with the tenant's AAA server.
5. Registration Accept returns allowed and rejected NSSAI; the UE uses only allowed S-NSSAIs in PDU session requests.

Procedure 2 • Commissioning an enterprise slice

1. The tenant's requirements are captured in a slice template (throughput, latency, devices, area, isolation, availability).
2. The operator's management system decomposes the slice into RAN, transport and core subnets and checks feasibility.
3. NFV orchestration instantiates slice-specific SMF/UPF (e.g. an on-premises UPF); SDN controllers set up transport paths; RAN policies reserve resource shares in the covered cells.
4. Subscriptions of the tenant's devices are updated with the new S-NSSAI; the NSSF configuration and NSAC quotas are updated.
5. The slice is activated and its performance is monitored against the SLA.

9 Worked examples

Worked example 1 • Decoding S-NSSAIs

Problem. A UE's allowed NSSAI contains (SST 1), (SST 2, SD `0x00A1B2`) and (SST 2, SD `0x00C3D4`). Interpret it.
Solution. One default eMBB slice without differentiator and two URLLC slices distinguished by their SDs – for example two different enterprise or industrial customers. The UE can open PDU sessions in all three slices simultaneously through the same AMF.

Worked example 2 • Isolation cost

Problem. Using Eq. (1), how much capacity does an operator save by sharing UPF capacity among the three slices of Section 6.1 instead of dedicating it, and what does the URLLC tenant risk?
Solution. $509.3 - 420.9 = 88.4$ units (17 % of the dedicated total). The URLLC slice then competes with the others during joint peaks; if its SLA requires guaranteed performance independent of others' traffic, its share must be reserved (logical isolation with priority) or dedicated (virtual isolation).

Worked example 3 • Dimensioning a quota

Problem. A logistics company expects 40 E of simultaneous sessions in its slice and accepts 1 % blocking. What NSAC quota should be configured and how does the blocking change if its load grows by 12.5 %?
Solution. $c = 53$ (Section 6.3). At $A = 45$ E with $c = 50$ the blocking would be 5.4 %; with $c = 53$ it is 3.1 % – still well above 1 % – the quota (and price) must be renegotiated as the tenant grows.

10 Technology comparison

Table 1. Ways of differentiating service in mobile networks.

	QoS classes (4G/5G)	Dedicated core (DECOR, 4G)	Network slicing (5G)
Granularity	Per flow/bearer	Per UE (core selection)	Per PDU session, end to end
Scope	Packet treatment	Core NFs	RAN, transport, core, management
Isolation	None (priorities)	Core instances	Logical to physical
Per-UE use	Many flows	One dedicated core	Up to 8 slices
Customer view	Not exposed	Not exposed	Product with SLA, tenant management

Slicing and QoS. Slices and QoS flows (Lecture 22) are complementary: a slice defines *which* network and resources serve a session; QoS flows define how packets *within* it are treated. A

URLLC slice still uses QoS flows with appropriate 5QIs.

11 Key takeaways

Key takeaways

- A slice is an end-to-end logical network identified by S-NSSAI = SST (8 bits) + optional SD (24 bits).
- NSSAI sets: configured, requested, allowed (max. 8), rejected; NSSF selects slices and AMF sets.
- Slices combine shared and dedicated NFs; RAN and transport must also be sliced for end-to-end guarantees.
- Isolation costs multiplexing gain: $G = \sum(\mu_i + z\sigma_i) / (\sum \mu_i + z\sqrt{\sum \sigma_i^2})$.
- RAN partitions give $R_i^{\min} = \lfloor \rho_i N \rfloor r_i$; NSAC quotas are dimensioned with Erlang B.
- NSSAA and NSAC let tenants control access and operators enforce contracted capacity.

12 Exercises

Exercise 1 • Level 1 – conceptual • Slices or QoS?

For each requirement, say whether a separate slice or a QoS flow within an existing slice is the better tool: (a) prioritising video calls over downloads for consumers; (b) a police network that must keep working when the public network is congested; (c) a factory whose data must not leave its premises.

Exercise 2 • Level 1 – conceptual • Isolation levels

Give an example of logical, virtual and physical isolation in the core and in the RAN.

Exercise 3 • Level 2 – numerical • Bursty slices

Repeat the calculation of Section 6.1 with $\sigma = 60$. What is the gain, and what does it mean for slices carrying bursty traffic?

Exercise 4 • Level 2 – numerical • RAN share

A URLLC slice is guaranteed 15 % of 273 resource blocks with a robust MCS of 0.2 Mbit/s per block. Compute its guaranteed rate. How many 64 kbit/s control streams fit, and what happens to unused blocks if the share is “minimum” rather than “dedicated”?

Exercise 5 • Level 3 – analytical • Quota design

Show that the smallest quota with $B \leq 1\,\%$ at $A = 40\,\mathrm{E}$ is $c = 53$ (use the recursion $B(A,k) = AB(A,k-1)/(k+AB(A,k-1))$). Then estimate the quota needed for $A = 80\,\mathrm{E}$ and compare the ratio c/A; explain the trend.

Exercise 6 • Level 3 – analytical • Designing a slice

A hospital wants a slice for 200 connected devices (monitors, imaging) with 10 ms latency, 99.99 % availability inside its campus, and data that must stay on site. Propose the S-NSSAI, the shared and dedicated components, the UPF placement, RAN measures and admission/authorisation features.

Solutions

Solution to Exercise 1

(a) QoS flows (different 5QIs) in the consumer slice. (b) A slice with reserved RAN resources, priority and preemption (and possibly dedicated core NFs) – isolation from public load is the requirement. (c) A slice with an on-premises UPF and local breakout (possibly NSSAA), since the requirement is about where the user plane terminates, not packet priority.

Solution to Exercise 2

Core: logical – shared SMF/UPF with per-slice policies; virtual – dedicated SMF/UPF instances on shared servers; physical – dedicated servers or site. RAN: logical – per-slice priority in a shared scheduler; virtual – reserved/dedicated PRB shares or a separate bandwidth part; physical – a dedicated carrier or dedicated small cells.

Solution to Exercise 3

Dedicated $3 \times (100 + 2.326 \times 60) = 718.7$; shared $300 + 2.326\sqrt{3} \times 60 = 541.7$; $G = 1.33$. The burstier the traffic, the more capacity isolation wastes: bursty best-effort slices should share; isolation should be reserved for slices whose SLA demands it.

Solution to Exercise 4

$\lfloor 0.15 \times 273 \rfloor = 40$ blocks (41 if rounded), $R^{\min} = 40 \times 0.2 = 8\,\mathrm{Mbit/s}$: 125 streams of 64 kbit/s (fewer with headroom). With a *minimum* share, unused blocks are lent to other slices and reclaimed when URLLC traffic appears (possible short scheduling delay); with a *dedicated* share they stay idle, giving the strongest guarantee at a capacity cost.

Solution to Exercise 5

The recursion gives $B(40,52) = 1.1\,\%$ and $B(40,53) = 0.82\,\%$, hence $c = 53$ ($c/A = 1.33$). For $A = 80\,\mathrm{E}$ the recursion gives $c = 96$ ($c/A = 1.2$): larger pools need relatively less headroom – the same economy of scale as in trunk dimensioning (Lecture 2) and the multiplexing gain of Section 6.1.

Solution to Exercise

S-NSSAI: SST 2 (URLLC) or an operator-specific SST, with an SD identifying the hospital. Shared: AMF, NSSF, UDM. Dedicated: SMF and a UPF on the hospital campus (local breakout, data on site), possibly redundant UPFs for 99.99 %. RAN: indoor small cells with a minimum or dedicated PRB share and slice priority; latency via 30 kHz numerology or mini-slots. Access: NSSAA with the hospital's AAA server; NSAC quota of about 200 UEs; transport path with guaranteed bandwidth. Monitoring of latency and availability against the SLA.

13 Further reading

Slicing in the 5G system is specified in TS 23.501 [5] (clause 5.15) with procedures in TS 23.502 [3]; S-NSSAI encoding in TS 23.003 [2]; management concepts in TS 28.530 [1]; service requirements in TS 22.261 [4]. Foukas et al. [6] survey slicing architectures and research questions; Rommer et al. [7] cover slicing in the 5G core.

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22

Lecture 22

Quality of Service (QoS) in 5G

کیفیت خدمات (QoS) در نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 7, 9, 13, 20 and 21; M/M/1 queue

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 5
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-22-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the 5G QoS model: QoS flows, QFI, QoS profiles, QoS rules and their mapping to radio bearers.
- **LO2** **Interpret** standardised 5QI characteristics (resource type, priority, packet delay budget, packet error rate, maximum data burst volume).
- **LO3** **Trace** how policy from the PCF becomes QoS flows enforced in UE, RAN and UPF.
- **LO4** **Compute** delay budgets, HARQ needs for a packet-error target and the effect of priority scheduling.
- **LO5** **Compare** the 5G flow-based model with LTE's bearer-based model.

2 Prerequisite knowledge

You need LTE scheduling and HARQ (Lecture 7), the EPS bearer and QCI model (Lecture 9), PFCP rules (Lecture 13), PDU session establishment (Lecture 20) and slicing (Lecture 21). The priority model extends the M/M/1 queue of Lecture 12.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
QoS flow	جریان کیفیت خدمات	Finest granularity of QoS differentiation in a PDU session.
QFI	شناسهٔ جریان QoS	6-bit identifier of a QoS flow, carried in N3 encapsulation.
5QI	شاخص QoS نسل پنجم	Pointer to a set of standardised QoS characteristics.
QoS profile	نمایهٔ QoS	Parameters given to the RAN per QoS flow (5QI, ARP, GFBR, MFBR, ...).
QoS rule	قاعدهٔ QoS	UE-side packet filters mapping uplink traffic to QoS flows.
GBR / non-GBR	با / بدون نرخ تضمین‌شده	Resource types; delay-critical GBR for URLLC.
PDB / PER	بودجهٔ تأخیر / نرخ خطای بسته	Upper bounds on delay and loss for a 5QI.
MDBV	بیشینهٔ حجم انفجار داده	Largest burst to be served within the PDB (delay-critical GBR).
SDAP	پروتکل سازگاری دادهٔ خدمت	RAN layer mapping QoS flows to data radio bearers.
Reflective QoS	بازتابی QoS	UE derives uplink QoS rules from downlink packets.

3 Introduction and motivation

Different applications need different treatment: a voice call needs low delay and tolerates some loss; a file transfer needs no loss but tolerates delay; a robot controller needs both low delay and extremely low loss; a background update needs neither. A mobile network cannot satisfy all of these with a single best-effort pipe, because the scarce resource – radio capacity – must be shared under load. QoS mechanisms decide which packets go first, which are guaranteed a rate, and which may be dropped.

LTE solved this with *bearers*: one per QoS class, each requiring signalling to set up along the whole path. 5G replaces bearers in the core with lightweight *QoS flows*, marked per packet and mapped flexibly to radio bearers by the RAN [5]. The resulting model is more scalable, supports URLLC with new delay-critical characteristics, and integrates with slicing (Lecture 21): a slice chooses the network, QoS flows the treatment inside it.

4 Fundamental concepts

4.1 QoS flows

A PDU session contains one or more QoS flows, each identified by a QFI (Figure 1). Every PDU session has a default QoS flow (non-GBR) that carries traffic not matching any other rule. Classification happens at the edges:

- **downlink**: the UPF classifies packets with PDRs (packet filters from PCC rules) and marks them with the QFI in the N3 encapsulation header;
- **uplink**: the UE classifies packets with *QoS rules* (packet filters with precedence) and the SDAP layer marks them;
- **RAN**: the gNB maps QoS flows to data radio bearers (DRBs) as it sees fit – several flows may share a DRB.

Because QFI marking is per packet, adding a QoS flow does not require a new tunnel: all flows of a PDU session share one N3 tunnel.

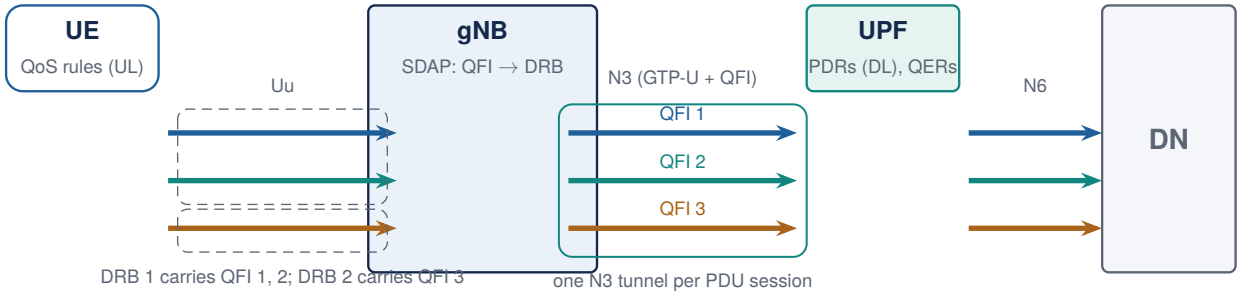


Figure 1. 5G QoS model: classification in UE and UPF, QFI marking on N3, flow-to-DRB mapping by SDAP in the gNB.

4.2 QoS parameters

Definition 1 • QoS profile and characteristics

The SMF gives the RAN a *QoS profile* per QoS flow: the 5QI, the allocation and retention priority (ARP), and for GBR flows the guaranteed and maximum flow bit rates (GFBR, MFBR), optionally notification control and a maximum packet loss rate. The 5QI points to *QoS characteristics*: resource type (non-GBR, GBR, delay-critical GBR), priority level, packet delay budget (PDB), packet error rate (PER), averaging window and, for delay-critical GBR, the maximum data burst volume (MDBV) [5].

Aggregate limits apply to non-GBR traffic: the *Session-AMBR* per PDU session (enforced by UPF and UE) and the *UE-AMBR* across a UE's sessions (enforced by the RAN).

4.3 Packet delay budget and packet error rate

The PDB is an upper bound on the delay between UE and the UPF's N6 termination, for a given confidence; it includes a core-network part and an access-network part. The PER is an upper bound on the rate of packets processed by the link layer but not delivered to the upper layer. For GBR flows, packets delayed beyond the PDB count as lost (for delay-critical GBR, they are counted in the PER), so PDB and PER together define the reliability target of a URLLC flow.

Table 1. Selected standardised 5QI values (TS 23.501, Table 5.7.4-1) [5]. Lower priority values mean higher priority.

5QI	Resource type	Priority	PDB	PER	Example services
1	GBR	20	100 ms	10^{-2}	Conversational voice
2	GBR	40	150 ms	10^{-3}	Conversational video (live)
3	GBR	30	50 ms	10^{-3}	Real-time gaming, V2X messages
4	GBR	50	300 ms	10^{-6}	Non-conversational video (buffered)
65	GBR	7	75 ms	10^{-2}	Mission-critical push-to-talk voice
5	Non-GBR	10	100 ms	10^{-6}	IMS signalling
6	Non-GBR	60	300 ms	10^{-6}	Buffered video, TCP applications
7	Non-GBR	70	100 ms	10^{-3}	Voice, live video, interactive gaming
9	Non-GBR	90	300 ms	10^{-6}	Default bearer traffic (web, e-mail)
82	Delay-critical GBR	19	10 ms	10^{-4}	Discrete automation (MDBV 255 bytes)
85	Delay-critical GBR	21	5 ms	10^{-5}	Electricity distribution, high voltage (MDBV 255 bytes)

5 Detailed technical explanation

5.1 From policy to QoS flows

QoS starts with policy (Figure 2). An application function (e.g. the IMS for a voice call) asks the PCF for treatment of a media flow; the PCF produces PCC rules with 5QI, ARP and bit rates and sends them to the SMF. The SMF binds each PCC rule to a QoS flow – an existing one with the same parameters or a new one with a new QFI – and installs: PDRs and QERs in the UPF (N4), the QoS profile in the gNB (N2), and the QoS rule in the UE (NAS). The gNB performs admission control for GBR flows and configures DRBs and the SDAP mapping.

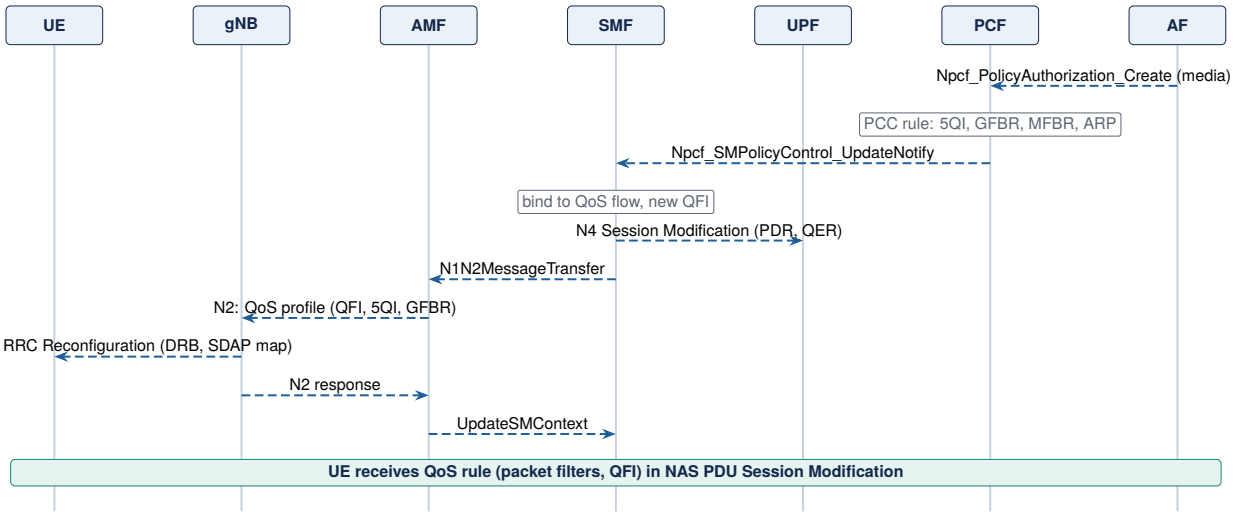


Figure 2. Establishing a GBR QoS flow triggered by an application function (simplified).

5.2 Reflective QoS

For applications with many short flows (e.g. web browsing with different priorities) signalling a QoS rule for each would be costly. With *reflective QoS* the UPF sets a reflective QoS indicator (RQI) in downlink packets; the UE then creates an uplink QoS rule mirroring the downlink packet's addresses and ports, with the same QFI, without NAS signalling.

5.3 Enforcement in the RAN

The gNB scheduler translates QoS profiles into scheduling decisions: priority levels order the queues, GFBR is guaranteed over the averaging window, PDB determines how long a packet may wait (and which numerology, mini-slots or configured grants are needed), and PER determines the target BLER and HARQ/repetition strategy. QoS notification control lets the gNB inform the SMF when a GFBR can no longer be guaranteed, so that the application can adapt (e.g. lower video resolution).

5.4 QoS monitoring

Since Release 16 the network can measure packet delay per QoS flow between UE, gNB and UPF and report it to the SMF and, via the NEF, to application functions – the basis for SLAs on URLLC slices and for applications that adapt to measured latency.

6 Mathematical models

6.1 Splitting the delay budget

Model. The PDB is shared between the core network (UPF to gNB, D_{CN}) and the access network (D_{AN}):

$$D_{\mathrm{AN}} = \mathrm{PDB} - D_{\mathrm{CN}}. \tag{1}$$

Example. 5QI 1 (100 ms) with an assumed core part of 20 ms leaves 80 ms for scheduling, HARQ and transmission – comfortable. 5QI 85 (5 ms) with a UPF 50 km away (0.25 ms one way in fibre plus processing, say 1 ms) leaves 4 ms: the UPF must be at the edge and the radio must use short slots.

6.2 Rate needed for a data burst

Model. For a delay-critical GBR flow, a burst of up to MDBV bytes must be delivered within the access-network budget. The instantaneous rate the scheduler must provide during the burst is

$$R_{\mathrm{burst}} \geq \frac{8\,\mathrm{MDBV}}{D_{\mathrm{AN}}}. \tag{2}$$

Example. 5QI 82 (MDBV 255 bytes, PDB 10 ms) with $D_{\mathrm{AN}} = 9$ ms: 227 kbit/s; 5QI 85 (255 bytes, PDB 5 ms) with $D_{\mathrm{AN}} = 4$ ms: 510 kbit/s. The rates are small; the challenge is delivering them *reliably* within milliseconds.

6.3 HARQ transmissions for a PER target

Model. If each transmission of a block fails independently with probability p (the target BLER), the residual error after n transmissions is p^n. To reach a target PER,

$$n \geq \frac{\ln \mathrm{PER}}{\ln p}, \tag{3}$$

and all n transmissions must fit into D_{AN}.

Example. PER $= 10^{-5}$: with $p = 0.1$, $n = 5$ transmissions (Figure 3); with a conservative MCS giving $p = 0.01$, $n = 3$. If one HARQ round trip takes 4 slots of 0.125 ms (120 kHz) = 0.5 ms, five transmissions need 2–2.5 ms – feasible within 4 ms; at 15 kHz (1 ms slots) they would need about 20 ms – impossible. URLLC therefore combines robust MCS (low p), short slots and sometimes blind repetitions.

Limitation. Errors are not independent when the channel is in a deep fade; diversity (frequency, antennas, multi-connectivity) is needed to make the independence assumption approximately valid.

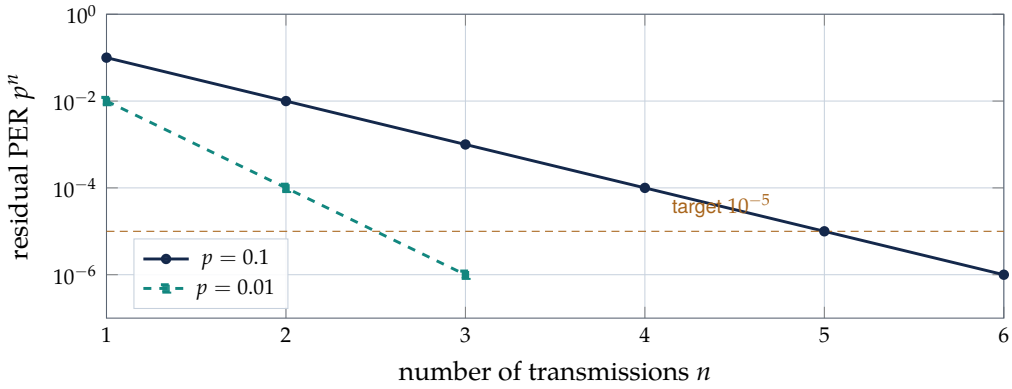


Figure 3. Residual packet error rate after n transmissions (Eq. (3)).

6.4 Priority scheduling

Model. Two classes share a link modelled as M/M/1 with service rate μ; class 1 (load ρ_1) has preemptive priority over class 2; total load ρ. The mean sojourn times are

$$T_1 = \frac{1}{\mu(1-\rho_1)}, \qquad T_2 = \frac{1}{\mu(1-\rho_1)(1-\rho)}, \tag{4}$$

compared with $T = 1/(\mu(1-\rho))$ for FIFO.

Example. $\mu = 1000$ packets/s, $\rho_1 = 0.2$: $T_1 = 1.25$ ms regardless of the total load; at $\rho = 0.9$, $T_2 = 12.5$ ms against 10 ms for FIFO (Figure 4).

Interpretation. Priority protects the high-priority class completely from the load of lower classes, at a moderate cost to the others – as long as the high-priority load itself stays small. This is why GBR and delay-critical flows need admission control: without it, too much "high-priority" traffic destroys the protection.

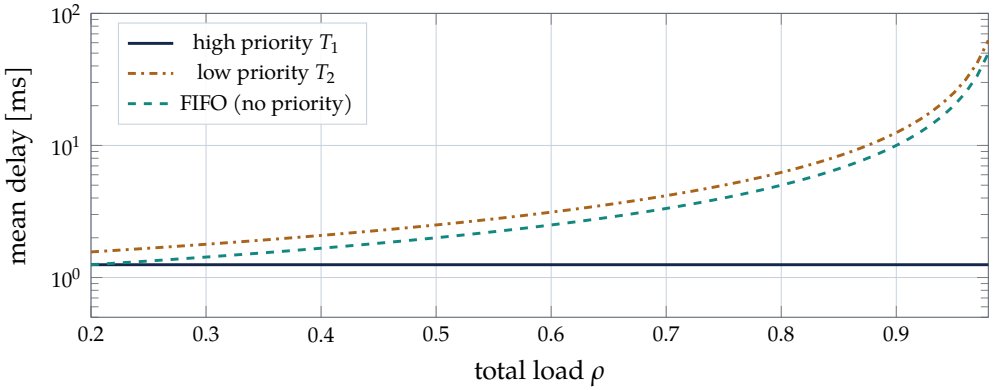


Figure 4. Mean delay with preemptive priority and with FIFO (Eq. (4), $\rho_1 = 0.2$, $\mu = 1000\,\mathrm{s}^{-1}$).

7 Network architecture and diagrams

QoS is enforced at three points: the UE (uplink classification and Session-AMBR), the gNB (scheduling per QoS flow and DRB, UE-AMBR, GFBR) and the UPF (downlink classification, gating, MFBR and Session-AMBR, usage reporting). The PCF decides; the SMF translates decisions into rules for each point (Figures 1 and 2). The transport network between gNB and UPF should map QFIs or 5QIs to DSCP classes so that it does not undo the differentiation.

8 Step-by-step procedures

Procedure 1 • Adding a GBR QoS flow for a voice call

1. The IMS (AF) sends the call's media description to the PCF (Npcf_PolicyAuthorization).
2. The PCF derives a PCC rule (5QI 1, GFBR/MFBR from the codec, ARP) and notifies the SMF.
3. The SMF allocates a QFI, installs a PDR and QER in the UPF and sends the QoS profile to the gNB and the QoS rule to the UE through the AMF.
4. The gNB admits the flow, maps it to a DRB with suitable RLC mode and scheduling, and confirms; voice packets now receive GBR treatment.
5. At call end the PCF removes the rule; the SMF releases the QoS flow in UPF, RAN and UE.

Procedure 2 • Checking a URLLC QoS design

1. From the 5QI, read PDB, PER and MDBV; subtract the core-network delay to obtain D_{AN} (Eq. (1)).
2. Compute the burst rate from MDBV (Eq. (2)) and the number of transmissions from PER and BLER (Eq. (3)).
3. Choose numerology, mini-slots or configured grants so that all transmissions fit into D_{AN}.

4 Check the priority load: the high-priority traffic must remain a small fraction of capacity (Eq. (4)); configure admission control.

9 Worked examples

Worked example 1 • Choosing 5QIs

Problem. Assign 5QIs to: VoNR voice, IMS SIP signalling, a live video call, web browsing, factory motion control.
Solution. Voice 1; SIP 5; live video 2 (or 7 if non-GBR); web 9 (default) or 6/8; motion control a delay-critical GBR value such as 82 (or 85 if 5 ms is required), with the UPF at the edge.

Worked example 2 • URLLC feasibility

Problem. A flow with 5QI 85 carries 200-byte bursts. The UPF is on site (core delay 0.5 ms). With 30 kHz SCS (0.5 ms slots), a HARQ round trip of 4 slots and $p = 0.01$, can the PER be met?
Solution. $D_{\mathrm{AN}} = 4.5$ ms; burst rate $8 \times 200/0.0045 = 356$ kbit/s; $n = \ln 10^{-5}/\ln 0.01 = 2.5 \to 3$ transmissions needing $3 \times 2 = 6$ ms – too long. Options: mini-slots or 120 kHz SCS (HARQ RTT 0.5 ms: 1.5 ms total), or blind repetitions within one slot window.

Worked example 3 • Priority protection

Problem. With Eq. (4), what happens to T_1 if the high-priority load grows from 0.2 to 0.6, and to T_2 at total load 0.9?
Solution. $T_1 = 1/(1000 \times 0.4) = 2.5$ ms (doubled); $T_2 = 1/(1000 \times 0.4 \times 0.1) = 25$ ms. High-priority traffic must itself be limited.

10 Technology comparison

Table 2. LTE bearer-based QoS versus 5G flow-based QoS.

	LTE (EPS bearers)	5G (QoS flows)
Unit of QoS	EPS bearer (radio + S1 + S5 bearer)	QoS flow, marked by QFI
Core tunnels	One per bearer	One per PDU session
Radio mapping	One radio bearer per EPS bearer	Flexible flow-to-DRB mapping (SDAP)
QoS index	QCI (1–9, 65–80)	5QI, including delay-critical GBR
Uplink classification	UL TFT, signalled	QoS rules, signalled or reflective
New flow	Dedicated bearer set-up end to end	QoS rule/profile update, no new tunnel
Delay monitoring	Limited	Per-flow QoS monitoring (Rel-16)

Same ideas, lighter mechanics. The QoS parameters (priority, PDB, PER, ARP, GBR/MBR, AMBR) are almost identical to LTE's; what changed is the mechanism – marking instead of tunnels, flexible RAN mapping, and reflective rules – which makes fine-grained QoS affordable.

11 Key takeaways

Key takeaways

- 5G QoS is flow-based: QFI marking per packet, one N3 tunnel per PDU session, SDAP maps flows to DRBs.
- 5QI gives resource type, priority, PDB, PER (and MDBV for delay-critical GBR); ARP and GFBR/MFBR complete the QoS profile.
- PCF → SMF → (UPF via N4, gNB via N2, UE via NAS): policy becomes enforceable rules.
- $D_{\text{AN}} = \text{PDB} - D_{\text{CN}}$; burst rate $\geq 8\,\text{MDBV}/D_{\text{AN}}$; transmissions $n \geq \ln \text{PER}/\ln p$.
- Priority protects high-priority traffic only if its own load is limited: $T_1 = 1/(\mu(1-\rho_1))$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Flows and bearers

A UE has an Internet PDU session with three QoS flows. How many N3 tunnels and how many DRBs does it need at least and at most?

Exercise 2 • Level 1 – conceptual • Reflective QoS

Explain why reflective QoS saves signalling and give one situation where it should not be used.

Exercise 3 • Level 2 – numerical • Budget split

For 5QI 3 (PDB 50 ms) and 5QI 82 (PDB 10 ms), compute D_{AN} when the UPF is 600 km away (5 µs/km, plus 2 ms processing) and when it is local (0.5 ms). Which 5QI forces an edge UPF?

Exercise 4 • Level 2 – numerical • HARQ

A flow needs PER 10^{-6}. Compute the number of transmissions for $p = 0.1$ and $p = 0.001$, and the time needed with a HARQ round trip of 1 ms. What does this imply for the choice of MCS?

Exercise 5 • Level 3 – analytical • Admission control

A cell carries 5QI 1 voice (GBR 40 kbit/s per call at the radio, assumed to cost 0.5 % of cell capacity per call). Using Eq. (4) with voice as class 1, how many calls can be admitted so that non-GBR traffic at total load 0.9 sees a mean delay at most twice its FIFO value? Interpret.

Exercise 6 • Level 3 – analytical • End-to-end QoS

A cloud-gaming application needs 30 ms RTT and 50 Mbit/s downlink. Propose 5QI(s), UPF placement, slice and transport measures, and explain which nodes enforce which parameter.

Solutions

Solution to Exercise 1

One N3 tunnel (per PDU session) in all cases. DRBs: at least one (all three flows mapped to one DRB) and at most three (one per flow); the gNB decides.

Solution to Exercise 2

The UE derives uplink rules from downlink packets marked with the RQI, so no NAS signalling is needed per flow – valuable for many short-lived flows. It should not be used where the uplink treatment must differ from what the downlink packet implies, or for security-sensitive policy where the network must control exactly which uplink packets get privileged treatment (a reflected rule could be abused by crafted downlink traffic).

Solution to Exercise 3

Remote UPF: $600 \times 5\,\mu\text{s} = 3\,\text{ms} + 2\,\text{ms} = 5\,\text{ms}$. 5QI 3: $50 - 5 = 45\,\text{ms}$ (remote) or 49.5 ms (local); 5QI 82: $10 - 5 = 5\,\text{ms}$ (remote) or 9.5 ms (local). 5QI 82 loses half its access budget with the remote UPF – it needs an edge UPF; 5QI 3 does not.

Solution to Exercise 4

$p = 0.1$: $n = 6$, about 6 ms; $p = 0.001$: $n = 2$, about 2 ms. A conservative MCS (low BLER) costs spectral efficiency but saves retransmissions and delay – the characteristic URLLC trade-off of capacity for latency and reliability.

Solution to Exercise 5

Requirement: $T_2 \le 2T_{\text{FIFO}} \Leftrightarrow 1/(1-\rho_1) \le 2 \Leftrightarrow \rho_1 \le 0.5$: at most 100 calls (0.5 % each). In practice operators limit GBR load much lower (e.g. 20–30 %) to keep non-GBR delay close to FIFO and to keep margin for channel variations that change the cost per call.

Solution to Exercise 6

5QI: a GBR video-type value (e.g. 3 for real-time gaming characteristics) for the game stream, with GFBR set to the needed rate or a high-priority non-GBR 5QI with sufficient Session-AMBR; 5QI 5/default for control. UPF at a regional edge so that the core part of the RTT is a few milliseconds; possibly an eMBB/gaming slice with RAN resource shares in busy cells; transport mapping of the flow to a priority class. Enforcement: UPF (classification, MFBR, AMBR), gNB (GFBR, scheduling priority, PDB), UE (uplink rules); the PCF authorises it based on the gaming provider's AF request.

13 Further reading

The 5G QoS model and the standardised 5QI table are in TS 23.501 [5] (clause 5.7), policy and charging control in TS 23.503 [3], QoS procedures in TS 23.502 [4], SDAP and RAN aspects in TS 38.300 [1]. The LTE counterpart is TS 23.203 [2]. Kleinrock [7] derives priority queueing results; Dahlman et al. [6] discuss URLLC techniques in the NR radio.

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23

Lecture 23

The 5G User Plane

صفحهٔ کاربر نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

DIFFICULTY
Advanced

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 6, 13, 16, 19 and 22

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-23-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G user-plane protocol stack from UE to data network, including the gNB's CU/DU split.
- **LO2** **Explain** the roles of SDAP, PDCP, RLC and MAC, and of GTP-U on F1-U, N3 and N9.
- **LO3** **Compare** RAN functional splits and **estimate** their fronthaul requirements.
- **LO4** **Describe** UPF packet processing with PFCP rules and its implementation options.
- **LO5** **Compute** header overhead, latency budgets, duplication reliability and sequence-number windows.

2 Prerequisite knowledge

You need the LTE radio protocol stack (Lecture 6), GTP-U and PFCP (Lecture 13), NFV data-plane techniques (Lecture 16), UPF topologies (Lecture 19) and the QoS model (Lecture 22). This lecture and the next one look at the 5G system "by plane": first the path of user data, then the signalling that controls it.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
User plane	صفحهٔ کاربر	Functions and protocols that carry user data.
CU / DU / RU	واحد مرکزی / توزیع‌شده / رادیویی	Parts of a split gNB.
F1-U / N3 / N9	واسط‌های صفحهٔ کاربر	GTP-U interfaces CU–DU, RAN–UPF, UPF–UPF.
Functional split	تقسیم کارکردی	Point in the protocol stack where the RAN is divided.
Fronthaul	فرانت‌هال	Transport between DU and RU (or baseband and radio).
PDCP duplication	تکرار PDCP	Sending the same PDCP PDU over two legs for reliability.
SN window	پنجرهٔ شمارهٔ ترتیب	Range of sequence numbers in flight for reordering.
PDR / FAR / QER / URR	قواعد PFCP	Detection, forwarding, QoS and usage rules in the UPF.
DPDK / SmartNIC	شتاب صفحهٔ داده	Software and hardware acceleration of packet processing.
Local breakout	خروج محلی	Exiting to a data network close to the RAN.

3 Introduction and motivation

Everything else in a mobile network exists to serve the user plane: the path that carries the subscriber's packets. Its performance – throughput, latency, reliability, cost per bit – is what users experience and what operators pay for. In 5G the user plane is more flexible than ever: the gNB can be split into centralised and distributed units, the core user plane is a programmable UPF that can be placed anywhere from a central data centre to a factory, and several UPFs can be chained in one session.

This flexibility creates design questions that this lecture equips you to answer quantitatively: where to split the RAN given the available fronthaul, where to place UPFs given a latency budget, how to reach URLLC reliability with duplication, and how much the protocol headers cost [3]–[5].

4 Fundamental concepts

4.1 The end-to-end stack

User data pass (Figure 1): in the UE through SDAP, PDCP, RLC, MAC and PHY; over the radio to the gNB-DU (RLC, MAC, PHY); over F1-U in GTP-U to the gNB-CU-UP (PDCP, SDAP); over N3 in GTP-U to the UPF, possibly over N9 to further UPFs; and out of N6 to the data network. The radio layers have these roles [4]:

- **SDAP**: maps QoS flows to data radio bearers and marks QFIs (Lecture 22).
- **PDCP**: header compression (ROHC), ciphering and optional integrity protection, reordering and duplicate removal, PDCP duplication.
- **RLC**: segmentation and, in acknowledged mode, ARQ retransmission; NR's RLC no longer concatenates or reorders.
- **MAC**: multiplexing of logical channels, HARQ, scheduling.

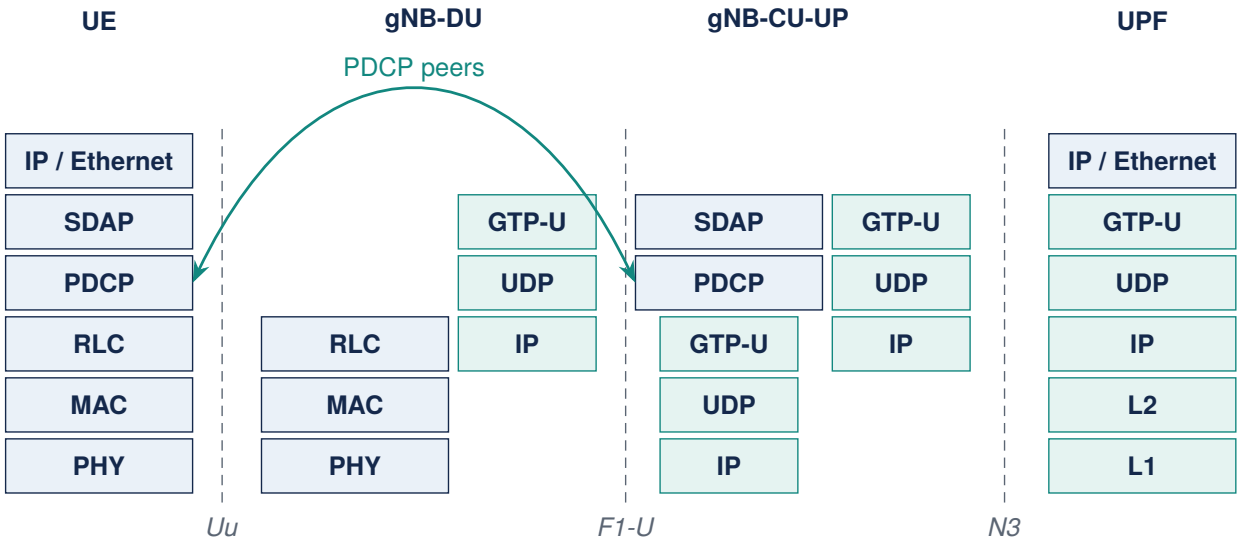


Figure 1. 5G user-plane protocol stack with a split gNB (CU-UP and DU).

4.2 Splitting the gNB

3GPP standardised a *higher-layer split* between a central unit (CU: RRC, SDAP, PDCP) and a distributed unit (DU: RLC, MAC, PHY), connected by F1 [3]; the CU is further split into control (CU-CP) and user plane (CU-UP), connected by E1. Lower-layer splits between the DU and a radio unit (RU) – in particular option 7.2, specified by the O-RAN Alliance – move part of the physical layer into the RU (Figure 2). The lower the split, the more functions can be centralised (simpler sites, coordination across cells), but the higher the fronthaul rate and the tighter the latency requirement (Section 6.2).

4.3 The UPF

The UPF processes every packet against the rules installed by the SMF (Figure 3) [2]: a *PDR* detects the packet (by TEID, UE address, SDF filter or QFI; the matching PDR with the highest precedence wins) and points to a *FAR* (forward with a new outer header, buffer, drop, duplicate), *QERs* (gating and rate limits) and *URRs* (counting and reporting volumes, time and events). Buffering for idle UEs follows a *BAR*. The UPF thus combines the S-GW and P-GW user-plane functions of

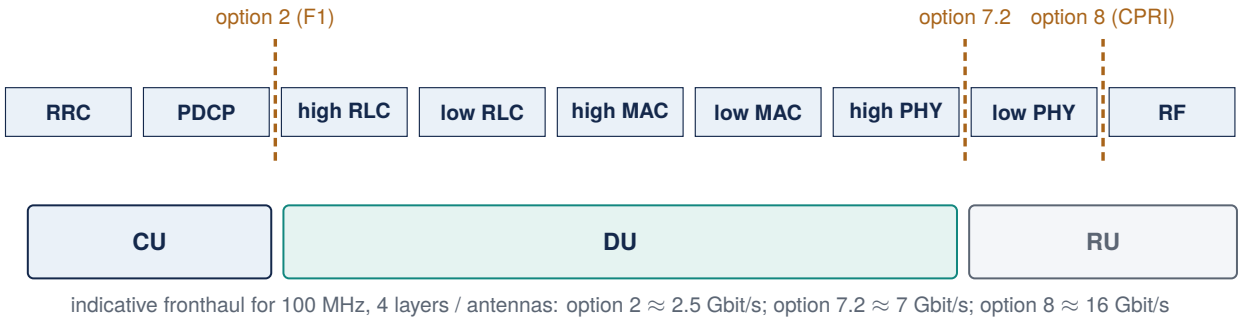


Figure 2. Selected RAN functional splits with indicative fronthaul rates (assumptions in Section 6.2).

the EPC (Lecture 13).

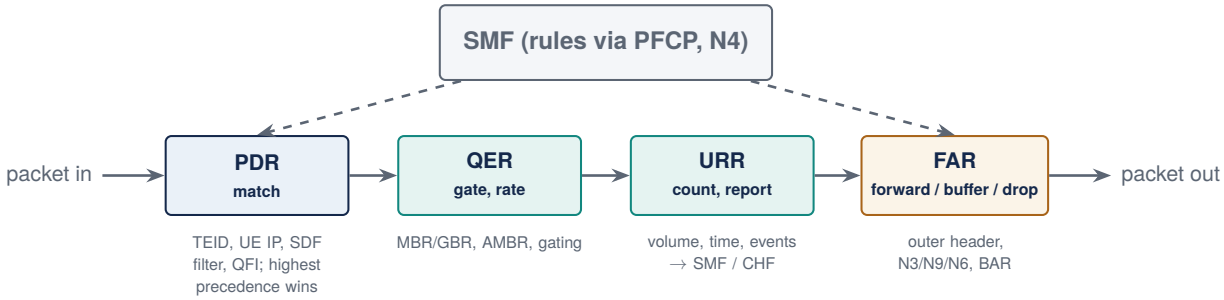


Figure 3. UPF packet processing with PFCP rules.

5 Detailed technical explanation

5.1 PDCP duplication and multi-connectivity

For URLLC, PDCP can duplicate packets and send the copies over two independent legs: two carriers of the same gNB (carrier aggregation) or two gNBs (dual connectivity). The receiver's PDCP discards the second copy. Duplication converts the independence of the legs into reliability (Section 6.4) and removes the interruption of a handover, at the cost of doubling the radio resources used for the duplicated flow. It can be activated dynamically by MAC control elements when the channel deteriorates.

5.2 Sequence numbers and reordering

PDCP numbers each SDU (12- or 18-bit SN); RLC in acknowledged mode numbers segments (12- or 18-bit SN), in unacknowledged mode 6- or 12-bit. The receiving PDCP delivers in order and waits up to a reordering timer for missing packets. The window of SNs that may be in flight is half the SN space; at high data rates a short SN wraps quickly, which is why NR introduced 18-bit SNs (Section 6.5).

5.3 F1-U and N3 transport

On F1-U the CU-UP and DU exchange PDCP PDUs in GTP-U with an NR user-plane protocol header that carries flow-control information (e.g. the DU's desired buffer size), so that the CU can pace the DU and avoid overflows when radio conditions change. On N3 and N9 the GTP-U header carries the QFI and the reflective QoS indicator in an extension header. Both run over IP transport networks that must preserve QoS by mapping QFIs/5QIs to DSCP values.

5.4 UPF implementation and placement

A UPF must classify, count and re-encapsulate every packet at line rate. Implementations range from software on servers with DPDK and SR-IOV (Lecture 16), through SmartNIC offload of encapsulation and classification, to programmable switches running P4 pipelines (Lecture 15) for the fast path while a CPU handles exceptions. Placement follows the latency budget (Section 6.3): central UPFs for Internet traffic, regional or edge UPFs for latency-sensitive and locally served traffic, on-premises UPFs for enterprise and industrial networks.

6 Mathematical models

6.1 Protocol overhead

Model. An IP packet of L bytes acquires SDAP, PDCP, RLC and MAC headers of $H = H_{\text{SDAP}} + H_{\text{PDCP}} + H_{\text{RLC}} + H_{\text{MAC}}$ bytes on the radio. The efficiency is

$$\eta = \frac{L}{L + H}. \tag{1}$$

Example. SDAP 1, PDCP 3 (18-bit SN), RLC AM 3, MAC subheader 3 bytes: $H = 10$ bytes. For $L = 1500$: $\eta = 99.3\,\%$; for a 60-byte voice packet (after ROHC compresses its 60-byte IP/UDP/RTP header to a few bytes, L is smaller still): $\eta = 85.7\,\%$. Integrity protection adds a 4-byte MAC-I to PDCP.

6.2 Fronthaul rate

Model. With split option 8 the fronthaul carries time-domain IQ samples: sampling rate f_s, b bits per I and per Q component, N_{ant} antennas and line coding c:

$$R_{\text{FH},8} = 2\, f_s\, b\, N_{\text{ant}}\, c. \tag{2}$$

With split 7.2 the fronthaul carries frequency-domain samples of the occupied resource elements per layer, approximately $R_{\text{FH},7.2} \approx 12\, N_{\text{PRB}}\, N_{\text{sym}}\, N_{\text{layers}}\, 2b'$ with N_{sym} symbols per second and compressed bit width b'.

Example. 100 MHz NR at 30 kHz: $f_s = 122.88$ Msample/s, $b = 16$, 4 antennas, $c = 66/64$: $R_{\text{FH},8} = 16.2$ Gbit/s; with 64 antennas 260 Gbit/s – impractical. Split 7.2 with 273 PRBs, 28 000 symbols/s, 4 layers, $b' = 9$: about 6.6 Gbit/s, scaling with layers rather than antennas. The higher-layer split (option 2) carries roughly the user data rate (about 2.5 Gbit/s for the carrier of Lecture 14).

Interpretation. Massive MIMO is only practical with splits that do beamforming in the RU; the choice of split is a trade-off between centralisation and transport cost.

6.3 Latency budget

Model. One-way user-plane latency between UE and an application server is the sum of its components:

$$T = T_{\text{radio}} + T_{\text{gNB}} + T_{\text{FH}} + T_{\text{BH}} + T_{\text{UPF}} + T_{\text{N6}}. \tag{3}$$

Example. With illustrative values (Figure 4): radio 2 ms, gNB processing 0.5 ms, fronthaul 0.1 ms, UPF 0.1 ms; an edge UPF with 0.2 ms transport and 0.1 ms to the server gives 3.0 ms, a central UPF 600 km away (3 ms transport) with 1 ms to the server gives 6.7 ms. Only the edge placement meets a 4 ms one-way target.

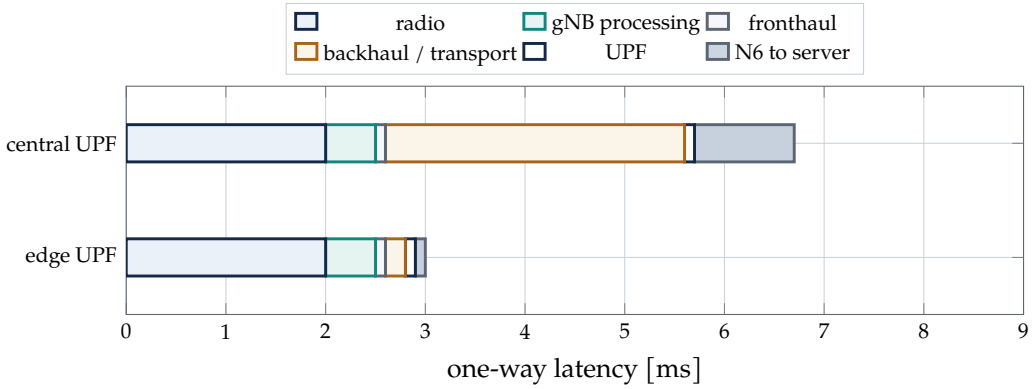


Figure 4. One-way user-plane latency budget with edge and central UPF (illustrative values, Eq. (3)).

6.4 Reliability with duplication

Model. If each of two independent legs loses a packet with probability p_1, p_2, duplication loses it only if both fail:

$$P_{\text{loss}} = p_1\, p_2. \tag{4}$$

Example. $p_1 = p_2 = 10^{-3}$ gives 10^{-6}, meeting a URLLC target that a single leg could reach only with a much more robust (and costly) MCS or more retransmissions – which might not fit the delay budget.

Limitation. Correlated losses (both carriers from the same site blocked by the same obstacle) break the independence assumption; dual connectivity with geographically separated nodes is more robust than carrier-level duplication.

6.5 Sequence-number window

Model. With an s-bit SN, at most 2^{s-1} PDUs may be in flight. At a packet rate P the window covers

$$T_{\text{win}} = \frac{2^{s-1}}{P}. \tag{5}$$

Example. 2 Gbit/s with 1500-byte packets: $P = 1.67 \times 10^5$ packets/s. A 12-bit SN covers 12.3 ms – shorter than HARQ and ARQ delays plus reordering; an 18-bit SN covers 0.79 s. High-throughput bearers therefore use 18-bit SNs.

7 Network architecture and diagrams

A typical 5G deployment combines RUs at the antenna sites, DUs at the cell sites or in nearby hubs (connected by 7.2 fronthaul), CU-UPs in regional data centres together with edge UPFs, and central UPFs for Internet traffic. Each element can be virtualised on common infrastructure (vRAN, Lecture 16), and the SMF chains UPFs per session (Figures 1 to 3).

8 Step-by-step procedures

Procedure 1 • Following a downlink packet

1. The packet arrives on N6 at the PSA UPF; a PDR matches it (UE address and SDF filter), the QER enforces rates, the URR counts it, and the FAR encapsulates it in GTP-U with the QFI towards the next hop (N9 or N3).
2. The CU-UP receives it on N3, SDAP maps the QFI to a DRB, PDCP compresses headers, ciphers and numbers it, and sends it in GTP-U over F1-U.
3. The DU's RLC segments it as scheduled by MAC; MAC multiplexes it into a transport block; PHY transmits it with HARQ.
4. The UE's layers reverse the process; PDCP reorders and delivers in sequence to SDAP and the IP layer.

Procedure 2 • Choosing a RAN split

1. Determine bandwidth, number of antennas and layers per cell and compute the fronthaul rate for candidate splits (Eq. (2)).
2. Compare with the available transport (fibre, microwave) and its latency; lower splits need sub-millisecond fronthaul latency.
3. Weigh the benefits of centralisation (coordination, pooling, simpler sites) against transport cost.
4. Choose the split per site type (e.g. 7.2 for urban massive MIMO, option 2 for remote sites with limited transport).

9 Worked examples

Worked example 1 • Overhead for small packets

Problem. IoT sensors send 20-byte payloads in IPv6/UDP without header compression. Compute the efficiency on the radio and with ROHC compressing IPv6/UDP to 4 bytes.
Solution. Without ROHC: $L = 20 + 48 = 68$, data efficiency $20/(68 + 10) = 25.6\,\%$. With ROHC: $L = 24$, $20/34 = 58.8\,\%$. Header compression matters most for small packets.

Worked example 2 • Fronthaul for massive MIMO

Problem. A 64-antenna RU with 100 MHz serves 16 layers. Compare option 8 and option 7.2 fronthaul (assumptions of Section 6.2).
Solution. Option 8: 260 Gbit/s. Option 7.2: $273 \times 12 \times 28\,000 \times 16 \times 18 = 26.4$ Gbit/s – ten times less, feasible over a 25 or 50 Gbit/s Ethernet link.

Worked example 3 • Duplication budget

Problem. Each leg has BLER 10^{-2} per transmission and can make 2 transmissions within the budget. What residual loss does a single leg reach, and what does duplication reach?
Solution. Single leg: $(10^{-2})^2 = 10^{-4}$. With duplication over two independent legs: $10^{-4} \times 10^{-4} = 10^{-8}$ (Eq. (4)) – beyond the 10^{-5} URLLC targets, leaving margin for correlation.

10 Technology comparison

Table 1. User plane in LTE/EPC and in NR/5GC.

	LTE / EPC	NR / 5GC
Radio layers	PDCP, RLC (with reordering), MAC	SDAP, PDCP, RLC (no reordering), MAC
RAN split	Monolithic eNB (CPRI to RRH)	CU/DU (F1), CU-CP/CU-UP (E1), lower-layer split 7.2
Core user plane	S-GW + P-GW (CUPS optional)	UPF chains (I-UPF, UL CL, PSA) controlled by PFCP
QoS on N3/S1-U	Tunnel per bearer	Tunnel per session, QFI per packet
Reliability	HARQ, ARQ	+ PDCP duplication, multi-connectivity
SN length	PDCP up to 18 bits (later)	PDCP 12/18, RLC AM 12/18 bits

Programmable all the way. From P4-programmed UPFs to virtualised CU/DU, the 5G user plane is software-defined end to end – which makes its performance a matter of engineering budgets (cycles per packet, microseconds per hop) rather than of fixed boxes.

11 Key takeaways

Key takeaways

- User data: UE (SDAP/PDCP/RLC/MAC/PHY) → DU (RLC/MAC/PHY) → F1-U → CU-UP (PDCP/SDAP) → N3 → UPF(s) → N6.
- Higher-layer split (option 2) needs about the user rate; option 8 scales with antennas ($2f_s b N_{\text{ant}} c$); 7.2 with layers.
- UPF processing: PDR match → QER → URR → FAR, programmed by the SMF over N4.
- Latency = $\sum$ components; edge UPFs remove milliseconds of transport.
- PDCP duplication gives $P_{\text{loss}} = p_1 p_2$; long SNs keep the window $2^{s-1}/P$ above reordering delays.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which layer?

Which layer performs: (a) ciphering; (b) segmentation; (c) QoS flow to DRB mapping; (d) HARQ; (e) duplicate removal; (f) ARQ?

Exercise 2 • Level 1 – conceptual • Why split?

Give two advantages and two disadvantages of centralising the CU for many DUs.

Exercise 3 • Level 2 – numerical • Fronthaul

A 40 MHz carrier at 30 kHz is sampled at 61.44 Msample/s. Compute the option-8 fronthaul for 8 antennas with 16-bit samples and 66/64 coding. Can it run over a 25 Gbit/s link?

Exercise 4 • Level 2 – numerical • SN window

A bearer carries 5 Gbit/s of 1200-byte packets. Compute the window covered by 12-bit and 18-bit PDCP SNs. Which is needed if the reordering timer can reach 100 ms?

Exercise 5 • Level 3 – analytical • Placing UPFs

An operator has one central data centre (800 km from the farthest cells) and can build regional sites at most 150 km from any cell. A service needs 5 ms one-way latency with radio 2.5 ms, gNB 0.5 ms, UPF and server 0.4 ms. Which placement works, using 5 µs/km for transport? Discuss the cost.

Exercise 6 • Level 3 – analytical • Correlated duplication

Two legs each lose packets with probability 10^{-3}, but with probability q they are blocked together (common cause), in which case both lose. Model the loss as $P = q + (1-q)p^2$ with $p = 10^{-3}$ and find the largest q for which $P \le 10^{-5}$. What does this imply for deployment?

Solutions

Solution to Exercise 1

(a) PDCP; (b) RLC; (c) SDAP; (d) MAC; (e) PDCP; (f) RLC (acknowledged mode).

Solution to Exercise 2

Advantages: pooling of PDCP/RRC processing (statistical multiplexing), easier coordination and mobility between cells of the same CU, simpler and cheaper cell sites, software upgrades in one place. Disadvantages: transport between CU and DU needed (bandwidth and latency), larger failure domain (one CU outage affects many cells), and latency-sensitive functions cannot be centralised beyond a distance.

Solution to Exercise 3

$2 \times 61.44 \times 10^6 \times 16 \times 8 \times 66/64 = 16.2\,\text{Gbit/s}$: fits a 25 Gbit/s link for one carrier, but not two such carriers.

Solution to Exercise 4

$P = 5 \times 10^9/9600 = 5.2 \times 10^5$ packets/s. 12-bit: $2048/P = 3.9$ ms; 18-bit: $131\,072/P = 0.25$ s. With a 100 ms reordering timer, only the 18-bit SN is safe.

Solution to Exercise 5

Fixed part $2.5 + 0.5 + 0.4 = 3.4$ ms, leaving 1.6 ms of transport: $D \leq 1.6\ \text{ms}/5\ \mu\text{s/km} = 320$ km. The central site (800 km, 4 ms) fails; regional sites (150 km, 0.75 ms) work. Cost: more UPF sites (hardware, operations), regional peering or caching to serve content locally – to be justified by the revenue of latency-sensitive services.

Solution to Exercise 6

$q + (1-q)10^{-6} \leq 10^{-5} \Rightarrow q \leq 9 \times 10^{-6}$ (approximately). Common-cause blocking must be extremely rare: the legs should come from different sites and directions (dual connectivity), not just different carriers of the same antenna.

13 Further reading

The NR user-plane architecture is described in TS 38.300 [4] and the split RAN in TS 38.401 [3]; PFCP in TS 29.244 [2]; GTP-U in TS 29.281 [1]; UPF functions and topologies in TS 23.501 [5]. Dahlman et al. [7] explain the NR layer-2 protocols; Bosshart et al. [6] introduce the P4 language used in programmable UPFs.

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[1] 3GPP, *General packet radio system (GPRS) tunnelling protocol user plane (GTPv1-U)*, TS 29.281, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29281.htm.

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[6] P. Bosshart, D. Daly, G. Gibb, *et al.*, "P4: Programming protocol-independent packet processors", *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 3, pp. 87–95, 2014.

[7] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic Press, 2020.

24

Lecture 24

The 5G Control Plane

صفحهٔ کنترل نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

DIFFICULTY
Advanced

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 4, 12, 18, 20 and 23

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1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G control-plane protocol stack (RRC, NAS, NGAP, SBI) and **explain** which entities terminate each protocol.
- **LO2** **Describe** the RRC states IDLE, INACTIVE and CONNECTED and their relation to the CM and RM states of the core.
- **LO3** **Trace** a service request from CM-IDLE and **estimate** the control-plane latency of state transitions.
- **LO4** **Compute** the signalling load on an AMF, the inactivity-timer trade-off and the optimal registration-area size.
- **LO5** **Explain** how 5G protects its control plane against overload and signalling storms.

2 Prerequisite knowledge

You need location management and paging (Lecture 4), the MME and its overload control (Lecture 12), the service-based architecture (Lecture 18), the registration and PDU-session procedures (Lecture 20) and the user plane of Lecture 23. This lecture looks at the same system from the signalling side: who talks to whom, in which state, and at what cost.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Control plane	صفحهٔ کنترل	Signalling that sets up, maintains and releases connectivity.
RRC	کنترل منابع رادیویی	Radio Resource Control between UE and gNB.
NAS (MM / SM)	لایهٔ غیردسترسی	Non-access stratum between UE and AMF (mobility) or SMF (sessions).
NGAP	پروتکل کاربرد NG	Application protocol on N2 between gNB and AMF, over SCTP.
RRC_INACTIVE	حالت غیرفعال RRC	State in which the UE context is kept but the radio connection is suspended.
CM / RM state	حالت مدیریت اتصال / ثبت	Core view of the UE: connected or idle, registered or deregistered.
Service request	درخواست خدمت	NAS procedure that moves a UE from CM-IDLE to CM-CONNECTED.
Registration area	ناحیهٔ ثبت	List of tracking areas in which an idle UE need not report its location.
RNA	ناحیهٔ اعلان مبتنی بر RAN	Area in which an inactive UE moves without notifying the network.
Signalling storm	طوفان سیگنالینگ	Burst of synchronised procedures that overloads control-plane nodes.

3 Introduction and motivation

The control plane does not carry a single user bit, yet it decides whether user bits can flow at all. Every time a smartphone wakes up to fetch a notification, it runs a small protocol exchange across the radio, the gNB, the AMF, the SMF and the UPF. Multiplied by millions of devices and dozens of transitions per hour, this signalling dominates the processing load of the core and a noticeable share of the radio resources – and its latency is felt by the user as the delay before the first packet.

5G reorganises the control plane in three ways: a new RRC state, RRC_INACTIVE, that avoids most core signalling for short bursts; a clean separation of mobility (AMF) and session (SMF) management; and service-based interfaces inside the core [3], [7]. This lecture explains the resulting protocol stack and state machines and gives the models to dimension them.

4 Fundamental concepts

4.1 The control-plane stack

Three protocol families make up the control plane (Figure 1):

- **RRC** between UE and gNB [4]: system information, connection setup, security activation on the radio, configuration of radio bearers, measurements and handover commands. It runs over PDCP, RLC and MAC on signalling radio bearers.
- **NAS** between UE and core [2]: NAS-MM (registration, service request, authentication, NAS security) terminates in the AMF; NAS-SM (PDU-session establishment and modification) is relayed by the AMF to the SMF. The gNB forwards NAS messages without reading them – they are integrity-protected and ciphered end to end between UE and AMF.
- **NGAP** between gNB and AMF on N2 [1]: UE context management, PDU-session resource setup, paging and handover signalling, over SCTP. Inside the core, the AMF and SMF use service-based interfaces (HTTP/2 with JSON, Lecture 18).

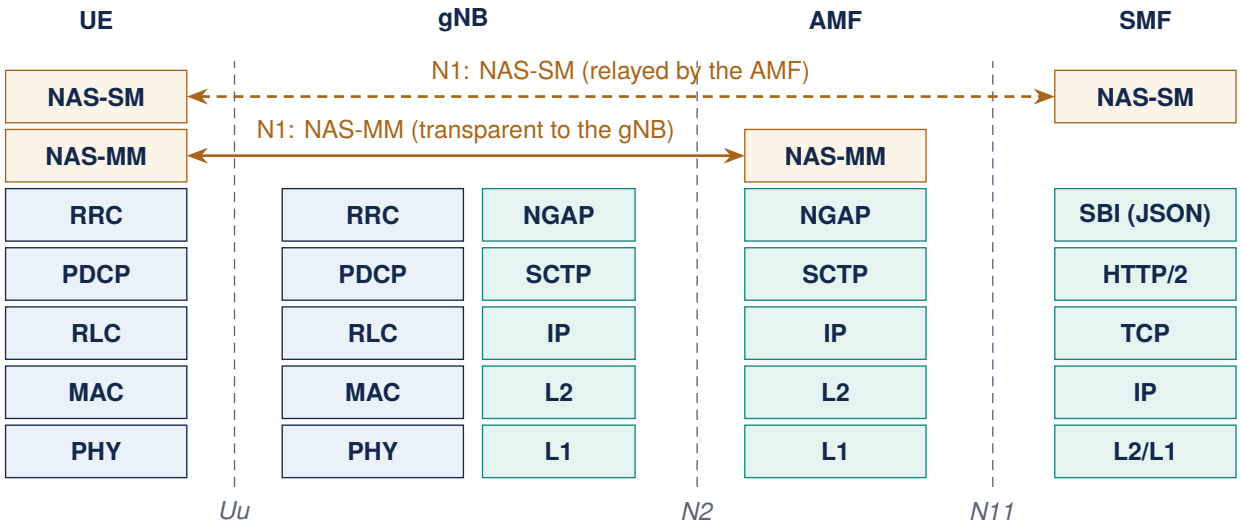


Figure 1. 5G control-plane protocol stack: RRC on Uu, NGAP on N2, NAS end to end between UE and AMF/SMF, and SBI in the core.

4.2 States of a UE

The network keeps three coupled state machines for every UE [4], [7]:

- **RM** (registration management) in the AMF: RM-DEREGISTERED or RM-REGISTERED.
- **CM** (connection management) in the AMF: CM-IDLE (no N2 connection for the UE; location known per registration area) or CM-CONNECTED (an N2 connection exists).
- **RRC** in the gNB and UE: RRC_IDLE, RRC_INACTIVE or RRC_CONNECTED (Figure 2).

In RRC_INACTIVE the UE behaves almost like an idle UE – it camps on cells, monitors paging and moves freely inside a RAN-based notification area (RNA) – but the gNB that suspended it keeps the UE context and the N2/N3 connections to the core remain up, so the core sees the UE as CM-CONNECTED. Returning to RRC_CONNECTED takes an RRC Resume exchange with no core signalling.

4.3 Reachability

Where the network can find the UE depends on its state: in CM-IDLE the AMF knows the registration area (a list of tracking areas) and must page there; in RRC_INACTIVE the anchor gNB knows the RNA and performs RAN paging itself; in RRC_CONNECTED the serving cell is known.

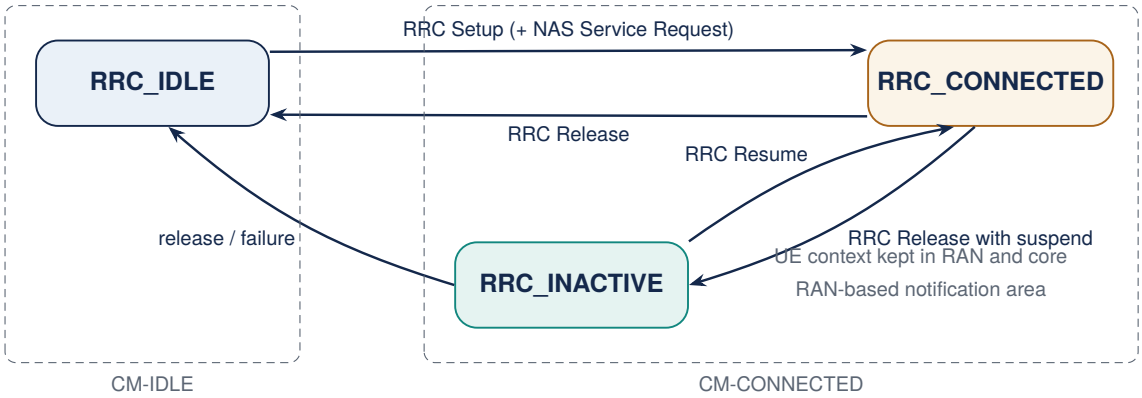


Figure 2. NR RRC states and transitions; RRC_INACTIVE belongs to CM-CONNECTED in the core.

The UE reports leaving its area by a mobility registration update (CM-IDLE) or an RNA update (RRC_INACTIVE), and refreshes it periodically.

5 Detailed technical explanation

5.1 Service request

When an idle UE has uplink data – or is paged for downlink data – it runs the service request (Figure 3) [5]. After random access the UE sets up an RRC connection and piggybacks the NAS Service Request on RRC Setup Complete; the gNB forwards it in an NGAP Initial UE Message to the AMF selected by the 5G-S-TMSI. The AMF checks NAS integrity, asks the SMF of each PDU session to be activated to re-establish its user plane (the SMF updates the UPF over N4), then sends an Initial Context Setup Request carrying the security key and the session resources. The gNB activates AS security, reconfigures the radio bearers and returns its downlink tunnel endpoints, which the SMF installs in the UPF.

5.2 Suspend and resume

With RRC_INACTIVE the gNB ends a connection by an RRC Release with suspend indication: the UE and the gNB keep the AS context and the security configuration, the UE receives a resume identity and an RNA. When the UE has data again it sends an RRC Resume Request; if it is in a cell of another gNB, that gNB retrieves the context from the anchor gNB over Xn and asks the AMF to switch the N2 path. In the common case of resuming in the same area, no NAS or core signalling is needed at all. Small amounts of data can even be sent during the resume procedure itself (small-data transmission, Release 17).

5.3 Separation of mobility and session management

In the EPC the MME handled both mobility and session signalling (Lecture 12). In 5G the AMF handles only mobility, access and NAS security, while the SMF handles sessions. A UE has one AMF but may have several SMFs (one per PDU session or slice). This separation allows non-3GPP access (Wi-Fi via N3IWF) to share the same AMF, keeps the AMF independent of session types, and lets slices have their own SMFs (Lecture 21).

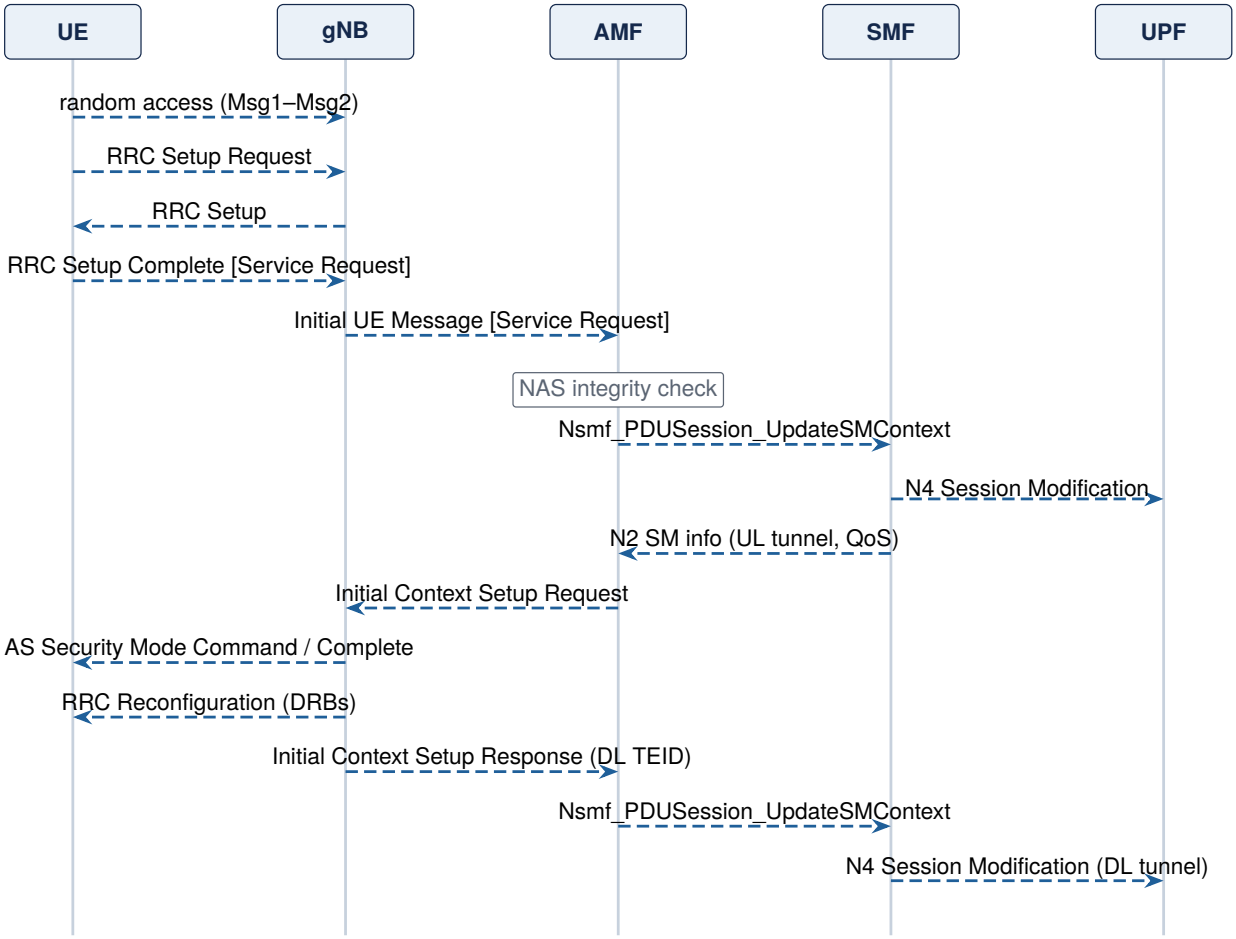


Figure 3. UE-triggered service request from CM-IDLE (simplified, after TS 23.502).

5.4 Overload and congestion control

Control-plane nodes protect themselves at several levels [2], [7]: unified access control in the UE bars access categories broadcast by the gNB before any message is sent; the AMF can ask gNBs to reject or reduce new RRC connections (NGAP overload start); NAS congestion control returns a back-off timer (e.g. T3346) in reject messages so that UEs retry later; and the NRF and SCP spread load over several AMF and SMF instances (Lecture 18).

6 Mathematical models

6.1 Signalling load

Model. If N UEs each perform procedure k at rate λ_k per hour and each procedure costs m_k messages at the AMF, the message rate in messages per second is

$$\Lambda = \frac{N}{3600} \sum_k \lambda_k \, m_k. \tag{1}$$

Example. $N = 2 \times 10^6$ with busy-hour rates per UE: 20 service requests (4 messages each), 2 mobility registrations (6), 3 handovers (6), 0.1 initial registrations (20) and 0.5 PDU-session establishments (8): $\sum_k \lambda_k m_k = 116$ messages per UE per hour and $\Lambda = 64\,444$ messages/s. If RRC_INACTIVE turns 80 % of the idle-to-connected transitions into resumes without core signalling, the sum falls to 52 and $\Lambda = 28\,889$ messages/s – a reduction of 55 %.

6.2 Inactivity timer

Model. A UE generates data bursts as a Poisson process of rate λ. After each burst the gNB keeps the connection for an inactivity time T before releasing (or suspending) it. A transition to idle happens after a burst only if the next burst comes later than T, so the transition rate and the mean connected "tail" per burst are

$$r(T) = \lambda \, e^{-\lambda T}, \qquad \bar{t}_{\text{tail}}(T) = \frac{1 - e^{-\lambda T}}{\lambda}. \tag{2}$$

The fraction of time spent in the tail is $\lambda \, \bar{t}_{\text{tail}} = 1 - e^{-\lambda T}$.

Example. One burst per minute ($\lambda = 1/60\,\text{s}^{-1}$): $T = 10$ s gives 50.8 transitions per hour and 15.4 % of the time in the tail; $T = 30$ s gives 36.4 transitions per hour and 39.3 % (Figure 4). A longer timer saves signalling but costs UE energy and radio resources; RRC_INACTIVE makes short timers affordable because each transition is cheap.

6.3 Registration-area size

Model. With the fluid-flow mobility model (Lecture 4), a UE moving at mean speed v in an area of perimeter P and size A crosses its border at rate $vP/(\pi A)$. For a square area of side a this is $R_u = 4v/(\pi a)$. If each update costs w_u and each paging event (rate λ_p) costs w_p in each of the $n = a^2/A_c$ cells of size A_c, the cost per UE and hour is

$$C(a) = w_u \frac{4v}{\pi a} + w_p \lambda_p \frac{a^2}{A_c}, \tag{3}$$

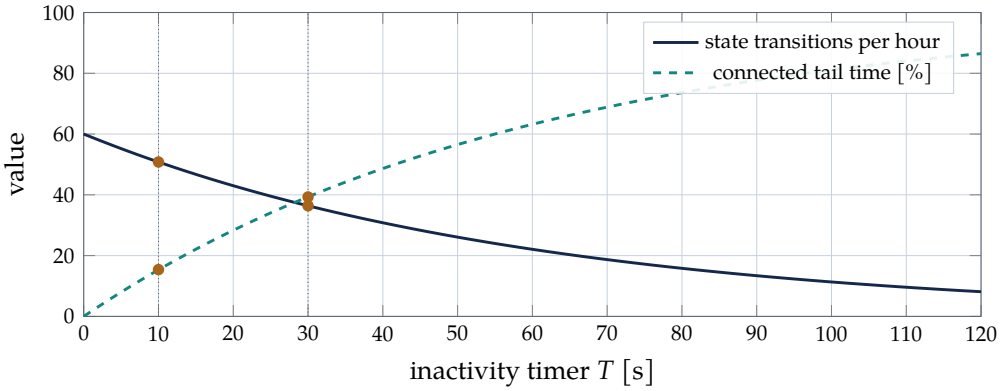


Figure 4. Inactivity-timer trade-off for one burst per minute (Eq. (2)).

minimised by

$$a^* = \left(\frac{2 w_u v A_c}{\pi w_p \lambda_p} \right)^{1/3}. \tag{4}$$

At the optimum the update cost is exactly twice the paging cost.

Example. $v = 20\,\text{km/h}$, $A_c = 1\,\text{km}^2$, $\lambda_p = 5$ per hour, $w_u = 10$, $w_p = 1$: $a^* = 2.94\,\text{km}$ (about 8.7 cells per area), with update cost 86.6 and paging cost 43.3 per UE and hour (Figure 5).

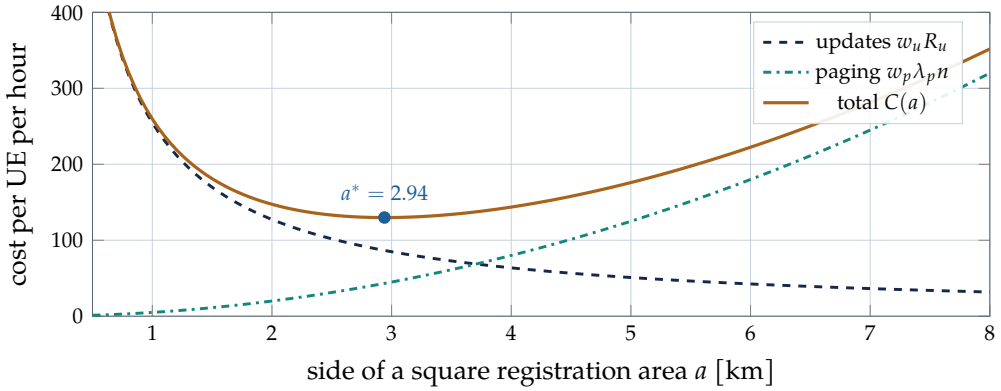


Figure 5. Registration-area size trade-off (Eqs. (3) and (4)).

6.4 Control-plane latency

Model. The time from a battery-efficient state to the start of data transfer is the sum of the steps of the transition:

$$T_{\text{CP}} = T_{\text{RA}} + T_{\text{RRC}} + T_{\text{core}} + T_{\text{ctx}}. \tag{5}$$

TR 38.913 sets a target of 10 ms [6]; ITU-R M.2410 requires 20 ms and encourages 10 ms [9].

Example. With illustrative values for 30 kHz subcarrier spacing – random access 2 ms, RRC exchange 4 ms, core signalling (N2, AMF, SMF and N4) 6 ms, context setup with AS security and reconfiguration 4 ms – the service request from RRC_IDLE takes 16 ms. A resume from RRC_INACTIVE needs only random access and the RRC exchange: 6 ms. Only the inactive state meets the 10 ms target.

7 Network architecture and diagrams

The control plane spans the UE, the gNB (CU-CP after a split, Lecture 23), the AMF set serving a region, and the SMFs, PCF, UDM and AUSF reached through the SBA. AMFs are deployed as sets of interchangeable instances sharing UE contexts in a storage function, so that a gNB can reach any instance of the set (a GUAMI identifies the set and pointer). Figures 1 to 3 show the protocol view, the state view and the procedure view of this architecture.

8 Step-by-step procedures

Procedure 1 • Mobile-terminated data for an idle UE

1. A downlink packet arrives at the UPF for a session without an N3 tunnel; the UPF buffers it (or forwards it to the SMF) and notifies the SMF.
2. The SMF asks the AMF to make the UE reachable (Namf_Communication_N1N2MessageTransfer).
3. The AMF pages the UE in all cells of its registration area through the gNBs (NGAP Paging).
4. The UE answers with a service request (Figure 3); the user plane is re-established and the buffered data are delivered.

Procedure 2 • Resuming from RRC_INACTIVE

1. The UE (with uplink data, or after RAN paging) performs random access and sends an RRC Resume Request with its resume identity.
2. If the context is held by another gNB, the new gNB retrieves it over Xn and later asks the AMF for an N2 path switch.
3. The gNB answers with RRC Resume; security is restored from the stored context with fresh keys.
4. The UE confirms with RRC Resume Complete and data flow on the existing N3 tunnels.

9 Worked examples

Worked example 1 • Dimensioning an AMF set

Problem. For the load of Section 6.1, how many AMF instances are needed if each processes $\mu = 20\,000$ messages/s and should run at no more than 70 % utilisation? Compare with and without RRC_INACTIVE.

Solution. Usable capacity per instance $0.7\mu = 14\,000$ messages/s. Without INACTIVE: $64\,444/14\,000 = 4.6$, so 5 instances. With INACTIVE: $28\,889/14\,000 = 2.06$, so 3 instances. Signalling efficiency in the RAN translates directly into fewer core instances.

Worked example 2 • Choosing the inactivity timer

Problem. With one burst per minute, what is the longest timer that keeps the UE in the tail for at most 20 % of the time, and how many transitions per hour result?
Solution. $1 - e^{-T/60} \leq 0.2 \Rightarrow T \leq -60 \ln 0.8 = 13.4\,\text{s}$. Transitions: $60\, e^{-13.4/60} = 60 \times 0.8 = 48$ per hour.

Worked example 3 • Registration area for fast users

Problem. Repeat the example of Section 6.3 for vehicular users with $v = 60\,\text{km/h}$.
Solution. $a^{*3} = 2 \times 10 \times 60 \times 1/(\pi \times 5) = 76.4$, $a^* = 4.24\,\text{km}$ (about 18 cells). Update cost $10 \times 4 \times 60/(\pi \times 4.24) = 180$, paging cost $5 \times 4.24^2 = 90$, total 270 per UE and hour. Faster users call for larger areas, which is why operators give TA lists along motorways and railways more cells.

10 Technology comparison

Table 1. Control plane in LTE/EPC and in NR/5GC.

	LTE / EPC	NR / 5GC
RRC states	IDLE, CONNECTED (suspend/resume for IoT only)	IDLE, INACTIVE, CONNECTED
Mobility and sessions	MME (both), S-GW/P-GW control	AMF (mobility), SMF (sessions)
Core interfaces	Diameter, GTPv2-C (point to point)	SBI over HTTP/2, NRF discovery
RAN–core protocol	S1AP over SCTP	NGAP over SCTP
Location areas	TA lists, paging by MME	TA lists (CM-IDLE) and RNA with RAN paging (INACTIVE)
Access control	ACB, EAB, back-off timers	Unified access control, back-off timers
Non-3GPP access	Separate ePDG path	Same AMF via N3IWF

Signalling is the price of efficiency. Each idle period saves energy and radio resources but must be paid for by a transition. The 5G control plane lowers that price (INACTIVE, fewer core messages, faster processing), which lets operators release connections earlier and serve many more devices per unit of core capacity.

11 Key takeaways

Key takeaways

- RRC (UE–gNB), NAS-MM (UE–AMF), NAS-SM (UE–SMF via AMF), NGAP (gNB–AMF over SCTP) and SBI in the core.
- RRC_INACTIVE keeps the context in the RAN and the core in CM-CONNECTED: resume needs no core signalling.

- Load $\Lambda = (N/3600)\sum_k \lambda_k m_k$; idle-to-connected transitions dominate it.
- Inactivity timer: transitions $\lambda e^{-\lambda T}$ versus tail fraction $1 - e^{-\lambda T}$.
- Registration area: $a^* = (2w_u v A_c/(\pi w_p \lambda_p))^{1/3}$; updates cost twice the paging at the optimum.

12 Exercises

Exercise 1 • Level 1 – conceptual • Where is the UE?

For each RRC state, give the CM state seen by the AMF and the granularity at which the network knows the UE's location.

Exercise 2 • Level 1 – conceptual • Transparent NAS

Why is NAS transparent to the gNB? Give a security reason and an architectural reason.

Exercise 3 • Level 2 – numerical • AMF load

An AMF set serves 5×10^5 UEs with busy-hour rates per UE of 30 service requests (4 messages each), 1 mobility registration (6) and 4 handovers (6). Compute the message rate.

Exercise 4 • Level 2 – numerical • Chatty applications

A messaging application produces one burst every 20 s on average. Compute the transitions per hour and the tail fraction for $T = 5$ s and $T = 10$ s.

Exercise 5 • Level 3 – analytical • Optimal area

Derive Eq. (4) from Eq. (3), show that the update cost is twice the paging cost at the optimum, and compute a^* for pedestrians ($v = 5$ km/h) with the other values of Section 6.3.

Exercise 6 • Level 3 – analytical • Signalling storm

After a core outage, 10^6 UEs must register again, each registration costing 20 messages at an AMF set of 5 instances with $\mu = 20\,000$ messages/s. If the UEs spread their attempts uniformly over a window W, what is the smallest W for 100 % and for 70 % utilisation? Which mechanisms enforce such a spread?

Solutions

Solution to Exercise 1

RRC_IDLE: CM-IDLE, registration area (list of TAs). RRC_INACTIVE: CM-CONNECTED, RNA known to the anchor gNB (the AMF knows the gNB). RRC_CONNECTED: CM-CONNECTED, serving cell.

Solution to Exercise 2

Security: NAS messages are protected end to end with keys shared only by UE and AMF, so a compromised or untrusted RAN node (including non-3GPP access) cannot read or alter them. Architecture: the RAN is independent of core procedures; new NAS features need no RAN change, and the same NAS runs over 3GPP and non-3GPP access.

Solution to Exercise 3

$\sum_k \lambda_k m_k = 120 + 6 + 24 = 150$ per UE and hour; $\Lambda = 5 \times 10^5 \times 150/3600 = 20\,833$ messages/s.

Solution to Exercise 4

$\lambda = 1/20\,\mathrm{s}^{-1}$, 180 bursts per hour. $T = 5\,\mathrm{s}$: $180\,e^{-0.25} = 140.2$ transitions per hour, tail $1 - e^{-0.25} = 22.1\,\%$. $T = 10\,\mathrm{s}$: $180\,e^{-0.5} = 109.2$, tail 39.3 %. Frequent small bursts are the case for which RRC_INACTIVE and small-data transmission were designed.

Solution to Exercise 5

$C'(a) = -4w_u v/(\pi a^2) + 2w_p \lambda_p a / A_c = 0 \Rightarrow a^3 = 2w_u v A_c/(\pi w_p \lambda_p)$. At the optimum $w_p \lambda_p a^2 / A_c = 2w_u v/(\pi a)$, i.e. paging cost $= 2w_u v/(\pi a)$, while update cost $= 4w_u v/(\pi a)$: twice as much. Pedestrians: $a^{*3} = 2 \times 10 \times 5/(5\pi) = 6.37$, $a^* = 1.85\,\mathrm{km}$.

Solution to Exercise 6

Total 2×10^7 messages; capacity 10^5 messages/s. 100 %: $W \geq 200\,\mathrm{s}$; 70 %: $W \geq 2 \times 10^7/(7 \times 10^4) = 286\,\mathrm{s}$. Mechanisms: randomised back-off timers in NAS rejects (e.g. T3346), unified access control barring broadcast by the gNBs, AMF overload indications to the RAN, and prioritisation of emergency and high-priority access.

13 Further reading

The RRC protocol and states are specified in TS 38.331 [4] and summarised in TS 38.300 [3]; NAS in TS 24.501 [2]; NGAP in TS 38.413 [1]; the CM and RM states in TS 23.501 [7] and the procedures in TS 23.502 [5]. The latency targets come from TR 38.913 [6] and ITU-R M.2410 [9]. Rommer et al. [10] explain the 5G core signalling; Dahlman et al. [8] the RRC states and small-data transmission.

References

[1] 3GPP, *NG-RAN; NG application protocol (NGAP)*, TS 38.413, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38413.htm.

[2] 3GPP, *Non-access-stratum (NAS) protocol for 5G system (5GS); stage 3*, TS 24.501, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/24501.htm.

[3] 3GPP, *NR; NR and NG-RAN overall description; stage-2*, TS 38.300, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38300.htm.

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[6] 3GPP, *Study on scenarios and requirements for next generation access technologies*, TR 38.913, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/38913.htm.

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[8] E. Dahlman, S. Parkvall, and J. Sköld, *5G NR: The Next Generation Wireless Access Technology*, 2nd ed. Academic Press, 2020.

[9] ITU-R, "Minimum requirements related to technical performance for IMT-2020 radio interface(s)", International Telecommunication Union, Report M.2410-0, 2017.

[10] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.

25

Lecture 25

Network Evolution: a Grand Synthesis

تکامل شبکه: یک جمع‌بندی بزرگ

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 1–24

In this lecture

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-25-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Trace** the evolution of radio access, core architecture, signalling, security and QoS from 2G to 5G.
- **LO2** **Map** network functions across generations (MSC/VLR, MME, AMF; HLR, HSS, UDM; GGSN, P-GW, UPF).
- **LO3** **Decompose** capacity growth into spectrum, spectral efficiency and densification.
- **LO4** **Quantify** long-term trends in peak rate, latency and cost per bit with simple growth models.
- **LO5** **Identify** the design principles that recur in every generation and will shape the next one.

2 Prerequisite knowledge

This lecture draws on the whole course: GSM, GPRS and the 2G/3G core (Lectures 1–4), LTE and the EPC (Lectures 5–13), and the 5G system, SDN, NFV and SBA (Lectures 14–24). No new protocol is introduced; the aim is to connect what you already know into one picture.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Generation	نسل	A family of standards meeting a set of ITU requirements (IMT-2000, IMT-Advanced, IMT-2020).
Flat architecture	معماری تخت	Fewer nodes on the user path; radio control in the base station.
All-IP	تمام‌IP	Voice and data carried as IP packets, no circuit-switched domain.
CP/UP separation	جداسازی صفحهٔ کنترل و کاربر	Independent scaling and placement of control and user plane.
Spectral efficiency	بازدهی طیفی	Throughput per hertz per cell (bit/s/Hz/cell).
Densification	متراکم‌سازی	More cells per unit area.
Area traffic capacity	ظرفیت ترافیکی ناحیه	Total throughput per unit area (bit/s/m^2).
Cost per bit	هزینه به ازای بیت	Network cost divided by carried traffic.
Softwarisation	نرم‌افزاری‌شدن	Network functions as software on general-purpose infrastructure.
Backward compatibility	سازگاری با گذشته	Interworking of a new generation with installed ones.

3 Introduction and motivation

Each generation of mobile networks was designed to fix the bottleneck of the previous one: 2G digitised voice and added security and roaming; 3G brought packet data at useful rates; 4G removed the circuit domain and flattened the network; 5G made the network programmable and service-specific. Seen one at a time, the generations look like a sequence of unrelated protocol stacks. Seen together, they show a small set of persistent ideas – separation of concerns, mobility anchors, subscriber databases, tunnelling, mutual authentication – implemented with ever more general technology [3], [7]. Recognising these ideas is what lets an engineer understand a new system quickly, and predict where it will go.

4 Fundamental concepts

4.1 Five dimensions of evolution

- **Radio access**: FDMA/TDMA (GSM), CDMA (UMTS), OFDMA with MIMO (LTE), flexible-numerology OFDM with massive MIMO and mmWave (NR).
- **Core**: circuit switching with MSC/VLR and SS7/MAP; an overlaid packet core (SGSN/GGSN); the all-IP EPC with Diameter and GTP; the service-based 5GC on cloud infrastructure.
- **Mobility**: hierarchical location and routing areas, anchors in the GGSN, P-GW and UPF; handover decisions moving from the BSC and RNC into the base station.
- **Security**: one-way GSM authentication (A3/A8) to mutual UMTS AKA, EPS-AKA with key hierarchy, and 5G-AKA/EAP-AKA′ with concealed identities (SUCI) and home control.
- **QoS**: from circuit guarantees through PDP contexts and EPS bearers to per-flow QoS with standardised 5QIs and slicing.

4.2 Flattening the path

The user-plane path has steadily shrunk (Figure 1). In GPRS and UMTS a packet passed through a base station, a radio controller (BSC/PCU or RNC), an SGSN and a GGSN. LTE moved radio control into the eNB and removed the controller; the S-GW and P-GW remained. In 5G a single UPF can serve as the whole core user plane, and it can be placed close to the radio.

4.3 Functions that survive

Names change, functions persist. The visitor location and mobility function (MSC/VLR, SGSN, MME, AMF), the subscriber database (HLR/AuC, HSS, UDM/AUSF), the mobility anchor and gateway to data networks (GGSN, P-GW, UPF with SMF) and policy control (PCRF, PCF) appear in every generation, each time with cleaner interfaces and better separation between control and user plane.

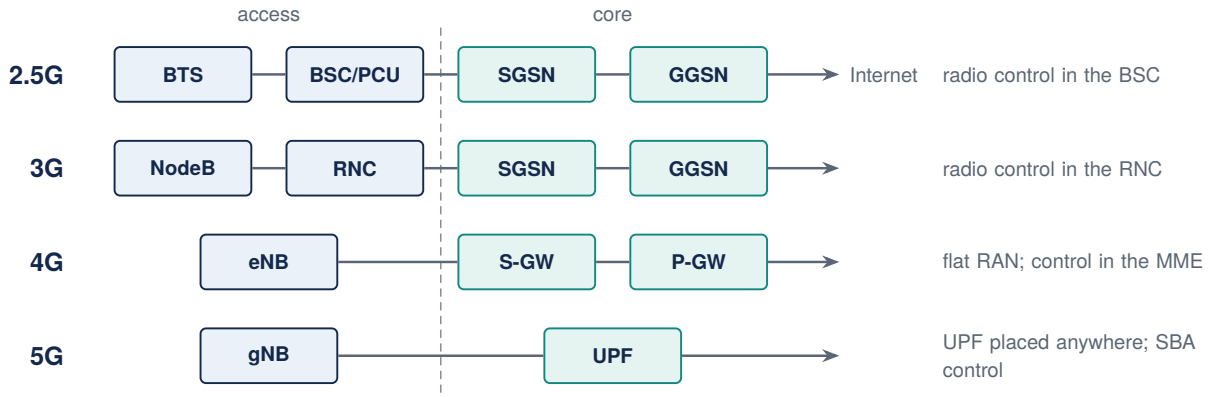


Figure 1. User-plane path of packet data from 2.5G to 5G (simplified).

5 Detailed technical explanation

5.1 From circuits to packets to services

2G was designed around a 64 kbit/s circuit per call. GPRS added a packet core in parallel, reusing the radio with dynamic time-slot allocation. UMTS kept two domains (CS and PS) with a common radio network. LTE dropped the CS domain: voice became an IMS service over VoLTE, with CS fallback and SRVCC for interworking. 5G keeps IMS for voice (VoNR) but restructures the core as services consumed over HTTP/2 – the network itself becomes a set of APIs.

5.2 Signalling protocols

SS7 and MAP connected the 2G/3G core; Diameter replaced MAP in the EPC (S6a, Gx, Rx); GTPv2-C carried session control; 5G uses REST-style service-based interfaces with OAuth-based authorisation, and PFCP for user-plane control. Each step moved towards protocols used in the wider IT industry, lowering development cost and enabling cloud deployment.

5.3 Security lessons

GSM authenticated only the subscriber, allowing false base stations; UMTS added network authentication and integrity protection; LTE separated keys per layer and per node; 5G encrypts the permanent identity (SUCI), gives the home network the final authentication decision and protects inter-operator signalling with the SEPP. Each fix answered an attack that became practical on the previous generation.

5.4 Coexistence

No generation replaces its predecessor at once. Networks run several in parallel for a decade or more, with interworking functions (N26 between AMF and MME, EN-DC in non-standalone 5G, CS fallback) and gradual spectrum refarming. Backward compatibility is a constant design constraint and a large part of the cost of every new system.

6 Mathematical models

6.1 Peak-rate growth

Model. An exponential trend $R(t) = R_0\, e^{k(t-t_0)}$ fitted through two points (t_0, R_0) and (t_1, R_1) has

$$k = \frac{\ln(R_1/R_0)}{t_1 - t_0}, \qquad T_2 = \frac{\ln 2}{k}. \tag{1}$$

Example. From GSM data (9.6 kbit/s, 1991) to the IMT-2020 requirement (20 Gbit/s, specified 2017–2018) [8]: $k = \ln(2.08 \times 10^6)/27 = 0.539$ per year, a factor 1.71 per year, doubling every 1.29 years (Figure 2).

Limitation. Peak rates are specification values under ideal conditions; user-experienced rates grow more slowly and depend on load and coverage.

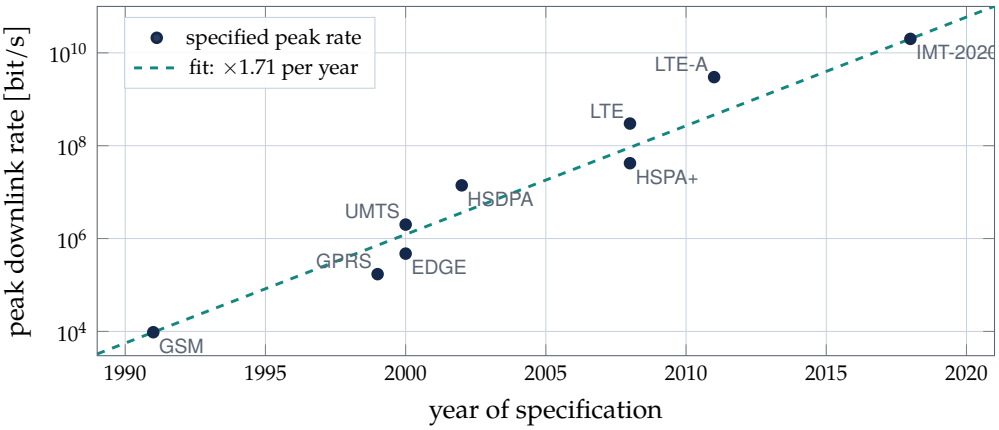


Figure 2. Specified downlink peak rates of 3GPP technologies and the exponential fit of Eq. (1).

6.2 Decomposing capacity

Model. The area traffic capacity is the product of cell density n (cells/km^2), bandwidth B and spectral efficiency η:

$$C_A = n\, B\, \eta, \tag{2}$$

so the growth between two deployments is the product of the three ratios.

Example. LTE baseline: $n = 4$, $B = 20$ MHz, $\eta = 1.5$ bit/s/Hz: $C_A = 120$ Mbit/s/km^2. 5G deployment: $n = 16$, $B = 100$ MHz, $\eta = 6$: $C_A = 9.6$ Gbit/s/km^2 – a factor $4 \times 5 \times 4 = 80$ (Figure 3). For comparison, IMT-2020 requires 10 Mbit/s/m^2 (10 Tbit/s/km^2) in the indoor hotspot scenario [8].

6.3 Latency along the path

Model. The one-way user-plane latency through h nodes is

$$T = \sum_{i=1}^{h} \left(t_{\mathrm{proc},i} + \frac{d_i}{v} \right), \tag{3}$$

with processing time $t_{\mathrm{proc},i}$, link length d_i and propagation speed $v \approx 2 \times 10^8$ m/s in fibre.

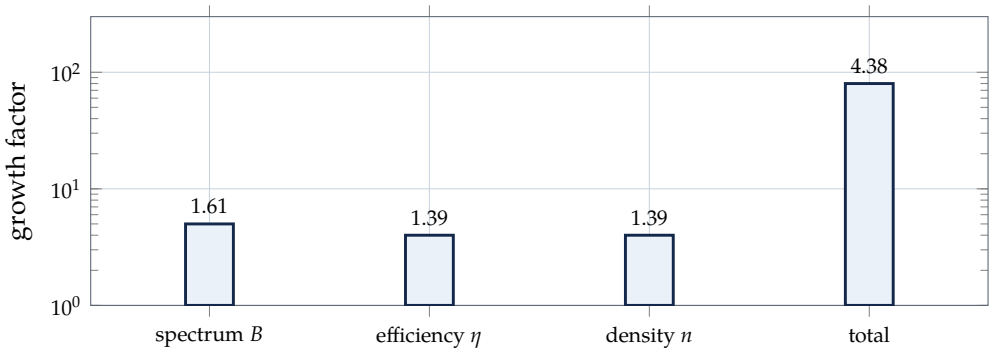


Figure 3. Area-capacity growth from the LTE baseline to the 5G deployment of Section 6.2.

Example. Four nodes with 0.5 ms each and 400 km of fibre (2 ms) give 4 ms; two nodes (gNB and edge UPF) with 0.25 ms each and 20 km give 0.6 ms. Flattening and edge placement together remove most of the core latency.

6.4 Cost per bit

Model. If traffic grows by a factor $1+g$ per year and revenue by $1+r$, the cost per bit must fall by $(1+r)/(1+g)$ per year to keep margins constant; it halves every

$$T_{1/2} = \frac{\ln 2}{\ln(1+g) - \ln(1+r)} \text{ years.} \tag{4}$$

Example. With $g = 30\,\%$ and flat revenue, $T_{1/2} = 2.64$ years. This pressure, more than peak rate, explains the industry's move to commodity hardware, automation and software-defined networks.

7 Network architecture and diagrams

Figure 1 shows the user path; the control side evolved in parallel from MSC/VLR and SGSN to the MME and then to the AMF and SMF of the service-based core. Table 1 maps the functions across generations.

Table 1. Corresponding functions across generations.

Function	2G/3G	4G (EPC)	5G (5GC)
Mobility and access control	MSC/VLR, SGSN	MME	AMF
Subscriber data and authentication	HLR/AuC	HSS	UDM, AUSF
Session control	SGSN, GGSN	MME, S-GW/P-GW control	SMF
User-plane anchor	GGSN	P-GW (S-GW)	UPF
Policy	– (static)	PCRF	PCF
Radio control	BSC, RNC	eNB	gNB (CU-CP)

8 Step-by-step procedures

Procedure 1 • Analysing a new mobile system

1. Identify the subscriber database, the mobility function, the session function and the user-plane anchor.
2. Draw the user path and count the nodes and tunnels (Eq. (3)).
3. Identify the authentication method and which identities travel in clear.
4. Identify the QoS unit (circuit, bearer, flow) and where it is enforced.
5. Locate the interworking with the previous generation.

Procedure 2 • Planning a capacity increase

1. Measure current traffic and forecast growth.
2. Compute the required factor and split it into spectrum, efficiency and density (Eq. (2)).
3. Price each lever (licences, new equipment, sites and backhaul).
4. Choose the cheapest combination and check the cost-per-bit target (Eq. (4)).

9 Worked examples

Worked example 1 • Densification needed

Problem. A hotspot must carry 10 Tbit/s/km^2 with $B = 400$ MHz and $\eta = 10$ bit/s/Hz. How many cells per km^2 are needed?
Solution. $n = 10^{13}/(4 \times 10^8 \times 10) = 2500$ cells/km^2, one cell per 400 m^2 – an inter-site distance of about 20 m, i.e. indoor small cells.

Worked example 2 • Extrapolating the trend

Problem. If the fit of Section 6.1 continued, when would peak rates reach 1 Tbit/s, a figure mentioned in 6G visions [10]?
Solution. $\ln(10^{12}/2 \times 10^{10})/0.539 = 7.3$ years after 2018, i.e. around 2025. That has not happened, showing that the trend is not a law: further growth needs wider bandwidths that exist only above 100 GHz (Lecture 26).

Worked example 3 • Cost pressure

Problem. Traffic grows 25 % per year and revenue 2 %. By how much must cost per bit fall each year, and when does it halve?
Solution. $1.02/1.25 = 0.816$: an 18.4 % fall per year; $T_{1/2} = \ln 2/(\ln 1.25 - \ln 1.02) = 3.41$ years.

10 Technology comparison

Table 2. Generations at a glance.

	2G (GSM)	3G (UMTS)	4G (LTE)	5G (NR)
Multiple access	TDMA/FDMA	WCDMA	OFDMA / SC-FDMA	OFDMA, flexible numerology
Core	CS (+ GPRS)	CS + PS	All-IP EPC	Service-based 5GC
Signalling	SS7/MAP	SS7/MAP	Diameter, GTPv2-C	HTTP/2 SBI, PFCP
Authentication	One-way	Mutual AKA	EPS-AKA	5G-AKA, EAP-AKA′, SUCI
QoS unit	Circuit	PDP context	EPS bearer	QoS flow, slice
Voice	CS	CS	VoLTE (IMS)	VoNR (IMS)

Same ideas, better tools. Every generation keeps the same functional roles but rebuilds them with more general technology: from dedicated switches to IP routers to cloud software.

11 Key takeaways

Key takeaways

- Functions persist across generations; their names, protocols and placement change.
- The user path flattened from four nodes (GPRS/UMTS) to one UPF in 5G.
- Capacity $C_A = nB\eta$: growth comes from spectrum, efficiency and densification together.
- Peak rate grew about 1.7 times per year from 1991 to 2018, but trends are not laws.
- Cost per bit must halve every $\ln 2 / (\ln(1+g) - \ln(1+r))$ years – the driver of softwarisation.

12 Exercises

Exercise 1 • Level 1 – conceptual • Mapping

Name the 4G and 5G counterparts of: (a) VLR; (b) HLR; (c) GGSN; (d) RNC.

Exercise 2 • Level 1 – conceptual • Why packets?

Give two reasons why mobile networks moved from circuit to packet switching for all traffic.

Exercise 3 • Level 2 – numerical • Growth rate

LTE Rel-8 offered 300 Mbit/s (2008) and LTE-Advanced 3 Gbit/s (2011). Compute k and the doubling time for this interval and compare with Section 6.1.

Exercise 4 • Level 2 – numerical • Capacity levers

An operator needs 20 times more area capacity. Spectrum can grow by 2.5 and efficiency by 2. What densification is needed?

Exercise 5 • Level 3 – analytical • Path latency

Compare with Eq. (3): (a) UMTS path with NodeB, RNC, SGSN and GGSN (1 ms each, 300 km fibre); (b) 5G with gNB and a UPF 30 km away (0.3 ms each).

Exercise 6 • Level 3 – analytical • The next bottleneck

Using the five dimensions of evolution, argue which bottleneck of 5G a sixth generation is most likely to address, with a quantitative argument.

Solutions

Solution to Exercise 1

(a) MME / AMF; (b) HSS / UDM (with AUSF); (c) P-GW / UPF with SMF; (d) none as a separate node – eNB / gNB (CU-CP).

Solution to Exercise 2

Statistical multiplexing uses capacity far better for bursty traffic, and a single IP network for voice and data lowers cost and allows Internet services to be offered directly.

Solution to Exercise 3

$k = \ln 10/3 = 0.768$ per year, $T_2 = 0.90$ years – faster than the long-term 1.29 years, because carrier aggregation and MIMO added bandwidth and layers in a single step.

Solution to Exercise 4

$20/(2.5 \times 2) = 4$: four times as many cells per unit area.

Solution to Exercise 5

(a) $4 \times 1 + 300\,\text{km}/(2 \times 10^5\,\text{km/s}) = 4 + 1.5 = 5.5\,\text{ms}$. (b) $2 \times 0.3 + 0.15 = 0.75\,\text{ms}$.

Solution to Exercise 6

Example argument: the peak-rate trend stalled because bandwidth below 100 GHz is exhausted; $C = B \log_2(1 + \text{SNR})$ grows linearly in B but only logarithmically in SNR, so the next tenfold step needs tens of gigahertz of spectrum (sub-THz), plus new ways to reduce cost per bit (AI-driven automation) and to cover the unconnected (non-terrestrial networks).

13 Further reading

ITU-R M.2083 [7] and M.2410 [8] define the IMT-2020 vision and requirements; TS 23.002 [2], TS 23.401 [1] and TS 23.501 [3] describe the architectures of the successive generations. Eberspächer et al. [5], Holma and Toskala [6], Sesia et al. [11] and Dahlman et al. [4] cover GSM, UMTS, LTE and NR; Rommer et al. [9] cover the 5G core.

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[1] 3GPP, *General packet radio service (GPRS) enhancements for evolved universal terrestrial radio access network (E-UTRAN) access*, TS 23.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23401.htm.

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26

Lecture 26

Future 6G Technologies

فناوری‌های آیندهٔ نسل ششم

2G — 3G — 4G — 5G — **6G**

COURSE PART
Part V – Planes and outlook (Lectures 23–26)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Advanced

PREREQUISITES
Lectures 14, 23 and 25; Shannon capacity and free-space propagation

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 5
7 Network architecture and diagrams . . . 7
8 Step-by-step procedures . . . 8
9 Worked examples . . . 8
10 Technology comparison . . . 9
11 Key takeaways . . . 9
12 Exercises . . . 9
13 Further reading . . . 11

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-26-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the IMT-2030 framework: its usage scenarios and overarching aspects.
- **LO2** **Explain** the main candidate technologies: sub-THz spectrum, cell-free massive MIMO, reconfigurable intelligent surfaces, integrated sensing and communication, non-terrestrial networks and AI-native design.
- **LO3** **Compute** free-space path loss at high frequencies and the capacity gain of wider bandwidth under a power limit.
- **LO4** **Estimate** RIS gain, sensing resolution and satellite propagation delay.
- **LO5** **Assess** which claims about 6G are established and which are still research.

2 Prerequisite knowledge

You need the 5G requirements and NR design (Lecture 14), the user-plane budgets of Lecture 23 and the evolution trends of Lecture 25, as well as the Shannon capacity formula and the free-space propagation (Friis) equation.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
IMT-2030	IMT-2030	ITU framework for the sixth generation (Recommendation M.2160).
Sub-THz	زیرتراهرتز	Carrier frequencies of roughly 100–300 GHz.
Cell-free massive MIMO	انبوه بدون سلول MIMO	Many distributed access points jointly serving all users.
RIS	سطح هوشمند بازپیکربندی‌پذیر	Surface of passive elements with controllable phase that shapes reflections.
ISAC	حسگری و ارتباط یکپارچه	Using the same signals and hardware for radar sensing and communication.
NTN	شبکهٔ غیرزمینی	Satellites and high-altitude platforms as part of the mobile network.
AI-native	هوش‌مصنوعی‌بنیاد	Designed with learned components and data pipelines from the start.
Range resolution	تفکیک‌پذیری فاصله	Smallest distance difference between two distinguishable targets.
Power-limited regime	رژیم محدود به توان	Where more bandwidth no longer increases capacity much.
Energy per bit	انرژی به ازای بیت	Consumed energy divided by delivered bits (J/bit).

3 Introduction and motivation

Each generation has been defined about ten years after the previous one, and 6G follows the same rhythm: the ITU published the IMT-2030 framework in 2023 [2], and 3GPP began 6G study work in Release 20, with normative specifications expected in later releases. As Lecture 25 showed, the next steps in capacity need much wider bandwidths, the cost per bit must keep falling, and large parts of the world are still unconnected. At the same time, the network is expected to do more than move bits: sense its environment, host AI and be sustainable and resilient [6].

Much of 6G is still research. This lecture therefore separates the agreed framework from candidate technologies, and gives simple physical models that let you judge claims about them yourself.

4 Fundamental concepts

4.1 The IMT-2030 framework

Recommendation ITU-R M.2160 [2] defines six usage scenarios (Figure 1). Three extend the 5G triangle: *immersive communication* (from eMBB), *hyper-reliable and low-latency communication* (HRLLC, from URLLC) and *massive communication* (from mMTC). Three are new: *ubiquitous connectivity*, *AI*

and communication and *integrated sensing and communication*. Four overarching aspects apply to all: sustainability, connecting the unconnected, ubiquitous intelligence, and security and resilience. Quantitative requirements are defined in a later ITU step, so numbers such as "1 Tbit/s" found in the literature are research targets, not requirements.

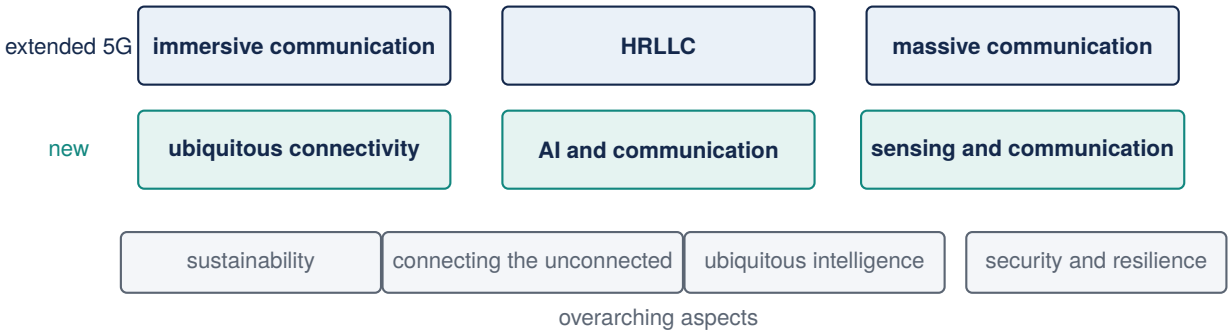


Figure 1. IMT-2030 usage scenarios and overarching aspects (after ITU-R M.2160).

4.2 Candidate technologies

- **New spectrum**: upper mid-band (around 7–24 GHz) for wide-area capacity and sub-THz bands for very high rates at short range [5].
- **Cell-free massive MIMO**: many access points coherently serving users without cell boundaries, giving uniform service [4].
- **Reconfigurable intelligent surfaces**: controllable reflectors that create paths around obstacles at low power [7].
- **Integrated sensing and communication**: the radio as a radar for positioning, mapping and gesture or object detection [3].
- **Non-terrestrial networks**: LEO satellites and high-altitude platforms integrated with the terrestrial system [1].
- **AI-native air interface and network**: learned receivers, beam management and automated operation.

5 Detailed technical explanation

5.1 Sub-THz communication

Above 100 GHz contiguous bandwidths of tens of gigahertz are available, but path loss grows with frequency for antennas of fixed gain, molecular absorption appears at specific frequencies, and blockage by walls, bodies and even foliage is severe. Very large arrays with narrow beams recover the loss (Section 6.1), which makes beam search and tracking the central problem. Hardware is also hard: power amplifiers are inefficient, phase noise grows with frequency, and sampling tens of gigahertz needs low-resolution converters.

5.2 Cell-free networks

In a cellular network users at cell edges suffer from interference and weak signals. In a cell-free network a central processing unit coordinates many distributed access points that all serve all users, with local precoding and fronthaul to the central unit. The result is more uniform rates; the costs are fronthaul and synchronisation – an extension of the CU/DU split and coordination ideas of Lecture 23.

5.3 Reconfigurable intelligent surfaces

An RIS is a panel of many sub-wavelength elements whose reflection phase can be set electronically. Properly configured, it adds a strong reflected path, for example around a blocking building. Its received power grows with the square of the number of elements (Section 6.3), but it suffers the product of two path losses (transmitter to surface, surface to receiver), so it helps most near the transmitter or the receiver.

5.4 Integrated sensing and communication

Wide bandwidths and large arrays are exactly what radars need. ISAC uses communication waveforms or shared hardware to measure range, angle and velocity of objects. Applications include positioning without dedicated infrastructure, traffic monitoring and environment mapping that in turn helps beam management. It also raises privacy questions, since the network could sense people who carry no device.

5.5 Non-terrestrial networks

Release 17 introduced NR over satellite (NTN) with adaptations for large delays and Doppler shifts. LEO constellations offer global coverage with tens of milliseconds of round-trip delay (Section 6.5); integrated with terrestrial networks they address "connecting the unconnected" and resilience after disasters.

5.6 AI-native networks

5G already uses data analytics (NWDAF) and AI in RAN optimisation. 6G aims to make learned components part of the design: AI-based channel state feedback, receivers and beam prediction, and network automation with closed loops. Open questions are standardisation of models, generalisation to unseen conditions, energy cost of the AI itself and explainability.

6 Mathematical models

6.1 Free-space path loss

Model. Between isotropic antennas at distance d and wavelength $\lambda = c/f$:

$$L_{\mathrm{FS}} = 20\log_{10}\left(\frac{4\pi d f}{c}\right)\ \mathrm{dB}. \tag{1}$$

With effective apertures A_t, A_r the Friis equation becomes $P_r = P_t A_t A_r/(\lambda^2 d^2)$, which *grows* with f^2.

Example. At 100 m: 83.3 dB at 3.5 GHz, 101.4 dB at 28 GHz and 115.4 dB at 140 GHz (Figure 2) – 32 dB more at 140 GHz than at 3.5 GHz. Arrays of the same physical size at both ends gain $20\log_{10}(140/3.5) = 32$ dB each, so the link budget can actually improve, at the price of very narrow beams.

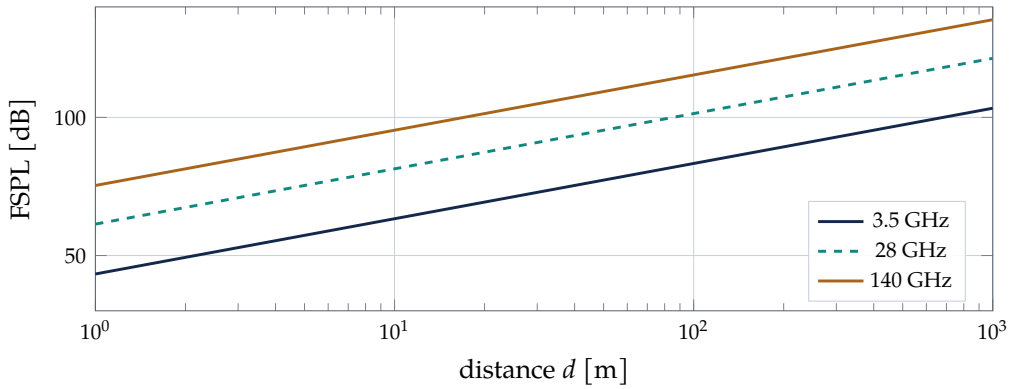


Figure 2. Free-space path loss (Eq. (1)).

6.2 Bandwidth under a power limit

Model. With received power P and noise density N_0:

$$C(B) = B\log_2\left(1+\frac{P}{N_0 B}\right) \xrightarrow{B\to\infty} \frac{P}{N_0 \ln 2}. \tag{2}$$

Example. If the SNR is 20 dB over 1 GHz: $C = 6.66$ Gbit/s; with 10 GHz, 34.6 Gbit/s; with 100 GHz, 100 Gbit/s; the limit is 144 Gbit/s (Figure 3). Wider bandwidth helps greatly only while the SNR stays high – hence the need for array gain.

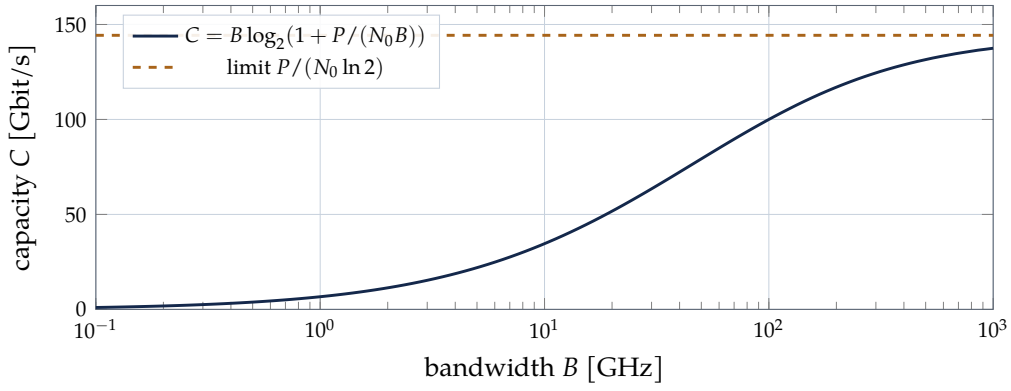


Figure 3. Capacity versus bandwidth at fixed received power (Eq. (2)).

6.3 RIS gain

Model. With N elements whose phases align the reflected signals at the receiver, the amplitudes add coherently:

$$P_r \propto N^2, \qquad G_{\text{RIS}} = 20\log_{10} N \text{ dB} \tag{3}$$

relative to a single element.

Example. $N = 1024$: 60.2 dB over one element (random phases would give only $10\log_{10} N = 30.1$ dB). The gain must overcome the product of the two path losses.

6.4 Sensing resolution

Model. A signal of bandwidth B separates two targets in range if

$$\Delta r = \frac{c}{2B}. \tag{4}$$

Example. $B = 1$ GHz: 15 cm; $B = 10$ GHz: 1.5 cm. Sub-THz bandwidths thus allow centimetre-level sensing.

6.5 Satellite delay

Model. For a satellite at altitude h, Earth radius R and elevation angle ε, the slant range is

$$d(\varepsilon) = \sqrt{(R+h)^2 - R^2\cos^2\varepsilon} - R\sin\varepsilon, \tag{5}$$

and the one-way delay d/c.

Example. $h = 600$ km: 600 km and 2.0 ms at zenith, 1076 km and 3.6 ms at 30°, 1932 km and 6.4 ms at 10° (Figure 4). A geostationary satellite (35 786 km) needs at least 119 ms.

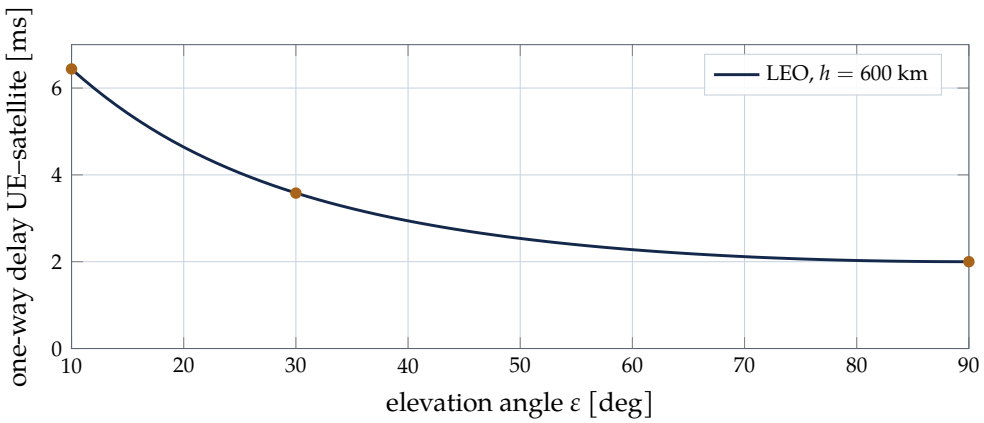


Figure 4. One-way propagation delay to a LEO satellite (Eq. (5)).

7 Network architecture and diagrams

A plausible 6G system keeps the service-based core of 5G (Lecture 18) and extends it with sensing and AI functions, integrates satellite and terrestrial access under one core, and deploys a RAN that mixes macro cells in the upper mid-band, dense sub-THz spots, cell-free clusters and RIS-assisted coverage. Figure 1 gives the service view; Figures 2 to 4 quantify the physical constraints.

8 Step-by-step procedures

Procedure 1 • Checking a sub-THz link budget

1. Compute the free-space loss at the carrier and distance (Eq. (1)); add absorption and blockage margins.
2. Add transmit and receive array gains from the array sizes.
3. Compute the noise power $N_0 B$ including the receiver noise figure, and the SNR.
4. Compute the capacity (Eq. (2)) and compare with the target rate.

Procedure 2 • Judging a 6G claim

1. Is it a requirement in an ITU or 3GPP document or a research target?
2. Which physical quantity limits it (bandwidth, SNR, delay, energy)? Check it with a model of this lecture.
3. What does it cost (spectrum, sites, energy, complexity)?
4. Which usage scenario of Figure 1 does it serve?

9 Worked examples

Worked example 1 • Short-range sub-THz link

Problem. Compute the free-space loss at 300 GHz over 10 m.
Solution. $20\log_{10}(4\pi \times 10 \times 3 \times 10^{11}/3 \times 10^{8}) = 102.0$ dB – about the same as 28 GHz over 100 m, which is why sub-THz suits short ranges.

Worked example 2 • More bandwidth

Problem. With the power of Section 6.2, what capacity does 50 GHz give?
Solution. SNR $= 100/50 = 2$; $C = 50\log_2 3 = 79.2$ Gbit/s: 50 times the bandwidth of the 1 GHz case gives only 12 times the capacity.

Worked example 3 • Satellite round trip

Problem. A UE and a gateway both see a 600 km LEO satellite at 30°. What is the round-trip propagation delay through the satellite (bent pipe)?
Solution. One way UE–satellite–gateway: $2 \times 3.58 = 7.2$ ms; round trip 14.3 ms – compatible with interactive services, unlike GEO (over 477 ms).

10 Technology comparison

Table 1. 5G and candidate 6G features.

	5G (IMT-2020)	6G (IMT-2030, candidates)
Usage scenarios	eMBB, URLLC, mMTC	Immersive, HRLLC, massive, ubiquitous, AI, sensing
Spectrum	Below 7 GHz and 24–71 GHz	+ upper mid-band and sub-THz
Radio architecture	Cells with massive MIMO	Cells + cell-free, RIS
Sensing	Positioning	Integrated sensing and communication
Non-terrestrial	NTN from Release 17	Integrated by design
AI	Analytics (NWDAF), RAN optimisation	AI-native air interface and operation
Status	Deployed	Research and early standardisation

Evolution, not revolution. As every generation before it, 6G will reuse the persistent functions of Lecture 25 and add new capabilities where physics and economics allow.

11 Key takeaways

Key takeaways

- IMT-2030 defines six usage scenarios and four overarching aspects; numerical requirements come later.
- Free-space loss grows as $20\log_{10} f$, but fixed-aperture arrays recover it: beams, not power, are the problem.
- Bandwidth helps only while SNR is high: $C \to P/(N_0 \ln 2)$.
- RIS gain scales as N^2; sensing resolution as $c/(2B)$; LEO delay is a few milliseconds.
- Separate requirements from research targets when reading about 6G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Scenarios

Name the three IMT-2030 usage scenarios that extend 5G and the three new ones.

Exercise 2 • Level 1 – conceptual • Blockage

Give two reasons why sub-THz links are more sensitive to blockage than links at 3.5 GHz, and one mitigation.

Exercise 3 • Level 2 – numerical • Radar resolution

Compute the range resolution of a 400 MHz NR carrier used for sensing.

Exercise 4 • Level 2 – numerical • RIS size

A 16-element RIS gives too weak a reflection. How many elements are needed for 20 dB more received power?

Exercise 5 • Level 3 – analytical • GEO versus LEO

Compute the minimum round-trip delay UE–satellite–gateway for a GEO satellite at 35 786 km and compare with Example 3. Which services are feasible?

Exercise 6 • Level 3 – analytical • Energy

Traffic grows 30 % per year while the network energy budget stays constant. By how much must energy per bit improve per year and over ten years?

Solutions

Solution to Exercise 1

Extended: immersive communication, HRLLC, massive communication. New: ubiquitous connectivity, AI and communication, integrated sensing and communication.

Solution to Exercise 2

Shorter wavelength means less diffraction around obstacles and higher penetration loss through materials; narrow beams are easily blocked. Mitigations: multi-connectivity to several access points, RIS-assisted paths, fast beam switching.

Solution to Exercise 3

$\Delta r = 3 \times 10^8 / (2 \times 4 \times 10^8) = 0.375\,\text{m}$.

Solution to Exercise 4

$(N/16)^2 = 100 \Rightarrow N = 160$ elements.

Solution to Exercise 5

Each hop at least 35 786 km (119.3 ms): one way 238.6 ms, round trip 477 ms, about 33 times the LEO value. Voice is degraded and interactive or real-time control services are not feasible; broadcasting and bulk data work well.

Solution to Exercise 6

Energy per bit must fall by a factor 1.3 each year (23 %), i.e. $1.3^{10} = 13.8$ times in ten years – hence the emphasis on sleep modes, efficient hardware and AI-driven energy saving.

13 Further reading

The IMT-2030 framework is Recommendation ITU-R M.2160 [2]. Saad et al. [6] present a 6G vision; Rappaport et al. [5] discuss frequencies above 100 GHz; Ngo et al. [4] introduce cell-free massive MIMO; Wu and Zhang [7] analyse intelligent reflecting surfaces; Liu et al. [3] survey ISAC; TR 38.811 [1] studies NR for non-terrestrial networks.

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